INTRODUCTION TO ELECTRICAL DISCHARGES IN GASES

WILEY SERIES IN PLASMA PHYSICS

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PREFACE

This book is written as a text for a one-semester introductory course in gas discharge physics at the advanced undergraduate or early graduate level. It makes no attempt to be complete but rather attempts to survey the areas of physics involved and to illustrate the types of problems and techniques used in this branch of physics. The approach is one of trying to explain the physical behavior of the electrons and ions in the ionized state rather than introduce mathematical completeness or sophistication into the discussion. This is, of course, not to de-emphasize the value of a proper mathematical approach but rather to lay the phenomenological foundations for a subsequent course in which the proper distribution function approach is dealt with in detail after the behavior of the average particles is understood physically.

As is true with most books that do not pretend to cover a field exhaustively, the book is somewhat biased in the direction of the author's own research contributions. Thus many of the examples used in the text are drawn from the interaction of high-frequency radiowaves and the ionized medium. On the other hand, these treatments are of a general enough nature so as to provide broad illustrations of the physical mechanisms involved.

Although this is designed as a textbook, no problems have been included.

As this material has been used in hectographic form for a number of years, it has been found that many more interesting problems can be drawn from the current periodical literature germane to classroom discussion, than is ever available in sets of printed questions at the end of every chapter Since this field is a very active one in present-day periodical literature, both teacher and student interest is greatly enhanced by this approach

Two further characteristics of the manner in which this material is presented grow directly out of the experience gained in using this material as a graduate text Experimental procedures and instrumental details are included only insofar as an understanding of the procedure is outlined Reliance is purposely placed on the student to go to the original sources to fill in the details of interest Furthermore, the lack of inclusion of all the details of mathematical derivations is arranged on purpose so that the student may think through the physics of many of the concepts by being assigned problems to fill in the intermediate steps for himself

I owe a debt of gratitude to the many graduate students who have helped me, through the years, to put this material into publishable form Professor Earl W McDaniel undertook a most careful and critical reading of the finished manuscript, and I am most grateful to him for his many suggestions for improvement of style and clarity My special thanks go to my secretary, Miss Doris McEntee, whose patience and good humor through many versions of the manuscript attest not only to her skill but to her very real contribution to this work

SANBORN C BROWN

Cambridge, Massachusetts
December 1965

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INTRODUCTION TO

ELECTRICAL DISCHARGES

IN GASES

Elastic Collision Phenomena

The phenomena associated with collision mechanisms govern a large fraction of the physics of the motions of charged particles in a gas. The simplest way to understand a collision would be to visualize the colliding particles as rigid spheres of radius r. Thus an electron coming close to an atom would either hit it or miss it and the collision cross section would be

$$Q_c = \pi r_c^2$$

where the collision radius r_c would be the sum of the radius of the atom and the radius of the electron $r_c = r_a + r_e$. The order of magnitude of r_c is an angstrom.

Probability of Elastic Collision for Electrons with Gas Atoms

Applying the kinetic theory model to the collision between electrons and atoms gives a much truer representation of reality than applying the rigid sphere model. Let us assume that a monoenergetic electron beam of cross-sectional area A passes through a gas at distance dx. The number density of the beam is n electrons per cubic centimeter, and each electron is traveling with velocity v. Let N be the number of atoms per cubic centimeter. Then the number of particles colliding per unit time is the probability of collision in a unit volume multiplied by the current density of incoming electrons.

$$\frac{dn}{dt} = -\left(\frac{NQ_cA\,dx}{A\,dx}\right)nv \tag{1.1}$$

$$\frac{dn}{n} = -NQ_cv\,dt. \tag{1.2}$$

This is true, of course, only when N is below the critical density where target atoms begin to obscure each other It is customary to define a probability of collision, P_c, as the average number of collisions which an electron makes in traveling 1 centimeter through a gas at 1 torr pressure at 0°C Thus

$$NQ_c = p_0 P_c \tag{1 3}$$

where the reduced pressure, p_0 is given by $p_0 = 273\,p/T$ Making this substitution as well as $v\,dt = dx$, we find

$$\frac{dn}{n} = -p_0 P_c\,dx \tag{1 4}$$

$$n = n_0 e^{-p_0 P_c x} \tag{1 5}$$

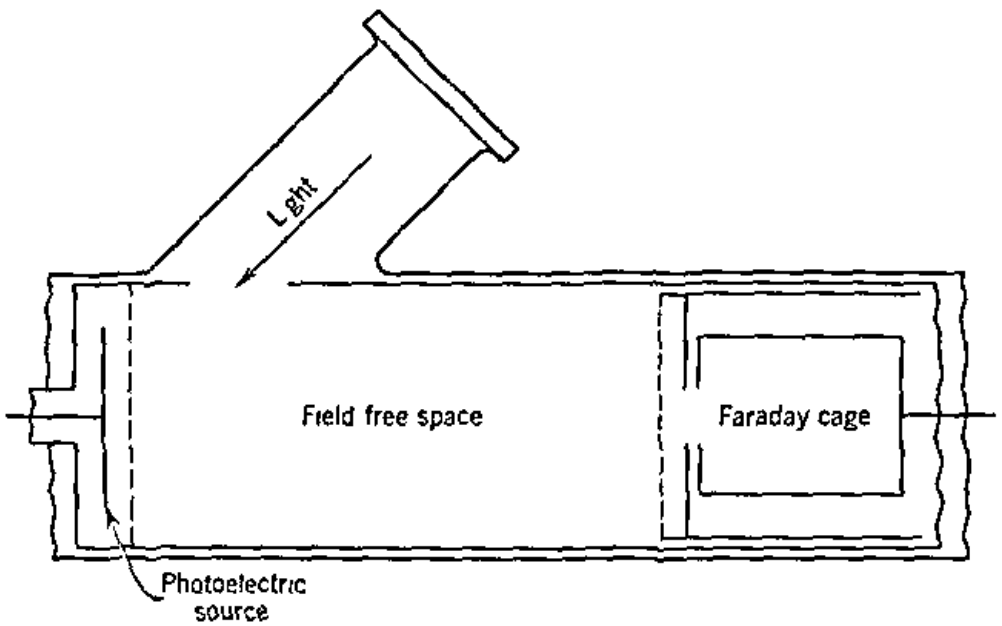


Fig 1 1 Sketch of Lenard s apparatus for measuring collision cross sections

The first measurements of the electron collision cross sections were carried out by Lenard[1] in 1903 These consisted of a series of experiments on the attenuation of a beam of monoenergetic electrons passing through a gas A sketch of his method is shown in Fig 1 1 An electron beam, emitted photoelectrically from a flat plate, passed through an accelerating screen into a field-free space where collisions took place The beam was then collected in a Faraday cage The difference in current measured as the beam entered and left the field free space arose because the electrons *scattered outward as a result of collisions with the gas atoms* Lenard called the number of collisions that occurred when an electron traveled a given distance at a standard pressure the "absorbing power," μ This is

an unfortunate term because true absorption (attachment or recombination) is a very rare event compared to the usual process of collision that results in the electrons being scattered out of the beam. On the assumption that the range of electrons is proportional to the electron energy, Lenard found confirmation in his data of the general law

$$I = I_0 e^{-\mu x}. \tag{1.6}$$

His curves for air and argon are given in Fig. 1.2.

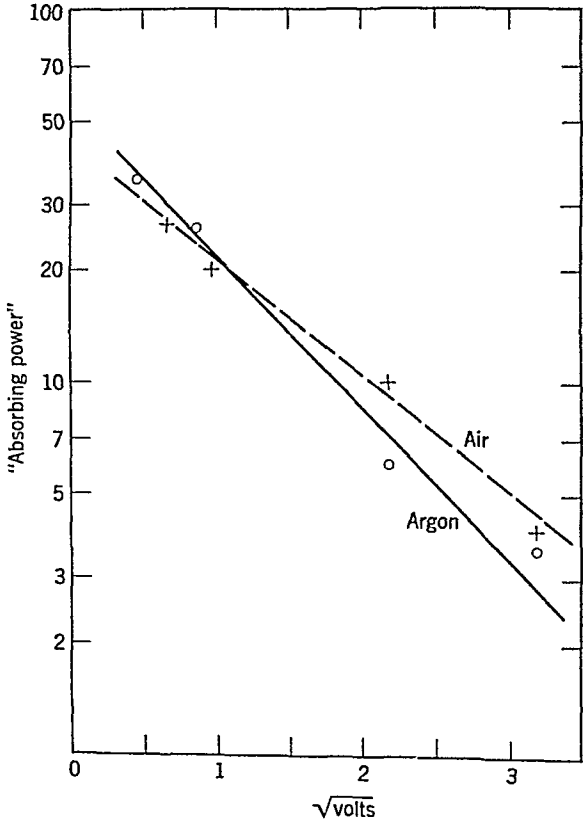


Fig. 1.2 Lenard's measurements of "absorbing power" of electrons in air and argon.

The nomenclature used for collision phenomena is not standardized. In very common use is the quantity ν_c, the collision frequency. If in Eq. (1.5) we substitute $x = vt$, we obtain

$$n = n_0 e^{-\nu_c t} \tag{1.7}$$

where

$$\nu_c = p_0 P_c v. \tag{1.8}$$

The reciprocal of ν_c is often used under the name of collision time, $\tau_c = 1/\nu_c$

The effective collision cross section for a single atom, Q, is defined as the probability of collision divided by the number of atoms per cubic centimeter at 1 torr pressure and 0°C

$$Q = 0\,283 \times 10^{-16} P_c$$

Since P_c has the units of $L^2L^{-3}p^{-1}$, that is, an area per unit volume per unit pressure, P_c is sometimes called the effective collision cross section, Q_c

Cross sections are sometimes given in terms of πa_0^2 where a_0 is the radius of the first Bohr orbit of the hydrogen atom

$$\pi a_0^2 = 8\,806 \times 10^{-17}\ \text{cm}^2$$

Another very common collision parameter, particularly in kinetic theory, is the mean free path, l_c, or the average distance a particle travels between collisions If n particles are moving through a gas in the x direction, in the distance dx there will be $n\,dx/l_c$ collisions If each collision removes an electron from the group, then the number lost in moving dx is

$$dn = -\frac{n\,dx}{l_c}$$

$$n = n_0 e^{-x/l_c} \tag{1 9}$$

The distribution of free paths is given in Fig 1 3 It shows that the number of free paths greater than l_c drops off very rapidly For example, only one particle in about 150 will have path lengths equal to $5l_c$

The relation between l_c and P_c is

$$l_c = \frac{1}{p_0 P_c} \tag{1 10}$$

In terms of the mean free path, the collision time, or as it is sometimes called, the mean free time, τ, can be written

$$\frac{1}{\tau} = \frac{v}{l_c} = 5\,93 \times 10^7 u^{1/2} p_0 P_c$$

where $u = mv^2/2e$ in volts

The difficulty with Lenard's simple apparatus was that electrons which suffered energy changes on collision were not necessarily removed from the beam Ramsauer[2] eliminated this difficulty by introducing a magnetic analyzer His apparatus is shown schematically in Fig 1 4 A photoelectric source emits electrons that are bent in a magnetic field A monoenergetic beam of electrons is thus focused on the entrance slit of a field-free region

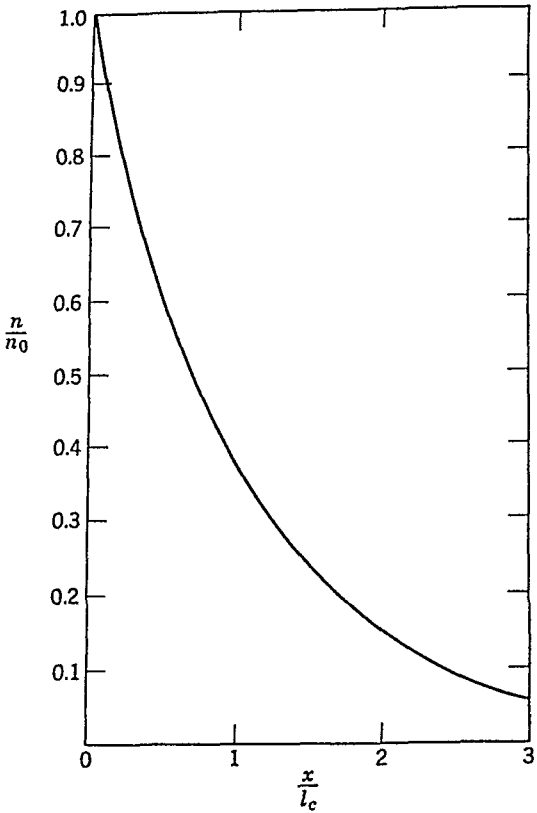


Fig. 1.3 Distribution of free paths.

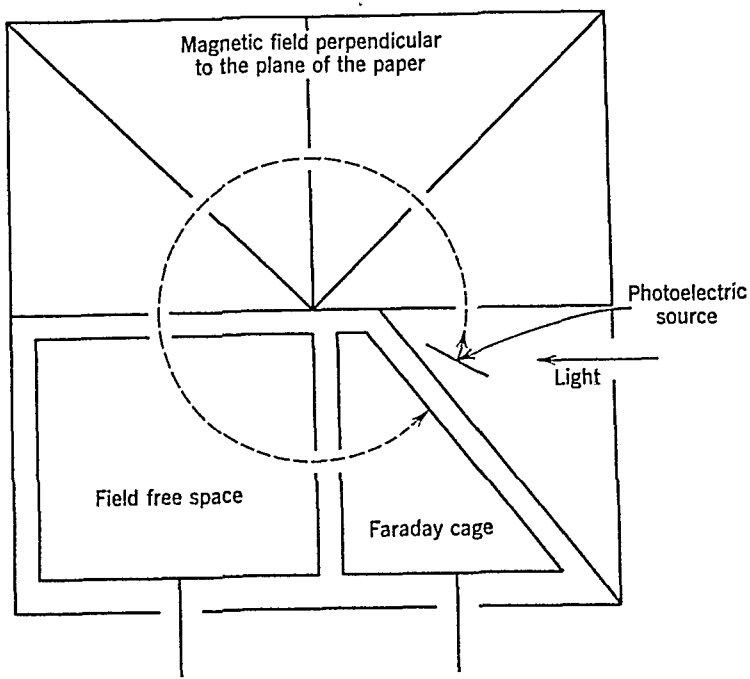


Fig. 1.4 A sketch of Ramsauer's magnetic analyzer.

where the collisions take place The electrons that remain in the beam are measured in a Faraday cage In this type of experiment a "total ' collision probability is measured Electrons are lost from the beam if they make any angular deflection greater than the angular aperture of the detector

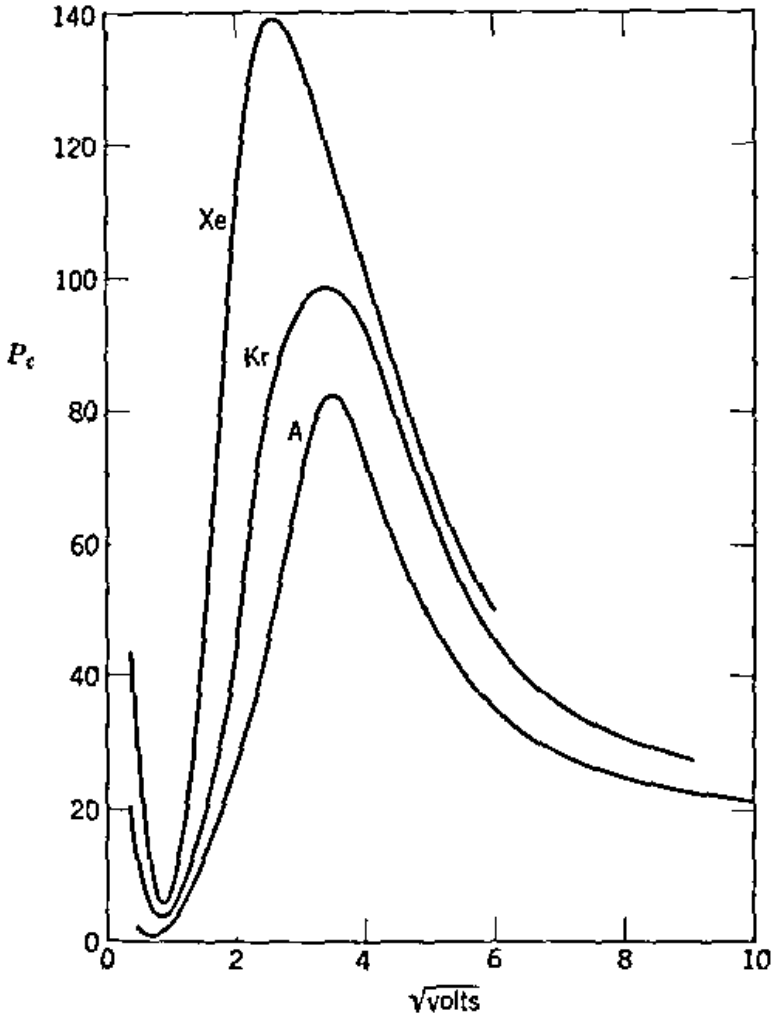


Fig 1.5a Typical elastic collision probabilities in atoms and molecules of similar electronic structure (From Ramsauer and Kollath [4])

Typical experimental results[3] for the noble gases are shown in Fig 1 5a At energies greater than a few volts, P_c decreases (as the electron velocity increases) inversely proportional to the ionization potential and directly proportional to the gas polarizability The scattering is controlled by the interaction of the free electron with the potential field of external electrons around the atom The magnitude and distortion of this field should, as observed, control the collision cross section This is particularly well illustrated in molecules of similar electronic structures, as shown in Fig 1 5b

The very low collision cross section at low energies was first observed by Lenard, but his observations were largely discounted as probably erroneous. Subsequently, Ramsauer studied this phenomenon in great detail. As a result, the great transparency of certain gases to low-energy

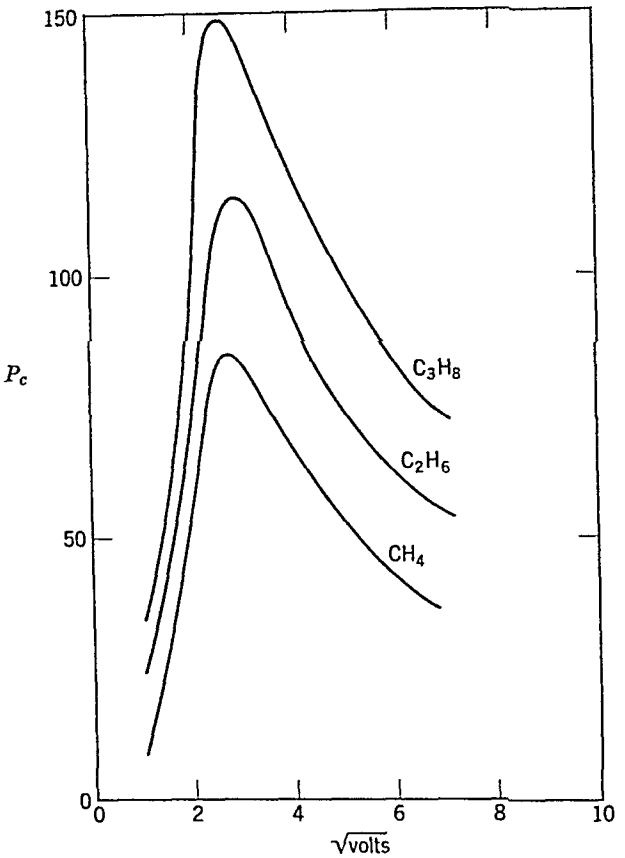


Fig. 1.5b

electrons is usually called the "Ramsauer effect." It was Allis and Morse[4] who first calculated this effect wave-mechanically to show that it results from the diffraction of the electron wave around the potential field of the target atom.

Let us now derive these cross sections in a more quantitative manner. Consider the trajectory of particles in a central force field colliding in a manner sketched in Fig. 1.6. Let particle 1 be an electron making an elastic collision with a neutral gas atom 2, and consider the collision with respect to the center of mass 0. About 0 the angular momentum

is always constant:

$$L = m_r r^2 \frac{d\alpha}{dt} \tag{1.11}$$

where m_r is the reduced mass

$$m_r = \frac{m_1 m_2}{m_1 + m_2}$$

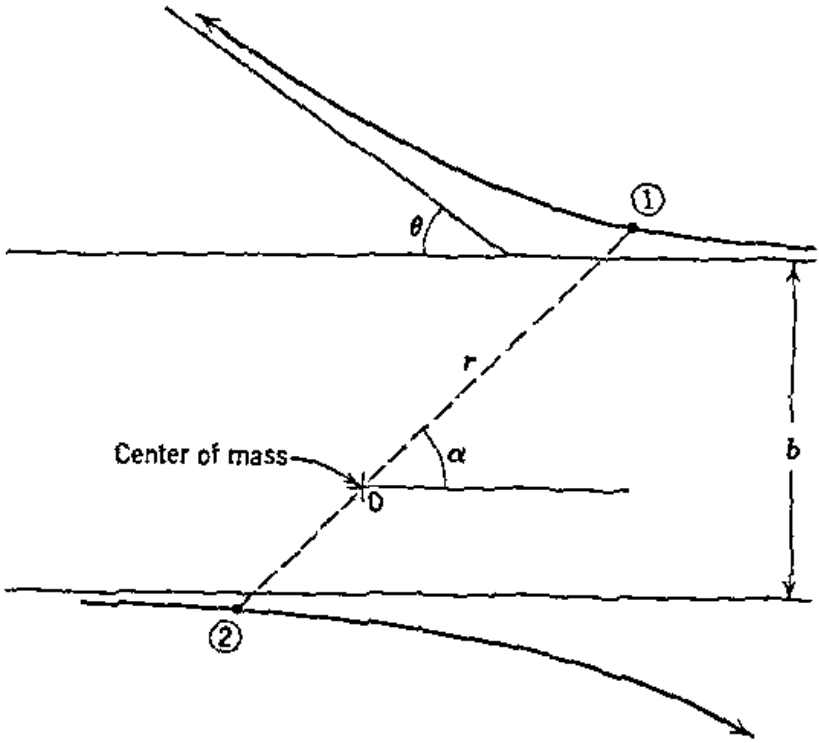


Fig. 1.6 Collision between particle 1 and particle 2

Consider the electron coming in from a very long distance with a relative velocity of v_0, and at such a far distance

$$\alpha_0 = \frac{b}{r}, \qquad \frac{d\alpha_0}{dt} = -\frac{b}{r^2}\frac{dr}{dt}\bigg|_0,$$

where b is called the "impact parameter" Substituting this value of $d\alpha_0/dt$ in Eq (1 11), and taking $v_0 = -dr/dt$, we find

$$L = m_r b v_0 \tag{1.12}$$

The total kinetic energy is given by

$$T = \tfrac{1}{2} m_r \left[\left(\frac{dr}{dt}\right)^2 + r^2 \left(\frac{d\alpha}{dt}\right)^2 \right] \tag{1 13}$$

which at a very far distance becomes T_0,

$$T_0 = \tfrac{1}{2} m_r v_0^2.$$

The total energy is the sum of the kinetic and potential energies

$$E = \tfrac{1}{2}m_r\left[\left(\frac{dr}{dt}\right)^2 + r^2\left(\frac{d\alpha}{dt}\right)^2\right] + \phi(r)$$

where $\phi(r)$ is an arbitrary potential function. Eliminating the time between the energy and the momentum gives the trajectory in polar coordinates

$$\frac{dr}{d\alpha} = \pm\frac{r^2}{b}\left(1 - \frac{\phi}{T_0} - \frac{b^2}{r^2}\right)^{1/2}, \tag{1.14}$$

with the minus sign applying to the particles when they are approaching, and the plus sign applying when they are receding.

At the angle of closest approach, α_m, corresponding to a minimum distance of separation r_m,

$$\frac{dr}{d\alpha} = 0, \qquad 1 - \frac{\phi}{T_0} - \frac{b^2}{{r_m}^2} = 0 \tag{1.15}$$

$$\alpha_m = \int_{r_m}^{\infty}\frac{d\alpha}{dr}\,dr.$$

The scattering angle θ is related to the angle α_m by

$$\theta = \pi - 2\alpha_m$$

so that, using Eq. (1.15)

$$\theta(b, v) = \pi - 2\int_{r_m}^{\infty}\frac{b\,dr/r^2}{\left(1 - \dfrac{2\phi}{mv^2} - \dfrac{b^2}{r^2}\right)^{1/2}}. \tag{1.16}$$

The parameter r_m is related to the collision cross section since $q(0) \approx \pi {r_m}^2$.

For a scattering potential which varies as r^{-n}, the cross section varies as $v^{-4/n}$. Substituting $\phi = ar^{-n}$ into Eq. (1.15), we find

$$1 - \frac{2a}{mv^2{r_m}^n} - \frac{b^2}{{r_m}^2} = 0.$$

We can always find a constant η such that $b = \eta r_m$, so we may write:

$$1 - \frac{2a}{mv^2{r_m}^n} - \eta^2\frac{{r_m}^2}{{r_m}^2}$$

or

$${r_m}^n = \frac{2a}{mv^2(1 - \eta^2)}$$

so that

$$r_m = \beta v^{-2/n}$$

$$q(\theta) \approx \pi {r_m}^2 \approx \pi\beta v^{2-4/n}.$$

Practically, this is an important result When scattering occurs as a result of polarization forces, the incident electron induces an electric dipole moment in the neutral atom, and the resulting induced dipole potential varies as r^{-4}, so that

$$q \sim 1/v$$

$$\nu_c = p_0 P_c v = \text{constant} \times p_0 \tag{1 17}$$

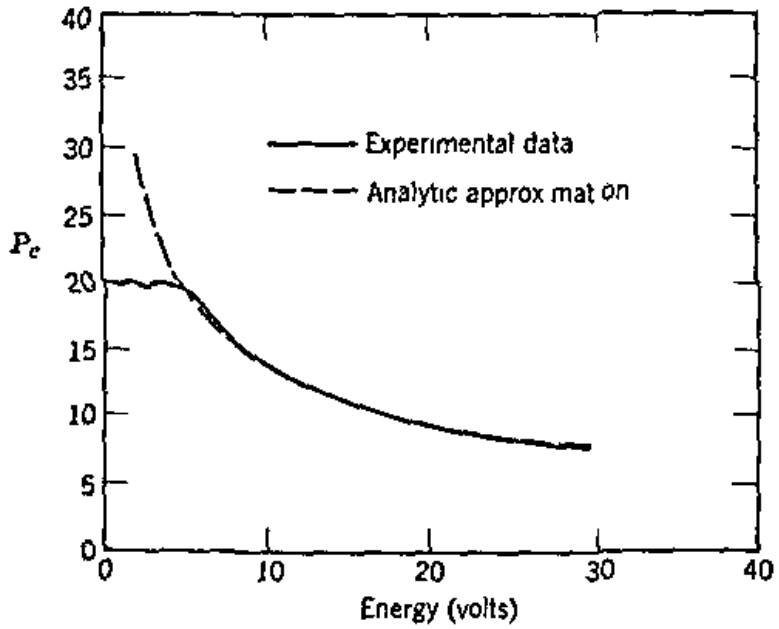


Fig 1 7 The probability of collision of electrons in helium as a function of energy

At energies above a few volts, this is a very good approximation for hydrogen and helium Here

$$(\nu_c)_{He} = 2\,37 \times 10^9 p_0$$

$$(\nu_c)_{H_2} = 5\,93 \times 10^9 p_0$$

The degree to which this approximation concurs with the observation for helium is illustrated in Fig 1 7

In terms of the collision sketched in Fig 1 6, we can define a differential cross section, $q(0)$, for scattering into the angle 0 Incident particles from an impact annulus of area $2\pi b\, db$ scatter into a solid angle $d\Omega = 2\pi \sin 0\, d0$ Then,

$$b\, db = q(0) \sin 0\, d0$$

and the differential cross section is given by

$$q(0) = \frac{b}{\sin 0} \frac{db}{d0} \tag{1 18}$$

The total cross section then becomes:

$$Q_c = \int q(\theta)\, d\Omega.$$

In the center-of-mass system, the change of momentum in the forward direction is $m_r v(1 - \cos\theta)$. This allows us to define a momentum transfer cross section

$$Q_m = \int q(\theta)(1 - \cos\theta)\, d\Omega.$$

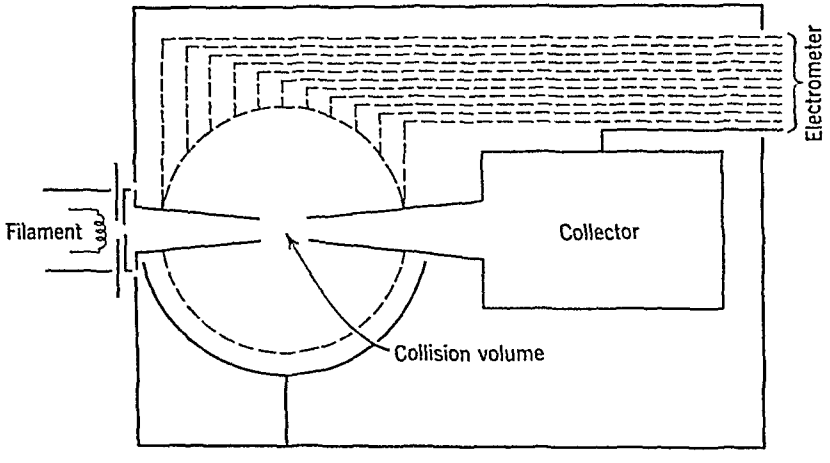


Fig. 1.8 A sketch of Ramsauer and Kollath's apparatus for measuring the angular dependence of electron scattering.

In a similar fashion, we may define a momentum transfer collision frequency

$$\nu_m = \nu_c\overline{(1 - \cos\theta)}.$$

In the elastic scattering of low-energy electrons one can consider a collision to have occurred when an average particle has been deflected through 90°. Here $\cos\theta = 0$ and $\nu_c = \nu_m$.

The momentum transfer cross section Q_m can be compared with Q_c if one knows the scattering is a function of angle and velocity. Electron scattering has been measured for a number of gases.[5] The apparatus that is used is illustrated schematically in Fig. 1.8. Typical results for argon are shown in Fig. 1.9. A direct comparison can be made between Q_m and Q_c when these measurements are available. The total cross section Q_c is converted into Q_m by using the measured scattering intensity and integrating over all the scattering angles from 0 to π. A comparison between these two cross sections[6] is shown in Fig. 1.10. In the low-energy range of electrons in helium, the momentum transfer cross section is

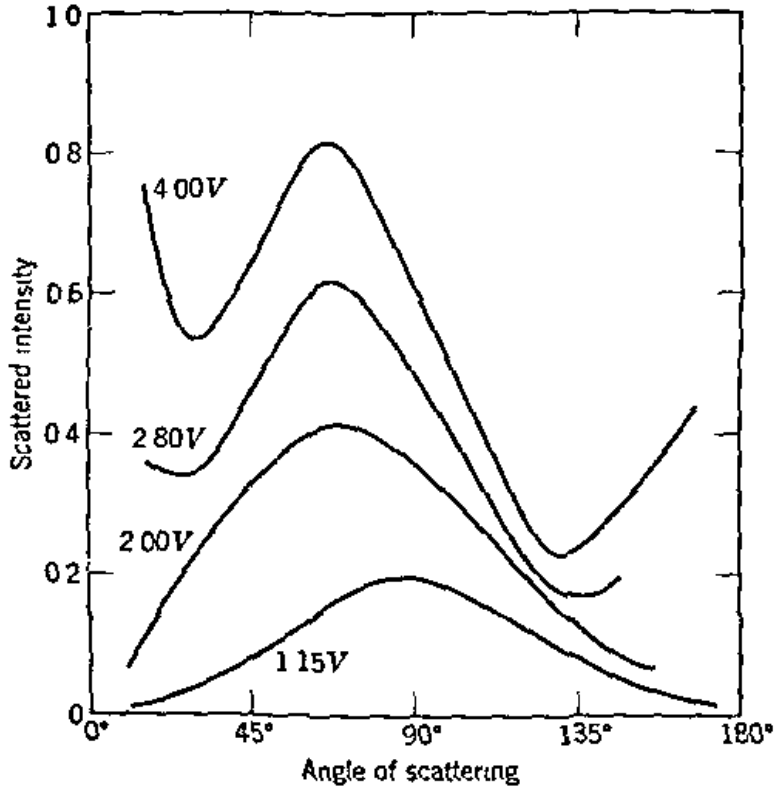


Fig 1 9 Typical measurements of electron scattering in argon

actually greater than the total cross section This means that for this gas, the electrons are more likely to be scattered backwards than forwards

Let us now refine our collision picture further by considering the wave nature of the electron We shall calculate the effect of the diffraction of electron waves by the potential field of outer-shell electrons in the target atom

In classical theory, the viewpoint is that the path of a particle will be deviated a certain definite amount by the presence of a potential field In the quantum theory formulation,[7] it is natural to define a probability $\alpha(b)$ that a particle with impact parameter between b and $b + db$ will suffer an observable deviation The effective collision cross section can then be expressed as

$$Q = 2\pi \int_0^\infty \alpha(b) b \, db \tag{1 19}$$

By using the relationship for the angular momentum L, $L = mvb$, Q can be expressed as

$$Q = \left(\frac{2\pi}{m^2 v^2}\right) \int_0^\infty L\beta(L) \, dL \tag{1 20}$$

where $\beta(L)$ is the probability that a particle with angular momentum between L and $L + dL$ will suffer an observable deviation According to

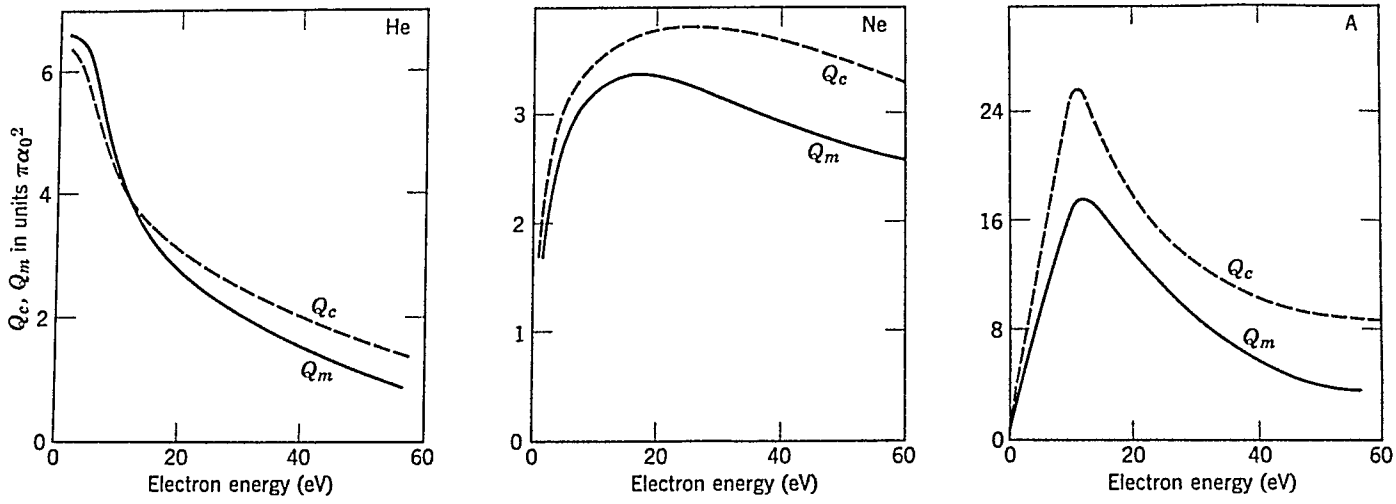


Fig. 1.10 Comparison between total and momentum transfer collision cross sections.

quantum theory however, L is quantized

$$L = [l(l+1)]^{1/2}\hbar \tag{1.21}$$

and thus Q must be a sum over L rather than an integral,

$$Q = \frac{\pi\hbar^2}{m^2v^2}\sum(2l+1)\gamma(l) \tag{1.22}$$

where

$$\gamma(l) = \beta[\{l(l+1)\}^{1/2}\hbar]$$

Let us now consider the form of the probability, $\gamma(l)$ We shall do this by considering for simplicity's sake, the scattering by a potential of the form

$$U(r) = \text{constant} \qquad r < a$$
$$U(r) = 0 \qquad r > a$$

In classical theory, we note that for head on collisions, where $l = 0$, $\gamma(0) = 1$

According to the quantum theory picture, the motion of particles making head-on collisions will be represented by a wavetrain of wavelength λ_1 for $r > a$, and of wavelength λ_2, for $r < a$ The two trains of different wavelength must join smoothly at the boundary $r = a$ This is accomplished by introducing a phase change in the train, $r > a$ It is relative to that which would exist in the absence of any potential This phase change will be observable at infinity and it alone indicates the occurrence of a collision However, we must remember that it is impossible, in principle, to avoid observing those waves that suffer a phase change which is a multiple of 2π Under these circumstances, the potential produces no observable effect on the particles with zero angular momentum, therefore, $\gamma(0) = 0$ This is true when the scattering results in introduction or elimination of a whole number of complete waves

Thus, we reason that $\gamma(0)$ will be a periodic function of the phase shift η_0 produced by the scattering potential Also, $\gamma(0)$ must have a maximum value of one, and never be less than zero The simplest function that satisfies all these conditions is $\sin^2\eta_0$ and an exact calculation shows that this is correct In a similar way, $\gamma(l) = \sin^2\eta_l$, where η_l is the phase shift produced in the waves of angular momentum $\{l(l+1)\}^{1/2}\hbar$ We now have

$$Q = \frac{\pi}{k^2}\sum(2l+1)\sin^2\eta_l \tag{1.23}$$

where the wave number $k = 2\pi/\lambda$, however, there still remains a factor 4 to be introduced We can argue that a factor of 2 arises to make the classical and quantum formulae become identical This occurs when the phase η_l are large for l between 0 and a large number l_0 The second factor of 2

arises because the effective area exposed by a rigid sphere of radius a to the scattering of waves is not πa^2 but $2\pi a^2$. Thus, finally, we obtain

$$Q = \frac{4\pi}{k^2} \sum (2l + 1) \sin^2 \eta_l. \tag{1.24}$$

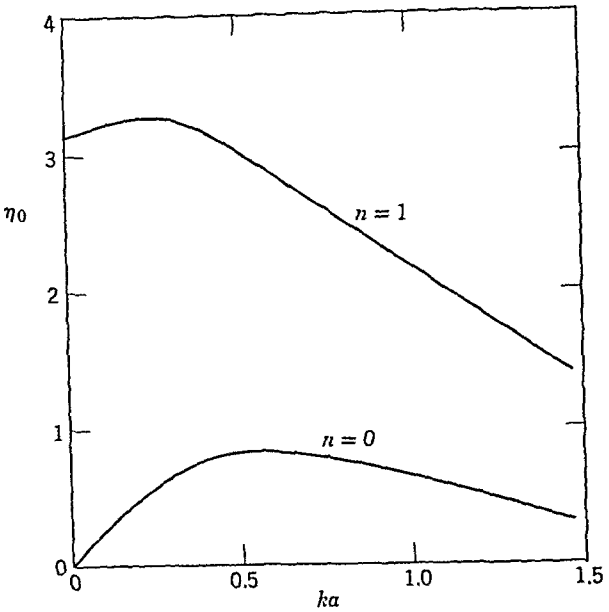


Fig. 1.11 Phase shifts for various atomic field strengths.

For low-energy electrons, L is very small, thus, we shall make the approximation that only the zero-order phase will be important,

$$Q_0 = \frac{4\pi}{k^2} \sin^2 \eta_0. \tag{1.25}$$

As $k \to 0$, $\eta_0 \to n\pi$ where n is a whole number determined by the strength of the atomic field. For the case of $n = 0$, η_0 as a function of energy (in terms of ka_0) is shown[8] in Fig. 1.11. Here we see that as the electron energy rises, the phase η_0 rises to a maximum, less than π (no added waves), and then falls. The cross section is shown in Fig. 1.12. This case would be for weak interaction fields characteristic of light elements like helium. The results for $n = 1$ are also shown in Fig. 1.11 and Fig. 1.12. Here we see the vanishingly small values of cross sections at low energies. Vanishing cross sections occur because $\phi(r)$ is strong enough to introduce whole numbers of additional waves within its range at the given energy.

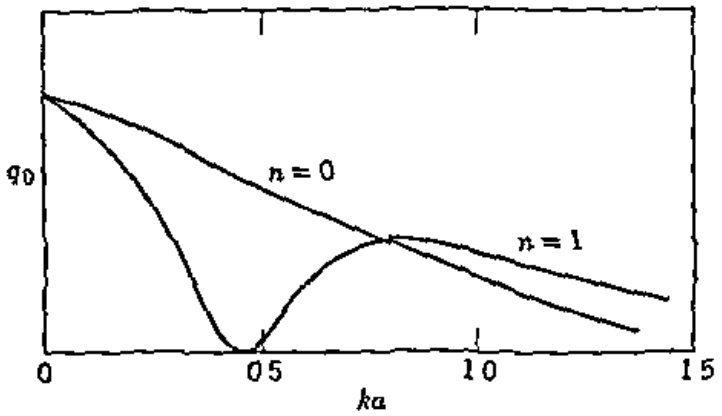


Fig 1 12 Partial cross sections corresponding to the cases illustrated in Fig 1 11

The complicated behavior of the electron cross section in regard to the heavier noble gas atoms can be explained in terms of the quasi periodic behavior of the partial cross sections q_0 as the atomic number of the atom changes where

$$Q_0 = \sum q_l$$

and

$$q_l = \frac{4\pi}{k^2}(2l + 1)\sin^2\eta_l$$

This is illustrated in Fig 1 13 for the case of krypton

The extremely low values of P_e for slow electrons in certain gases characterize what is known as the Ramsauer effect As was mentioned previously the theoretical explanation of this effect was first given by Allis and Morse[4] in terms of the diffraction of an electron wave as it passes an atom The remarkable agreement between this theory and experiment (illustrated in Fig 1 13), shows the validity of the wave mechanical approach to the scattering problem [9] Since the original theory was proposed,[10] elastic-collision cross sections have been calculated for several gases and they all substantiate the explanation of the Ramsauer effect in terms of the diffraction of the electron waves

At energies below a few volts, experimental measurements are not in agreement Several ac methods have been tried which are particularly well suited to measure collision frequencies in the low energy region where the dc method, similar to that used by Ramsauer, becomes difficult The principal advantage of an ac technique results from the fact that the energy imparted to the electron by the electric field in making the measurements, is less than the thermal energy, and hence does not modify the result in the course of the measurements

When electrons are acted upon by an electric field, and are oscillating with a radian frequency ω, and $E = E_0 e^{j\omega t}$ in the presence of a gas in

which collisions are assumed to act as a continuous viscous damping force, then Newton's second law of motion takes the form

$$m\frac{dv}{dt} + mv_m v = -eE_0 e^{j\omega t}. \tag{1.26}$$

Solving this equation for the velocity

$$v = \frac{-eE_0 e^{j\omega t}}{j\omega m + mv_m}, \tag{1.27}$$

the conductivity $\sigma = J/E$ where $J = -nev$, hence,

$$\sigma = \frac{ne^2}{m}\left\langle \frac{v_m - j\omega}{v_m^2 + \omega^2} \right\rangle_{\text{ave}}, \tag{1.28}$$

where the average must be taken over the velocity distribution of the electrons.

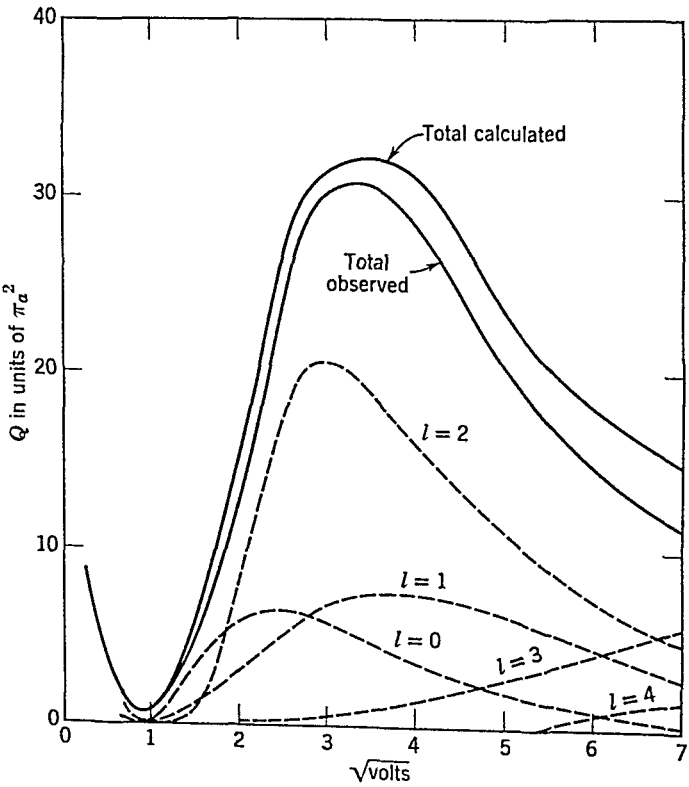


Fig. 1.13 Calculated and observed collision cross sections for electrons in krypton.

In the absence of collisions, an electron oscillates with its velocity 90° out of phase with the applied electric field, and hence constitutes a pure imaginary conductivity Collisions correspond to a real component of the conductivity and

$$\frac{\sigma_r}{\sigma_i} = \frac{\nu_m}{\omega}, \tag{1 29}$$

where

$$P_m = \frac{\nu_m}{p_0 v}$$

Thus in principle, the probability of collision, P_m, can be determined by measuring the complex conductivity of the discharge, the gas pressure, and the frequency of the applied field

Typical of these ac measurements are the experiments by Bekefi and Brown[11] on the collision cross section of slow electrons in hydrogen Measurements of conductivity were made in a rectangular parallelepiped microwave cavity The gas was enclosed in a small cubic quartz bottle situated at the center of the cavity in a region of nearly uniform microwave field A gas discharge was produced periodically by a pulsed microwave signal that was fed into one of the three fundamental modes of the cavity The plasma was studied in the afterglow by a microwave field in a second mode The real and imaginary parts of the conductivity were determined from the change in the loaded Q value of the cavity and from the change of the resonant frequency of the cavity, respectively The low intensity probing signal produced negligible perturbation of the decaying plasma The electrons were heated by feeding microwave power into the third mode of the cavity, the electric field strengths used in heating the electrons were computed from the power input and from the Q of this mode

The details of the microwave method are outlined in a series of articles[12] based on Slater's[13] formulation for the change in resonant frequency caused by a plasma within a microwave cavity

$$\frac{\Delta\omega}{\omega_0} = \frac{1}{2\omega_0\epsilon_0} \frac{\int_I \sigma_i E^2\, dV}{\int_I E^2\, dV} = \frac{e^2}{2m\epsilon_0(\nu_m^2 + \omega_0^2)} \frac{\int_V n_0 E^2\, dV}{\int_V E^2\, dV}$$

$$= \frac{n_0 e^2 \eta}{2m\epsilon_0(\nu_m^2 + \omega_0^2)} \tag{1 30}$$

Here ω_0 is the resonant frequency, ϵ_0 the permittivity of free space, n_0 the peak electron density and η a geometrical factor depending on the spacial

distribution of both the electrons and the electric field in the cavity. The integrations have to be carried out over the whole volume of the cavity and the assumption is made that the electric currents are small compared to the displacement current, that is, $\sigma_i \ll \omega_0 \epsilon_0$.

The ratio of the real to the imaginary part of the microwave conductivity[14] can be obtained by writing Eq. (1.28) in its complete form:

$$-\frac{\sigma_r}{p_0\sigma_i} = \frac{\int_0^\infty \frac{P_m/\omega}{1+a(v)} v^4 \frac{\partial f}{\partial v}\, dv}{\int_0^\infty \frac{1}{1+a(v)} v^3 \frac{\partial f}{\partial v}\, dv} \tag{1.31}$$

where $a(v) = (P_m p_0 v/\omega)^2$, and f is the electron velocity distribution function. For gas pressures greater than 1 torr, thermal equilibrium between the electrons and the hydrogen molecules is established within 10^{-4} seconds after the discharge has ceased. Henceforth the electrons have a Maxwellian distribution

$$f \sim e^{-mv^2/2kT_e},$$

which corresponds to an electron temperature T_e equal to the gas temperature T_g.

The collision probability, P_m, is obtained by solving the integral equation Eq. (1.31). It can be seen from Eq. (1.31) that, in the limit of very low pressures [$a(v) \ll 1$], and of very high pressures [$a(v) \gg 1$], $\sigma_r/p_0\sigma_i$ is independent of p_0; at intermediate pressures, $\sigma_r/p_0\sigma_i$ varies with p_0 in a manner governed by the magnitude and velocity dependence of P_m. The integrals in Eq. (1.31) are difficult to evaluate unless P_m is taken to vary with velocity in a simple manner. The usual assumption is that P_m can be approximated by a power law of the form $P_m \sim (v/v_0)^h$, where v_0 is the most probable electron velocity at the assumed gas temperature.

The experimental measurements for hydrogen are best fitted by the assumption that $P_m = 28(v/v_0)^{0.6}$.

One of the basic assumptions made in the development of the collision cross section equations was that the form of the potential function remained constant. As plasmas are extended to higher and higher degrees of ionization, this assumption needs to be examined as the collisions change from elastic collisions between electrons and neutral atoms to electrons and ions. There is a limit above which the conductivity equation

$$\sigma = \frac{ne^2}{m(\nu_m + j\omega)} \tag{1.32}$$

does not hold. This limit is approached when the electron-ion interaction begins to affect the momentum of the average electron. The upper limit

to the plasma conductivity may be obtained from the dc conductivity for a Lorentzian gas (a fully ionized plasma, devoid of electron-electron interactions and with all the positive ions considered at rest) To show this we can proceed as follows Ordinary Coulomb scattering is given by what is often called the Rutherford scattering law

$$q(\theta) = \frac{1}{4}\left(\frac{e^2}{mv^2}\right)^2 \frac{1}{\sin^4(\theta/2)}$$

where

$$Q_c = \int_{\theta_{\min}}^{\pi} q(\theta)\, d\Omega$$

Here $\theta_{\min}$ occurs at the minimum distance of approach, considered to be a constant

$$Q_m = 2\pi \int_{\theta_{\min}}^{\pi} q(\theta)(1 - \cos\theta)\sin\theta\, d\theta$$

$$= \frac{2\pi}{4}\left(\frac{e^2}{mv^2}\right)^2 \int \frac{1}{\sin^4(\theta/2)}(1 - \cos\theta)\sin\theta\, d\theta$$

Solving for Q_m we get

$$Q_m = 2\pi\left(\frac{e^2}{mv^2}\right)^2 \ln\left[\frac{2}{(1 - \cos\theta_{\min})}\right]$$

Using

$$\sigma = \frac{ne^2}{m\nu_m}, \qquad v = \left(\frac{3kt}{m}\right)^{1/2}$$

and the relation between Q_m and ν, we arrive at the approximate answer that [15]

$$\sigma \approx 2 \times 10^3 \left(\frac{kT}{e}\right)^{3/2} \text{ mhos/meter} \tag{1 33}$$

where k is Boltzmann's constant, T is the kinetic electron temperature, and e is the electronic charge The dc conductivity of a fully ionized plasma is given by an equation analogous to Eq (1 32) with ν_m replaced by an effective collision frequency ν_{ei} arising from electron-ion interaction Thus,

$$2 \times 10^3 \left(\frac{kT}{e}\right)^{3/2} = \frac{ne^2}{m\nu_{ei}}$$

or

$$\nu_{ei} = \frac{Ne^2}{(2 \times 10^3)(kT/e)^{3/2}} \tag{1 34}$$

where N is the density of a fully ionized plasma For a partially ionized

plasma, the collision frequencies arising from electron-atom and electron-ion interaction can be assumed to add, leading to an effective collision frequency:

$$\nu_{\text{eff}} = \nu_m(1 - f) + \frac{Ne^2 f}{(2 \times 10^3)(kT/e)^{3/2}} \tag{1.35}$$

where f is the degree of ionization. The effective collision frequency can be

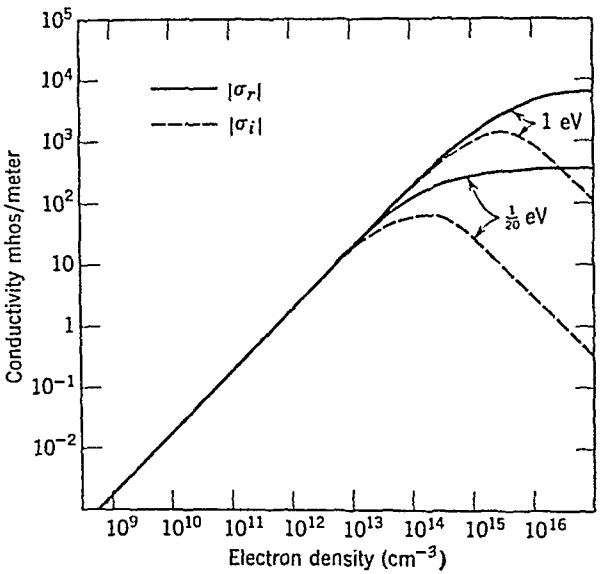


Fig. 1.14 Conductivity as a function of electron density.

used in Eq. (1.32) to obtain the ac conductivity of a plasma for all degrees of ionization.

$$\sigma = \frac{ne^2}{m(\nu_{\text{eff}} + j\omega)}. \tag{1.36}$$

This equation is approximate in that ν_{eff} is not independent of electron energy. It can be used, however, to calculate the limits of validity of Eq. (1.32). Figure 1.14 is a plot of the real and imaginary components of the conductivity given by Eq. (1.36), as a function of electron density. The plasma is assumed to be produced in hydrogen at a pressure of 4 torr. The radian frequency, $\omega = 2 \times 10^{10}$ rad/sec, so that $\nu_m \simeq \omega$. Extensive calculations of the electrical conductivity of a hydrogen plasma with variations of ionization degree as a function of average electron energy have been carried out by Dreicer.[16]

Probability of Elastic Collision for Ions with Gas Atoms

The probability of elastic collision between ions and gas atoms has been studied by various methods similar to those used for electrons Some examples of these methods are the studies of Muschlitz[17] and his coworkers using a Nier-type mass spectrometer for the momentum analysis of an ion beam feeding into a scattering chamber Their work suggests that polarization forces can be responsible for collision phenomena and that inverse

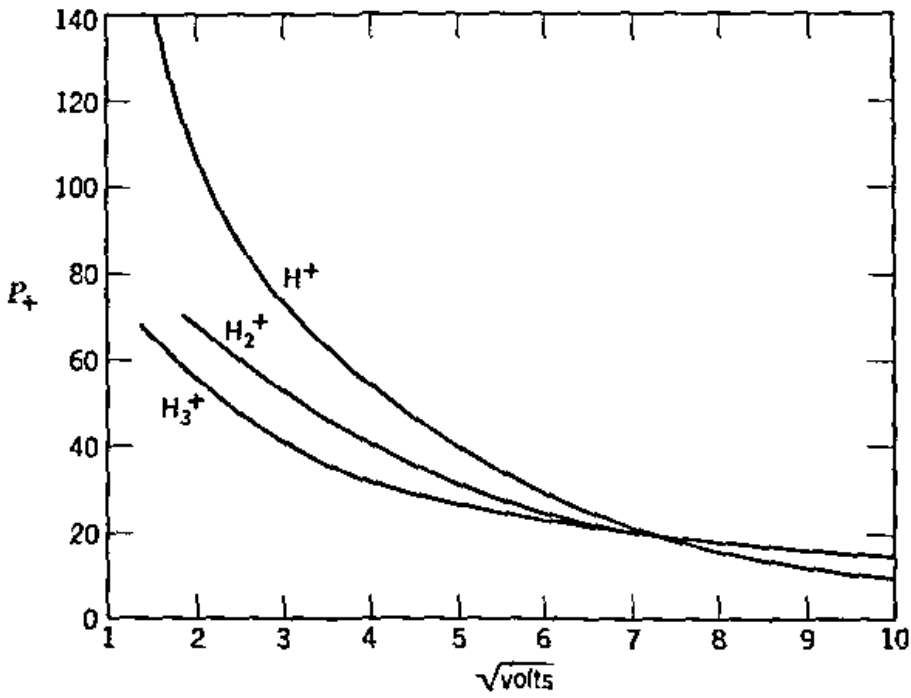


Fig 1 15 Probability of collision for hydrogen ions with hydrogen

fourth power repulsion can explain the form of the collision probabilities as a function of energy Results of these measurements are illustrated in Fig 1 15

Collision probabilities in a given gas increase with the mass of the ion and decrease to an approximately constant value with increasing kinetic energy of the ion For a given ion, the collision probability increases with increasing atomic mass of the vehicular gas These conclusions are illustrated in Figs 1 16 and 1 17, from the measurements of Ramsauer and Beeck [18]

The width of a cyclotron resonance line is largely determined by the collision that the charged particle makes with the surrounding gas atoms Although several attempts have been made to use this effect for determining electron elastic collision cross sections, results have not been superior to other methods [19] On the other hand, for ions, considerable

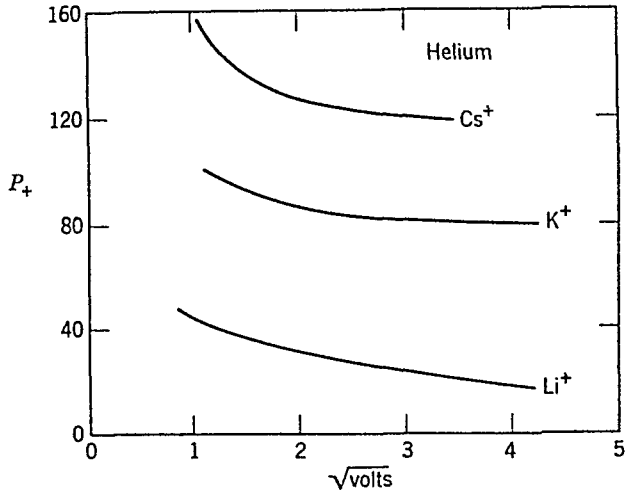


Fig. 1.16 Probability of collision of alkali metal ions in helium.

accuracy has been achieved,[20] and the method is worthy of discussion. The method may be understood physically by considering the motion of the plasma particles in crossed ac electric and dc magnetic fields. In the absence of collisions, the phase of the electric current with respect to the electric field is determined solely by the magnetic field. For those values

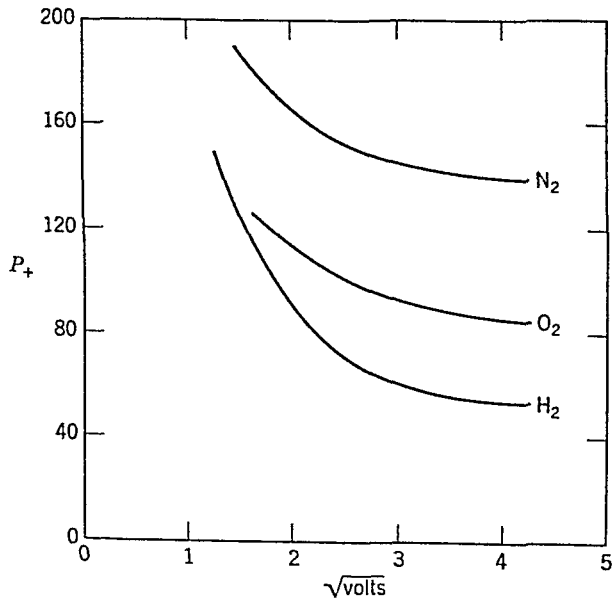


Fig. 1.17 Probability of collision of potassium ions in various gases.

of the magnetic field that are lower than the cyclotron value, for a given radian frequency, the current lags behind the electric field, for higher values, it leads Introducing collisions serves to shift the position of no phase difference, just as the addition of damping to any oscillator shifts its natural frequency

Writing Newton's second law for singly charged positive ions with an applied magnetic field, we find

$$m\frac{dv}{dt} + m_r\nu_c v = e[E_0 \sin \omega t + \mathbf{v} \times \mathbf{B}] \tag{1 37}$$

where m_r is the reduced mass of the ion

$$m_r = \frac{mM}{m + M}$$

and M is the mass of the bombarded atom When we introduce the cyclotron frequency $\omega_b = eB/m$, and a reduced collision frequency,[20] $\nu_0 = \nu_c m_r/m$, as the collision frequency an ion would have if it lost all its nonrandom velocity on each collision,

$$\frac{dv}{dt} + \nu_0 v = \frac{eE_0}{m} \sin \omega t + v_y\omega_b$$

where the magnetic field is applied in the z direction and the electric field in the x direction

The power absorbed by the electron is $P = eEv$ and the velocity is

$$v = \left(\frac{-e/m}{j\omega + \nu_c}\right)E_0 = -\frac{eE_0}{m\nu_c} \tag{1 38}$$

so that the total power absorbed is

$$P_0 = n\frac{e^2E_0^2}{4m\nu_0} \tag{1 39}$$

and the effective electric field for the case of a transverse magnetic field is

$$E_0^2 = E^2\left[\frac{\nu_0^2}{\nu_0^2 + (\omega - \omega_b)^2} + \frac{\nu_0^2}{\nu_0^2 + (\omega + \omega_b)^2}\right] \tag{1 40}$$

At cyclotron resonance, $\omega = \omega_b$, and the lines are narrow so that $(\omega + \omega_b)^2 \gg \nu_0^2$, the second term in this equation vanishes and the line shape is Lorentzian For this case, the half width of the line at half-maximum is exactly equal to the reduced collision frequency

$$\nu_0 = \Delta\omega^{1/2}$$

and a direct measure of the collision frequency is possible

Typical results of this method for N_2^+ in N_2 are illustrated in Fig 1 18

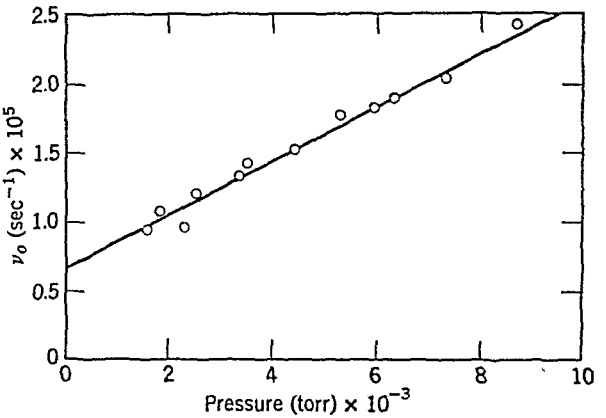


Fig. 1.18 Measurements of N_2^+ collision frequencies with nitrogen gas.

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Chapter 2

Free Diffusion

Consider particles diffusing in a gas whose density is essentially uniform. The number density, n, of the particles is assumed to be much smaller than that of the molecules. Suppose the flow is one-dimensional and takes place along the x axis, where a density gradient dn/dx has been established. Assume that a uniform flow of particles has been established, so that steady-state conditions prevail. Let v_x be the x component of the instantaneous velocity of a particle, undergoing random collisions with the gas molecules, whose velocity distribution is taken to be isotropic. The particles are considered to be in energy equilibrium with the molecules, and v_x is independent of the coordinates. Let ν be the average frequency with which a particle collides with molecules, θ be the scattering angle in the laboratory system, and m be the particle mass. The number density of the particles with velocity between v and $v + dv$ is $n_v\, dn_v$. The time rate of change of the momentum of these particles is

$$\frac{d(mv_x n_v\, dn_v)}{dt} = -mv_x \nu\overline{(1 - \cos\theta)} n_v\, dn_v.$$

Let $\overline{\nu_m} = \nu\overline{(1 - \cos\theta)}$ be the average "collision frequency for momentum transfer." The left side of the above equation can be written, after cancellation of the factor m, as

$$\frac{d(n_v\, dn_v v_x)}{dt} = \frac{v_x\, d(n_v\, dn_v v_x)}{dx} = \frac{d(n_v\, dn_v v_x^2)}{dx}$$

and the right side becomes

$$-n_v\, dn_v v_x \overline{\nu_m}.$$

When we average over v_x, integrate over n_v, substitute the particle current density

$$\Gamma_x = n\overline{v_x}, \quad \text{and put} \quad \overline{v_x^2} = \overline{v^2}/3,$$

we find

$$\frac{d\left[\left(\int n_v\, dn_v\right)\overline{(v_x^2)}\right]}{dx} = -\left[\int (n_v\, dn_v)\right]\overline{v_x \nu_m}$$

$$\frac{d(n\overline{v^2}/3)}{dx} = -n\overline{v_x \nu_m},$$

so

$$\Gamma_x = n\overline{v_x} = -\frac{d[n\overline{v^2}/3\overline{\nu_m}]}{dx}$$

or

$$\Gamma_x = -\frac{d[(v^2/3\nu_m)_{\text{ave}} n]}{dx}$$

Generalizing to three dimensional diffusion, we have

$$\boldsymbol{\Gamma} = -\nabla[(v^2/3\nu_m)_{\text{ave}} n]$$

which is called Fick's law of diffusion,

$$\boldsymbol{\Gamma} = -\nabla(Dn) \tag{2.1}$$

Here D is the diffusion coefficient, given by

$$D = (v^2/3\nu_m)_{\text{ave}} = (v^2 l_m/3v)_{\text{ave}} = (v l_m/3)_{\text{ave}} \tag{2.2}$$

The mean free path for momentum transfer, l_m, is given by the equation

$$l_m = l/\overline{(1 - \cos\theta)}$$

If we discuss the particle motion before a steady flow is established, the time variation of the number density, with no sources present, may be written as the equation of continuity

$$\frac{\partial n}{\partial t} + \nabla \cdot \boldsymbol{\Gamma} = 0$$

which leads directly to the diffusion equation

$$\frac{\partial n}{\partial t} = D\,\nabla^2 n \tag{2.3}$$

if we assume that D is independent of position Assuming the particle to diffuse exponentially such that the particle density is given by

$$n(x, y, z, t) = n_0(x, y, z)e^{-\nu t},$$

the diffusion equation becomes

$$D\nabla^2 n = -nv = -\left(\frac{v}{D}\right) Dn$$
$$\nabla^2 n = -\left(\frac{v}{D}\right) n. \tag{2.4}$$

For example, if we solve for n in one dimension, $n = Ae^{-i(x/r)} + Be^{i(x/r)}$. In general, the solution of this equation is a characteristic value problem that is usually solved in terms of a diffusion length, Λ, where

$$\Lambda^2 = D/v.$$

As an illustration of a solution for a particular geometry, for a cylinder of height h with flat ends of radius r, we find

$$\frac{1}{\Lambda^2} = \frac{v}{D} = \left(\frac{\pi}{h}\right)^2 + \left(\frac{2.405}{r}\right)^2. \tag{2.5}$$

Undertaking the problem in greater detail, we shall consider the diffusion of electrons of density n per cubic centimeter through a background gas of atoms of density N per cubic centimeter. When $N \gg n$ and when $|\nabla n|$ is sufficiently small, such a diffusion is given by a law that states that the particle current density Γ (the number of particles crossing a unit area per second), is opposite in direction to the density gradient ∇n and is directly proportional to it.

The calculation just discussed assumes only a single fundamental diffusion mode, which is not always a sufficiently good approximation to take care of the physical behavior of a plasma particularly if there is a time variation in the concentration. Writing the diffusion equation as

$$\nabla^2 n = -n/D\tau \tag{2.6}$$

we find that the solution for infinite parallel plates can take the form

$$n(x) = \sum_k n_k \cos(x/\Lambda_k) \tag{2.7}$$

where $\Lambda_k^2 = D\tau_k$.

Corresponding to each Λ_k there is a decay time τ_k. The average particle density n can be made up of many terms:

$$n_{\text{ave}} = A_1 e^{-t/\tau_1} + A_2 e^{-t/\tau_2} + A_3 e^{-t/\tau_3} + \cdots. \tag{2.8}$$

Taking a cubical cavity as an example, the ratio of time constants is found to be

$$\frac{\tau_1}{\tau_2} = 3.67; \qquad \frac{\tau_1}{\tau_3} = 6.33; \qquad \frac{\tau_1}{\tau_4} = 9.$$

If only the fundamental mode of diffusion is operative, measurements

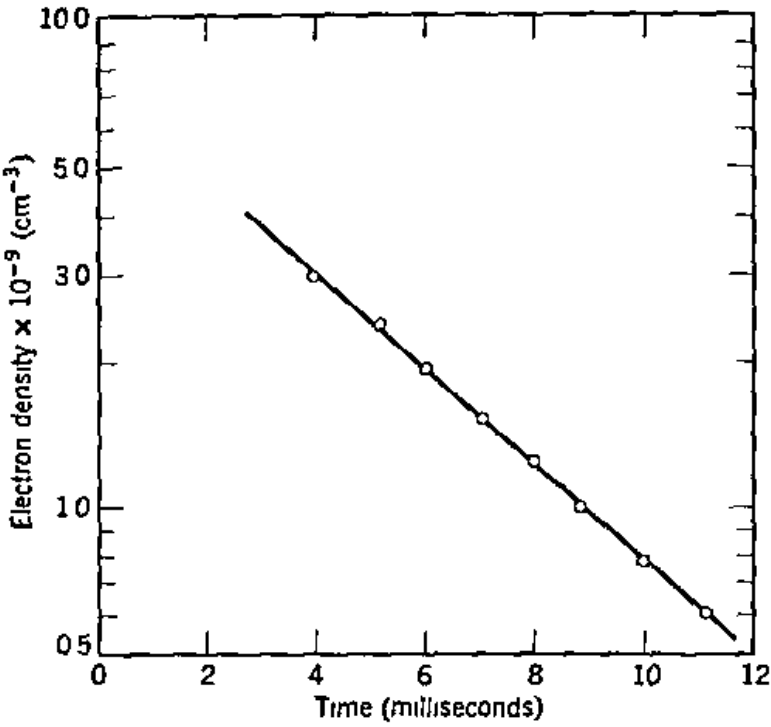


Fig 2 1 Electron density decay for diffusion

of electron loss in a resonant cavity after a discharge has been turned off will exhibit a logarithmic decay of electron density as a function of time This is illustrated in Fig 2 1 If the initial spatial electron distribution is asymmetric, higher modes will be excited and will decay at a higher rate than the fundamental so that a semilog plot of density as a function of time will not be a straight line Under some conditions, a clear indication of the mixture of modes can be demonstrated experimentally [1] Such a case is illustrated in Fig 2 2

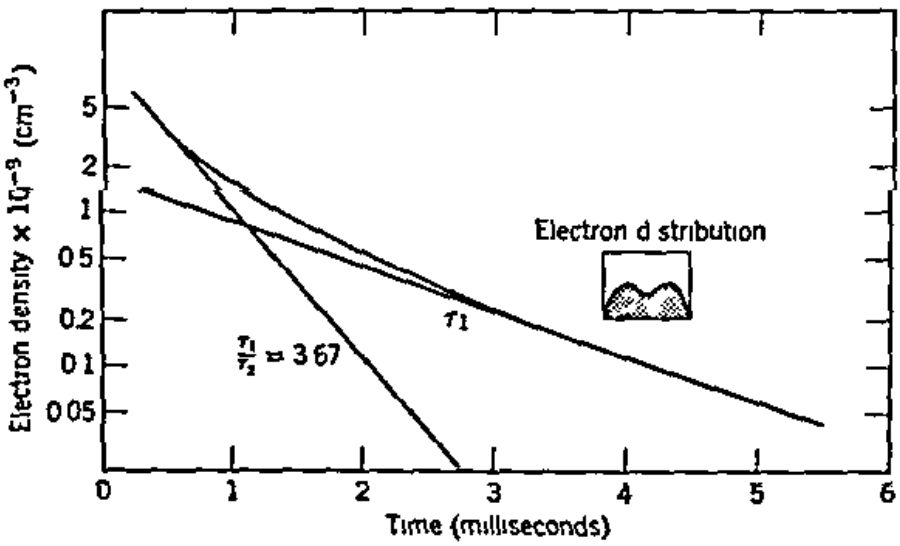


Fig 2.2 Electron density decay with higher mode diffusion

Let us derive the law of diffusion and the expression for the random current density Γ_r.

For this purpose we will use the concept of free paths and assume the familiar law for the distribution of free paths which states that the probability that a particle will travel a distance r without being scattered is

$$P = e^{-r/l_c}. \tag{2.9}$$

The necessary assumptions are:

1. We have a uniform background of gas atoms of density N per centimeter^3 and a nonuniform distribution of electrons of density $n(r)$ per centimeter^3 such that $N \gg n(r)$.
2. We assume that the scattering of electrons by the gas atoms is isotropic.
3. The electron density $n(r)$ is a smooth slowly varying function of r.

To compute the net transport of electrons across a unit area, we choose a unit area dS, normal to the z axis in the xy plane and centered at the origin for convenience. If we could sit at the origin and count particles, we would count how many particles in question (electrons) crossed from the upper plane through the unit area into the lower plane, regardless of their particular velocity components. The number of counts per unit time is the current Γ_{-z}. The corresponding number crossing per unit time in the upward sense is Γ_{+z}. The net current Γ is the difference between Γ_{+z} and Γ_{-z}.

The "counting" process is as follows: we choose an incremental volume dV at a distance r from the origin. We then find how many particles that originate in dV cross the unit area dS and add the contributions from all such volume elements in the upper plane to compute Γ_{-z}. A similar process is used to compute Γ_{+z}.

Referring to Fig. 2.3 we consider first the number of particles crossing dS in the $-z$ sense; that is, those particles which originate in the upper plane. We assume that owing to collision with the background gas, the electron velocity distribution is isotropic. The fraction leaving dV and heading toward dS compared to the total number of electrons which have their "origin" in dV is simply the ratio of the solid angle subtended by dS to the total solid angle. The solid angle subtended by dS is $dS \cos\theta/r^2$ and the total solid angle is 4π. Not all of the electrons leaving dV arrive at dS; some will collide with the gas molecules and their velocities will again be made isotropic. The number leaving dV is then attenuated by e^{-r/l_c}. It would seem that we are overlooking those particles that make collisions. This is not true because they are considered when we move our variable r up and down in the summing process. This brings up the

point of what exactly we mean when we say that certain electrons have their "origin" in dV The particles originating in dV are not $n\,dV$ but rather those particles that have made a collision in dV at the time interval in question Diffusion is a transport phenomenon, we are interested in tracing how many of the particles that made their last collision in dV pass through dS We consider the average particle Its collision probability per unit length is $1/l_e$, and the frequency of collision is $\bar{v}/l_e$ The volume

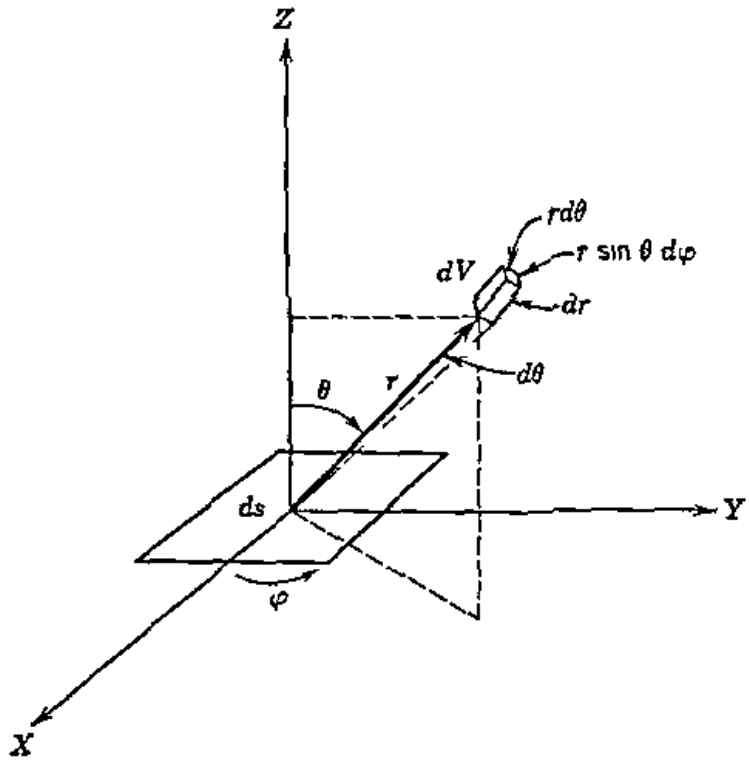


Fig 2.3

rate of collisions is $n\,dVv/l_e$ This represents the number of particles originating in dV per second The number of particles arriving at dS per second that originate in dV is

$$\Gamma_{-z}\,dS = \frac{nv}{l_e}\,dV\frac{dS\cos\theta}{4\pi r^2}\,e^{-r/l_e}$$

$$= \frac{nv}{l_e}\,r^2\sin\theta\,dr\,d\theta\,d\phi\,\frac{dS\cos\theta}{4\pi r^2}\,e^{-r/l_e}$$

$$= \frac{nv}{4\pi l_e}\sin\theta\cos\theta\,dr\,d\theta\,d\phi e^{-r/l_e}$$

To obtain the total current Γ_{-z} we integrate over the upper plane, that is, ϕ goes from 2π to 0, θ from $\pi/2$ to 0, and r from ∞ to 0 We assume that the dimensions of the box are much larger than the mean free path

to justify this last range of r values,

$$\Gamma_{-z} = \frac{\bar{v}}{4\pi l_c} \int_0^\infty dr \int_0^{\pi/2} d\theta \int_0^{2\pi} d\phi n(r) \sin\theta \cos\theta e^{-r/l_c}.$$

We now assume that $n(r)$ is a smooth function of position about the origin so that a Taylor series expansion for $n(r)$ might be written

$$\begin{aligned} n = n_0 &+ \left(\frac{\partial n}{\partial x}\right)_0 x + \left(\frac{\partial n}{\partial y}\right)_0 y + \left(\frac{\partial n}{\partial z}\right)_0 z + \frac{1}{2}\left(\frac{\partial^2 n}{\partial x^2}\right)_0 x^2 \\ &+ \frac{1}{2}\left(\frac{\partial^2 n}{\partial y^2}\right)_0 y^2 + \frac{1}{2}\left(\frac{\partial^2 n}{\partial z^2}\right)_0 z^2 + \cdots + \frac{1}{k!}\left(\frac{\partial^k n}{\partial x^k}\right)_0 x^k \\ &+ \frac{1}{k!}\left(\frac{\partial^k n}{\partial y^k}\right)_0 y^k + \frac{1}{k!}\left(\frac{\partial^k n}{\partial z^k}\right)_0 z^k + \cdots. \end{aligned}$$

The reason for the assumption of the slow variation of $n(r)$ is that the Taylor series expansion be valid for all values of r in the integral. Although the largest value of r is infinite, the value of r for which a substantial contribution is made to the integral is considerably smaller, because of the exponential term. At a distance of three mean free paths, the integral is diminished by $\frac{1}{20}$ and as a result, a criterion might be established that $n(r)$ should not change by more than 10% over a distance of $3l_c$.

The form of n, as it stands, cannot be evaluated in the integral. The coordinates (x, y, z) must be transformed into (r, θ, ϕ). The values of $(\partial^k n/\partial x^k)_0$, $(\partial^k n/\partial y^k)_0$, etc., are constants and need not be transformed.

$$\begin{aligned} x &= r \sin\theta \cos\phi \\ y &= r \sin\theta \sin\phi \\ z &= r \cos\theta. \end{aligned}$$

We now proceed with integration breaking up the terms according to their order.

(0) *Zeroeth Order*

$$\begin{aligned} \Gamma_{-z} &= \frac{\bar{v}}{4\pi l_c} \int_0^\infty dr \int_0^{\pi/2} d\theta \int_0^{2\pi} d\phi n_0 \sin\theta \cos\theta e^{-r/l_c} \\ &= \frac{\bar{v}}{4\pi l_c} \int_0^{2\pi} d\phi \int_0^{\pi/2} \sin\theta \, d(\sin\theta) \int_0^\infty n_0 e^{-r/l_c} \, dr \\ &= \frac{\bar{v}}{4\pi l_c} (2\pi)(\tfrac{1}{2})(n_0 l_c) = \frac{n_0 \bar{v}}{4}. \end{aligned}$$

This is a rather familiar term representing thermal fluctuations of the electrons in the container

(1) *First-Order Terms*

(a) x component

$$\Gamma_{-x} = \frac{v}{4\pi l_e}\int_0^\infty \left(\frac{\partial n}{\partial x}\right)_0 re^{-r/l_e}\,dr \int_0^{\pi/2} \sin^2\theta\cos\theta\,d\theta \int_0^{2\pi}\cos\phi\,d\phi$$

$$= 0 \quad \text{since} \quad \int_0^{2\pi}\cos\phi\,d\phi = \sin\phi\Big]_0^{2\pi} = 0$$

(b) y component

$$\Gamma_{-y} = \frac{\bar{v}}{4\pi l_e}\int_0^\infty \left(\frac{\partial n}{\partial y}\right)_0 re^{-r/l_e}\,dr \int_0^{\pi/2} \sin^2\theta\cos\theta\,d\theta \int_0^{2\pi}\sin\phi\,d\phi$$

$$= 0 \quad \text{since} \quad \int_0^{2\pi}\sin\phi\,d\phi = \cos\phi\Big]_0^{2\pi} = 0$$

(c) z component

$$\Gamma_{-z} = \frac{\bar{v}}{4\pi l_e}\int_0^\infty \left(\frac{\partial n}{\partial z}\right)_0 re^{-r/l_e}\,dr \int_0^{\pi/2} \sin\theta\cos^2\theta\,d\theta \int_0^{2\pi} d\phi$$

$$= \frac{v}{4\pi l_e}\, l_e^2\left(\frac{\partial n}{\partial z}\right)_0 \frac{1}{3}\ 2\pi$$

$$= \frac{\bar{v} l_e}{6}\left(\frac{\partial n}{\partial z}\right)_0$$

We may continue this process indefinitely, but to summarize we give here a few terms for Γ_{-z}

$$\Gamma_{-z} = \frac{n_0 v}{4} + \frac{v l_e}{6}\left(\frac{\partial n}{\partial z}\right)_0 + \frac{\bar{v} l_e^2}{16}\left[\left(\frac{\partial^2 n}{\partial x^2}\right)_0 + \left(\frac{\partial^2 n}{\partial y^2}\right)\right]_0 + \frac{\bar{v} l_e^2}{8}\left(\frac{\partial^2 n}{\partial z^2}\right)_0$$

$$+ \quad + \frac{\bar{v}}{2}\, l_e^m \frac{1}{m+2}\left(\frac{\partial^m n}{\partial z^m}\right)_0$$

$$+ \frac{\bar{v}}{4\pi}\frac{l_e^m}{m+2}\left[\left(\frac{\partial^m n}{\partial x^m}\right)_0 F(\cos^k\phi) + \left(\frac{\partial^m n}{\partial y^m}\right)_0 F(\sin^k\phi)\right] \qquad (2\,10)$$

$$+ \frac{\bar{v}}{2}\frac{l_e^{m+1}}{m+3}\left(\frac{\partial^{m+1} n}{\partial z^{m+1}}\right)_0 + \qquad m \text{ even}$$

By the same process, we can compute the flow across dS in the posit sense, and find for the positive flow

$$\Gamma_{+z} = \frac{n_0\bar{v}}{4} - \frac{\bar{v}l_c}{6}\left(\frac{\partial n}{\partial z}\right)_0 + \frac{\bar{v}l_c^2}{16}\left[\left(\frac{\partial^2 n}{\partial x^2}\right)_0 + \left(\frac{\partial^2 n}{\partial y^2}\right)_0\right]$$

$$+ \frac{\bar{v}l_c^2}{8}\left(\frac{\partial^2 n}{\partial z^2}\right)_0 + \cdots + \frac{\bar{v}}{2}\frac{l_c^m}{m+2}\left(\frac{\partial^m n}{\partial z^m}\right)_0$$

$$+ \frac{\bar{v}}{4\pi}\frac{l_c^m}{m+2}\left[\left(\frac{\partial^m n}{\partial x^m}\right)_0 F(\cos^k\phi) + \left(\frac{\partial^m n}{\partial y^m}\right)_0 F(\sin^k\phi)\right] \quad (2.$$

$$- \frac{\bar{v}}{2}\frac{l_c^{m+1}}{m+3}\left(\frac{\partial^{m+1} n}{\partial z^{m+1}}\right)_0 + \cdots.$$

The net flow of current $\Gamma_z = \Gamma_{+z} - \Gamma_{-z}$ can be evaluated now:

$$\Gamma_z = -\frac{\bar{v}l_c}{3}\left(\frac{\partial n}{\partial z}\right)_0 - \bar{v}\sum_{k=1}^{\infty}\frac{l_c^{2k+1}}{2k+3}\left(\frac{\partial^{2k+1} n}{\partial z^{2k+1}}\right) + \cdots \quad (2.$$

$$k = 1, 2, 3 \cdots.$$

Therefore, we see that the x and y gradients of the electron distribut do not affect the net transport of electrons across a unit area in the c tainer although they do affect the one-dimensional flow. A sec observation is that the net flow depends only on odd terms in the Fou expansion, a result which is quite obvious from a physical standpo Even functions are symmetric with respect to z and $-z$ and they do create a difference necessary for long-range transport.

To generalize these results, we can state the following: the origin point about which we made our expansion was arbitrary. The result t applies to any point in the container except those in boundary lay The choice of the z axis was arbitrary; it could have been the x or y a Therefore, without any additional labor, we obtain the following expressions:

$$\Gamma_x = -\frac{\bar{v}l_c}{3}\left(\frac{\partial n}{\partial x}\right)_0 - \bar{v}\sum_{k=1}^{\infty}\frac{l_c^{2k+1}}{2k+3}\left(\frac{\partial^{2k+1} n}{\partial x^{2k+1}}\right) \quad (2$$

$$\Gamma_y = -\frac{\bar{v}l_c}{3}\left(\frac{\partial n}{\partial y}\right)_0 - \bar{v}\sum_{k=1}^{\infty}\frac{l_c^{k+1}}{2k+3}\left(\frac{\partial^{2k+1} n}{\partial y^{2k+1}}\right). \quad (2$$

These expressions are rigorously correct under the approximations we have made; however, they are cumbersome to use as they sta

Approximate expressions correct to the second order are as follows

$$\Gamma_x = -\frac{\bar{v}l_c}{3}\left(\frac{\partial n}{\partial x}\right)_0$$

$$\Gamma_y = -\frac{\bar{v}l_c}{3}\left(\frac{\partial n}{\partial y}\right)_0$$

$$\Gamma_z = -\frac{\bar{v}l_c}{3}\left(\frac{\partial n}{\partial z}\right)_0$$

From these it is obvious that if we define

$$\begin{aligned}\boldsymbol{\Gamma} &= \boldsymbol{\Gamma}_x + \boldsymbol{\Gamma}_y + \boldsymbol{\Gamma}_z \\ &= -\frac{\bar{v}l_c}{3}\left[\left(\frac{\partial n}{\partial x}\right)_0 \mathbf{I} + \left(\frac{\partial n}{\partial y}\right)_0 \mathbf{J} + \left(\frac{\partial n}{\partial z}\right)_0 \mathbf{K}\right] \qquad (2.15) \\ &= -\frac{\bar{v}l_c}{3}(\bar{\nabla}n)_0 = -D\bar{\nabla}n_0\end{aligned}$$

Therefore, we obtain the result that D, the diffusion constant, is equal to $\bar{v}l_c/3$

The equation we have just derived applies to steady-state conditions, that is, we have explicitly stated that the electron density is time independent Time-dependent solutions are now obtained by letting the electron density be a very gradual function of time A net accumulation or depletion of electrons occurs in a given volume element if the current flowing in is not balanced by the current flowing out Let us choose a volume element centered at $(x_0\ y_0, z_0)$ with sides dx, dy, and dz, as indicated in Fig 2 4

For convenience let us consider the y component of the flux Γ_y The number of particles flowing through the area $dAy_0 = dx\,dz$ into the volume element $dV = dx\,dy\,dz$ is $\Gamma_{y_0}\,dAy_0$ The number of particles flowing through $dA(y_0 + dy) = dx\,dz$ out of the volume element is $\Gamma_{y_0+dy}\,dx\,dz$ The net accumulation of particles per unit time is

$$\frac{d(n\,dV)}{dt} = (\Gamma_{y_0})\,dx\,dz - (\Gamma(y_0 + dy))\,dx\,dz$$

$$\frac{dn}{dt}\,dV = (\Gamma_{y_0} - \Gamma_{y_0+dy})\,dx\,dz$$

We now assume that Γ_y can be written as a Taylor series expansion about y_0 If we keep only first-order terms, the expression for dn/dt

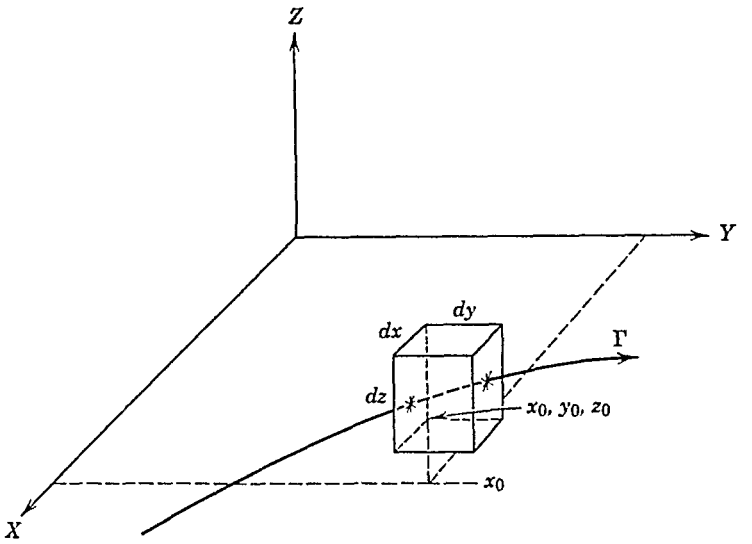


Fig. 2.4

simplifies considerably:

$$\frac{dn}{dt}\,dx\,dy\,dz = \left(\Gamma_{y_0} - \Gamma_{y_0} - \frac{\partial \Gamma}{\partial y}\,dy\right) dx\,dz$$

$$\frac{dn}{dt} = -\left(\frac{\partial \Gamma}{\partial y}\right)_{y_0}.$$

The negative sign is physically valid; if $(\partial\Gamma/\partial y)_0$ is positive, then $\Gamma_{y_0+dy} > \Gamma_{y_0}$ and the net flow is outward resulting in a time decrease of n. Similar expressions can be written for the x and z components from which it should be obvious that the rate of change in the electron density owing to the electron flow is given by the negative of the gradient of the current density

$$\frac{dn}{dt} = -\nabla \cdot \mathbf{\Gamma}.$$

The electron density can also be changed by the presence of sources or sinks, and a term Q is added to the flow term giving the continuity equation

$$\frac{dn}{dt} = -\nabla \cdot \mathbf{\Gamma} + Q. \tag{2.16}$$

The final form of the time-dependent diffusion equation is realized by substituting the result $\mathbf{\Gamma} = -D\,\overline{\nabla} n$ into this equation:

$$\frac{dn}{dt} = D\nabla^2 n + Q. \tag{2.17}$$

In the discussion of free diffusion of electrons in a background gas of uniform density N, we have assumed that the only type of interaction in the entire system is the collisional short-range interaction between the electrons and the atoms only In an actual plasma, ions as well as electrons will exist along with the neutral atoms, and hence there will be long-range Coulomb interactions between the charged particles Also, the density of the atoms of the background gas may not be uniform In such a case it is obvious that the diffusion coefficient will be a function of position In these cases, one should keep the diffusion coefficient within the gradient operator since $\nabla Dn \neq D \nabla n$ If one is careful, the result of this analysis could be modified to suit the physics of any given diffusion problem

Motion of a Charged Particle on a Magnetic Field

The equation of motion for a charged particle in a magnetic field is given by Newton's second law

$$m \frac{d^2\mathbf{r}}{dt^2} = q\left(\frac{d\mathbf{r}}{dt} \times \mathbf{B}\right)$$

We assume that the magnetic field, B, is uniform and directed parallel to the z axis The three scalar equations of motion are

$$mx = eyB, \qquad my = -exB, \qquad mz = 0$$

or

$$x = \omega_b y, \qquad y = -\omega_b x, \qquad z = 0$$

where

$$\omega_b = \frac{qB}{m} = \frac{eB}{m} \tag{2 18}$$

is called the "cyclotron" frequency when $q = e$ is the electronic charge The sign convention we will use makes Eq (2 18) positive for electrons and negative for positive ions Cyclotron resonance will then occur for an electric field rotating clockwise around the magnetic field when electrons are responsible

These equations can be integrated to find that the motion along the magnetic field is unaffected by the magnetic field, whereas the motion in the plane perpendicular occurs with the frequency ω_b The analytical result is

$$\begin{aligned}
v_x &= v_x(0) \cos \omega_b t + v_y(0) \sin \omega_b t \\
v_y &= v_y(0) \cos \omega_b t - v_x(0) \sin \omega_b t \\
v_z &= v_z(0) \\
x &= x(0) + 1/\omega_b[v_x(0) \sin \omega_b t + v_y(0)(1 - \cos \omega_b t)] \\
y &= y(0) + 1/\omega_b[v_y(0) \sin \omega_b t - v_x(0)(1 - \cos \omega_b t)] \\
z &= z(0) + v_z(0)t
\end{aligned}$$

With these results we may now proceed to consider the effects of the magnetic field on diffusion.

In the macroscopic approach to diffusion and the action of electromagnetic fields, Newton's second law is written down for the average electron and the equation is solved for the motion. The implied assumption is that the average properties used in the equation of motion are the "proper" averages, that is, they are the averages that yield results compatible with the results of the microscopic approach,

If we consider only uniform fields, we do not have to make a distinction between the drift velocity, the average velocity of the particles contained in a small volume V, and the time average of the velocity of an individual particle.

A heuristic approach to this problem is now considered. First, we restrict ourselves to steady-state problems, thus eliminating $\mathbf{dv}/dt$. We then naïvely consider the forces on the particles to be composed of: (a) the external forces, (b) the pressure gradient forces, and (c) the rate of change of the average momentum due to collisions. Thus it is seen that this approach is not simply a fluid dynamic approach but is actually the equation of motion for a hypothetical "average" particle. Using these assumptions, we write Newton's second law as

$$mv_c\mathbf{v} = \mathbf{F} - \frac{\nabla p}{n}. \tag{2.19}$$

For the case in point, we have

$$\mathbf{F} = e\mathbf{v} \times \mathbf{B}. \tag{2.20}$$

From kinetic theory, we have

$$p = \tfrac{1}{3}nm\langle v^2\rangle. \tag{2.21}$$

We now combine Eqs. (2.19), (2.20), and (2.21) to obtain:

$$\mathbf{v} = \frac{1}{mv_c}\left\{e\mathbf{v} \times \mathbf{B} - \tfrac{1}{3}m\langle v^2\rangle\frac{\nabla n}{n}\right\} \tag{2.22}$$

$$\mathbf{v} = \frac{e}{mv_c}\mathbf{v} \times \mathbf{B} - \frac{\langle v^2\rangle}{3v_c}\frac{\nabla n}{n}$$

$$\mathbf{v} = \frac{1}{v_c}(\boldsymbol{\omega}_b \times \mathbf{v}) - D\frac{\nabla n}{n}$$

where

$$D = \frac{\langle v^2\rangle}{3v_c}$$

so that

$$-D\frac{\nabla n}{n} = \left[-\frac{\boldsymbol{\omega}_b}{v_c}x + 1\right]\mathbf{v}. \tag{2.23}$$

In the above steps we have used $\omega_b = -eB/m$ We again assume that the magnetic field is along the z direction and write Eq (2 23) in matrix form [2]

$$-\frac{D}{n}\begin{Bmatrix}\partial n/\partial x\\ \partial n/\partial y\\ \partial n/\partial z\end{Bmatrix} = \begin{bmatrix}1 & \omega_b/\nu_c & 0\\ -\omega_b/\nu_c & 1 & 0\\ 0 & 0 & 1\end{bmatrix}\begin{Bmatrix}v_x\\ v_y\\ v_z\end{Bmatrix} \tag{2 24}$$

To find the desired relation for Γ in terms of ∇n, we must invert the above matrix equation We find

$$D = \frac{\langle v^2\rangle}{3}\begin{vmatrix}\dfrac{\nu_c}{\nu_c^2+\omega_b^2} & \dfrac{\omega_b}{\nu_c^2+\omega_b^2} & 0\\ \dfrac{-\omega_b}{\nu_c^2+\omega_b^2} & \dfrac{\nu_c}{\nu_c^2+\omega_b^2} & 0\\ 0 & 0 & \dfrac{1}{\nu_c}\end{vmatrix} \tag{2 25}$$

To clarify the physics of this matrix, we write Eq (2 25) in the form

$$D = \frac{\langle v^2\rangle}{3}\begin{vmatrix}D_\top & D_\perp & 0\\ -D_\perp & D_\top & 0\\ 0 & 0 & D_\parallel\end{vmatrix}$$

Here the term $D_\parallel$ is the coefficient for diffusion parallel to the direction of the magnetic field, and is clearly unaffected by B $D_\top$ and $D_\perp$ could be called the transverse and perpendicular diffusion coefficients They lie in a plane perpendicular to the direction of the magnetic field and are obviously modified by it $D_\perp$ points in the direction of the Hall effect and for this reason is sometimes called Hall diffusion

Let us now discuss the diffusion tensor in the limits of high and low magnetic fields

When $\omega_b \gg \nu_c$, we can see from Eq (2 25) that $D_\top$ varies as $1/\omega_b^2$ and hence as $1/B^2$ This is experimentally important, of course, since it allows one to inhibit radial diffusion of a plasma column by means of an axial magnetic field It is so difficult to arrange an experiment to test this conclusion without introducing space charge complications that it is doubtful whether this free diffusion coefficient has ever been measured by itself Hence, the transverse diffusion in strong magnetic field is subject to active study

With this same assumption of $\omega_b \gg \nu_c$, if one keeps B constant, $D_\top \sim \langle v^2\rangle/\omega_b^2 \sim \varepsilon^2 m^2$, leading to the somewhat surprising result that ions will

diffuse faster than electrons across the B field although slower than electrons parallel to the magnetic field.

When we have small magnetic fields that imply small ω_b, we can neglect the off-diagonal terms in the diffusion tensor in Eq. (2.25), that is,

$$\frac{\omega_b}{\omega_b^2 + \nu_c^2} \ll \frac{\nu_c}{\omega_b^2 + \nu_c^2} \qquad \text{for } \omega_b \ll \nu_c.$$

We then have a symmetric diffusion tensor[3]

$$(D)_{B \text{ small}} = \frac{\langle v^2 \rangle}{3} \begin{bmatrix} \dfrac{\nu_c}{\nu_c^2 + \omega_b^2} & 0 & 0 \\ 0 & \dfrac{\nu_c}{\nu_c^2 + \omega_b^2} & 0 \\ 0 & 0 & \dfrac{1}{\nu_c} \end{bmatrix}. \tag{2.26}$$

The effective diffusion length corresponds to a solution of the diffusion equation of the form

$$n = n_0(x, y, z)e^{-Dt/\Lambda_b^2}.$$

We can conveniently think of the effect of the magnetic field as increasing the size of the container in the directions perpendicular to the field. A cavity with a diffusion length which is normally Λ, where

$$\frac{1}{\Lambda^2} = \frac{1}{\Lambda_x^2} + \frac{1}{\Lambda_y^2} + \frac{1}{\Lambda_z^2},$$

will, when a magnetic field **B** is impressed along the z axis, have an effective diffusion length Λ_b that is given by

$$\frac{1}{\Lambda_b^2} = \left(\frac{\nu_c^2}{\nu_c^2 + \omega_b^2}\right)\left(\frac{1}{\Lambda_x^2} + \frac{1}{\Lambda_y^2}\right) + \frac{1}{\Lambda_z^2}. \tag{2.27}$$

The effect of the magnetic field is equivalent to expanding the cavity in the ratio

$$\left(\frac{\nu_c^2 + \omega_b^2}{\nu_c^2}\right)^{1/2}$$

in all directions perpendicular to the magnetic field.

In considering some examples of Eq. (2.27) when applied to specific cases, we shall first discuss a cylindrical cavity whose height is greater than the radius and where the axis of the cylinder and the magnetic fields are in the same direction. Here we have

$$\frac{1}{\Lambda_b^2} = \frac{\nu_c^2}{\nu_c^2 + \omega_b^2}\left(\frac{2.405}{R}\right)^2 + \left(\frac{\pi}{L}\right)^2. \tag{2.28}$$

If we take a flat cylindrical cavity with the magnetic field transverse to the axis, the solution of the diffusion equations comes out in terms of an elliptical cylinder whose diffusion length is given by

$$\frac{1}{\Lambda_b^2} = \frac{\nu_e^2}{\nu_e^2 + \omega_b^2}\left[\left(\frac{\pi}{L}\right)^2 + \frac{1}{2}\left(\frac{2.405}{R}\right)^2\right] + \frac{1}{2}\left(\frac{2.405}{R}\right)^2 \qquad (2.29)$$

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Chapter 3

Mobility

Electrons

In the presence of collisions, Newton's second law may be written,

$$m\frac{dv}{dt} + (m\nu_m)v = -eE_0e^{j\omega t} \tag{3.1}$$

and the velocity is

$$v = \left(\frac{-e/m}{j\omega + \nu_m}\right)E \tag{3.2}$$

where $E = E_0e^{j\omega t}$. We define the proportionality constant between v and E as the mobility, μ, so that

$$\mu = \frac{e/m}{j\omega + \nu_m}\,. \tag{3.3}$$

In the case of dc

$$\mu = \frac{e}{m\nu_m}\,. \tag{3.4}$$

In gases where the collision frequency is independent of energy, $\nu_m =$ (constant) p, the electron drift velocity will be directly proportional to E/p. This approximation is very good for hydrogen and helium, and the agreement is illustrated for hydrogen in Fig. 3.1.

A large fraction of the early data on electron mobilities came from the studies of Bradbury and Nielsen,[1] who used an electron-shutter method. The principle of the method can be described with reference to Fig. 3.2. Photoelectrons emitted from plate P are accelerated by the electric field maintained between P and A and because of collisions quickly attain a

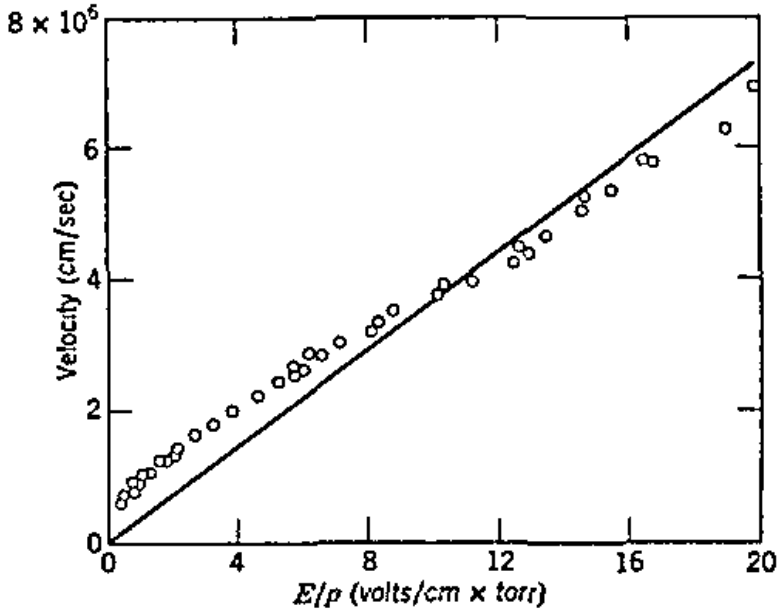


Fig. 3.1 Electron drift velocity in hydrogen

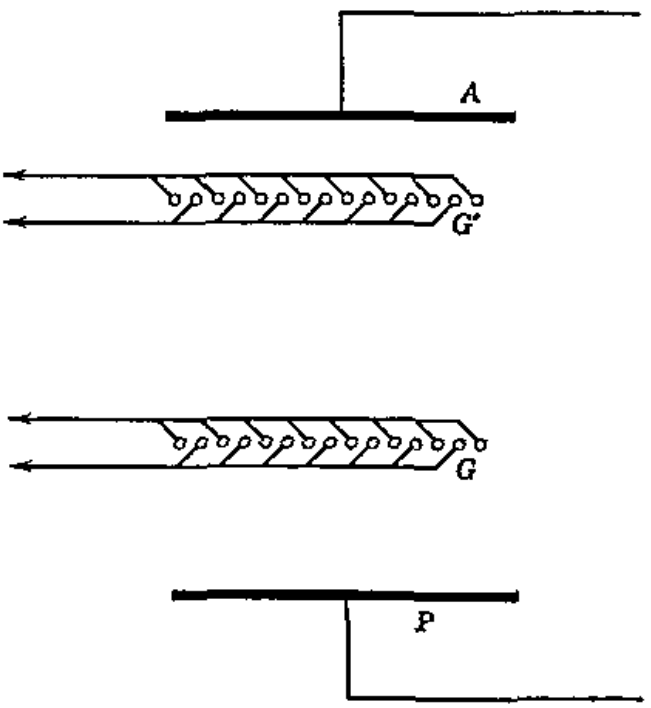


Fig. 3.2 Sketch of the ac shutter method of measuring electron mobilities

constant drift velocity. In passing through the tube, which contains the gas under consideration, the electrons pass through two sets of grids G and G'. Alternate wires of these grids are connected to an ac potential of such a magnitude that only those electrons that arrive at the grids when the electric field passes through zero are transmitted. The electron bunch that is thus allowed to pass into the uniform field region between the grids drifts across the space. The second grid is driven in phase with the first grid so that only if the electron transit time is equal to one or more full half-cycles of the grid frequency will current be collected at A. Thus when we know the distance between the grids and the frequency at which

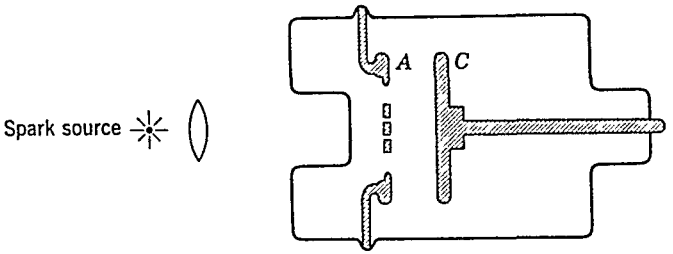


Fig. 3.3 Apparatus for measuring electron mobilities.

current is observed, the drift velocity of the electrons may be determined. Typical results from this method are shown by the data points in Fig. 3.1.

Another method of measuring electron mobilities is illustrated by the studies of Hornbeck.[2] A diagram of his apparatus is shown in Fig. 3.3. Light from a spark source falls on plate C, releasing a very short burst of photoelectrons, and they move under the action of an electric field to plate A. With the sudden creation of the electrons by the external light pulse, the current rises abruptly from zero and remains constant, while the packet of electrons crosses from cathode to anode. The current drops discontinuously to zero at one electron transit time, except for diffusion effects and the finite time duration of the light pulse. The electron velocity may be calculated from the distance between C and A and the transit time by using either of the following equations:

$$v = \frac{x}{t} \quad \text{or} \quad v = \frac{ix}{A}$$

where A is the integrated area under the initial photoimpulse, i is the current, and t is the transit time. Hornbeck's results are given in Fig. 3.4.

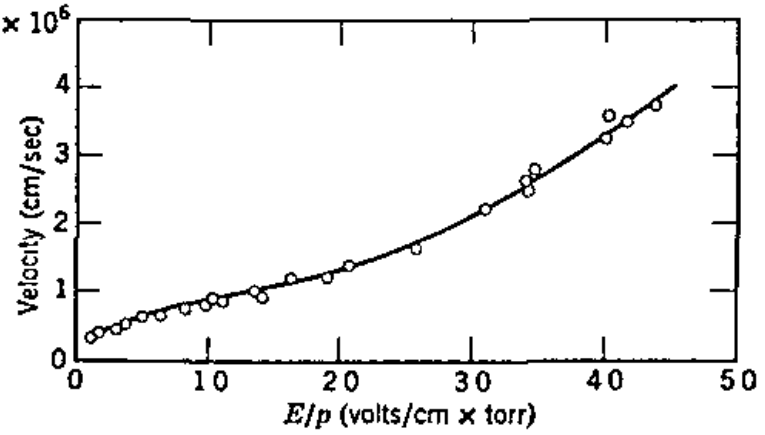


Fig 3 4 Drift velocity of electrons in helium as a function of E/p

The Boltzmann Equation

Although deriving the equation for the mobility by the simple calculation from Newton's second law gives a straightforward definition of the mobility, as in Eq (3 3), it is obvious that this useful "average electron" picture could be greatly improved by a kinetic theory approach The usual starting point for kinetic theory calculations is the Boltzmann equation It describes the effect of applied forces and collisions on the velocity distribution function F of the particles Basically it has the form of a continuity equation for the flow of particles in phase space and is written as

$$\frac{\partial F}{\partial t} + \mathbf{v} \cdot \nabla_r F + \frac{e\mathbf{E}}{m} \cdot \nabla_v F = \left(\frac{\partial F}{\partial t}\right)_{\text{coll}} \tag{3 5}$$

where ∇_r is the gradient operator in three-dimensional configuration space and ∇_v the equivalent operator in velocity space

If we assume that there is no spatial variation in F and that the electric field is in a single direction, Eq (3 5) can be simplified to be

$$\frac{\partial F}{\partial t} + \frac{eE}{m}\frac{\partial F}{\partial v} = \left(\frac{\partial F}{\partial t}\right)_{\text{coll}} \tag{3 6}$$

Writing the electric field as

$$E = E_0 + E_1(j\omega t)$$

and assuming $E_1 \ll E_0$, we see that E_1 produces a negligible perturbation of the velocity distribution established by the stationary field E_0 We may therefore write the total distribution function as the sum of the pseudo-stationary state distribution, f_0, and a perturbation term f_1

$$F = f_0 + f_1$$

where f_0 is considered to be the distribution function for small dc fields, is spherically symmetrical and is nearly Maxwellian.

Rewriting Eq. (3.6) with the second-order terms eliminated, we find

$$\frac{\partial f_1}{\partial t} + \frac{eE_0}{m}\frac{\partial f_0}{\partial v} + \frac{eE_1}{m}\frac{\partial f_0}{\partial v} = \left(\frac{\partial f_0}{\partial t}\right)_{\text{coll}} + \left(\frac{\partial f_1}{\partial t}\right)_{\text{coll}}. \tag{3.7}$$

Separating this into two equations in E_0 and E_1, we find

$$\frac{eE_0}{m}\frac{\partial f_0}{\partial v} = \left(\frac{\partial f_0}{\partial t}\right)_{\text{coll}}$$

$$\frac{\partial f_1}{\partial t} + \frac{eE_1}{m}\frac{\partial f_0}{\partial v} = \left(\frac{\partial f_1}{\partial t}\right)_{\text{coll}} = -\nu_m(v) f_1. \tag{3.8}$$

Turning our attention to the second of these equations and assuming f_1 has a time variation $e^{j\omega t}$, we may write

$$j\omega f_1 + \frac{eE_1}{m}\frac{\partial f_0}{\partial v} = -\nu_m(v) f_1$$

$$f_1 = -\frac{-\left(\frac{e}{m}\right)E_1\frac{\partial f_0}{\partial v}}{j\omega + \nu_m(v)}.$$

The conductivity is

$$\sigma = \frac{J}{E_1} = \frac{ne}{E_1}\int vF\, d^3v$$

and substituting our values of E_1 and F, we find

$$\sigma = -\frac{ne^2}{m}\,4\pi\int_0^\infty \frac{v^3\frac{\partial f_0}{\partial v}\,dv}{j\omega + \nu_m(v)}. \tag{3.9}$$

If ν_m is independent of velocity,

$$\sigma = \frac{ne^2}{m}\frac{1}{j\omega + \nu_m} \tag{3.10}$$

in agreement with Eq. (3.3).

Average Energy of Electrons

Since both mobility and diffusion are collision-controlled phenomena, there should be an intimate connection between them. As a matter of fact, the ratio of the diffusion coefficient to the dc mobility of electrons is an important quantity as it measures the average electron energy.

Since $D = v^2/3\nu_m$ and $\mu = e/m\nu_m$,

$$\frac{D}{\mu} = \frac{2}{3e}(\tfrac{1}{2}mv^2) = \tfrac{2}{3}u_{ave} \tag{3 11}$$

where u_{ave} is the average electron energy in volts. The numerical factor of $\frac{2}{3}$ results from the type of averaging used and is correct for a Maxwellian velocity distribution.

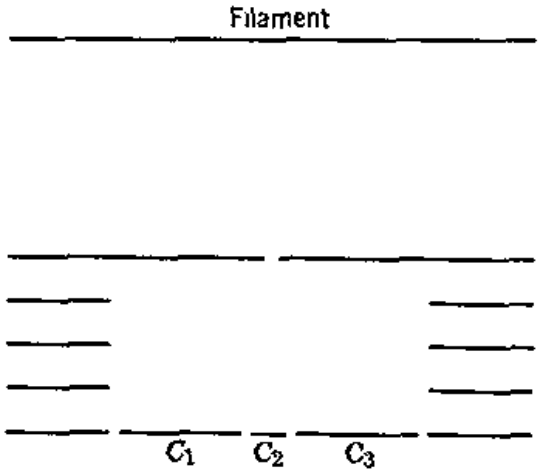


Fig 3 5 Schematic diagram of Townsend's method for measuring mobility and diffusion

If we write the particle flow equation for electrons including both diffusion and mobility, we have

$$\Gamma = -D\,\nabla n - n\mu\mathbf{E} \tag{3 12}$$

When we assume that the electric field is applied in the z direction and that a steady-state condition prevails, $\partial n/\partial t = 0$, the continuity equation gives

$$\nabla^2 n = -\frac{\mu}{D}E\frac{\partial n}{\partial z} \tag{3 13}$$

Townsend[3] designed an experiment to determine μ/D for which he was able to solve this differential equation for the geometric conditions of his apparatus. He used the apparatus sketched in Fig 3 5. In the absence of any magnetic field, the beam collected at $C_1C_2C_3$ will be spread out by the diffusion of the electrons while they are moving under the action of the electric field E. This beam diffusion may be correlated with the experimental observations by solving the differential equation in terms of $\mu E/D$ and R, where $R = C_2/(C_1 + C_2 + C_3)$ and the C's correspond to the currents drawn by the plates shown in Fig 3 5. The solution is shown

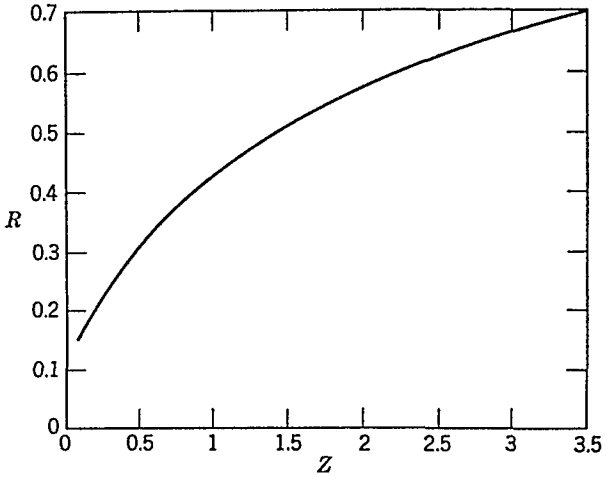


Fig. 3.6 Townsend's solution to Eq. (3.13).

graphically[4] in Fig. 3.6 as a plot of R versus Z, where $Z = \mu E/40.3D$, the numerical factor being introduced so that the abscissa may represent the quantities in the usual units. Thus, when the value of the applied electric field E and the gas pressure p are known, the average energy is determined as a function of E/p. The results of this type of measurement are shown for electrons in hydrogen in Fig. 3.7.

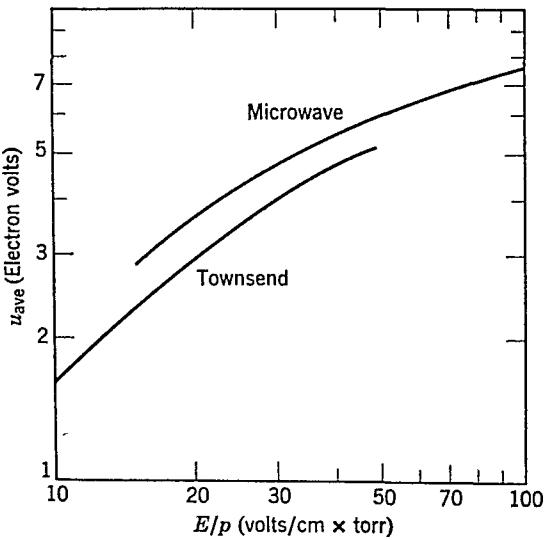


Fig. 3.7 Average energy of electrons in hydrogen.

Another method of determining the average electron energy is by measuring a microwave gas discharge breakdown modified by an applied dc field [5] A gas will break down when the losses of electrons to the walls of a container are replaced by ionization in the body of the gas When an ac field alone is applied, electrons are lost by diffusion When a small dc sweeping field is applied, electrons are lost both by diffusion and mobility, thus changing the breakdown condition In this case the flow of electrons is given by

$$\Gamma = -D\,\nabla n - n\mu \mathrm{E}_{\mathrm{dc}} \tag{3 14}$$

When the electrons lost are replaced by new ones resulting from ionization,

$$\nabla \cdot \Gamma = \nu_i n \tag{3 15}$$

If the applied dc field is too small to affect appreciably the frequency of ionization ν_i, and E_{dc} is directed along the z axis, the density at breakdown is the solution of

$$\nabla^2 n = -\frac{\mu}{D} E_{\mathrm{dc}} \frac{\partial n}{\partial z} - \frac{\nu_i}{D} n \tag{3 16}$$

This equation may be solved for the case of a cylinder of axial height L and axial coordinate z, and radius R and radial coordinate r, by separation of variables to yield

$$n = \text{constant}\, J_0\left(\frac{2\,4r}{R}\right)\left[\sin\frac{\pi}{L}z\right]\exp\left(-\frac{\mu}{D}\frac{E_{dc}}{2}z\right) \tag{3 17}$$

where J_0 is the zero order Bessel function This solution is subject to the breakdown conditions

$$\frac{\nu_i}{D} = \frac{1}{\Lambda_{\mathrm{dc}}{}^2} \tag{3 18}$$

where Λ_{dc} defines a modified diffusion length

$$\frac{1}{\Lambda_{\mathrm{dc}}{}^2} = \frac{1}{\Lambda^2} + \left(\frac{\mu}{D}\frac{E_{\mathrm{dc}}}{2}\right)^2, \tag{3 19}$$

and Λ is the characteristic diffusion length previously defined for cylindrical geometry The only difference between the breakdown condition in the ac-dc case and the pure ac case is the substitution of a modified diffusion length Λ_{dc} for the characteristic diffusion length Λ Having found $1/\Lambda_{dc}$ experimentally and knowing the geometrical $1/\Lambda$, we can calculate D/μ from Eq (3 19) The experimental determinations of the average electron energy u_{ave} in hydrogen by Townsend and Bailey[4] by the Townsend method, and by Varnerin and Brown[5] by the microwave method, are both given in Fig (3 7)

Ions

On the average, an ion moves with a constant drift velocity so long as the velocity gained from the field is small compared to the random velocity resulting from the thermal motion of the ions themselves. Thus the ion mobility is defined as

$$\mu_i = \frac{e}{M_i \nu_c} . \tag{3.20}$$

Since the collision frequency $\nu_c = vpP_c$ and the mean free path $l = 1/p_0 P_c$, the average drift velocity may be written as

$$\mu \sim \frac{el}{M_i v_r} \tag{3.21}$$

where v_r is the average random velocity and the constant of proportionality results from the type of averaging. This particular formulation comes from assuming perfectly elastic collisions and equal masses for ion and atom. Langevin[6] derived a more exact equation for mobility taking into account the momentum changes on impact. He also took care of the fact that after a collision the ion may have an initial velocity in the direction of the electric field. In terms of unequal masses of ions and atoms he obtained

$$\mu \sim \frac{el}{M_i v_{ri}} \left(1 + \frac{M_i}{M_g}\right)^{1/2} . \tag{3.22}$$

Since the kinetic energy of the gas atoms and ions are equal, Eq. (3.22) may also take the form

$$\mu \sim \frac{el}{M_g v_{rg}} \left(1 + \frac{M_g}{M_i}\right)^{1/2} . \tag{3.23}$$

The numerical factor arises from the particular process of averaging. Equation (3.23) predicts the variation of μ with mass as

$$\mu \sim \left(1 + \frac{M_g}{M_i}\right)^{1/2} . \tag{3.24}$$

This is shown to be the case[7] in Fig. 3.8.

Although the form of Eq. (3.23) predicts the variation of μ with mass correctly, it does not predict its magnitude. This led Langevin to carry his theoretical work further and assume that a gas atom could be polarized by an ion when the two approached each other. This assumes that there can occur a momentum exchange between ion and atom caused by the induced dipole in the absence of a close impact.

To calculate the effect in terms of the dielectric coefficient, let us review briefly the physics of the polarization of a dielectric. Assume a parallel

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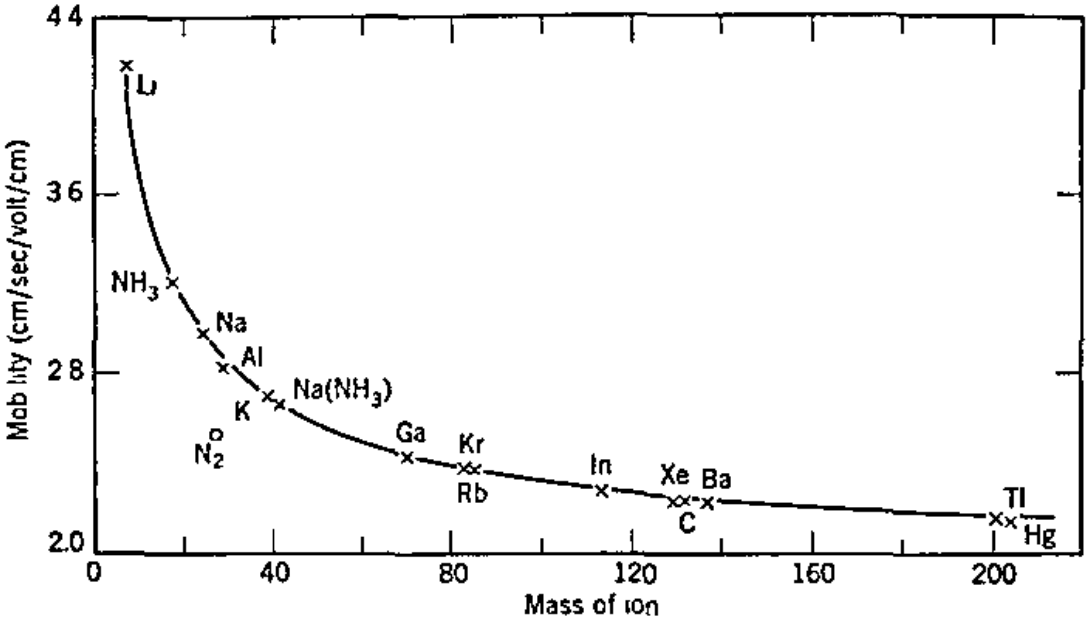


Fig 3 8 Variation of mobility with ionic mass, at 1 atmosphere pressure and room temperature

plate geometry with plates of area A, surface charge density σ separated by a distance d In the absence of a dielectric the field between the plates is $E = 4\pi\sigma$ Introducing between the plates a dielectric of coefficient K reduces the field to the value $E' = E/K$ The surface charge density induced on the dielectric is then σ' and

$$E - E' = 4\pi\sigma'$$

$$\frac{E - E'}{E'} = \frac{4\pi\sigma}{E'} = K - 1$$

The surface charge density is just equal to the magnitude of the polarization vector, P, of the dielectric

$$\sigma' = \frac{(K-1)E'}{4\pi} = P$$

If N is the number of molecules per cubic centimeter, the dipole moment per molecule

$$\frac{P}{N} = \frac{(K-1)E'}{4\pi N}$$

which in a Coulombian field $E' = e/r^2$ may be written as

$$\frac{P}{N} = \frac{(K-1)}{4\pi N}\frac{e}{r^2}$$

The force with which a dipole moment is attracted by a point charge e at a distance r is

$$F = \frac{2Pe}{Nr^3} = \frac{(K-1)e^2}{2\pi N r^5}$$

and the potential energy is

$$V = -\frac{(K-1)e^2}{8\pi N r^4}. \tag{3.25}$$

Langevin, in developing his theory, assumed solid elastic sphere impacts for repulsive forces and the inverse fifth-power attractive polarization force to obtain an expression for mobility in the form

$$\mu = \frac{g(\lambda)}{\sqrt{\rho(K-1)}}\left(1 + \frac{M_g}{M_i}\right)^{1/2} \tag{3.26}$$

where

$$\lambda^2 = \frac{8\pi p\sigma^4}{(K-1)e^2}. \tag{3.27}$$

p is pressure and σ is the sum of the radii of the ions and gas atoms. The term $g(\lambda)$ is calculated[7] from λ by taking into account all possible orbits involved in solid elastic impacts, the actual calculation of which is a monumental labor which involves such integrals as:

$$\int_0^\infty y e^{-z^2} z^4\, dz$$

where

$$y = \int_0^\infty \cos^2\theta\, d(\beta)^2$$

where $\pi - 2\theta$ is the angle through which the relative paths of ion and atom are deflected in the encounter and β^2 is a variable connected to θ by an equation obtained from the dynamics of the encounter.

The z term involves the momentum and energy conservation in the collision in terms of the masses and their relative velocity v before the encounter such that

$$z = \frac{v}{\sqrt{2kT}}\left(\frac{M_i M_g}{M_i + M_g}\right)^{1/2}.$$

The form of the relation between $g(\lambda)$ and λ is shown in Fig. 3.9.

To emphasize the physical nature of λ, let us write $p = NkT$ and rewrite Eq. (3.27)

$$\lambda^2 = \frac{8\pi N\sigma^4}{(K-1)e^2} kT = \frac{KE}{PE}.$$

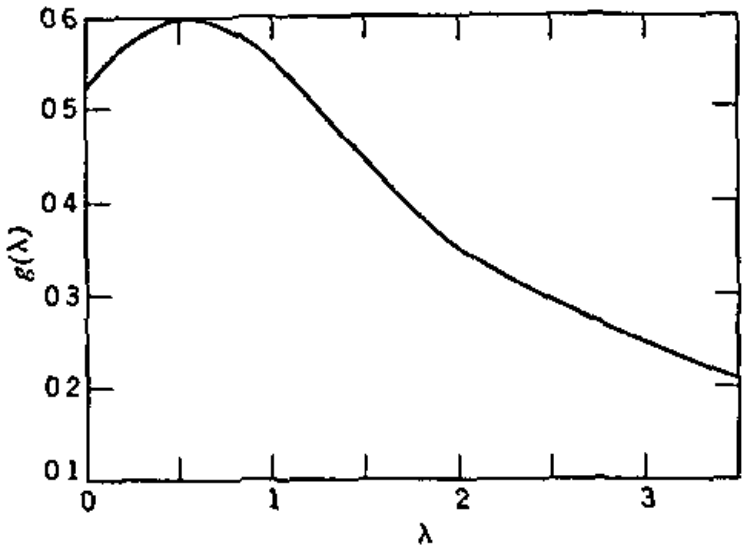


Fig 3 9 The calculation of $g(\lambda)$ as a function of λ

Thus λ^2 can be considered as the ratio of the ion kinetic energy to the energy of attraction when the ions and atoms are in contact

As one can see from Eq (3 27), λ measures the relative importance of elastic sphere and polarization scattering Thus as K increases and/or σ decreases λ approaches zero, and, as seen in Fig 3 9, $g(\lambda)$ approaches 0 51

Thus the polarization or small ion limit of Eq (3 11) becomes

$$\mu = \frac{0\,51}{\sqrt{\rho(K-1)}}\left(1+\frac{M_g}{M_i}\right)^{1/2} \tag{3 28}$$

In the case where the polarization effect is negligible compared to the elastic hard sphere collision, it is useful to write Eq (3 26) eliminating the polarization term From Eq (3 27)

$$\sqrt{K-1} = \frac{\sqrt{8\pi p}}{\lambda}\frac{\sigma^2}{e}$$

so that

$$\mu = \frac{\lambda g(\lambda)}{\sqrt{8\pi p\rho}}\frac{e}{\sigma^2}\left(1+\frac{M_g}{M_i}\right)^{1/2}$$

The form of $\lambda\, g(\lambda)$ as a function of λ is given in Fig 3 10 From here we see that as $\lambda \to \infty$, $\lambda\, g(\lambda) \to 0\,75$ Thus for the elastic sphere limit

$$\mu = \frac{0\,75e}{\sqrt{8\pi p\rho\sigma^2}}\left(1+\frac{M_g}{M_i}\right)^{1/2} \tag{3 29}$$

Based on the Boltzmann transport equation, G H Wannier[8] developed a theory considering both rigid sphere repulsion and polarization attraction For the rigid sphere case he calculated isotopic scattering in the

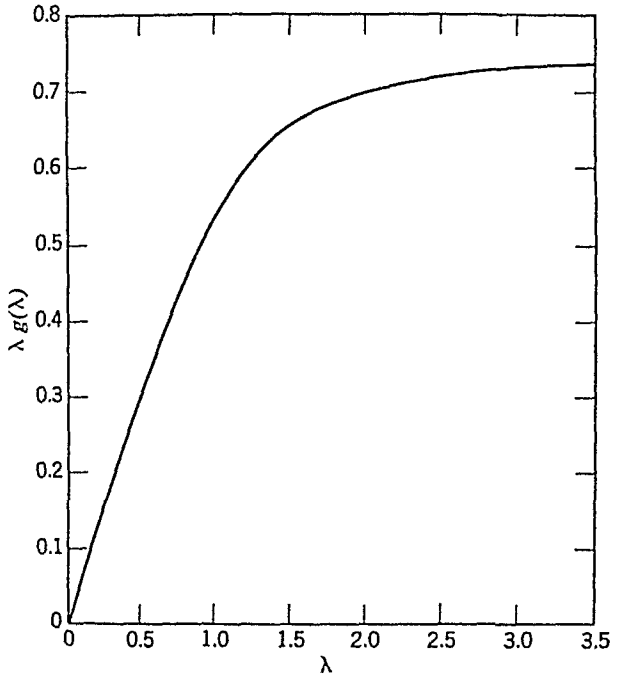


Fig. 3.10 The calculation of $\lambda g(\lambda)$ as a function of λ.

center-of-mass system using constant mean free path and constant collision cross-section assumptions. In the case of polarization attraction he found the proper assumption to be constant mean free time. He concluded that when the mobility is controlled by the polarization force, μ is proportional to $1/p$ so that at low E/p, v should be proportional to E/p. At high values of E/p, the energy gained between collisions is proportional to the kinetic energy of the ion and hence the drift velocity is proportional to $(E/p)^{1/2}$. These variations are illustrated in Fig. 3.11 for neon ions in neon.[2]

The elastic sphere model gives a collision cross section independent of the velocity of approach of the colliding particles. The cross section for polarization attraction is inversely proportional to the velocity of approach, producing a constant mean free time between collisions. Thus from Eq. (3.20), the mobility is independent of E/p for low E/p, but decreases at high E/p as illustrated in Fig. 3.12. To within the accuracy of experimental observation, Frost[9] has shown that for atomic ions of He, Ne, and A in parent gas

$$\mu = \mu^* \left[1 + a\left(\frac{E}{p}\right)\right]^{1/2}. \tag{3.30}$$

Values of μ^* and a are given in Table 3.1, where μ^* is taken at 1 torr and 300°C.

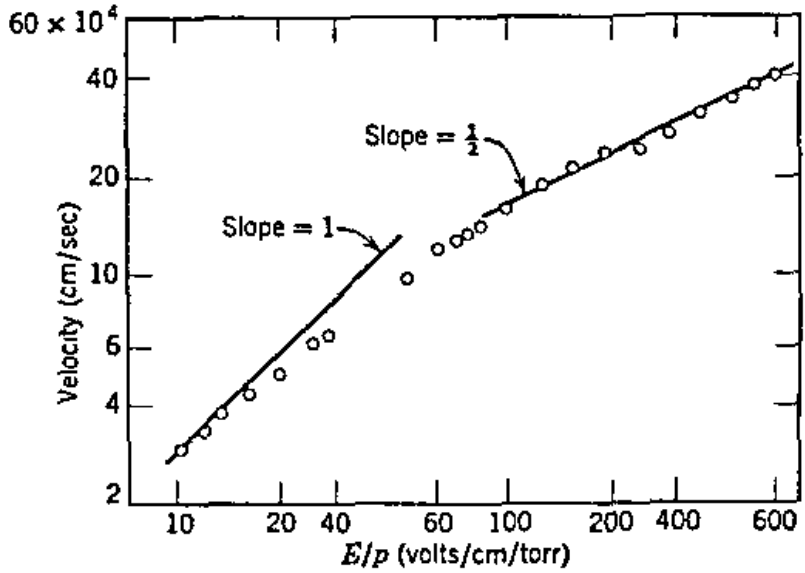


Fig 3 11 Drift velocity of neon ions in neon

One of the useful methods of measuring mobility of ions was worked out by Townsend [10] The apparatus is the same as shown schematically in Fig 3 5 A magnetic field is applied to the apparatus at right angles to the plane of the paper The experimental procedure is to change the value of the magnetic field keeping the electric field E and the pressure in the apparatus constant The magnetic field is first adjusted for a value of $+B$ such that the current to the collector C_3 as measured by the electrometer equals the sum of the current measured by the other two collecting electrodes, $C_3 = C_1 + C_2$ The value of the magnetic field is then reversed and a value $-B$ adjusted so that $C_1 = C_2 + C_3$ The center of the beam

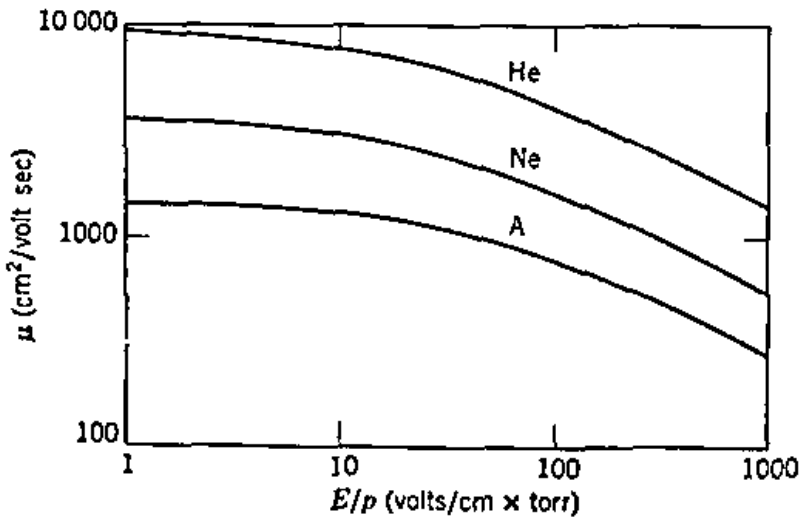


Fig 3 12 Ionic mobility in helium, neon and argon at a pressure of 1 torr and 300°K

Table 3.1. Values of μ^* and a in Eq. (3.15)

Ion	Gas	μ^*	a
He^+	He	9200	0.040
Ne^+	Ne	3500	0.040
A^+	A	1460	0.0264

is then displaced the width of C_2 by the sum of the two fields. If the average of the fields is denoted by B, the beam is deflected a distance of $a/2$ centimeters where a is the width of C_2. If one assumes that the curvature of the beam is small, the angle of deflection of the beam is related to the fields and the drift velocity of the ions as follows: $\tan \theta = a/2d = Bev/Ee$. Thus, by measuring B and E and knowing the deflection, one may determine the mobility constant.

Experimental determinations of ions of a gas moving through the same gas under the action of an electric field were also measured by Hornbeck.[2] His apparatus is again illustrated in Fig. 3.3. The experiment was arranged to cause the release of a 0.1-μsec pulse of photoelectrons from the cathode of the tube and to observe the current transient that follows. Ionization and excitation of the gas molecules occurred by electron impact during the light pulse. Because the electron drift velocity in the field is high compared with the positive-ion drift velocity, an initial transient occurs that is attributable to the initial photoelectrons followed by a smaller current lasting the order of 10 μsec. The slower component originates primarily from the motion of positive ions in the field. Hornbeck's drawing of the idealized transient current as a function of time following the formation at $t = 0$ of an exponential distribution of positive ions between the electrodes is shown in Fig. 3.13. The experiment provides a direct, time-of-flight method for measuring the drift velocity of positive ions in a gas as a function of E/p up to very high E/p.

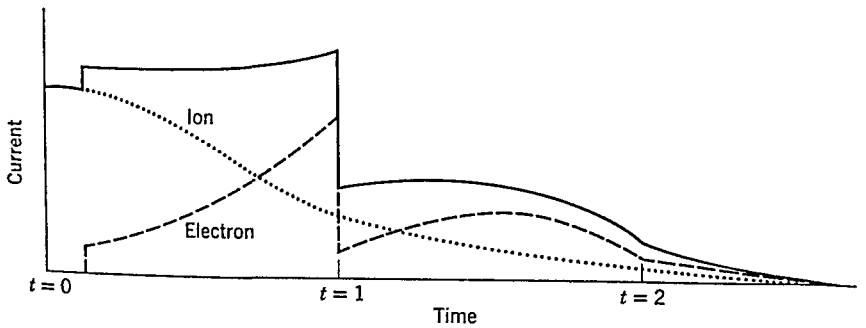


Fig. 3.13 Hornbeck's idealized transient current measurements.

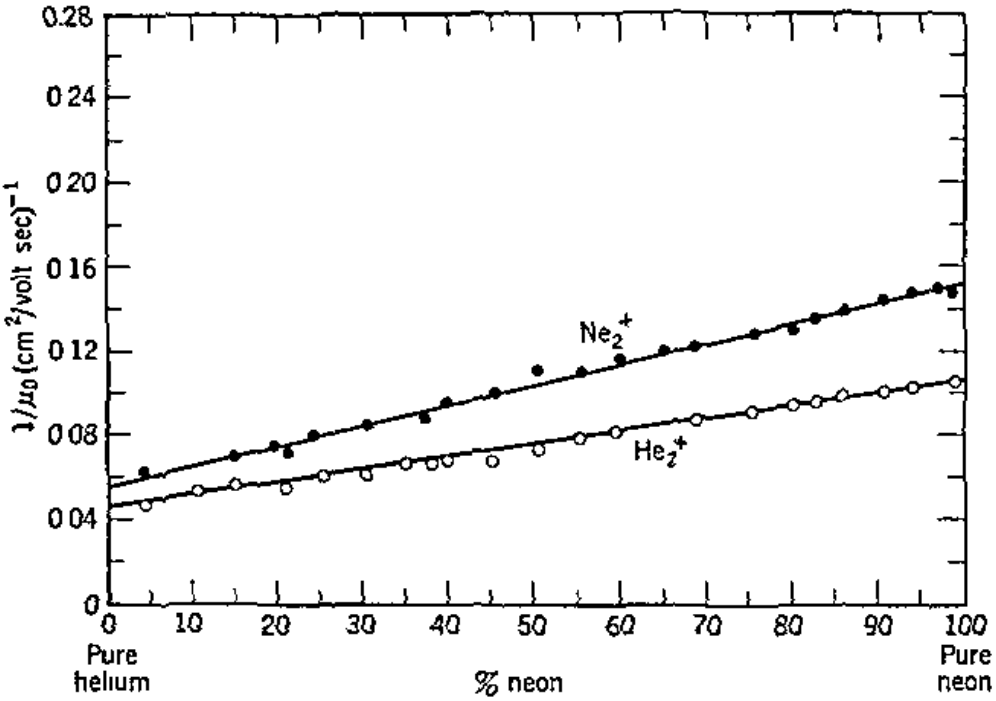


Fig 3 14 Blanc s law plot for Ne_2^+ and He_2^+ ions

The mobility of ions through a gas mixture depends in a straightforward manner on the relative concentrations of the mixed gases

$$\frac{1}{\mu} = \frac{f_1}{\mu_1} + \frac{f_2}{\mu_2} \tag{3 31}$$

where f_1 and f_2 are the fractional concentrations of gas 1 and gas 2 ($f_1 + f_2 = 1$), and μ_1 and μ_2 are the mobilities of the given ionic species in the pure gases The assumption is made that ν_e of each species has the same velocity dependence Since Blanc in 1908 first showed[11] that the reciprocal of the mobility was linearly proportional to the relative concentrations in a binary gas mixture, this relationship is often called *Blanc's law* Recent experimental verification of this relationship has been demonstrated by McDaniel and Crane[12] and by Biondi and Chanin [13] Data from the latter paper are reproduced in Fig 3 14

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Chapter 4

Ambipolar Diffusion

When the density of electrons and ions becomes large enough, their mutual Coulomb fields affect their free diffusion This modification can be discussed in terms of the space charge field E_s The particle flow equation for positive ions may be written as

$$\Gamma_+ = n_+ v_+ = -D_+ \nabla n + n_+ \mu_+ E_s \tag{4 1}$$

which for positive ions and electrons give the velocities as

$$v_+ = -\frac{D_+}{n_+} \nabla n_+ + \mu_+ E_s$$

$$v_- = -\frac{D_-}{n_-} \nabla n_- - \mu_- E_s$$

Eliminating E_s and making the assumptions that $n_+ = n_- = n$, $\nabla n_+ = \nabla n_- = \nabla n$ and $v_- = v_+ = v$

$$v = -\left(\frac{D_+ \mu_- + D_- \mu_+}{\mu_+ + \mu_-}\right) \frac{\nabla n}{n} \tag{4 2}$$

This is the defining equation for the ambipolar diffusion coefficient

$$D_a = \frac{D_- \mu_+ + D_+ \mu_-}{\mu_+ + \mu_-} \tag{4 3}$$

The space charge field E_s may be obtained from Eq (4 1) and

$$\Gamma = -D_a \nabla n$$

and the result is that

$$E_s = -\frac{(D_- - D_+)}{(\mu_- + \mu_+)} \frac{\nabla n}{n} \tag{4 4}$$

The magnitude of the ambipolar diffusion coefficient in terms of free diffusion may be easily determined approximately. Since

$$\frac{D_+}{\mu_+} = \frac{D_-}{\mu_-} = \frac{2}{3} u_{\text{ave}},$$

$$D_a = \frac{D_-\mu_+ + D_+\mu_-}{\mu_+ + \mu_-} = \frac{2D_+\mu_-}{\mu_+ + \mu_-} \approx 2D_+. \tag{4.5}$$

In terms of temperatures, we may write

$$\frac{D_a}{2\mu_+} = \frac{kT}{e}$$

or

$$\mu_+ = \frac{eD_a}{2kT} = \frac{5.9 \times 10^3 D_a}{T}$$

where D_a is in units of cm^2 sec^{-1} and μ in units of cm^2 volts^{-1} sec^{-1}.

Since $D = v^2/3\nu_c$ and we can often approximate $\nu_c = (\text{constant})p$, we find that as a usual thing

$$D_a p = \text{constant}$$

at constant energy, independent of pressure. This has been demonstrated experimentally by Biondi and Brown.[1] Actually this constancy is observed to break down at pressures below 0.1 torr, due to the phenomenon of diffusion cooling. In any distribution of electron velocities it always tends to be the faster electrons that escape, thus the average energy decreases. Although we may write

$$D_a \approx D_+\left(1 + \frac{T_-}{T_g}\right) \tag{4.6}$$

this should be written in terms of the average velocities as

$$D_a \approx D_+\left(1 + \frac{A\langle v_-^2\rangle}{\langle v_g^2\rangle}\right)$$

where A is a parameter close to unity. Thus if the electron velocity is very much less than that of the gas atoms

$$D_a \approx D_+$$

which is just one half of the value calculated under the assumption that the electron and atom velocities are equal. Diffusion is usually ambipolar if $n_- \approx n_e \gtrapprox 10^8/\text{cm}^3$.

Debye Length

It is useful to define a characteristic length in a plasma that measures the shielding effect of positive ions that can be considered to be surrounding the electrons with a space charge shield This can be calculated in an approximate fashion by considering a plasma between parallel plate conductors where the space charge field, E_s, is produced by the separation

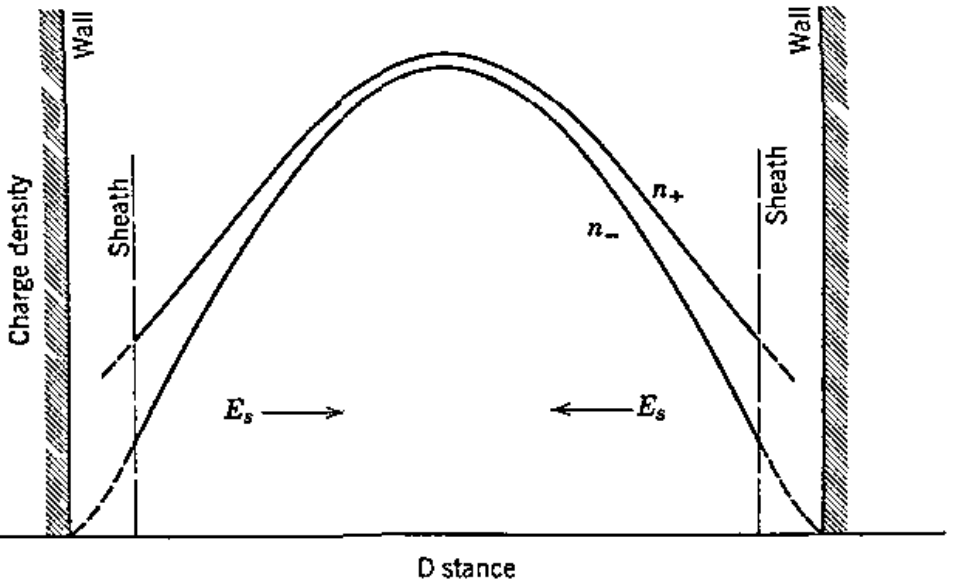


Fig 4 1 Space charge field produced by separation of ions and electrons at the walls

of ions and electrons at the wall, as shown in Fig 4 1 If we take L as the distance from the center of the plasma to the wall,

$$E_s \approx \frac{en\ L}{\epsilon_0}$$

and

$$\nabla n_- \approx \frac{n_-}{L}$$

Assuming ambipolar diffusion is taking place so that the electron flow is greatly reduced

$$D_- \nabla n_- \approx \mu_- n_- E_s,$$

then

$$L^2 \approx \frac{\epsilon_0}{en_-} \frac{D}{\mu_-}$$

This defines what is customarily called the Debye length

$$\lambda_D{}^2 = \frac{\epsilon_0}{en_-} \frac{D_-}{\mu_-} \tag{4 7}$$

Since D/μ is a measure of the energy, one can illustrate the characteristics of various types of gas discharge phenomena in terms of the electron energy, density, and Debye length. This is shown in Fig. 4.2.

This is a very useful concept. For example, in considering the effects of long-range forces in collision theory, it can be shown that the force of

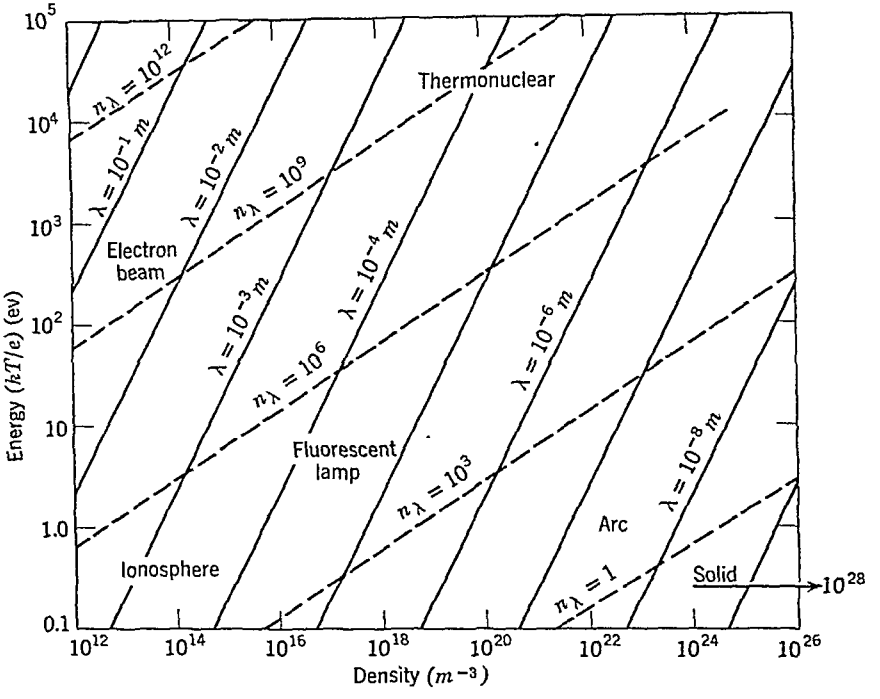


Fig. 4.2 Typical plasmas characterized by their energy, density, and Debye length.

interaction would be cut off at the Debye length. Also in laboratory-produced plasmas, the Debye length is important in that it gives roughly the thickness of the sheath that develops wherever the plasma is in contact with a surface. Without such a sheath, a plasma in the absence of a magnetic field would lose electrons much more rapidly than positive ions, owing to the greater electron velocity.

The concept of Debye length as a characteristic of the plasma state is basic. A plasma is defined as a neutral collection of electrons and ions in which the Debye length is smaller than any boundaries surrounding the plasma. If this condition is not fulfilled, one cannot correctly call the ionized medium a plasma.

Transition Diffusion

Having shown the effects of free electron diffusion and ambipolar diffusion, it is worth while to discuss briefly the transition between these two cases If we produce a discharge that is controlled by ambipolar diffusion in its fundamental mode in plane parallel plate geometry, we expect a cosinusoidal electron spatial distribution In the low power discharges however, the ion density throughout the discharge is not high enough to produce ambipolar diffusion throughout the volume, and we can have a case in which the discharge is ambipolar at the center but controlled by the free diffusion coefficient at the walls

To calculate this effect, we write the flow equations

$$\Gamma_+ = -D_+ \nabla n_+ + E_s \mu_+ n_+ \tag{4 8}$$

$$\Gamma \ = -D_- \nabla n_- - E_s \mu_- n_- \tag{4 9}$$

Taking the flow of ions and electrons to be the same and eliminating E_s, we get

$$(\mu_+ n_+ + \mu \ n \ )\Gamma = -(\mu_+ D_- + \mu_- D_+) n_+ \nabla n_-, \tag{4 10}$$

putting into this equation the ambipolar diffusion coefficient from Eq 4 3 and defining the space charge density as

$$\rho = (n_+ - n_-)e \tag{4 11}$$

and the conductivity as

$$\sigma = (\mu_+ n_+ + \mu \ n_-)e \tag{4 12}$$

we obtain

$$\Gamma = -D_a \left[1 + \frac{\mu_- \rho}{\sigma} \right] \nabla n_- = -D_s \nabla n_-, \tag{4 13}$$

which defines the effective diffusion coefficient

$$D_s = D_a \left[1 + \frac{\mu_- \rho}{\sigma} \right] \tag{4 14}$$

The space charge field can be obtained by substituting Eq (4 13) in Eq (4 9)

$$E_s = \frac{D_s - D_-}{\mu_-} \frac{\nabla n}{n_-} \tag{4 15}$$

Poisson s equation $\nabla \ \epsilon_0 E_s = (n_+ - n_-)e$ gives

$$\rho = \epsilon_0 \frac{D_- - D_s}{\mu_-} \left[\left(\frac{\nabla n}{n} \right)^2 - \frac{\nabla^2 n}{n} \right], \tag{4 16}$$

where n_- is put equal to n If we consider the center of the plasma the point where $\nabla n = 0$ and substitute $1/\Lambda^2$ for $-\nabla^2 n/n$, we find the space

charge at the center to be

$$\rho_0 = \frac{D_- - D_s}{\mu_-}\frac{\epsilon_0}{\Lambda^2}. \tag{4.17}$$

We can express the effective diffusion coefficient D_s in terms of the central conductivity σ_0 by combining Eq. (4.14) with Eq. (4.17), to obtain

$$D_s = D_a \frac{D_- + \Lambda^2\sigma_0/\epsilon_0}{D_a + \Lambda^2\sigma_0/\epsilon_0}. \tag{4.18}$$

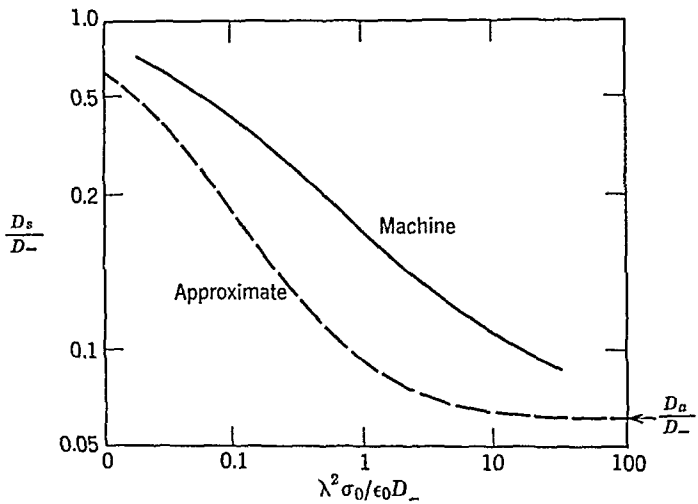


Fig. 4.3 The solution for the approximate Eq. (4.18) and the machine solution of Allis and Rose[2] for the *unsimplified equations describing the transition from* ambipolar to free diffusion.

The solution of Eq. (4.18) is shown in Fig. 4.3. A machine solution without the approximations inherent in Eq. (4.18) has been carried out by Allis and Rose,[2] and their results are also shown in Fig. 4.3. From Eqs. (4.17) and 4.18 we see that $D_s = D_-$ when $\rho_0 \approx 0$, and that D_s decreases as the central space charge increases and saturates when $D_s = D_a$.

Using D_s as given by

$$\frac{\partial n}{\partial t} = \frac{D_s}{\Lambda^2} n,$$

we may define an effective diffusion length Λ_s such that

$$\frac{\partial n}{\partial t} = \frac{D_s}{\Lambda^2} n \equiv \frac{D_-}{\Lambda_s^2} n.$$

Equation (4 16) can be rearranged to take into account the space charge effect, leading to

$$\Lambda_s^2 = \Lambda^2 \frac{1 + \Lambda^2 \sigma_0/\epsilon_0 D_a}{1 + \Lambda^2 \sigma_0/\epsilon_0 D_-}. \tag{4 19}$$

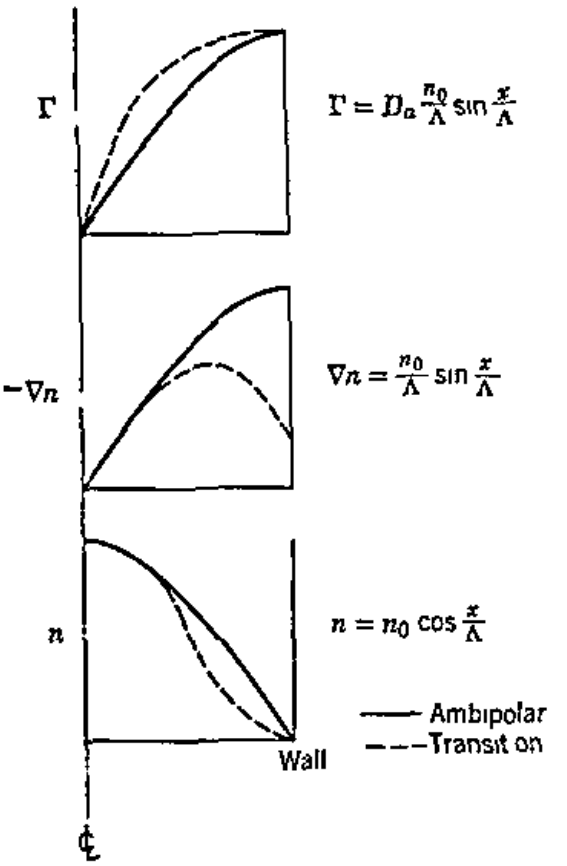


Fig 4 4 The effects of ambipolar and transient diffusion on the flow, the gradient, and the density between parallel plates

Thus we can see that the effect of space charge is to increase the effective size of the volume occupied by the plasma

The variation of the diffusion coefficient will change the cosinusoidal spatial distribution that would obtain if the diffusion coefficient were constant The situation can best be described with reference to Fig 4 4

The rate of production of ionization stays approximately constant across the discharge, and thus the total current to the walls remains essentially unchanged However, as we move from the center of a discharge whose effective diffusion coefficient is increasing with radius, more current will flow because of the higher diffusion coefficient than for the case of pure ambipolar diffusion Since the effective diffusion coefficient increases all the way to the walls, and since the current reaching the walls is limited,

∇n has to decrease below the ambipolar value near the walls. If the gradient of n decreases, the spatial distribution of the density departs from a cosinusoidal distribution for ambipolar diffusion, to a bell-shaped distribution for a transition case.

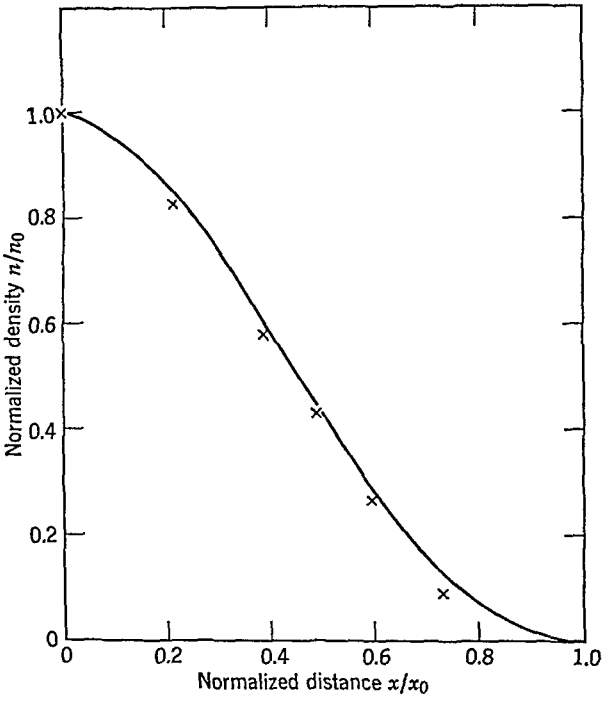


Fig. 4.5 Experimental check of the theory for the transition from ambipolar to free diffusion. The crosses indicate experimental data, the solid curve the machine solution of Allis and Rose.[2]

Allis and Rose[2] in carrying out their theoretical study of the transition from free to ambipolar diffusion, examined the current equations determining the process and obtained approximate solutions that can be compared directly with experiment. As a result of a microwave discharge having been set up in a resonant cavity, probe measurements were taken of the electron density as a function of position.[3] The agreement between theory and experiment is evident in Fig. 4.5.

Diffusion of Several Species

Equation (4.6) shows us that if we have a mixture of electrons and positive ions, we expect a single value of D_a. However, if the plasma is made up of electrons and two types of positive ions, there will be three different

diffusion coefficients

$$(D_a)_e \approx \frac{(n_+)_1}{n_e}(D_a)_1 + \frac{(n_+)_2}{n_e}(D_a)_2$$
$$(D_a)_1 \approx (D_+)_1\left(1 + \frac{T_e}{T_g}\right) \tag{4 20}$$
$$(D_a)_2 \approx (D_+)_2\left(1 + \frac{T_e}{T_g}\right)$$

This shows that although the diffusion of one ion is unaffected by the other, the D_a for electrons $(D_a)_e$ can change depending on the relative concentration of each ionic species compared with n_e

Similar diffusion coefficients may be written down for a mixture of electrons, negative ions, and positive ions and these take the form

$$(D_a)_e \approx \left(1 + \frac{n_-}{n_e}\right)D_+\left(1 + \frac{T_e}{T_g}\right) + \frac{n_-}{n_e}D_-\left(\frac{T_e}{T_g} - 1\right)$$
$$(D_a)_- \approx 2\left(1 + \frac{n_-}{n_e}\right)\frac{D_+D_-}{D_e} - D_-\left(\frac{T_e}{T_g} - 1\right) \tag{4 21}$$
$$(D_a)_+ \approx D_+\left(1 + \frac{T_e}{T_g}\right)$$

Ambipolar Diffusion with Variable Ion Mobility

The ambipolar-diffusion coefficient was derived on the assumption that the positive ion mobility is a constant independent of energy In certain ranges of energy this is not a valid assumption Frost[4] has calculated this effect Starting with the relation

$$E_s = -\frac{D_- - D_+}{\mu_- + \mu_+}\frac{\nabla n}{n}$$

and assuming $\mu_- \gg \mu_+$ and $D_- \gg D_+$, we find that

$$E_s = -\left(\frac{D_-}{\mu_-}\right)\frac{\nabla n}{n} \approx u_{\text{ave}}\frac{\nabla n}{n}$$

We rewrite Eq (4 1) for positive ions in the form

$$\Gamma = -(\mu_+ u_{\text{ave}} + D_+)\nabla n$$
$$\Gamma = -\mu_+ u_{\text{ave}}\nabla n = \mu_+ En$$

Using the continuity equation for a going discharge, we find

$$\frac{\partial n}{\partial t} + \nabla \cdot \Gamma = \nu_i n$$

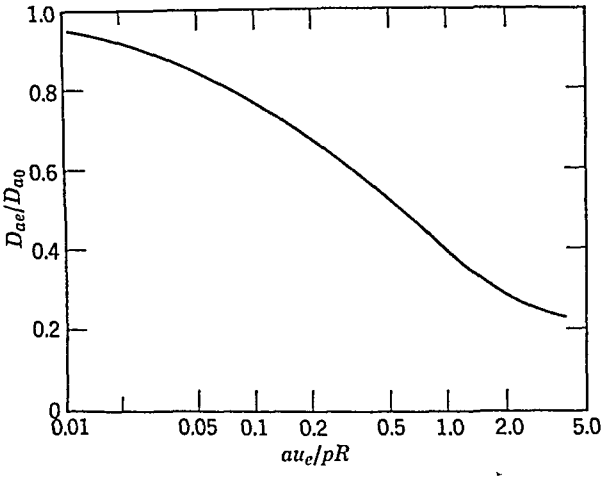


Fig. 4.6 Effective ambipolar diffusion coefficient for a long cylindrical tube.

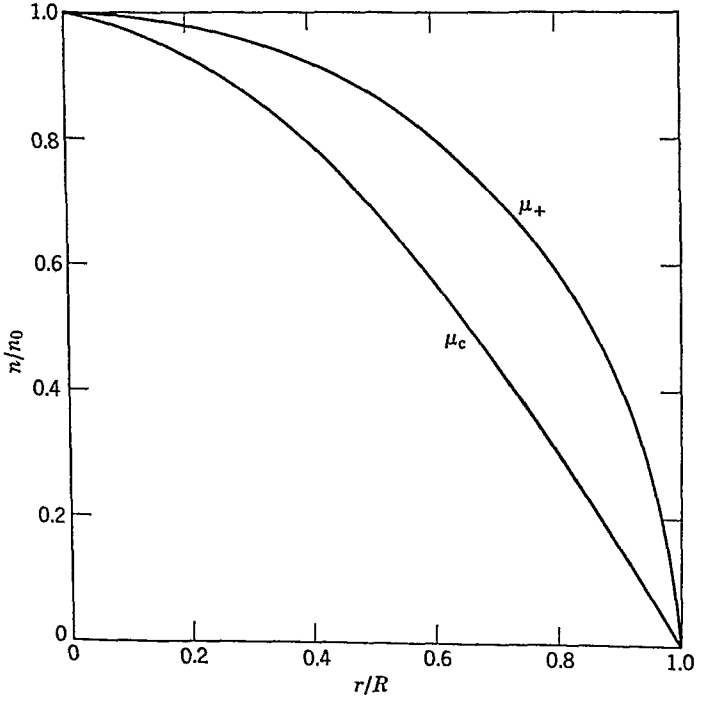


Fig. 4.7 Variation of charged particle density for a long cylindrical tube.

where ν_i is the frequency of ionization This leads to the diffusion length $(D_a/\Lambda^2) = \nu_i$, which we may particularize for a long cylindrical discharge as $\Lambda = R/2\,405$

Frost solved the flow equation for a long cylindrical tube for the variation of an effective ambipolar diffusion coefficient D_{ae} His results are shown in Figs 4 6 and 4 7 In Fig 4 6 is plotted the ratio of D_{ae}/D_{a0} where D_{a0} is the coefficient given by Eq (4 3) as a function of au_{ave}/pR (a is given in Table 3 1) Figure 4 7 shows the variation of density with radius for a constant and variable μ

Ambipolar Diffusion in a Magnetic Field

Let us now consider the modifications of the diffusion equations due to the presence of a magnetic field applied along the z axis

The particle flow is given by

$$\Gamma_\pm = \pm n_\pm \mu_\pm E_s \pm \mu_\pm(\Gamma_\pm \times B) - D_\pm \nabla n_\pm \tag{4 22}$$

We may write the three equations for the components of $\Gamma_\pm$

$$(\Gamma_\pm)_r \mp \mu_\pm(\Gamma_\pm)_\theta B = -\frac{\partial}{\partial r}(D_\pm n) \pm \mu_\pm(E_s)_r n$$

$$(\Gamma_\pm)_\theta \pm \mu_\pm(\Gamma_\pm)_r B = -\frac{1}{r}\frac{\partial}{\partial 0}(D_\pm n) \pm \mu_\pm(E_s)_\theta n$$

$$(\Gamma_\pm)_z = -\frac{\partial}{\partial z}(D_\pm n) \pm \mu_\pm n(E_s)_z$$

In the second of these equations both terms on the right hand side vanish the first term because we assume cylindrical symmetry the second because $\nabla \times E_s = 0$ Thus we may eliminate $(\Gamma_\pm)_\theta$ from the first equation by using the second equation which is now

$$(\Gamma_\pm)_\theta = \mp \mu_\pm(\Gamma_\pm)_r B$$

and

$$(\Gamma_\pm)_r[1 + \mu_\pm{}^2 B^2] = -\frac{\partial}{\partial r}(D_\pm n) \pm \mu_\pm n(E_s)_r \tag{4 23}$$

We now make what has been called the "congruence assumption"

$$\left.\begin{aligned}(\Gamma_+)_r &= (\Gamma_-)_r \\ (\Gamma_+)_z &= (\Gamma_-)_z\end{aligned}\right\} \tag{4 24}$$

This assumption corresponds to the case where the walls are perfect insulators This assumption is suggested but not required by our implicit

assumption of no time dependencies which leads to

$$\nabla \cdot \mathbf{\Gamma}_+ = \nabla \cdot \mathbf{\Gamma}_-. \tag{4.25}$$

Using Eq. (4.24) we may write (4.23) as the two equations which follow:

$$\Gamma_r[1 + \mu_+{}^2B^2] = -\frac{\partial}{\partial r}(D_+ n) + \mu_+ n(E_s)_r$$

$$\Gamma_r[1 + \mu_-{}^2B^2] = -\frac{\partial}{\partial r}(D_- n) - \mu_- n(E_s)_r.$$

Multiplying the first of these equations by μ_-, the second by μ_+, and adding, we obtain

$$\Gamma_r[1 + \mu_+\mu_-B^2] = -\left[\frac{\mu_- D_+ + \mu_+ D_-}{\mu_+ + \mu_-}\right]\frac{\partial n}{\partial r}$$

or

$$\Gamma_r = -\frac{D_a}{(1 + \mu_+\mu_-B^2)}\frac{\partial n}{\partial r}$$

$$\Gamma_r = -\frac{D_s \partial n}{\partial r}. \tag{4.26}$$

D_a is the usual ambipolar diffusion coefficient and D_s is the effective diffusion coefficient for finite values of B $(0 < B < \infty)$. We clearly see from Eq. (4.26) that, as in the case of free diffusion, the effective diffusion coefficient when $B = 0$ is larger than for $B \neq 0$. Using the same technique, we obtain for the diffusion current in the z direction

$$\Gamma_z = -\frac{D_a \partial n}{\partial z} \tag{4.27}$$

which is what we expected since the motion along the field is unaffected by the field. Defining

$$b_a = 1 + \mu_+\mu_-B^2$$

we have

$$\left.\begin{aligned}\Gamma_r &= -\frac{D_a}{b_a}\frac{\partial n}{\partial r}\\ \Gamma_z &= -\frac{D_a \partial n}{\partial z}\end{aligned}\right\}. \tag{4.28}$$

The requirement that the particles lost by diffusion must be replaced by ionization in order to maintain the discharge is given by

$$\nabla \cdot \mathbf{\Gamma} = n\nu_i.$$

Using Eq (4 28) we have

$$\frac{1}{r}\frac{\partial}{\partial r}(r\Gamma_r) + \frac{\partial}{\partial z}\Gamma_z = n\nu_i$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left[r\left(-\frac{D_a}{b_a}\frac{\partial n}{\partial r}\right)\right] - D_a\frac{\partial^2 n}{\partial z^2} = n\nu_i$$

or

$$\frac{1}{b_a}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial n}{\partial r}\right)\right] + \frac{\partial^2 n}{\partial z^2} = -\frac{\nu_i}{D_a}n$$

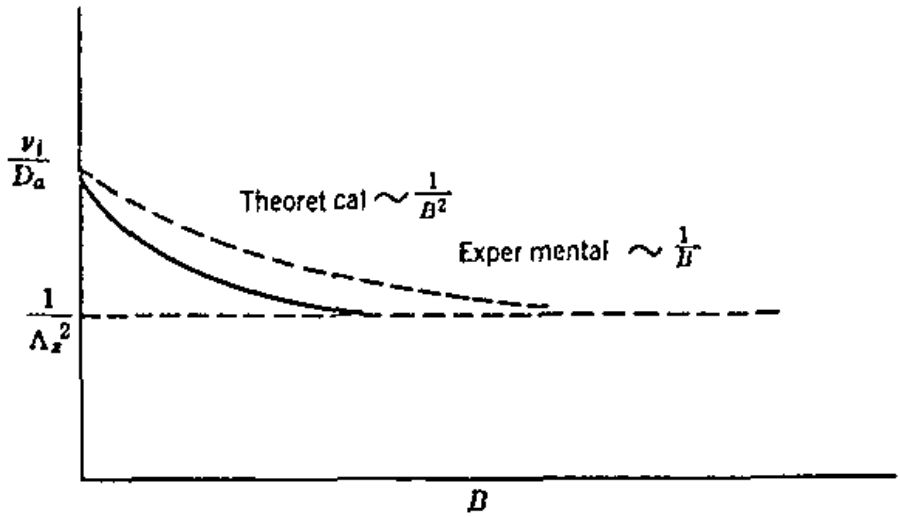


Fig 4 8 Diffusion in a magnetic field

The solution is

$$\left(\frac{\nu_i}{D_a}\right)_{cond} = \left(\frac{1}{b_a}\frac{1}{\Lambda_r^2} + \frac{1}{\Lambda_z^2}\right) \tag{4 29}$$

which shows that the ionization frequency necessary to maintain a steady-state discharge is less in this case than in the case of no magnetic field This effect can, of course, be attributed to the decreased rate of diffusion in the r direction as shown by Eq (4 26)

A qualitative plot of Eq (4 29) appears in Fig 4 8 Also shown are the experimental results that have been obtained There is a basic discrepancy between the theoretical and experimental results which as yet has not been explained

Another general case of interest is the case of conducting walls Here Γ_+ is not necessarily equal to Γ_- because the plasma can neutralize itself either through currents in the walls or in an external circuit

It would seem that a valid assumption here might be that $(\Gamma_-)_r = 0$ Its validity is much improved if we assume a cylinder of small length to diameter ratio The main reason for hypothesizing this is that since the

electrons move much faster than the ions, they will tend to have much tighter cyclotron orbits than the ions and will thus tend to diffuse less in the radial direction.

Using Eq. (4.23) as evaluated for the electrons ($(\Gamma_-)_r$) to eliminate $(E_s)_r$ from the other terms of Eq. (4.23), we obtain

$$(\Gamma_+)_r = -\frac{\partial n}{\partial r} D_a \left(\frac{\mu_+ + \mu_-}{\mu_-}\right) \frac{1}{b_+}$$

where

$$b_+ = 1 + \mu_+^2 B^2.$$

Setting $\nabla \cdot \Gamma_+ = n\nu_i$, we obtain

$$\left(\frac{\nu_i}{D_a}\right)_{(\Gamma_-)_r=0} = \frac{1}{\Lambda_z^2} + \frac{1}{b_+ \Lambda_r^2}. \tag{4.30}$$

Similarly if $(\Gamma_+)_r$ is assumed equal to zero, we obtain (for high B and $L \ll R$)

$$\left(\frac{\nu_i}{D_a}\right)_{(\Gamma_+)_r=0} = \frac{1}{\Lambda_z^2} + \frac{1}{b_- \Lambda_r^2} \tag{4.31}$$

where

$$b_- = 1 + \mu_-^2 B^2.$$

In the same spirit we make the only two remaining assumptions to find:

$$\left(\frac{\nu_i}{D_a}\right)_{(\Gamma_+)_z=0} = \frac{1}{b_a \Lambda_r^2} + \frac{b_-}{b_a \Lambda_z^2} \tag{4.32}$$

and

$$\left(\frac{\nu_i}{D_a}\right)_{(\Gamma_-)_z=0} = \frac{1}{b_a \Lambda_r^2} + \frac{b_+}{b_a \Lambda_z^2}. \tag{4.33}$$

For comparison, a qualitative plot of all these relations and the relation obtained by making the congruence assumption is shown in Fig. 4.9. For small B the frequency of rotation of the charged particles around the field lines is small. Since we have assumed constant temperature, we must conclude that at low values of B the orbits have large radii and hence the permissible values of ν_i/D_a fall between $\Gamma_{+z} = 0$ and $\Gamma_{-z} = 0$, whereas at high values of B the orbits decrease in size and thus the permissible values of ν_i/D_a fall between $\Gamma_{+r} = 0$ and $\Gamma_{-r} = 0$.

We note that the general effect of a magnetic field upon diffusion has been seen to be a reduction in diffusion rate and hence a partial containment of the plasma. To emphasize the effect of the magnetic field as a modifier of the effective diffusion rate, it is instructive to replot Fig. 4.9 in terms of the effective $1/\Lambda$ as a function of $\mu_+\mu_- B^2$ as is shown in Fig. 4.10. Since Allis[5] worked out the limiting case for $(\Gamma_+)_z = 0$ and Simon's[6]

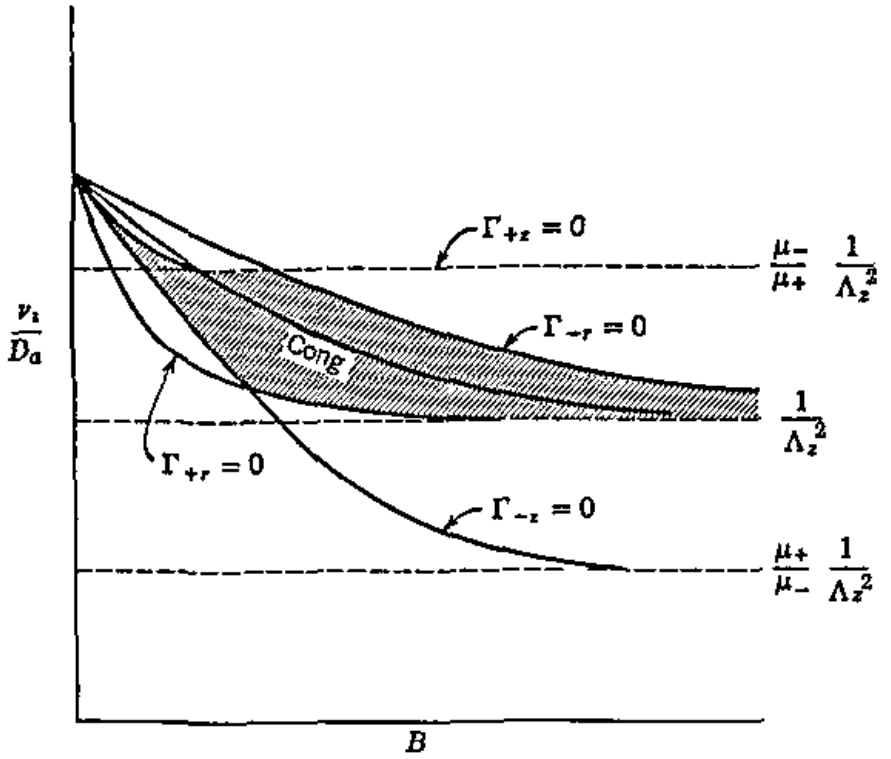


Fig 4 9 General effects of magnetic field upon diffusion

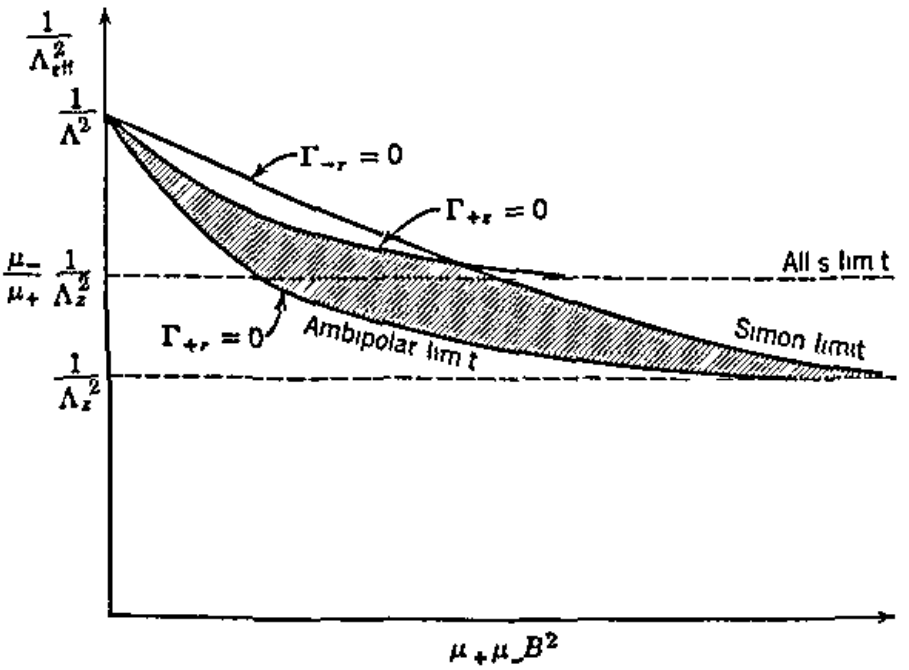


Fig 4 10 Various types of diffusion limits in a magnetic field

name is associated with the case for $(\Gamma_-)_r = 0$, these are often called the Allis limit and the Simon limit. The case for $(\Gamma_+)_r = 0$ corresponds to ambipolar diffusion. Being more explicit about these definitions we may define the various Λ's as:

$$\frac{1}{\Lambda_B^2} = \left(\frac{2.4}{R}\right)^2 \frac{1}{1 + \mu_+\mu_- B^2} + \left(\frac{\pi}{L}\right)^2$$

$$\frac{1}{\Lambda_S^2} = \left(\frac{2.4}{R}\right)^2 \frac{1}{1 + \mu_+^2 B^2} + \left(\frac{\pi}{L}\right)^2$$

$$\frac{1}{\Lambda_A^2} = \left(\frac{2.4}{R}\right)^2 \frac{1}{1 + \mu_+\mu_- B^2} + \left(\frac{\pi}{L}\right)^2 \frac{(1 + \mu_-^2 B^2)}{(1 + \mu_+\mu_- B^2)}$$

$$\Lambda_S = \Lambda_A \text{ at } \left(\frac{\pi R}{2.4L}\right)^2 = \frac{\mu_-}{\mu_+} \frac{1}{1 + \mu_+^2 B^2}.$$

Experimentally one measures ν_i and from the known B and the presumably known $\mu_+\mu_-$ and Λ, the diffusion coefficient is deduced from

$$D_a = \nu_i \Lambda^2.$$

Our plot then says, in effect, that we know D_a only as well as we can determine the boundary conditions to apply, within the hatched area. The hatched area represents the possible region for diffusion.

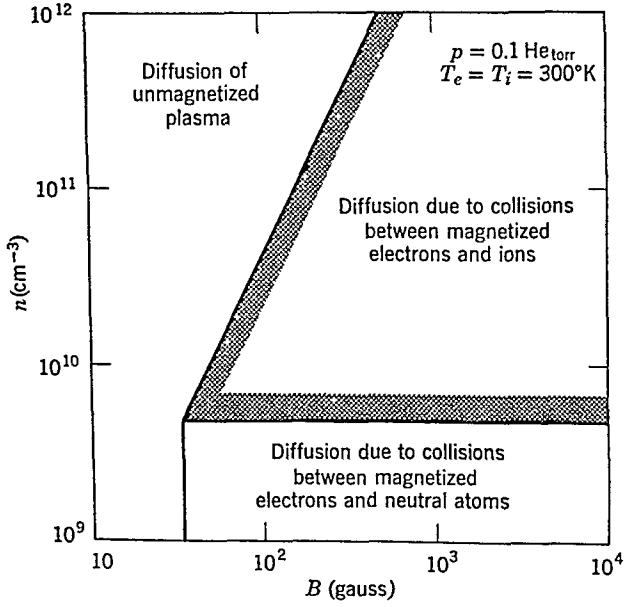


Fig. 4.11 Types of diffusion in a three-component plasma.

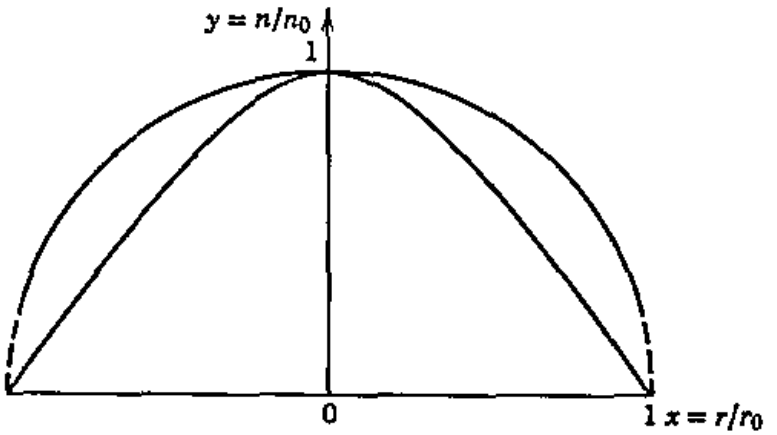


Fig 4 12 Radial distribution of density across a magnetic field

A great deal of work has been done on the diffusion of fully ionized plasmas, or equivalently, on plasmas at low degrees of ionization when Coulomb collisions are important Figure 4 11 shows the various diffusion regions resulting from Coulomb collisions in a helium plasma of electrons, ions, and neutral atoms The lower limit of density corresponds to equal collision frequency for electron-ion and electron-atom collisions The upper limit separates Coulomb diffusion from ion-atom collisions This transition occurs when the Coulomb diffusion coefficient across the magnetic field, $(D_{ei})_\perp$ becomes comparable with $(D_{ia})_\parallel$ with no magnetic field

The ambipolar diffusion of charged particles in the presence of a magnetic field has so far been discussed only for the case where the diffusion coefficient $D_a/(1 + \mu_+\mu\ B^2)$ was independent of the electron density and a steady state was achieved such that $\nabla \cdot \Gamma = n\nu_e$ One can, however, obtain a solution without making the assumption that D_a and $(1 + \mu_+\mu_-\beta^2) = b_a$ are independent of position This has been done for the case of cylindrical coordinates by Alikhanov, Zakharov, and Khorasanov[7] who arrive at a radial distribution of density in a plasma across a magnetic field illustrated by the upper curve in Fig 4 12 The lower curve is the Bessel function solution assuming D_a/b_a = constant independent of position

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Chapter 5

Inelastic Collisions

Excitation

An electron can make only elastic collisions with a gas atom when its energy is insufficient to excite any levels characteristic of the target atom. Consider, for example, the lowest few levels of the helium atom illustrated in Fig. 5.1. Providing the electron energy is less than the 19.82 volts of the 2^3S state of this atom, the electron will be elastically scattered. If the electron energy is higher than this, however, inelastic collisions will take place, which if they lead to excitation of bound energy levels in the atom, allow us to define a probability of excitation. The probability of excitation P_x, is defined as the number of atoms that are excited per electron per centimeter path length at a pressure of 1 torr and at 0°C.

The probability of excitation can be measured by the classical Franck and Hertz experiment. Electrons are accelerated through a uniform field to a grid with a plate directly behind the grid. A retarding potential applied between the grid and the plate enables one to measure the number of electrons that have made inelastic collisions. By varying the accelerating potential, the dependence of the number of inelastic collisions on velocity may be estimated. Although the usual form of tube used in a Franck and Hertz experiment is not well-suited for accurate determinations, Maier-Leibnitz[1] used an improved type of apparatus to measure inelastic collision probabilities in many gases, shown in Fig. 5.2.

To illustrate the experimental measurements of excitation cross sections in relation to structure of the target atom, let us take the case of helium, the pertinent energy level diagram which is shown in Fig. 5.1. Schulz and Fox[2] have taken good measurements on this gas, and we shall describe their method briefly.

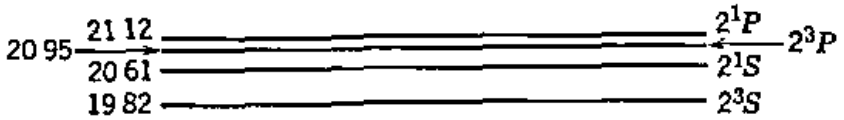




Fig. 5.1 Energy levels of helium

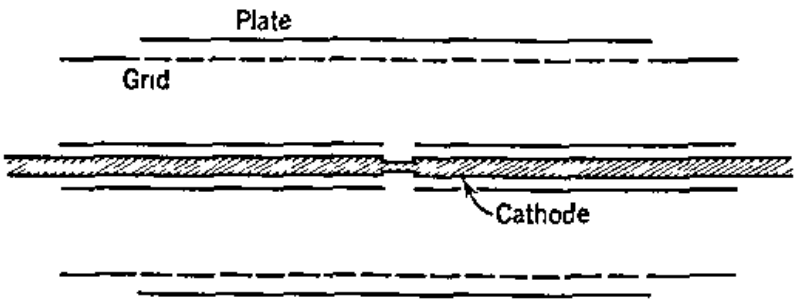


Fig. 5.2 Diagram of the Maier-Leibnitz apparatus for measuring inelastic collision probabilities

To make an accurate measurement of excitation probabilities, one needs to have control of both the energy of the bombarding electrons and those that have lost energy in inelastic collisions. Schulz and Fox used an electron gun which employed a retarding potential difference method so that the effect of nearly monoenergetic electrons was measured. A sketch of the electron gun is shown in Fig. 5.3. The electrons coming from the filament travel through three electrodes. The first, P_1, is at a positive potential of about 3 volts with respect to the filament and serves to draw electrons from the filament. The second, P_2, is slightly negative, V_R,

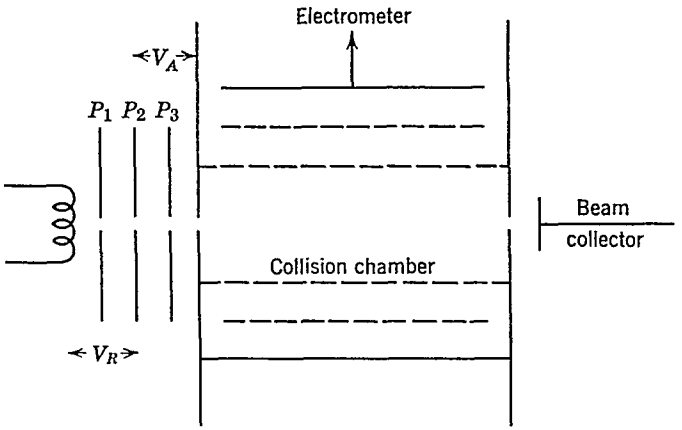


Fig. 5.3 The Schulz and Fox apparatus.

with respect to the filament. An axial magnetic field of 100 gauss is applied to keep the electron beam together. The electron distribution in the forward direction is chopped off by the retarding potential V_R. Only the high-energy electrons in the distribution can penetrate through the potential barrier of the second plate. The third plate, P_3, is at the same potential as P_1. The collision chamber is at the potential V_A with respect to the retarding electrode P_2. If we change the retarding potential V_R by a small amount δV_R, the distribution of electrons will be cut off sharply at a slightly different energy. The difference between retarding by V_R and $V_R + \delta V_R$ is due solely to electrons in the energy band δV_R. In order to obtain the excitation function resulting from electrons in the small-energy range δR_R, one takes two readings for each value of V_A. The difference between the two readings gives the excitation from electrons in the energy interval δV_R. By measuring the "difference current," one obtains the excitation resulting from electrons in a small-energy band without actually creating a beam of monoenergetic electrons. This is illustrated schematically in Fig. 5.4.

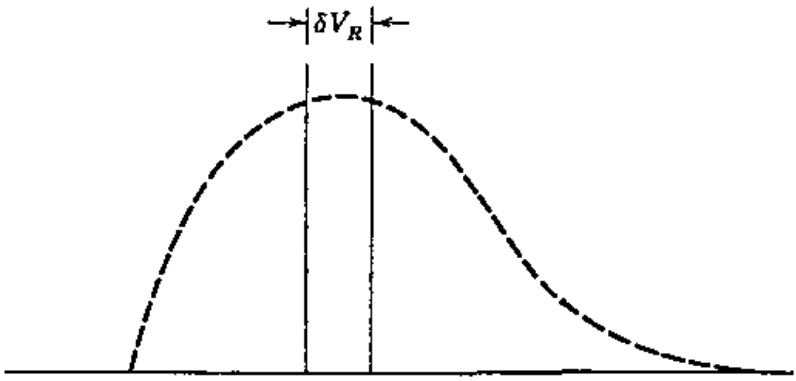


Fig. 5 4 Electron energy distribution in the retarding potential difference method of obtaining the effect of nearly monoenergetic electrons

To increase the sensitivity of measuring low-energy electrons near the excitation thresholds, Schulz[3] devised what he calls a "trapped-electron method" A simplified diagram of his tube is shown in Fig 5 5. The electrons from the gun pass through the collision chamber C and are collected at E When electrons in the beam lose nearly all of their energy in an inelastic collision, they will be trapped in the axial direction Since they can reach neither of the collision chamber electrodes, they will stay in the tube until they diffuse to the positive electrode M Only those electrons that have lost an amount of energy between V_A and $V_A + W$ (W = well depth) will be collected at M

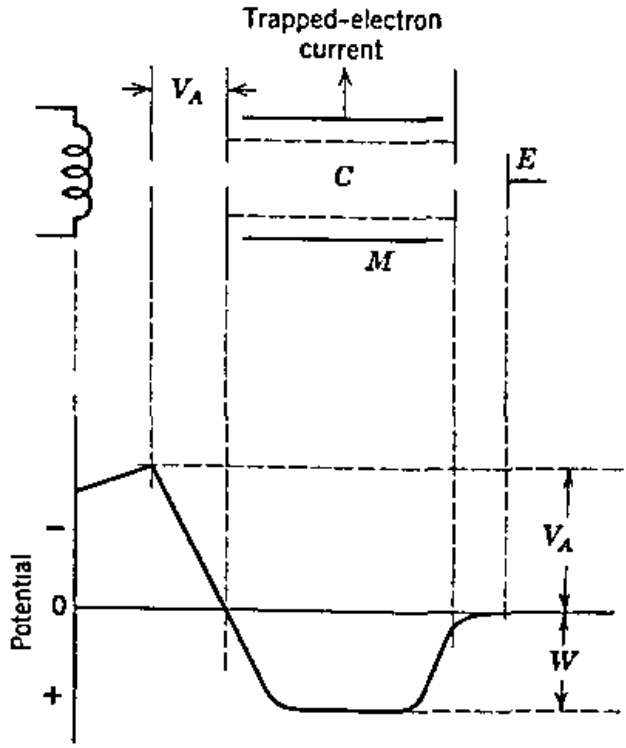


Fig. 5 5 Schulz's trapped-electron method

The excitation spectrum is obtained by measuring the slow-electron current as a function of the accelerating voltage V_A at a constant well depth. The effective cross section Q_x can be given by

$$i_{-} = i_0 N Q_x L \tag{5.1}$$

where i_{-} is the peak current of slow electrons, i_0 is the electron beam current, L is the length of the collision chamber, and N is the gas density.

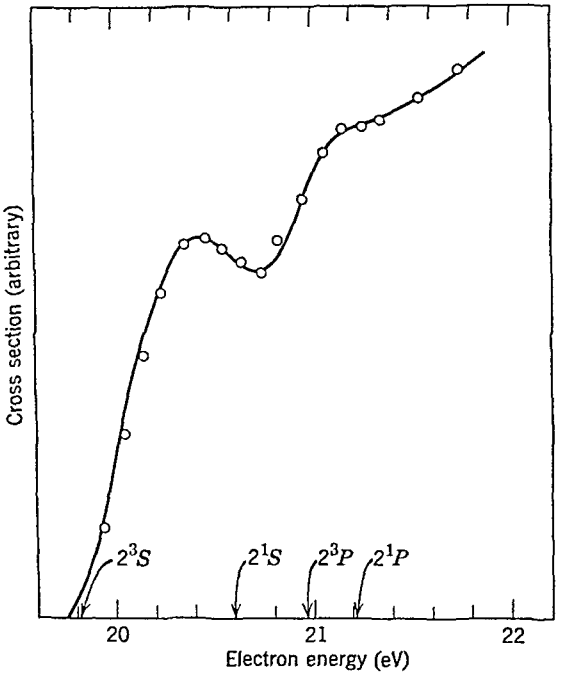


Fig. 5.6 Excitation cross section for helium.

Schulz's measurements of the excitation cross section for helium are shown in Fig. 5.6. The correlation between these measurements and the helium energy level diagram of Fig. 5.1 is very obvious.

Schulz also applied his trapped-electron method to molecular electronic excitations and found particularly interesting results in N_2 and CO. Let us discuss the nitrogen case in some detail. Figure 5.7 shows his measurements of the low-energy inelastic collisions between electrons and nitrogen. The large peak in the cross section at 2.3 electron volts does not result from any known electronically excited states of the nitrogen molecule, and Schulz suggests that this results from a temporary negative ion state. Figure 5.8 shows a hypothetical diagram of such a temporary negative ion state, and the ground state of nitrogen. On this model, the incident electron

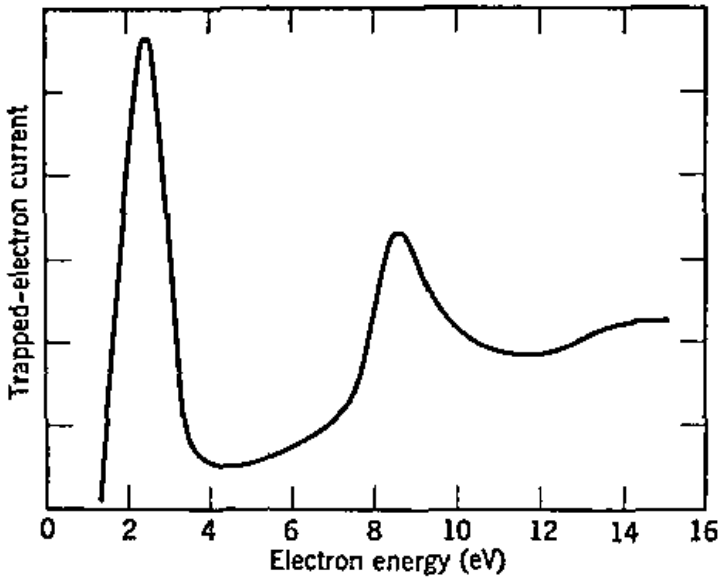


Fig 5 7 Electron inelastic collisions in nitrogen

excites the negative ion state and the compound molecule decays to the various vibrational state at the neutral molecule

A sharp increase in the total cross section at 2 3 electron volts in nitrogen was observed in Ramsauer-type experiments, and Fisk's wave-mechanical calculation,[4] shown in Fig 5 9, seemed to confirm the elastic nature of the process Since the resonance behavior in elastic and inelastic processes is related, it is possible that Fisk's elastic resonance process is a manifestation of the existence of the nitrogen negative ion state From the experimental point of view, since the incident electron does not suffer

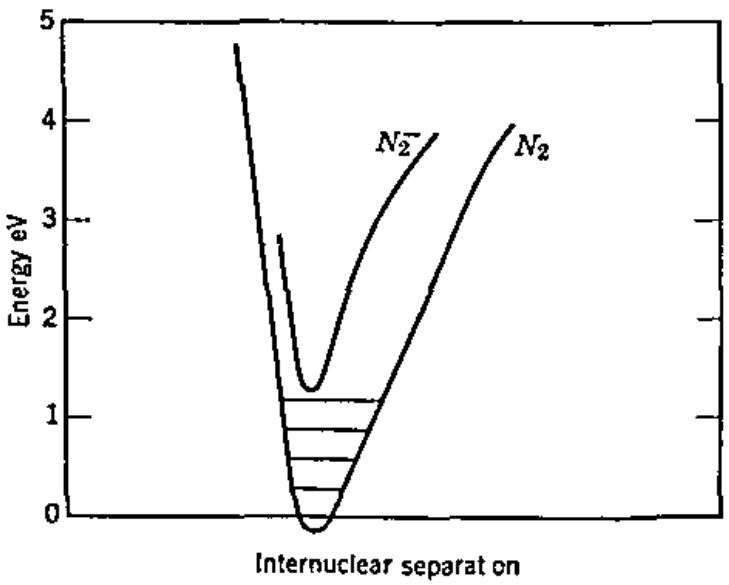


Fig 5 8 Potential energy diagram for nitrogen

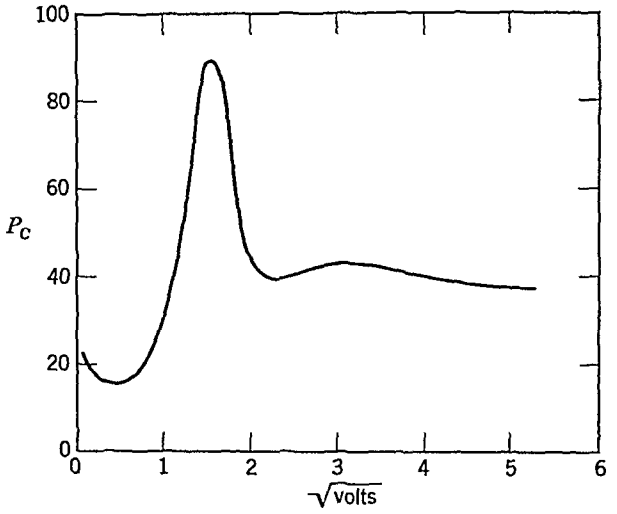


Fig. 5.9 Probability of collision of electrons in nitrogen.

any energy loss if the negative ion relaxes to the ground vibrational state, this type of collision is indistinguishable from an elastic collision.

Stimulated Emission

One of the most dramatic outgrowths of studying electronic excitations is the development of the gas laser. An example of this is the helium-neon laser mechanism. In Fig. 5.1 we show the energy levels in helium. To the same scale let us add appropriate neon levels, as illustrated in Fig. 5.10. Since the 2^3S level in helium is metastable, direct transitions to the ground state are forbidden. It happens that the top level of the $2S$ state in neon is very close to and slightly below this metastable helium level so that the $2S$ state in neon may be efficiently populated by resonance transfer on collision with a helium excited atom. In the neon atom ordinary radiative transitions occur from the four $2S$ levels to the ten $2p$ levels, but the lifetime of the $2S$ levels is about an order of magnitude greater than that of the $2p$ so that a population inversion is obtained which leads to negative absorption and hence the possibility of laser action. Whether the inversion will actually take place[5] will depend on the relative concentration of helium and neon in the gas mixture and the temperature of the exciting electrons.

Ionization

The probability of ionization, P_i, has attracted the attention of many workers, although most of the earlier measurements were of a qualitative

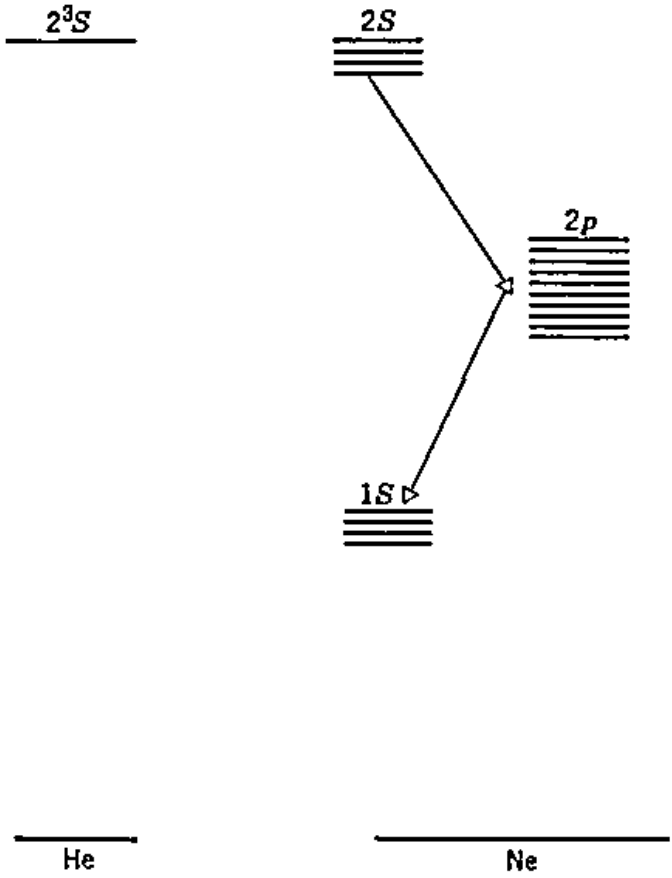


Fig 5 10 Energy levels in helium and neon

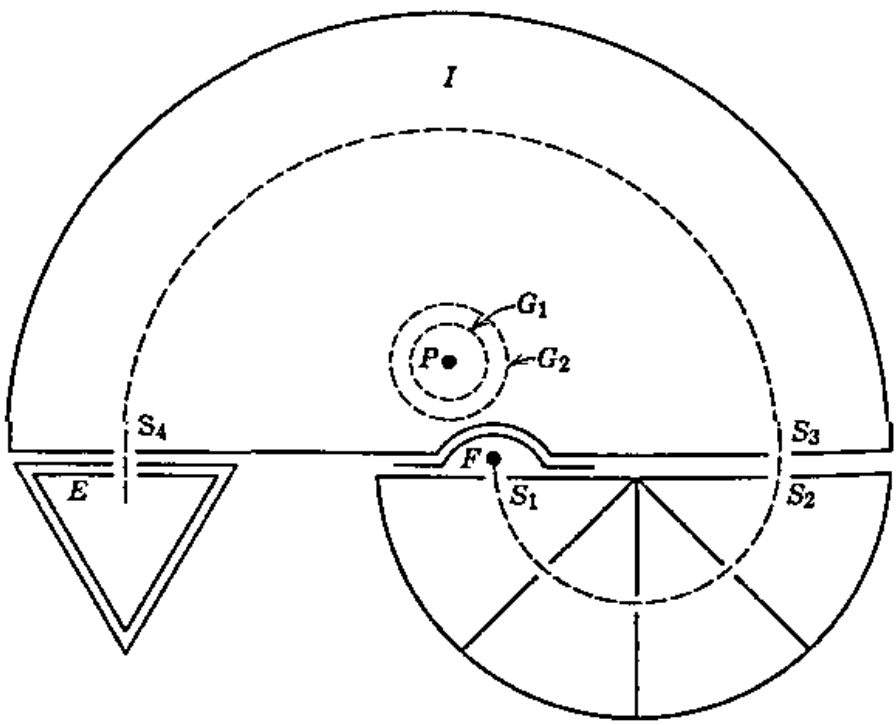


Fig 5 11 Nottingham s apparatus for measuring the probability of ionization

nature only. Maier-Leibnitz used his method for measuring P_i for a number of gases, but the best of the early work was done on mercury by Nottingham.[6] His work, carried out with a magnetically analyzed electron beam under conditions of high purity, stands as a classic in experimental excellence. A sketch of his apparatus is shown in Fig. 5.11. Electrons from the filament F were accelerated into the analyzer through slit S_1. Those of

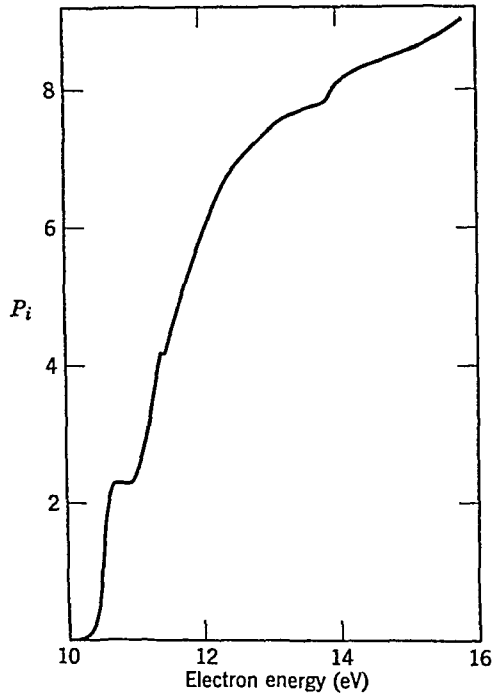


Fig. 5.12 Probability of ionization in mercury.

the desired energy followed a circular path to slit S_2 after which they were accelerated into the ion chamber I at S_3. The main beam passed through the exit slit S_4 into the electron cage E to be measured. The positive ion collector P is surrounded by two grids G_1 and G_2. Typical measurements which Nottingham obtained are reproduced in Fig. 5.12. A large number of the most recent measurements on P_i have been made by the Westinghouse group using the retarding potential difference method discussed in detail in connection with the measurements of P_x.

In any use of the P_i function in terms of its energy dependence, details of the low-energy behavior can be of great importance. Real differences are found between spectroscopic measurements and electron beam

experiments in the determination of the ionization potential Typical electron beam data are shown in Fig 5 13 It is seen that the most probable transition, corresponding to the onset of strong ionization, requires about 16 0 volts whereas the spectroscopic value is 15 4 volts As the energy spread of the bombarding electron beam was made less and less, it became evident that this tailing off of the lower end of the P_t curve could not all be explained from the point of view of the experimental beam spread A

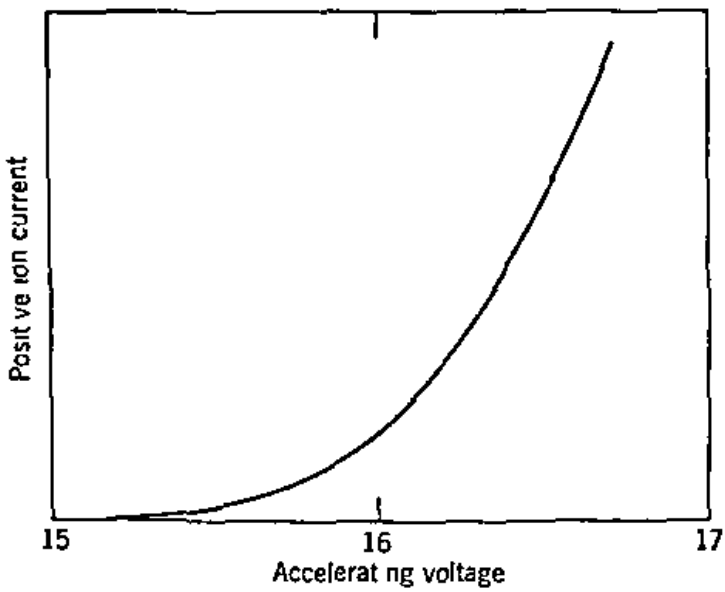


Fig 5 13 Electron beam measurements of ionization in hydrogen

study of the hydrogen case shows that there can arise an interesting physical explanation to this low energy tail

In a collision between an electron and a hydrogen molecule, the momentum transferred to the heavy nucleus is very small and the position of the nuclei remains unchanged (the Franck Condon principle) In hydrogen the molecule and its ion have appreciably different equilibrium nuclear separations, and the removal of an electron by impact will result in the formation of an ion with vibrational energy The potential energy curves for the ground states of H_2 and H_2^+ obtained spectroscopically are shown in Fig 5 14 The first process producing ionization will be excitation to the stable $^2\Sigma_g^+$ state of H_2^+ The equilibrium internuclear distance for H_2 is less than that of H_2^+ so that the most probable transition is not to the lowest vibrational level of H_2^+ As seen in Fig 5 14, the most probable transition, corresponding to the onset of strong ionization, requires about 16 volts, and this is the value obtained for the ionization potential of H_2 as observed in Fig 5 13

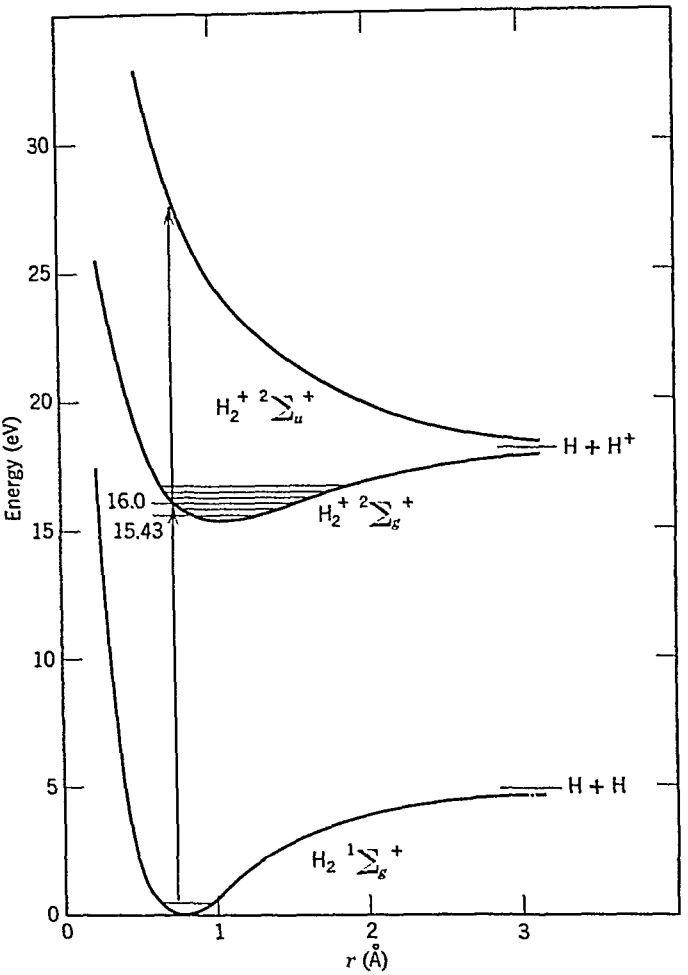


Fig. 5.14 Potential energy curves for hydrogen.

Resonance Charge Transfer

Collision can occur between an ion and a neutral atom or molecule in which the charge is transferred from the ion to the atom. This results in the formation of slow ions with atoms having an energy practically equal to the initial energy of the ions. This process may be of extreme importance on a thermonuclear plasma because what started as an energetic ion may escape from the plasma as a fast neutral atom leaving a low-energy ion behind.

The slow mobility of ions moving among atoms or molecules of the same species has been known for a long time. It is illustrated in Fig. 3.5. In this plot, where the mobility of a large number of ions passing through

nitrogen is shown, it is obvious that the nitrogen ion is the only one that does not fall on the curve This deviation results from the resonant transfer of charge from the nitrogen ion to a nitrogen molecule

The effective collision cross section between two atoms in which electronic transitions occur goes through a maximum when the difference in internal energy, ΔE, is a minimum The maximum cross section will occur when the energy, $h\nu$ is equal to the energy difference ΔE Massey[7] introduced what he called an "adiabatic criterion" for the maximum cross section of the charge transfer mechanism If, on collision, the relative motion of the atoms is so small that the electron motion of transition between atoms can adjust itself to small changes in the internuclear distance, the transition is as unlikely as it would be if there were no energy of relative motion However as the energy rises out of this "adiabatic" region, the transition time becomes comparable to the collision time and the transition probability increases rapidly

The time of collision may be written as $1/v$ and is also equal to a/v where a is the range of interaction and v is the relative velocity

$$v = \left(\frac{2e}{m}\frac{V}{300}\right)^{1/2} \tag{5 2}$$

Here V is the energy of incident ions in volts and m is the ion mass

Thus

$$\frac{a}{v} = \frac{1}{\nu} \quad \text{and} \quad \frac{a\nu}{v} = 1$$

Since

$$\Delta E = h\nu, \qquad \nu = \frac{\Delta E}{h}$$

and we have the result that

$$\frac{a\,\Delta E}{hv} = 1 \tag{5 3}$$

This means that when $v \ll a\,\Delta E/h$, charge transfer does not occur since this corresponds to the Bates adiabatic region and at $v \sim a\,\Delta E/h$, the electron starts to transfer efficiently

The cross section for an ion traveling through its own gas is greatest when its kinetic energy is the least This is shown[9] in Fig 5 15 For ions interacting with atoms of a different species, the cross section passes through a maximum, this maximum occurring at higher energies as the difference in the ionization potentials increases This is shown[9] in Fig 5 16

The measurements of the charge transfer for protons in hydrogen is

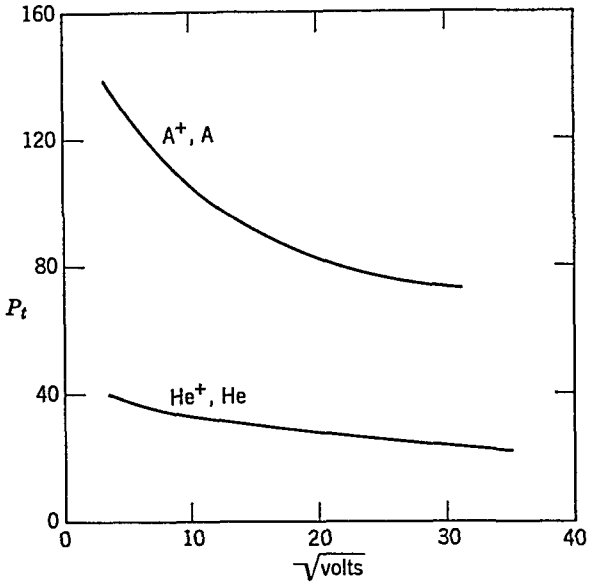


Fig. 5.15 Charge transfer probabilities for ions traveling in their own gas.

interesting. In a very low-energy plasma, the vehicular gas is mainly H_2 whereas in a high-energy plasma, the molecules are dissociated into atoms. Thus one can have a mixture of the two types of charge transfer collisions, one in which the bombarding particle is the same as the vehicular gas, and one in which they differ. These cross sections are

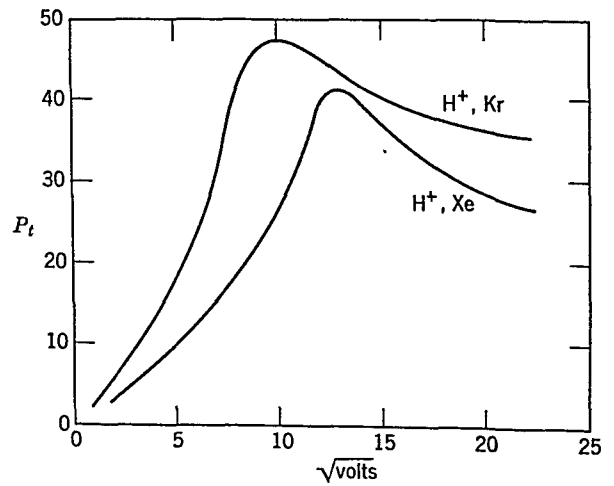


Fig. 5.16 Probabilities of charge transfer for ions interacting with a different species.

shown[8] in Fig 5 17, where to be able to superimpose the two target particles, the cross sections are plotted per atom (as if H_2 were dissociated) Above an energy of 10 keV, the cross sections per atom in H and H_2 become the same

Although many of the charge-transfer cross sections appear to behave in a normal fashion, there are some anomalous cases[9] such as 0^+ in A and

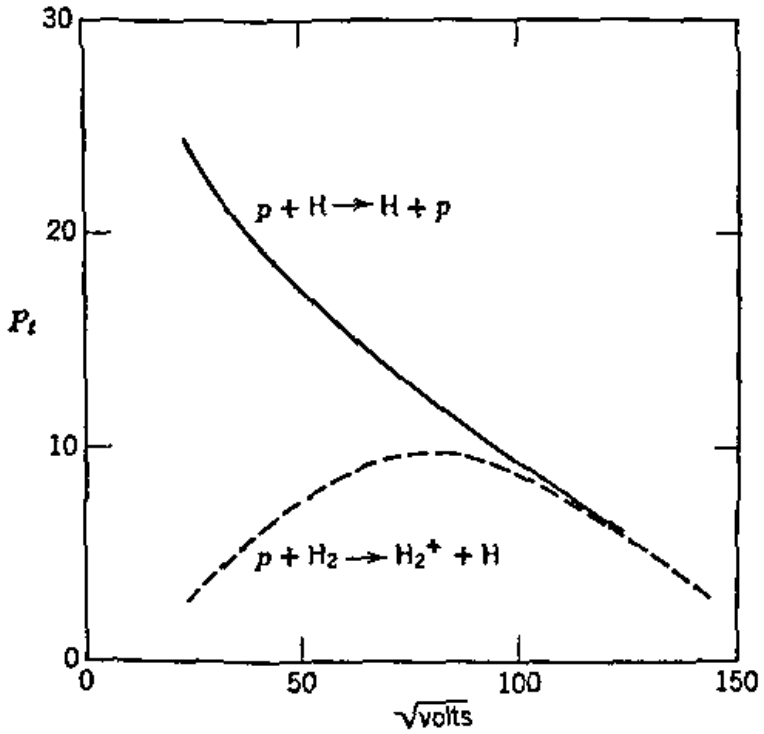


Fig. 5 17 Charge transfer cross sections for protons in H and H_2

N^+ in A and 0^+ in He, given in Fig 5 18 In these cases it seems likely that other inelastic collisions contribute to the measured cross section For the molecular case, there are a large number of possibilities involving vibrational and rotational states as well as electronic metastable states of the oxygen ion, the nitrogen ion, and the oxygen atom Some of these may well have low values of ΔE, a fact that would account for the observed *maximum cross section at very low energies*

When the difference in velocities between the ion and the atom of the same gas becomes large, the charge-transfer cross sections asymptotically approach a constant value that depends upon the ionization potential u_i Sena[10] has worked out an approximate theory for this, which gives for a value of this limiting cross section

$$P_t = \frac{1\,88 \times 10^4}{u_i^2} \tag{5 4}$$

Holstein[11] has worked out a general theory applicable to any charge-transfer collision between an atomic ion and atom for which the wave functions are known. He shows that the charge-transfer collision region may be considered as a sphere of radius r_c surrounding the atom. Inside r_c the probability of charge transfer is $\frac{1}{2}$, and outside r_c it is zero.

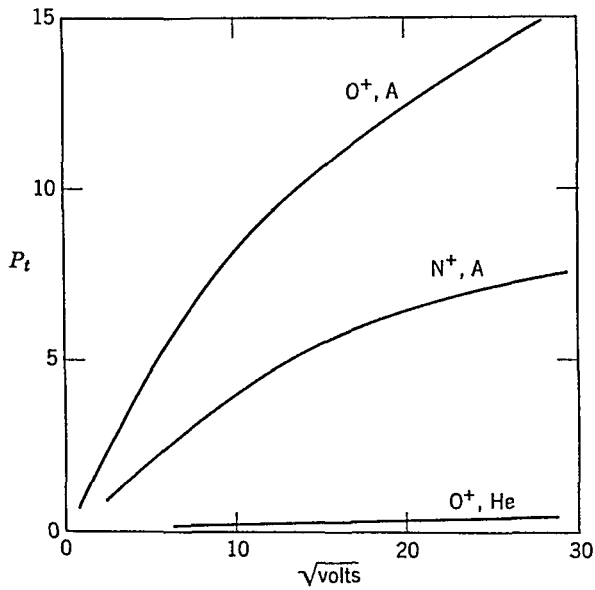


Fig. 5.18 Anomalous charge transfer collision cross sections.

As an ion approaches an atom, it will be deflected toward the atom as a result of the dipole moment it induces in the atom. If the distance of closest approach is greater than r_c, the only deflection of the ion results from scattering. If the ion enters the charge-transfer sphere, half the time it emerges as a neutral atom and the other half remains an ion. The momentum-transfer collision cross section resulting from the charge transfer is

$$P_t = \pi b_c^2 \tag{5.5}$$

where b_c is the critical impact parameter for charge transfer and is related to r_c by

$$\left(\frac{b_c}{r_c}\right)^2 = 1 + \frac{\phi_c}{E}. \tag{5.6}$$

Here E is the initial kinetic energy of relative motion between ion and

atom

$$E = \tfrac{1}{2}m(v_{\text{atom}} - v_{\text{ion}})^2 \tag{5.7}$$

and ϕ_c is the potential energy at r_c resulting from the polarization attraction

$$\phi_c = \frac{e^2 a}{2r_c^4} \tag{5.8}$$

where a is the polarizability of the gas atom

Holstein's theory also gives a relationship between the charge-transfer

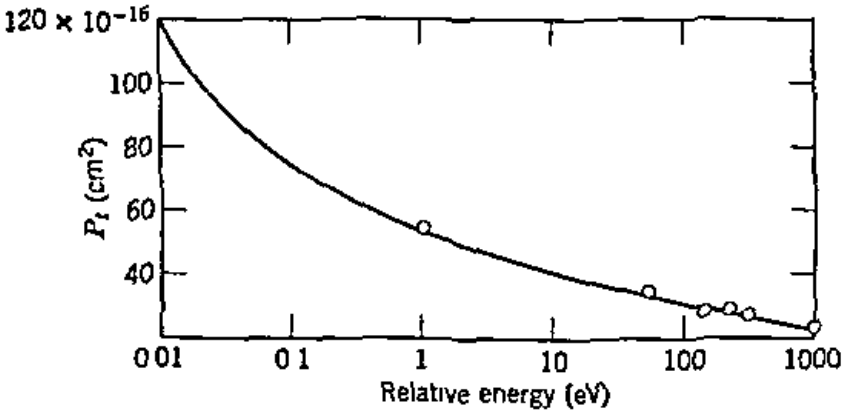


Fig 5 19 Charge transfer cross section for argon ions in argon

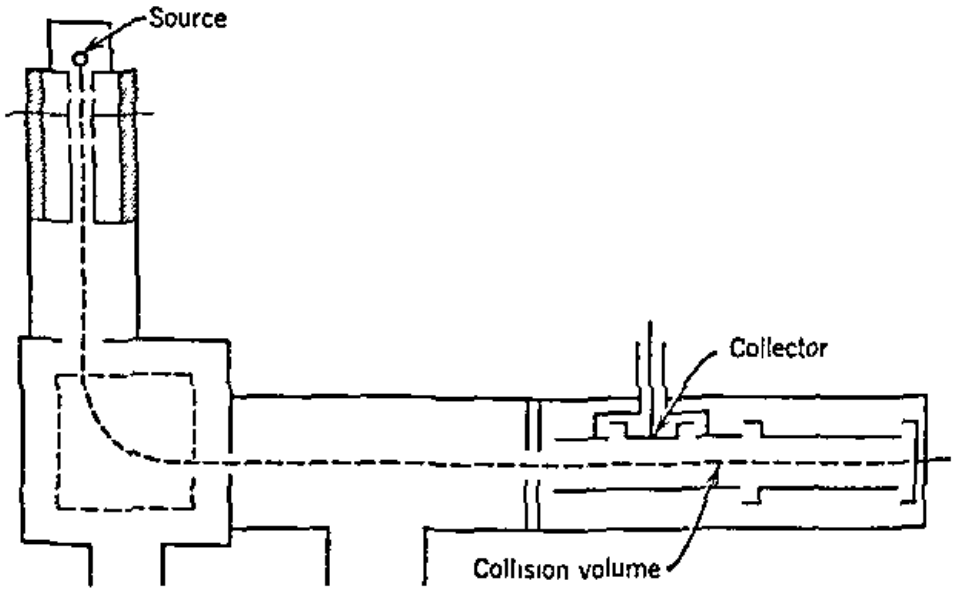


Fig 5 20 Schematic diagram of apparatus for measuring charge transfer phenomena

cross section and the relative energy between the ion and atom. This relationship takes the form[12]

$$E = K_1 P_t^{1/2} e^{-K_2 P_t^{1/2}} \tag{5.9}$$

where K_1 and K_2 are functions of ϕ_c. Fitting the form of this equation to available data[13] by Kushnir, Palyukh and Sena, and Ziegler demonstrates the agreement evident in Fig. 5.19.

A substantial number of measurements have been made of the probability of charge-transfer collisions. In principle, these measurements are carried out by placing in a collision chamber, through which a beam of ions is traveling, an electrode maintained at a potential of a few volts negative relative to the walls of the chamber. It is assumed that the low-energy ions produced in the gas arise from charge transfer and that ions elastically scattered by the gas molecules are scattered only through small angles and without much loss of energy, so that they will not in general be collected. A diagram of such an arrangement for the experimental determination of the charge-transfer phenomenon is shown in Fig. 5.20.

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Chapter 6

Ionization

A great deal of the early work on the passage of electricity through gases was carried out by Townsend and his students Rather than ionization probabilities, the measured current and voltage characteristics were of major concern A typical type of apparatus used in such studies was a parallel plate electrode structure arranged so that a beam of light could fall on the cathode If photoelectrons are released from the cathode plate, we find that as the potential between anode and cathode is increased, a saturation value is reached which corresponds to the collection of all the initially produced photoelectrons As the electric field is further increased, a rapid rise of current corresponds to the production of ionization by collision of the initial photoelectrons with the gas atoms Such behavior is illustrated[1] in Fig 6 1 The data are actually obtained by plotting current

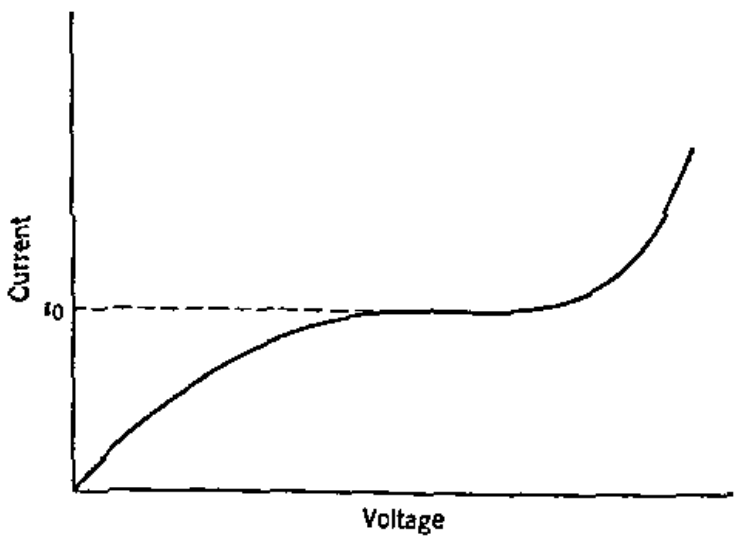


Fig 6 1 Current-voltage characteristics of a typical Townsend tube

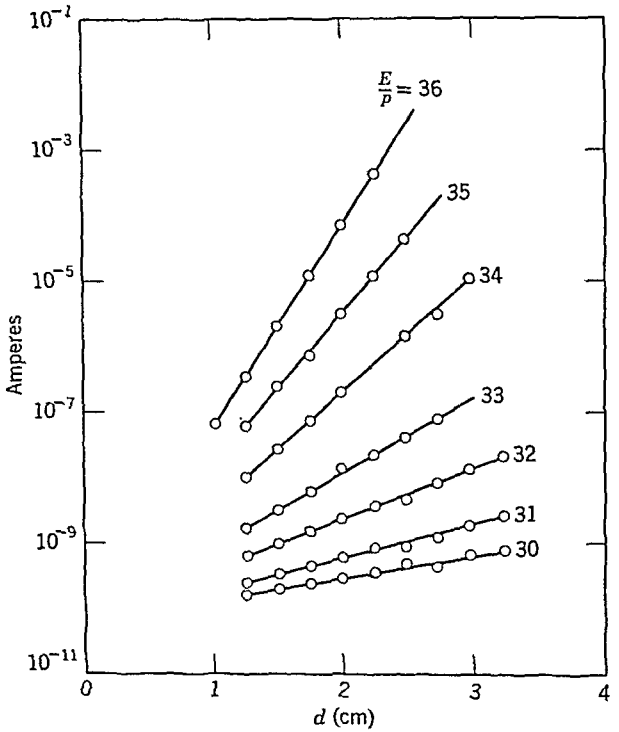


Fig. 6.2 Experimental measurements in air from which to determine Townsend's first ionization coefficient.

as a function of plate separation as illustrated in Fig. 6.2 for a given field strength and pressure.

DC Coefficients

If an electron creates α new electrons in a path length of 1 cm in the field direction, the increase of electrons produced by n electrons in a distance dx is

$$dn = \alpha n \, dx$$

so that

$$n = n_0 e^{\alpha x} \tag{6.1}$$

where n_0 is the initial electron concentration. This corresponds to the observed current variation $i = i_0 e^{\alpha x}$. The quantity α is called the first Townsend coefficient. Alpha itself is not a function of E/p, but α/p is. Data of α/p as a function of E/p have been measured for many gases, and typical curves are shown in Fig. 6.3. Since $\log \alpha/p$ against $\log p/E$ is so nearly linear, it is often convenient to plot them in this way, as illustrated in Fig. 6.4.

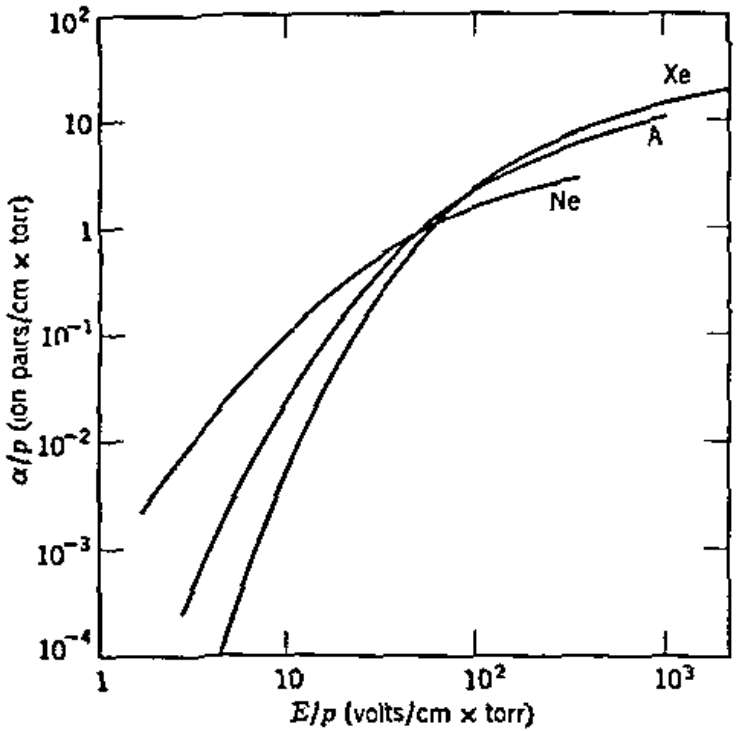


Fig 6 3 Townsend s first ionization coefficients for neon, argon, and xenon

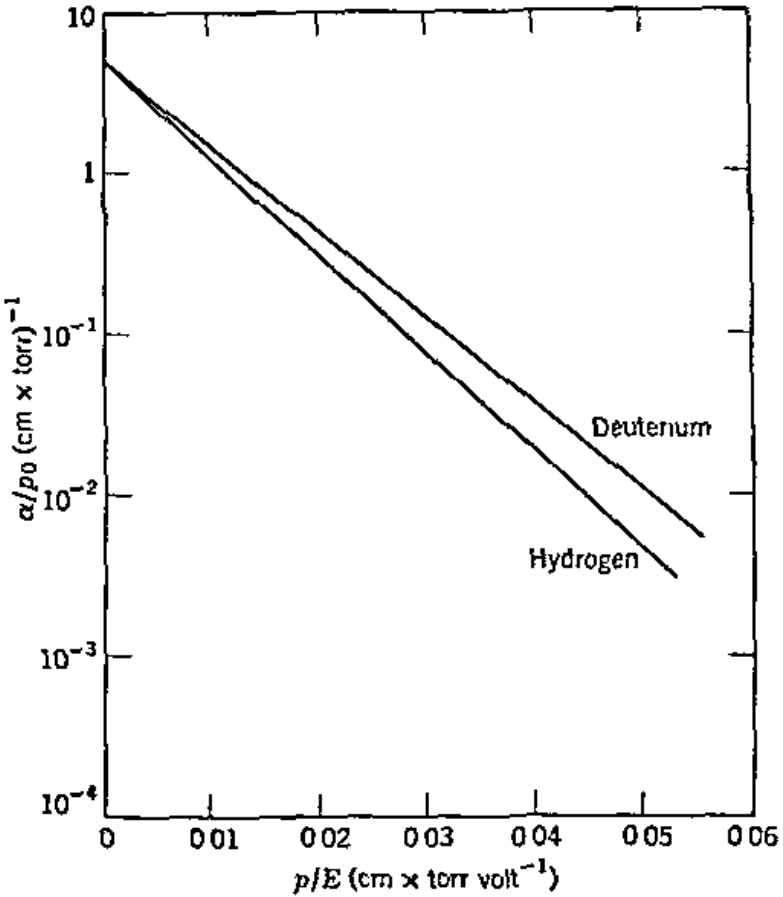


Fig 6 4 α/p as a function of p/E

Another very useful ionization coefficient is that defined in terms of the ionization per volt[2] (rather than per centimeter). In this case

$$n = n_0 e^{\eta(V-V_0)} \tag{6.2}$$

where V_0 is a constant that is necessary because the energy distribution becomes steady only after the electrons have traveled a certain distance from the cathode. We realize that η is actually an average which makes us

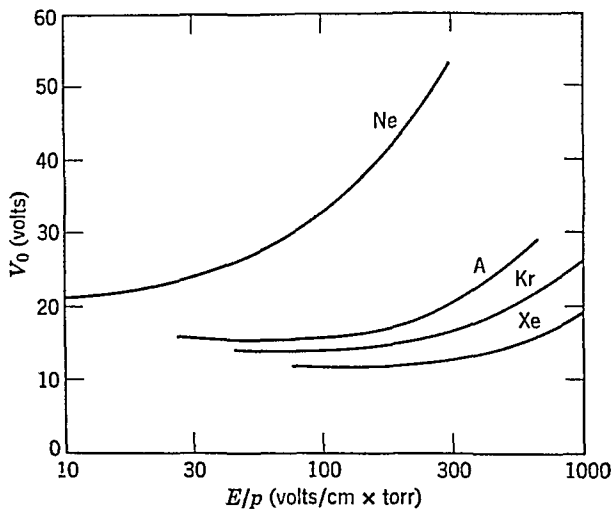


Fig. 6.5 Values of V_0 in Eq. (6.2).

take into account the fact that at shorter distances from the cathode than V_0, η will be a function of the voltage. Thus

$$dn = n\eta(V)\, dV$$

$$n = n_0 \exp\left[\int_0^V \eta\, dV\right].$$

The current increase is determined by this integral. Since there is no increase for voltages less than V_0, the average may be defined as

$$\bar{\eta} = \frac{\int_0^V \eta\, dV}{V - V_0}$$

leading to $n = n_0 e^{\bar{\eta}(V-V_0)}$. V_0 as a function of E/p for the noble gases is shown in Fig. 6.5.

The ionization coefficient η is related to α by

$$\eta = \frac{\alpha}{E} \tag{6.3}$$

Studies of the first ionization coefficients for mixed gases show some interesting physical phenomena The mixtures of neon and argon,[3] illustrating some of these effects, are depicted in Fig 6 6 The very large

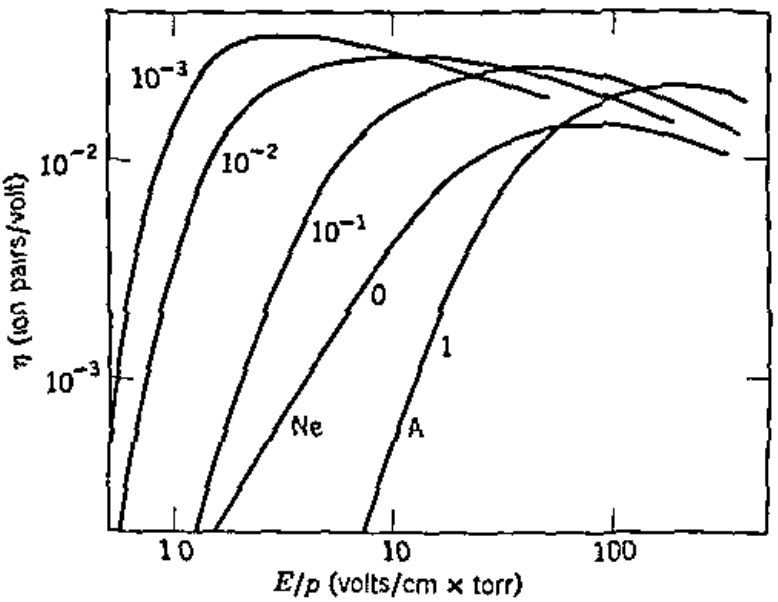


Fig 6 6 Illustration of the Penning effect Ionizations per volt per torr at 0°C for neon argon mixtures The numbers on each curve give the ratio of the argon pressure to the total pressure of the mixture

changes in the values of η result from the interaction of the metastable neon atoms and the normal argon atoms The neon metastable level lies at 16 6 volts, whereas the argon ionization potenial is 15 7 volts A collision between these two results in the ionization of the argon and the return of the neon to its ground state The increase in ionization resulting from this cause is obvious from the very large values of η at low E/p This phenomenon is called the *Penning effect*, and the gas mixtures are called Penning mixtures

One specific Penning mixture that has some very special theoretical advantages is helium gas containing small admixtures of mercury vapor [4] Helium has a metastable level at 19 8 volts, and transitions from this level to the ground state by radiation are forbidden Since the metastable states have mean lives of the order of thousands of microseconds, practically every helium atom that reaches an energy of 19 8 volts collides with a mercury atom and loses its energy by ionizing the mercury Therefore, each inelastic collision produces an ionization, and the effective

ionization potential is $u_i = 19.8$ volts. Since the main gas is helium, this mixture, which is called *Heg*, behaves as a gas with collision frequency ν_c, as one having independence of energy, and as one which has no excitation levels—a true Maxwellian gas.

Table 6.1

Gas	C_1 ion pairs / cm × torr	C_2 V / cm × torr	Range of Validity E/p V / cm × torr	V_i in V
H_2	5	130	150–600	15.4
N_2	12	342	100–600	15.5
O_2	—	—	—	12.2
CO_2	20	466	500–1000	13.7
Air	15	365	100–800	—
H_2O	13	290	150–1000	12.6
HCl	25	380	200–1000	—
He	3	34 (25)	20–150 (3–10)	24.5
Ne	4	100	100–400	21.5
A	14	180	100–600	15.7
Kr	17	240	100–1000	14.0
Xe	26	350	200–800	12.1
Hg	20	370	200–600	10.4

It is often convenient to develop an expression which can be empirically fitted to the form of the α/p versus E/p data. A very common one is due to von Engel and Steenbeck,[5] and takes the form

$$\frac{\alpha}{p} = C_1 \exp\left[-C_2\left(\frac{p}{E}\right)\right]. \tag{6.4}$$

The constants C_1 and C_2 can be fitted empirically over fairly wide ranges of E/p for a large number of gases. For various gases, C_1 has values of a few tens of ion pairs per centimeter × torr, and C_2 has values of a few hundred volts per centimeter × torr. Typical values are given in Table 6.1.

The proper way to derive a theory for the variation of α/p with E/p is not from the behavior of an average electron but from knowledge of the electron energy distribution function. A frequency of ionization per electron may be defined as $\nu_i = v/l_i = vp_0P_i$ where p_0 is the reduced pressure. If we represent the distribution function as $f(u)$, we find

$$\nu_i = p_0 \int_{u_i}^{\infty} P_i v f(u)\, du \tag{6.5}$$

where u_i is the ionization potential, and the electron velocity $\tilde{v}$ is given as

a function of the average electron energy $\bar{u}$ in suitable units as

$$v = \left(\frac{2e}{300m}\right)^{1/2} \bar{u}^{1/4}$$

The first ionization coefficients may be written in terms of an "ionization" collision frequency Since α is the number of electrons produced on collision of the primary electron traveling 1 cm, we can also write $\alpha = \nu_i/v$,

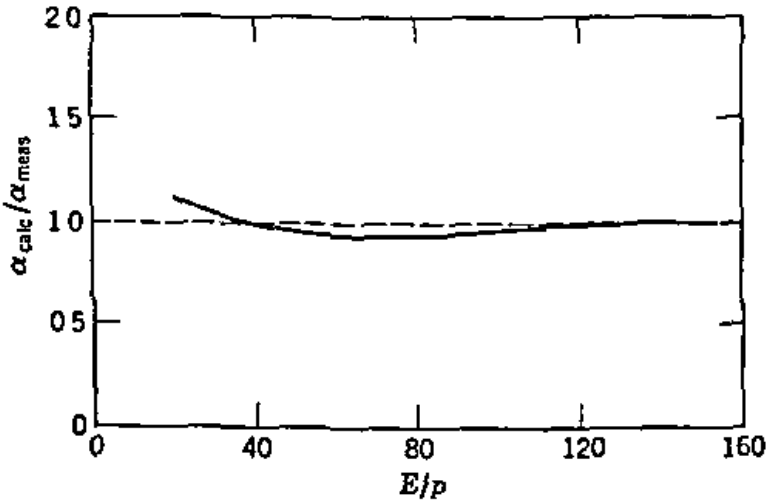


Fig 6 7 Comparison of theory and experiment for the first ionization coefficient in hydrogen

where ν_i is the frequency of ionization on collision and v is the average drift velocity of the electron in the field The average drift velocity $v = \mu E$,

$$\alpha = \frac{\nu_i}{\mu E}, \quad \text{and} \quad \eta = \frac{\nu_i}{\mu E^2} \tag{6 6}$$

Since $\alpha = \nu_i/v$, we find

$$\frac{\alpha}{p_0} = \left(\frac{e}{150m}\right)^{1/2} \frac{1}{v} \int_{u_i}^{\infty} P_i u^{1/2} f(u)\, du \tag{6 7}$$

For a Maxwellian distribution Emeleus, Lunt, and Meek[6] found

$$\frac{\alpha}{p} = \frac{1\,23 \times 10^8}{v} (u)^{-3/2} \int_{u_i}^{\infty} P_i u e^{-(3\,1/2u)}\, du \tag{6 8}$$

By using this equation for hydrogen, Blevin and Haydon[7] found excellent agreement with experimental measurements[8] of the first ionization coefficient This comparison is shown in Fig 6 7

AC Coefficients

When a high frequency electric field is applied to a gas primary ionization is caused by the electron motion, and if we calculate the maximum kinetic energy in the oscillatory motion of an electron at the

minimum field intensity to produce ionization experimentally determined, we find that this energy corresponds to about 10^{-3} volts, a figure very far below the ionization potential.

It is well known that a free electron in a vacuum under the action of an alternating electric field oscillates with its velocity 90° out of phase with the field, and thus on the average takes no power from the applied field. The electron can gain energy from the field only by suffering collisions with the gas atoms. It does so by having its ordered oscillatory motion changed to random motion on collision. The electron gains random energy on each collision until it is able to make an inelastic collision with a gas atom, thus leading to ionization.

When an ultrahigh-frequency field is applied to a gas, the electron loss is controlled by diffusion. As we saw previously, we can discuss this in terms of the characteristic diffusion length Λ where $\nu_i/D = l/\Lambda^2$. By analogy with the first Townsend coefficient for dc ionization where the electron loss is controlled by mobility, we may define a coefficient for high-frequency discharge where the loss is by diffusion.

$$\zeta = \frac{\nu_i}{DE^2} = \frac{1}{\Lambda^2 E^2}. \tag{6.9}$$

Measurements of this ionization are shown as a function of E/p in a typical example[9] reproduced in Fig. 6.8. The ratio of the two ionization coefficients may be written as

$$\frac{\eta}{\zeta} = \frac{D}{\mu}, \tag{6.10}$$

which in Chapter 5 we saw was a measure of the average electron energy. Thus in principle we could determine the dc coefficient η from the ac coefficient ζ if we knew the average energy as a function of E/p. This can be done correctly only for those cases where it can be shown that the electron energy distribution functions are essentially the same. For ultrahigh-frequency breakdown in hydrogen in which a small dc field could be superimposed on the ac field, η was thus determined from ζ by Varnerin and Brown[10] and in Fig. 6.9 is shown to be in agreement with the results of dc measurements[11] of Ayers and of Hale.

Magnetic Fields

A proper variable is a combination of variables that is convenient because it is dimensionless or can be made dimensionless by multiplying by a constant factor.

The proper variable for discussing the effect of the magnetic field is B/p as can be seen from the following argument: $\omega_b/\nu_c = eB/m\nu_c$. As we

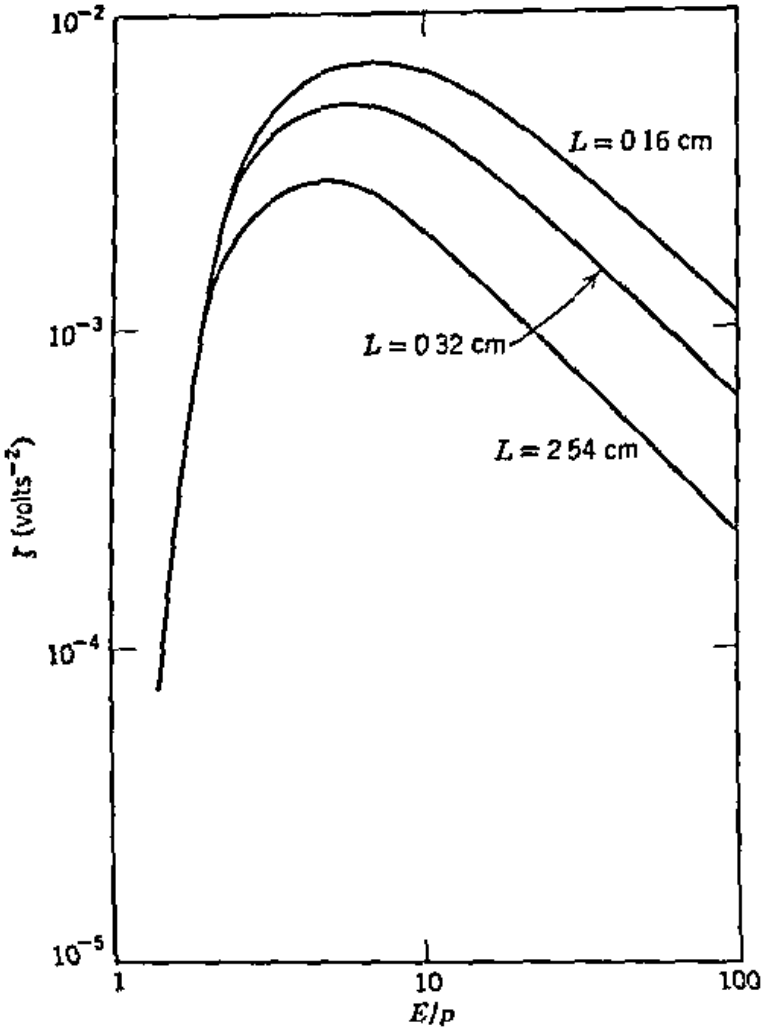


Fig 6 8 AC ionization coefficient for helium

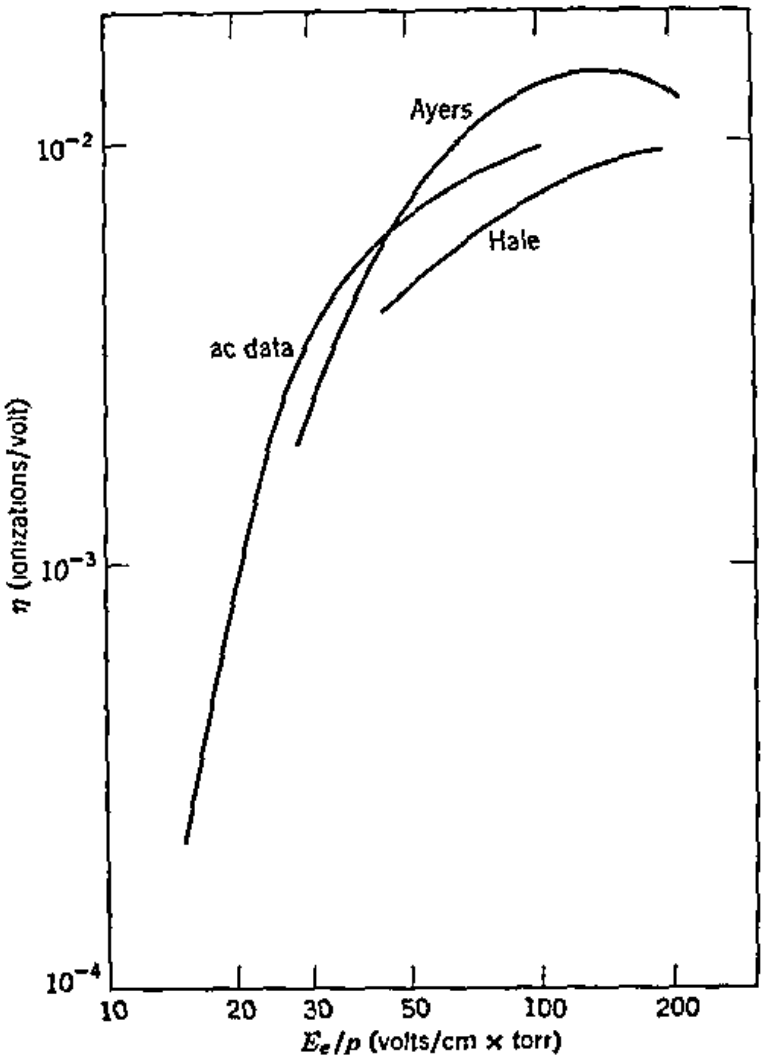


Fig 6 9 Comparison of ac and dc measurements of η

have seen previously, $\nu_c = Kp$ so that $\omega_b/\nu_c = (e/mK)(B/p)$ so that the variable is in fact B/p.

The first ionization coefficient α is modified by the addition of a magnetic field B insofar as the electric field producing the ionization E should be replaced by the effective field E_b, as discussed in the section on free diffusion in a magnetic field. Thus

$$\left(\frac{E}{p}\right)_b = \left(\frac{E}{p}\right)\left(\frac{\nu_c^2}{\nu_c^2 + \omega_b^2}\right)^{1/2}. \tag{6.11}$$

Thus we can modify the empirical Eq. (6.4) to read:

$$\frac{\alpha}{p} = C_1\left(\frac{\nu_c^2 + \omega_b^2}{\nu_c^2}\right)^{1/2} \exp\left[-C_2\left(\frac{p}{E}\right)\left(\frac{\nu_c^2 + \omega_b^2}{\nu_c^2}\right)^{1/2}\right]. \tag{6.12}$$

If we substitute the values for C_1 and C_2 for hydrogen from Table 6.1 into Eq. (6.12) that correspond to various measured values of B/p, this gives a family of theoretical lines which have been checked experimentally by Bernstein[12] with the excellent results illustrated in Fig. 6.10.

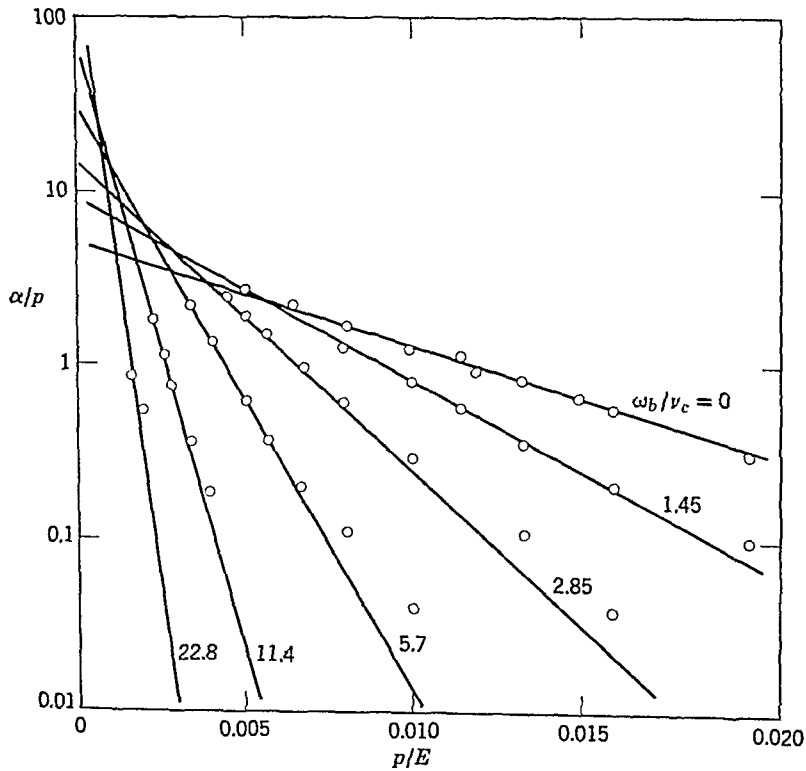


Fig. 6.10 Agreement between theory and experiment for α/p *versus* p/E in hydrogen with various of magnetic field.

Photoionization

Light passing through a gas is absorbed according to the usual equation $I = I_0 e^{-\mu x}$, where x is related to the path length L by $x = L(p/760)(273/T)$ At standard temperature and pressure μ is proportional to Loschmidt's number $n_0 = 2.69 \times 10^{19}$ molecules per cubic centimeter The proportionality constant Q_p has the units of cm^2 and hence is called a cross section $\mu = Q_p n_0$

The methods of measuring the photoionization cross sections involve the following steps A monochromatic light source is used to measure the ratio I/I_0 and the optical depth of the gas x to determine the total absorption coefficient μ The number of ions created per second by the light must also be measured along with the absolute intensity of the source I_0, at a given wavelength in terms of the number of photons per second impinging on the gas

From the total absorption coefficient μ and the absolute intensity I_0, we can calculate the number of photons removed from the light beam per second The ratio of the ion current collected to the number of

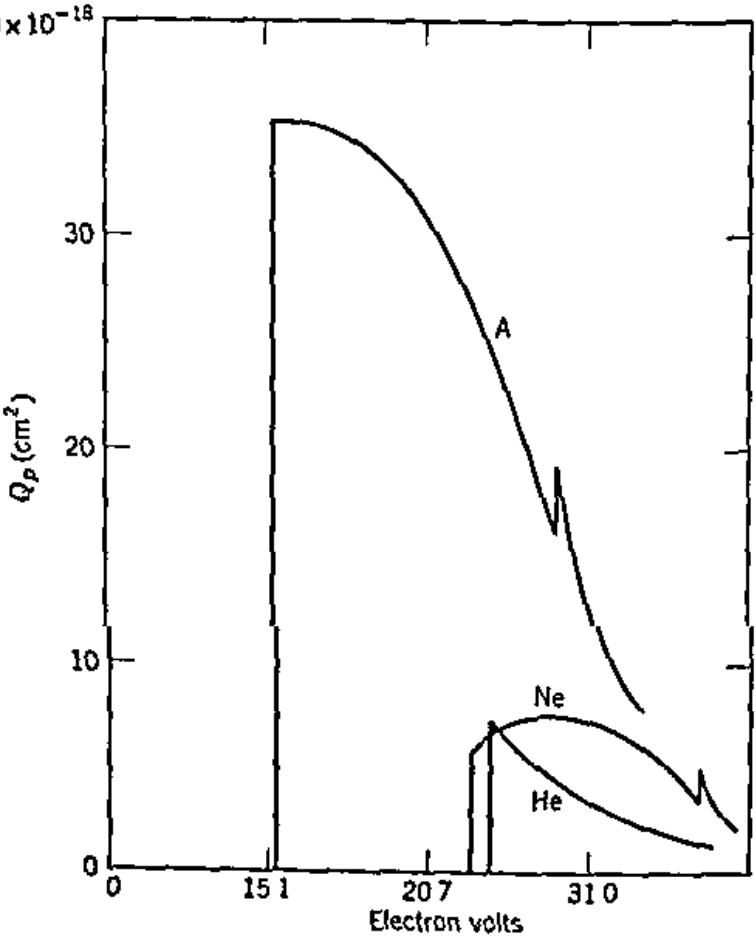


Fig 6 11 Photoionization cross sections in argon, neon and helium

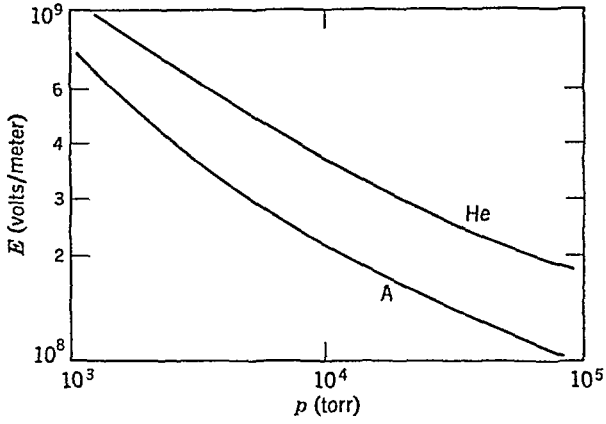


Fig. 6.12 Laser light breakdown in helium and argon as a function of pressure.

photons absorbed gives the photoionization efficiency. This number multiplied by the total absorption cross section μ gives the photoionization cross section Q_p for the given wavelength of incident light. Typical curves[13] as a function of photon energy are shown in Fig. 6.11.

With a sufficiently powerful beam of light, such as a laser beam, direct ionization can result from an electron absorbing sufficient energy from the photon beam. This occurs in a multiple-stage process with visible light where low-energy photons of between 1 and 2 electron volts are absorbed by an electron in the vicinity of an atom (to conserve momentum), thereby increasing the electron's random kinetic energy. After multiple collisions between photon, electron, and gas atoms, the kinetic energy of the electron exceeds the ionization potential of the gas atoms, and ionization occurs. Meyerand and Haught[14] have achieved particle densities of 10^{17} electrons per cubic centimeter using a 30-megawatt ruby laser with a pulse length of 50 nsec. Because of the necessity for the three-body collision involved in this mechanism, ionization has only been observed at high pressures. The ionization becomes easier, the higher the pressure, and the lower the ionization potential of the gas. This is illustrated in Fig. 6.12.

Thermal Ionization

The usual method for dealing with the problem of thermal ionization is by a thermodynamic treatment of dissociation equilibrium[15], a treatment which is good only in thermodynamic equilibrium. From the law of mass action

$$\frac{p_1 p_2}{p_0} = CT^{5/2} e^{-W/kT}$$

where the p's are the partial pressures of the different species, C is a

constant derivable from the Nernst heat theorem and takes the form

$$C = \left(\frac{2\pi m_e k}{h^2}\right)^{3/2} k$$

and W is the heat of reaction or the enthalpy per mole associated with the reaction In our notation, $(p_1 p_2)/p_0 = (n_+ n_-/n_0)kT$, so that

$$\frac{n_- n_+}{n_0} = CT^{3/2} e^{-eu_i/kT} \tag{6 13}$$

This equation, often called the Saha equation, is usually expressed in terms of the degree of ionization, $f = n_-/n_0 = n_+/n_0$ If p is the actual gas pressure,

$$\frac{f^2}{1-f^2} p = 2\,4 \times 10^{-4} T^{5/4} e^{-eu_i/kT} \tag{6 14}$$

When the degree of ionization is very low, $f \sim e^{-eu_i/2kT}$ and proportional to $l/\sqrt{p}$, also $n_- \sim \sqrt{n_0}$

A typical plot of Eq (6 13) for three different values of the ionization potential is shown in Fig 6 13 For example, using this plot for a real gas

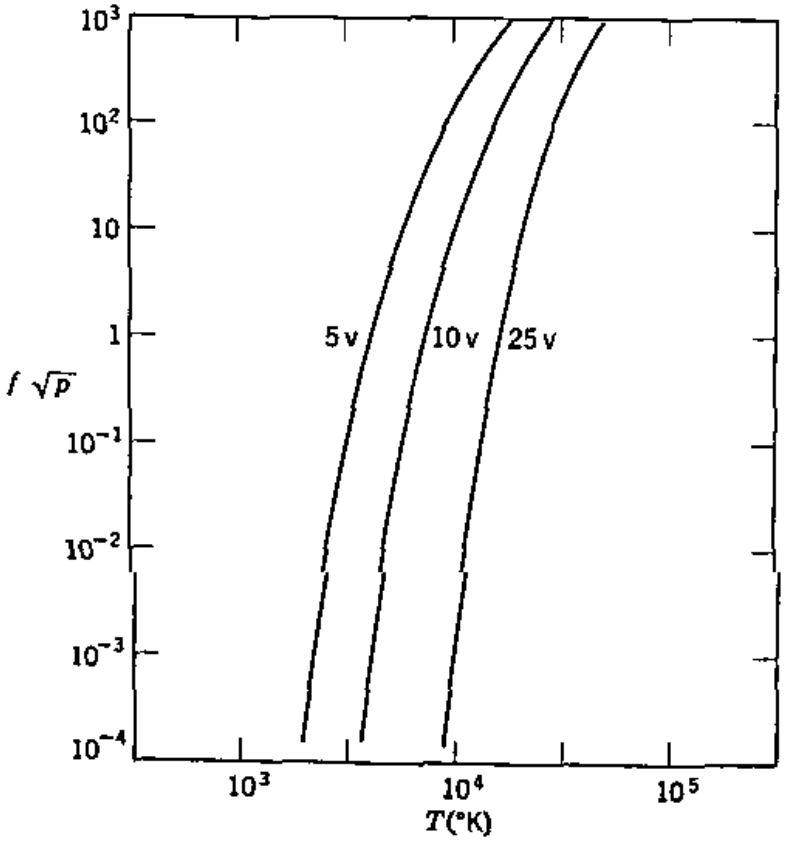


Fig 6 13 Degree of ionization in a gas as a function of temperature for gases of different ionization potentials

like nitrogen ($u_i = 15$ ev.), at one atmosphere pressure the gas would be 10% ionized at 13,000°K.

Shock Wave Ionization

Although in the steady state it is difficult to achieve temperatures high enough to produce thermal ionization predicted by Eq. (6.14), it is a common phenomenon in shock tubes. The average speed of particles in a perfect gas is given by kinetic theory

$$\langle v \rangle = \left(\frac{8kT}{\pi m}\right)^{1/2}.$$

For a monatomic gas, the velocity of propagation of sound is

$$C = \left(\frac{5kT}{3m}\right)^{1/2}.$$

As long as the source producing the pressure amplitude variations does not move at speeds equal to or greater than the propagation velocity, a sound wave that travels at the propagation velocity is produced. If the sound source velocities exceed the normal propagation velocity, the waves produced travel at a higher velocity and are termed shock waves. A shock wave, unlike a sound wave, is not associated with a vibratory source, but is usually a sudden compression like the explosion of a diaphragm, or the expansion of a spark discharge.

The essential effect of the shock wave is to convert the basically parallel particle velocity components into random velocities having a net drift velocity much smaller than the initial parallel velocities. Thus the mechanism appears to be collisions of particles with resulting scattering and randomizing of velocities. Thermodynamic equilibrium is reached by a process of relaxation immediately following the shock wave, and here some of the kinetic energy is absorbed by inelastic collisions including ionization following Eq. (6.14).

The construction of a conventional shock tube is about as simple as a piece of apparatus can be, consisting of two lengths of straight tubing separated by a thin diaphragm. The two sections can be filled and evacuated separately, one section being used as a driver section and the other as the shock tube itself. Customarily the pressure in the driver section is increased until the diaphragm bursts and a shock wave then travels down the shock tube. The pressure in the driver section may be increased slowly by a compressor or, particularly in the production of strong shocks, the driver section may be filled with a combustible mixture like oxygen and hydrogen and the mixture ignited.

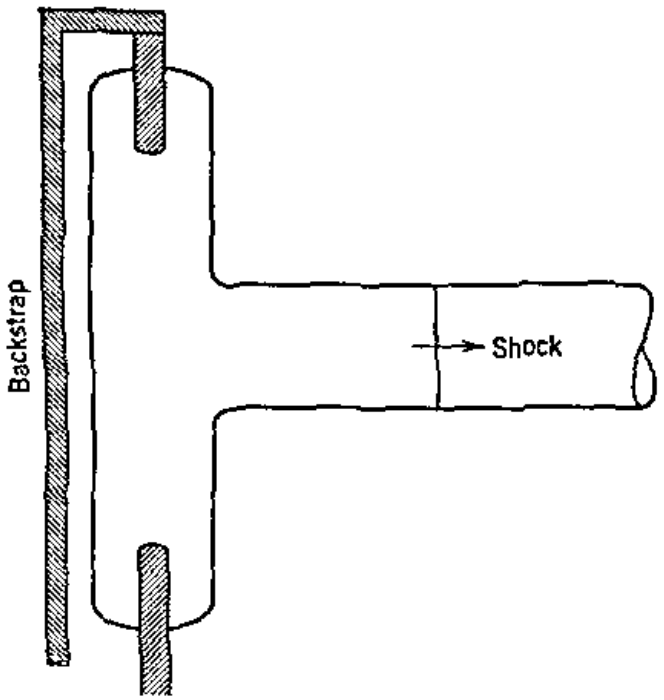


Fig 6 14 Magnetically driven shock tube

Electrically driven shock waves are produced experimentally by discharging a capacitor through an ionized gas, the heat produced by the spark initiating the shock A variation of this method is called a *magnetically driven* shock which is illustrated in Fig 6 14 Here the shock wave is produced by the discharge between the electrodes, and the plasma is accelerated down the shock tube by the magnetic field caused by the current flowing through the *backstrap*

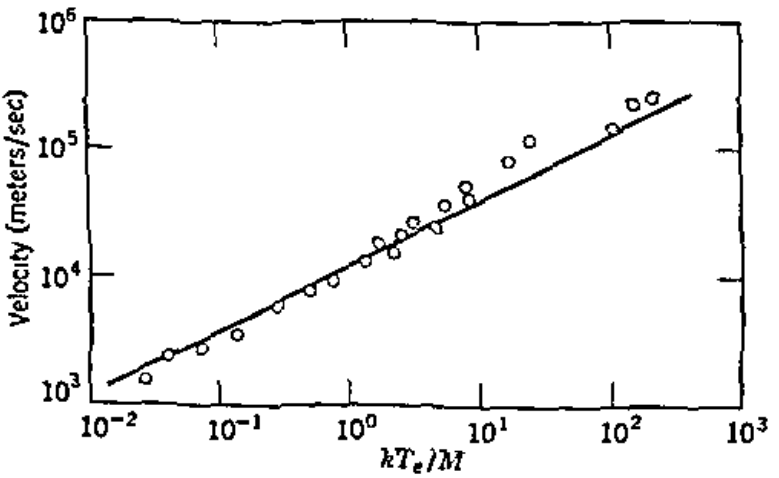


Fig 6 15 Velocity of an electron driven shock wave as a function of electron temperature The data points correspond to various gases (helium hydrogen deuterium and xenon)

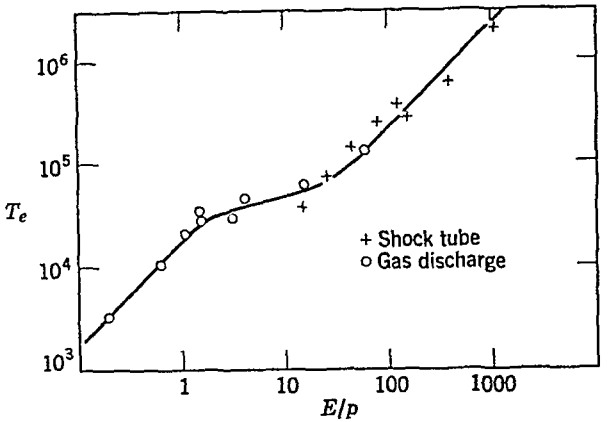


Fig. 6.16 Comparison of electron temperatures as a function of E/p in electrical and shock wave ionization.

Electron-driven shock waves travel with a velocity given by the relation $v^2 \sim kT_e/M$, where M is the mass of the ion.

Fowler, Paxton, and Hughes[16] have derived a semiempirical relation between the velocity of the electron shock and the energy per mean free path (E/p) of the discharge producing the shock, taking into account the diffusion loss of electrons in cylindrical shock tubes. Their relation is

$$\frac{kT_e}{M} = \frac{e}{3(3Mm)^{\frac{1}{2}}} \frac{E/p}{P_c\left[1 + \left(\frac{2.405}{a}\right)\Big/ P_c p\right]} \tag{6.15}$$

where a is the radius of the shock tube. Using the temperature–E/p relationship of Eq. (6.15), we can compare this with the gas discharge measurements of T_e versus E/p, with the result illustrated in Fig. 6.15. The difference between the low E/p and high E/p curves results from the difference in collision probability between electrons and neutrals and electrons and ions as illustrated[17] in Fig. 6.16.

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Chapter 7

Secondary Emission from Surfaces

The secondary electron emission from surfaces is not strictly a gas discharge phenomenon. A great many of the observed effects associated with the ionized state, however, are controlled by the secondary emission of electrons when the photons, electrons, and ions created in the discharge hit the walls or electrodes of the containing vessel. For these reasons, a great deal of study has been aimed at understanding these secondary effects.

Secondary Electron Emission Produced by Electron Impact

When a surface is bombarded by electrons, secondary electrons can be emitted. Typical of the experimental method of studying this effect is the method of Farnsworth,[1] whose apparatus is sketched in Fig. 7.1. The essential feature of the apparatus is the conducting sphere at whose center the target is placed. The primary electrons enter the sphere through *D*,

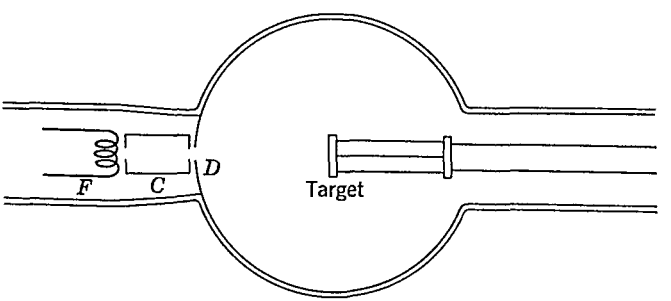


Fig. 7.1 Farnsworth's apparatus for measuring secondary electron emission from surfaces under electron bombardment.

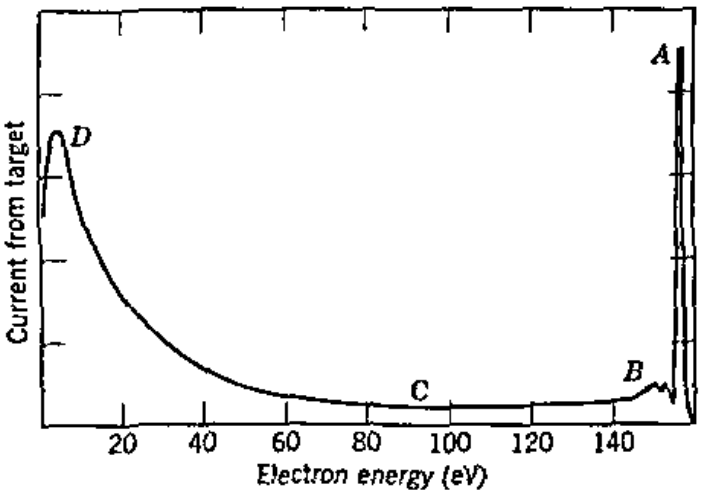


Fig 7 2 Secondary electron velocity distribution from gold

they come from the filament F, and are accelerated between F and C The target may be moved into the side tube by a magnet where new surfaces can be sputtered on by a high-frequency induction heater The current to the sphere measures the secondary electron effect, while the total current to the sphere plus the target give the primary current By adjusting a retarding potential between the target and the conducting sphere, the energy distribution of the secondary electrons may be determined A typical distribution of this sort[2] is given in Fig 7 2 for a bombarding energy of 150 volts The large peak at A measures the direct reflection of the primary beam The fine structure at B represents single inelastic scattering where the structure observable in this peak is given by the distribution of energy bands for electrons in the metal of the target The large low-energy peak D, results from the true secondary electron emission The intermediate region C is a mixture of the long high-energy tail of the secondary electrons and beam electrons which have been scattered within the target and directed backwards into the vacuum

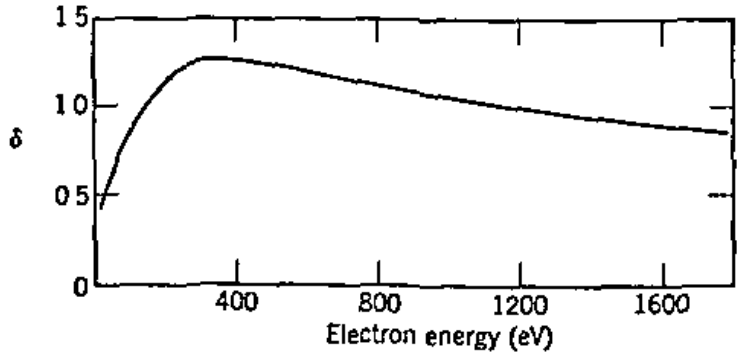


Fig 7 3 Secondary electron yield curve for electrons on molybdenum

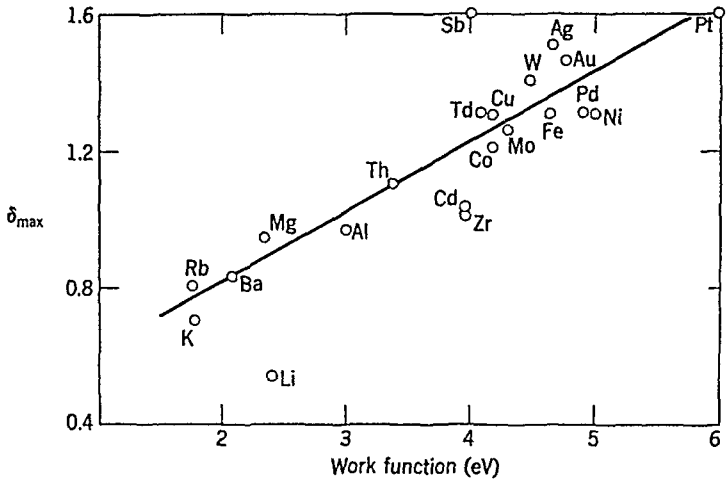


Fig. 7.4 The maximum secondary electron emission yield plotted against the work function.

The most commonly reported quantity in secondary electron emission studies is the yield of secondaries, given the symbol δ, which is defined as the ratio between the total number of emitted electrons and the total number of primary electrons. A typical yield curve[3] is shown in Fig. 7.3. A correlation between the work function of the target and the maximum yield is illustrated in Fig. 7.4. Although the work function probably plays an unimportant role in secondary electron emission, other properties of the metal that do depend upon the work function play a dominant role in determining the yield, and thus lead to the result shown in the figure.

Multipactor Effect

A number of workers have studied the case of electron multiplication attributed to secondary electron emission by direct bombardment of the electrodes in a high-frequency experiment. Typical of these studies are those of Hatch and Williams[5] who have used parallel plate electrodes in an evacuated bell jar. Electrode separation was adjusted by means of a sliding plunger passing through an O-ring seal. The usable electrode separation varied from 1 to over 5 cm. Power was supplied at frequencies of 70 and 140 megacycles/sec. Breakdown was indicated by the appearance of a glow and the sudden lowering of the applied voltage.

In considering the behavior of electrons in these experiments, not only is the magnitude of the electron motion important, but also the phase. An electron crossing the electrode space will collide with the opposite electrode and release secondary electrons. If this occurs as the electric field passes through zero, the reversed electric field will accelerate the

secondaries back across the gap When the electric field has such a value that the transit time across the gap equals an odd integral multiple of one-half cycle of the ac field, the secondary electrons formed by the initial electron become primary electrons for the next half-cycle to form another group of secondary electrons In this way, large electron densities may be built up in the electrode gap The phenomenon is called the multipactor effect

To describe this behavior analytically let us write down Newton's second law including the phase angle ϕ

$$m\frac{dv}{dt} = -eE_p \sin(\omega t + \phi) \tag{7 1}$$

By solving for the velocity of arrival of the primary electron v in terms of the velocity with which it was emitted as a secondary v_0, we find

$$v = v_0 - \frac{2eE_p}{m\omega}\cos\phi \tag{7 2}$$

and by taking $k = v/v_0$ as a constant,

$$v = \frac{k}{k-1} - \frac{2eE_p}{m\omega}\cos\phi, \tag{7 3}$$

and integrating Eq (7 2) again to determine L (the plate separation) evaluated for a one half cycle transit time, and solving for E_p, we get

$$E_p = \frac{\omega^2 L}{e/m\Phi} \tag{7 4}$$

where

$$\Phi = (2n-1)\frac{k+1}{k-1}\pi\cos\phi + 2\sin\phi$$

Here n signifies the number of half-cycle transit times of the electrons across the gap If we write the velocity in terms of the energy $eu = mv^2/2$, we find

$$fL = \frac{(k-1)\Phi}{\pi k\cos\phi}\left(\frac{eu}{8m}\right)^{1/2} \tag{7 5}$$

In simple experiments where the breakdown of the gap is determined by merely raising the field until the gap becomes conducting at a constant frequency, data as shown in Fig 7 5 are obtained The vertical rise of breakdown field at a given frequency is called the *cut off* For a constant k, Φ, and u this cut off is determined from Eq (7 5) as

$$f_{co}L = \text{constant} \tag{7 6}$$

Experimental verification of this relation is illustrated in Fig 7 6

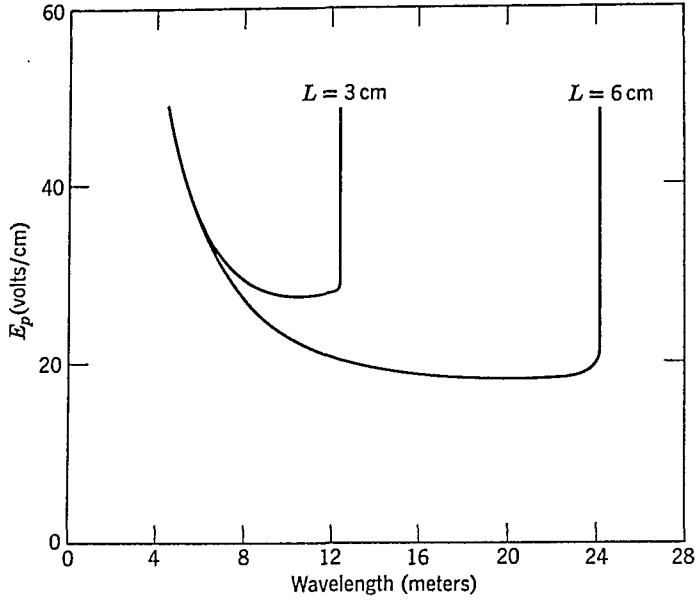


Fig. 7.5 Breakdown field strength as a function of wavelength for flat-ended cylindrical tubes at 3-cm and 6-cm length with axes parallel to the electric field.

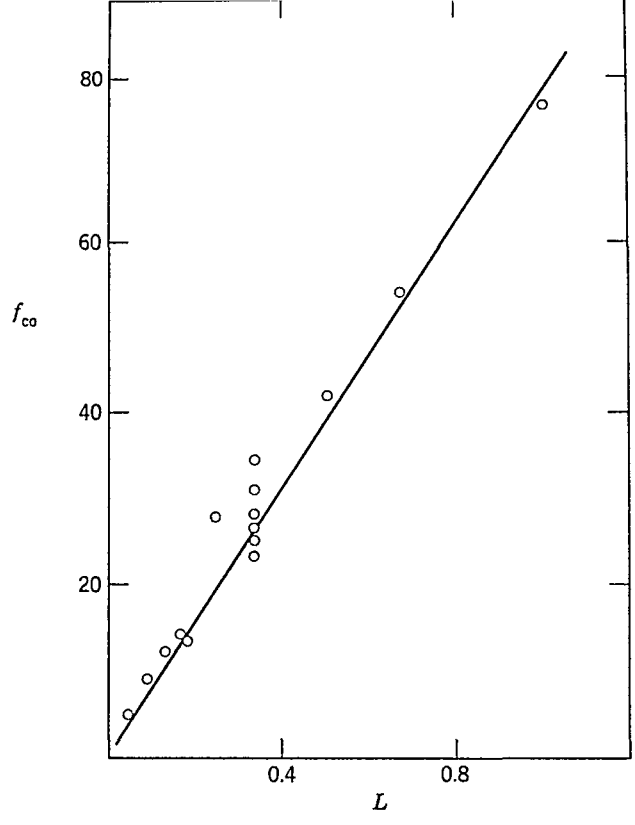


Fig. 7.6 Cut-off frequency as a function of transit distance.

On more comprehensive experiments, not only the minimum field for the secondary emission yield $\delta = 1$ was utilized but also the maximum field for $\delta = 1$ Measurements of this sort by Hatch and Williams[5] are reproduced in Fig 7 7 Data on the lower branch of the curve were taken

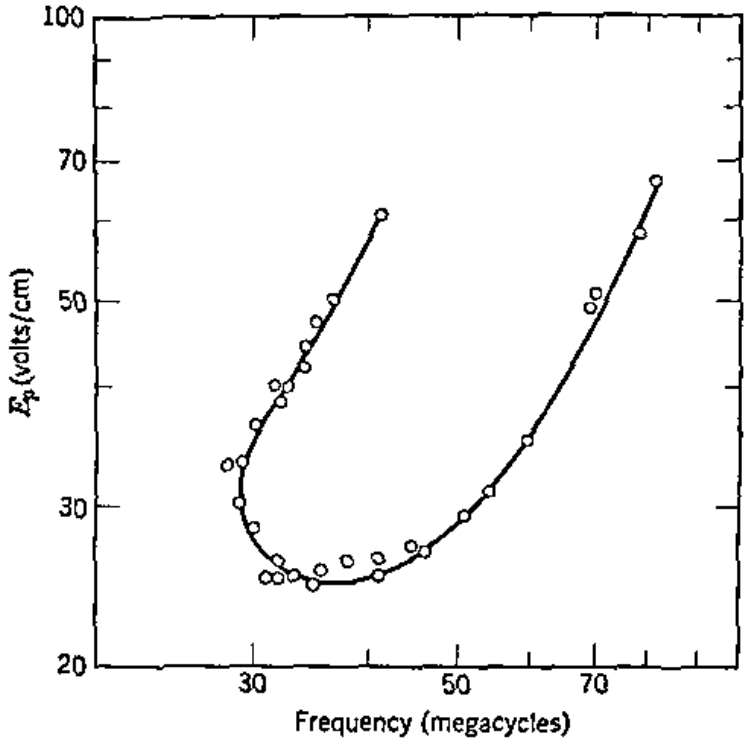


Fig 7 7 Breakdown field strength as a function of wavelength for hydrogen at a pressure of 10 ^{3} torr

by increasing the electric field strength until breakdown occurred By suddenly applying a high field and then lowering it slowly, the upper breakdown curve was observed

Effect of a Magnetic Field

Electron multiplication can occur in a number of ways in a magnetic field, and one phenomenon depending upon secondary electron emission at the end walls of a quartz tube serve as an illustration Experiments have been carried out[7] in a cylindrical microwave cavity oscillating in a TE_{111} mode at a wavelength of approximately 10 cm The gas is contained in a quartz tube of 1 0 cm radius coaxial with the axis of the cavity The magnetic field is parallel to the axis of the cavity

The equation of motion of an electron in crossed uniform E and B fields is given by

$$m\frac{dv}{dt} = -(e\mathbf{E} + e\mathbf{v} \times \mathbf{B}) \tag{7 7}$$

An important quantity is the average time during which an electron can keep gaining energy. This quantity is controlled by collision events that disrupt the electron-energy gain. First, there are collisions with the gas atoms. In hydrogen the collision frequency for electrons with gas atoms is almost independent of electron energy above 4 electron volts, given by $\nu_m = 5 \times 10^9 p \text{ sec}^{-1}$. For a representative pressure of 10^{-4} torr, the collision time τ_c is approximately 2 μsec. Thus, at a value 0.5% off cyclotron resonance, the electron will gain and lose its energy approximately 70 times before it collides with an atom. Second, there are collisions with the walls of the quartz tube. The time taken between collisions with the radial walls is much longer than 2 μsec, since a representative 20-electron-volt electron has a cyclotron radius of only 0.01 cm.

At low pressures, when electron collisions with the gas are infrequent, the electrons gain energy from the microwave field only through the cyclotron-resonance effect. If we assume that the electron enters the field at time $t = 0$ with zero velocity, we find that its energy in a direction perpendicular to the B-field $u_\perp$ is obtained from Eq. (7.7) as follows:

$$u_\perp = \frac{e^2E^2}{2m(\omega - \omega_b)^2} \sin^2 [\tfrac{1}{2}(\omega + \omega_b)t]. \tag{7.8}$$

Equation (7.8) gives the electron energy for all values of ω_b. At $\omega = \omega_b$, the condition for cyclotron resonance, $u_\perp$, is given by

$$u_\perp = \frac{e^2E^2}{8m} t^2. \tag{7.9}$$

Since many collisions occur before particles reach the radial wall diffusion theory holds, at least in the radial direction. The effective diffusion length in the radial direction is given by

$$\Lambda_e = \frac{R}{2.405} \frac{\omega_b^2 + \nu_m^2}{\nu_m^2}, \tag{7.10}$$

which, for our representative case, is approximately 100 m for a pressure of 10^{-4} torr. Thus, as a first approximation, we can neglect collisions with the radial walls, as compared with collisions with the end walls.

The time of a collision with the end walls, τ_w, is determined by the axial velocity v_z. The average axial velocity for electrons should be between 10^7 and 10^8 cm/sec. If we take a value of 5×10^7 cm/sec and a tube length of 6 cm, we have $\tau_w \approx 0.1$ μsec. Thus $\tau_c \approx 20\tau_w$. This implies that the electron-energy gain, at least at cyclotron resonance, will be determined by collisions with the end walls rather than by collisions with atoms.

With uniform magnetic fields at cyclotron resonance, conduction can be produced and maintained with surprisingly low microwave powers (milliwatts), in spite of an expected large electron loss at the end walls of the quartz tube Although mechanisms at the quartz end walls are not well understood, it is a fact that the loss of electrons to the end walls is small, if it exists This loss occurs because the net flow of charge to the end walls must be zero in equilibrium If the end walls charge negatively, the loss of electrons to the end walls is determined by the ion drift velocity If the

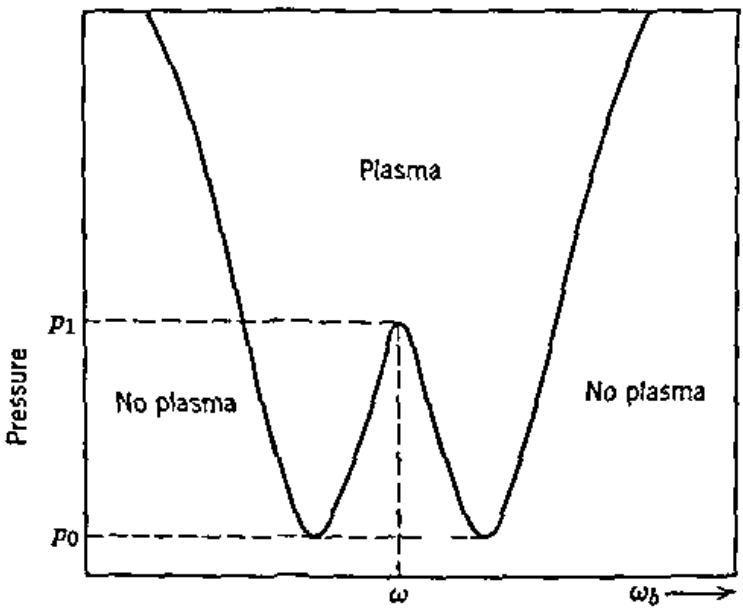


Fig 7 8 Low-pressure microwave breakdown in a magnetic field

end walls charge positively, the ions cannot reach the wall, and no electrons can be lost in the axial direction Whether the end wall charges negatively or positively is determined by the energy of the impinging electrons The yield is unity for energies between 30 eV and 50 eV

The characteristic behavior of this low-pressure gas-discharge breakdown in a magnetic field is illustrated in Fig 7 8 in which the necessary pressure for breakdown at a given value of electric field is plotted as a function of the magnetic field The graph of Fig 7 8 is symmetrical because the electron energy is a function of $\omega - \omega_b$ Thus deviation from $\omega - \omega_b$ in either direction causes the electron energy to decrease

The surprising feature of the plots of Fig 7 8 is that for large electric-field strengths, it is easier to break down the gas with a frequency different from that corresponding to cyclotron resonance than at cyclotron cyclotron resonance This can be explained by referring to Fig 7 9, in which the ionization frequency ν_i is plotted as a function of electron energy for several pressures In Fig 7 9, ν_L, the frequency of electron loss from

the discharge, is also plotted. Although the exact nature of ν_L is uncertain, we can assume that it is independent of gas pressure and that it increases with electron energy. The intersection of the ν_i and ν_L curves gives the energy and pressure at which a discharge can be maintained. There are two such points of intersection for each pressure above a minimum pressure $p_{\min}$. Thus, the surprising feature of Fig. 7.8 is qualitatively

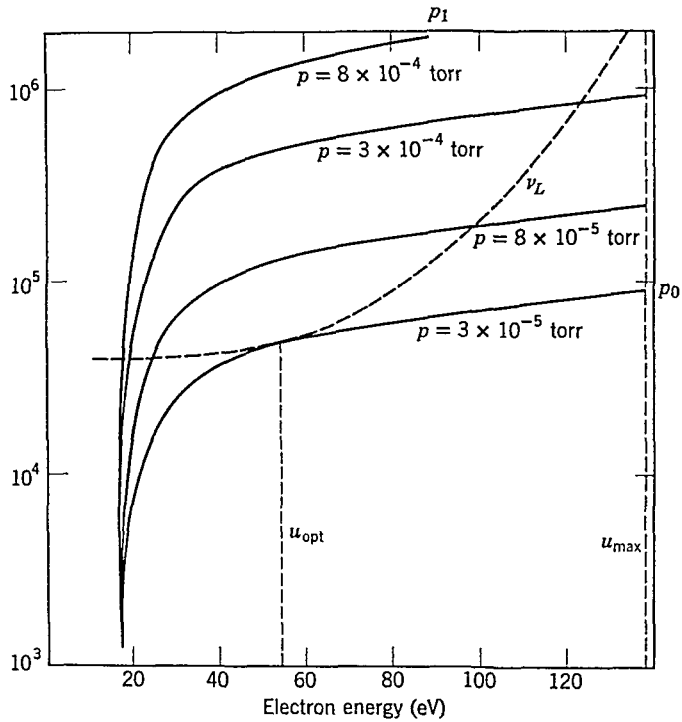


Fig. 7.9 Frequency of ionization as a function of electron energy.

explained. At cyclotron resonance (high mean electron energy), the frequency of electron loss increases faster with energy than does ν_i. As a result, it is easier to maintain the plasma at lower electron energies, that is, off cyclotron resonance. There is an optimum energy u_{opt} that corresponds to easiest breakdown.

Secondary Electron Emission Produced by Heavy Particle Impact and Photons: the Second Townsend Coefficient

When a beam of positive ions strikes a surface, secondary electron emission is likely to occur. Townsend measured this effect of the secondary processes in the same experiment that he measured α/p. He showed that

the current as a function of plate separation for constant E/p did not increase as a simple logarithm for high current but increased more rapidly at high values, as shown[8] in Fig 7 10 These data were described in terms of the current carried by the discharge as

$$i = \frac{i_0 e^{\alpha x}}{1 - \gamma(e^{\alpha x} - 1)} \tag{7 11}$$

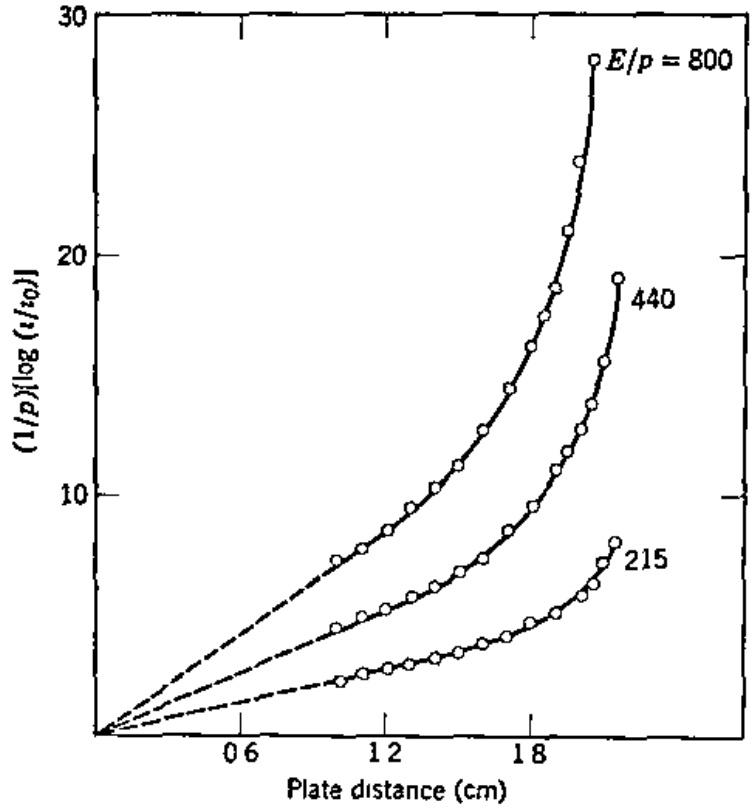


Fig 7 10 Townsend type measurements to determine the second ionization coefficient

In this expression, γ is the efficiency of production of secondary electrons per ion pair formed in the gas It is this quantity which is called Townsend's second ionization coefficient

A simplification of Eq (7 11) results from the fact that $e^{\alpha x} \gg 1$ and

$$i = \frac{i_0 e^{\alpha x}}{1 - \gamma e^{\alpha x}} \tag{7 12}$$

When the value of x approaches a value d for which the denominator approaches zero, the current tends to infinity independent of i_0 This condition defines the breakdown of a gas into a conductor For this condition

$$\gamma e^{\alpha d} = 1 \quad \text{or} \quad \gamma e^{\eta l \, g} = 1 \tag{7 13}$$

where V_B is the breakdown voltage. The physical picture of breakdown is that a sufficient number of ion pairs are formed by the α mechanism so that through the γ process at least one electron is produced to maintain the current.

The original Townsend method for measuring γ did not discriminate between secondaries produced by positive ions, γ_i, and the secondary production by photons, γ_p. In a number of more recent experiments these two types of secondary processes have been studied separately. Among the

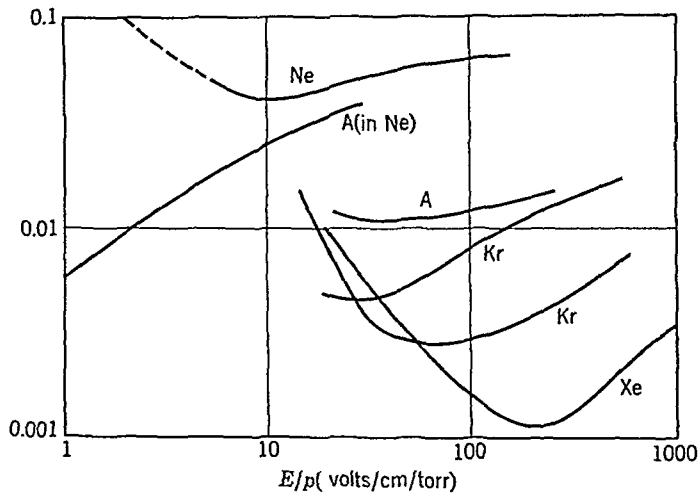


Fig. 7.11 Second Townsend coefficient for copper in the rare gases.

early measurements that separated the various types of γ's were those of Kruithof[9] reproduced in Fig. 7.11. At high values of E/p, the mechanism is that of γ_i. The value of γ_i decreases slowly with decreasing E/p as a result of back diffusion of the electrons liberated at the cathode. At lower values of E/p the second Townsend coefficient increases with decreasing E/p as a result of the secondary emission of electrons from the cathode by photon bombardment, γ_p, or by bombardment by metastable atoms, γ_m.

The most careful work on the ejection of electrons from metals under positive ion bombardment has been done by Hagstrum.[10] His apparatus is shown in Fig. 7.12. The mass spectrometer produced a homogeneous ion beam whose kinetic energy could be varied from 10 to 10^3 electron volts. It was essential for the success of the experiment that no ions strike the edges of the apertures in electrodes M_1, M_2, N, and S, and that all ions strike only the target T inside the sphere. The state of the target surface was investigated by observation of the rate of absorption of gas upon it,

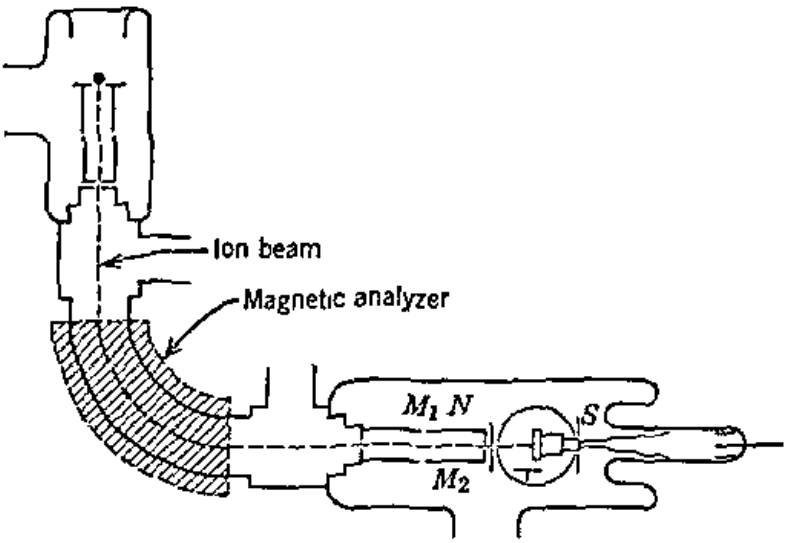


Fig 7 12 Sketch of Hagstrum s apparatus for measuring secondary electron emission by positive ion bombardment

and data with and without absorbed gas layers were reported The results of some of Hagstrum's measurements are given in Fig 7 13 Here the plot of experimental data gives total electron yield γ_i as a function of ion kinetic energy for the ions He^+ (curves 1 and 2), He^{++} (curves 3 and 4), and He_2^+ (curve 5) on Mo Curves 1 and 3 are atomically clean Mo, curves 2, 4, and 5 for Mo covered with a monolayer of gas As a result of his outstanding work, we are able to choose, from several theoretical

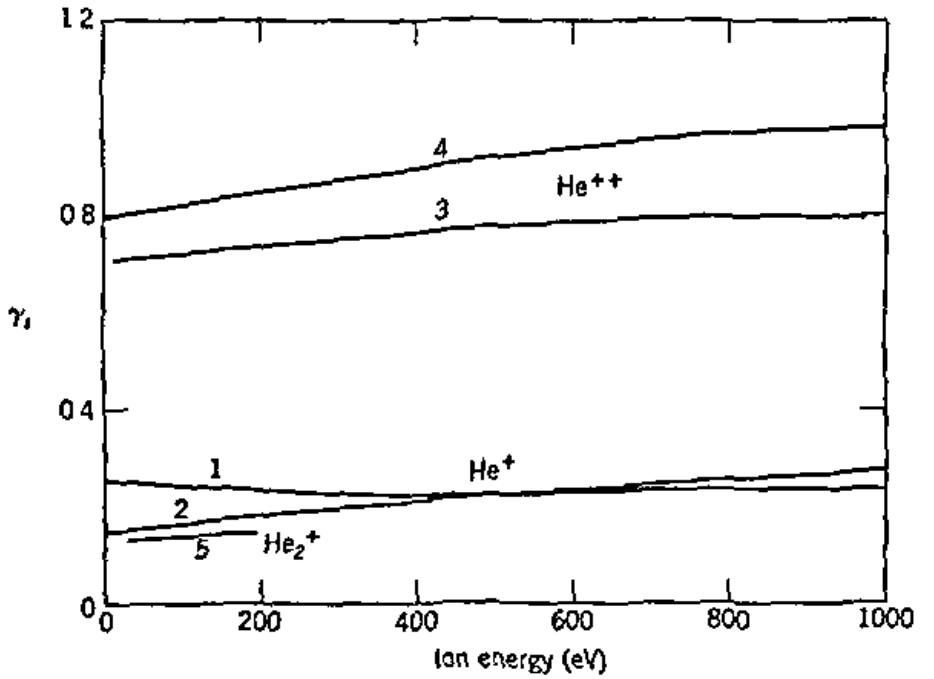


Fig 7 13 γ_i as a function of helium ion energy

possibilities, the detailed mechanism controlling these phenomena. The process is a two-stage one called *potential* ejection. The electronic transition characteristic of the mechanism is illustrated for He^+ incident on Mo in Fig. 7.14. The first step consists of the resonance capture of a conduction electron (1) by tunneling through the barrier into the ^3S metastable state of the He atom. In step 2, the excited atom thus formed is

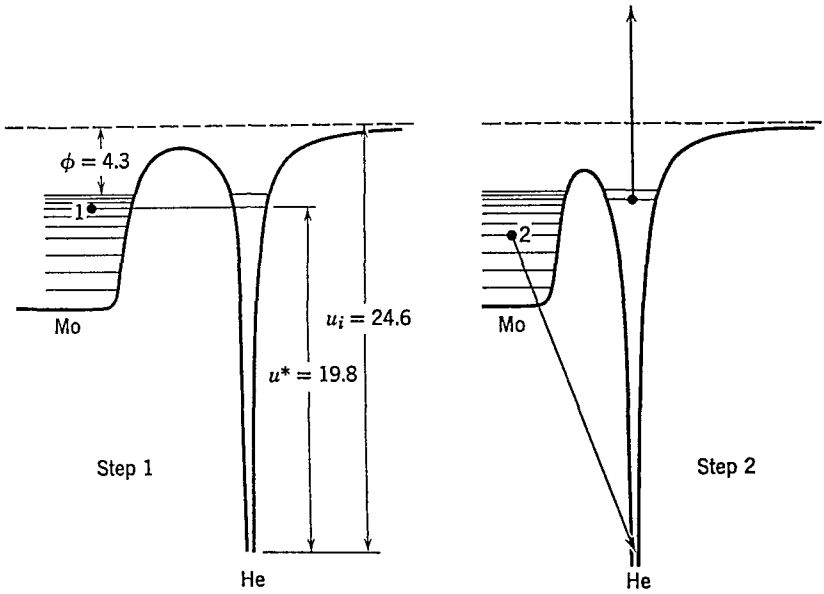


Fig. 7.14 Electronic transitions characteristic of electron emission from Mo under He^+ bombardment.

de-excited by capture of a second metal electron (2) with the simultaneous excitation of the electron in the metastable state into the continuum.

Photoelectric Effect

Turning now to γ_p, we may discuss the photoelectric emission from solids briefly as follows. In order for an electron to escape from a metal, it must get an energy sufficient to raise it over the surface barrier. Since the electrons of the conduction band have a maximum energy W_m associated with the direction normal to the surface (see Fig. 7.15), light quanta must provide an additional amount $e\phi = W_a - W_m$ (ϕ is the work function of the metal) in order to let an electron escape over the potential barrier W_a, such that

$$h\nu_0 = W_a - W_m \tag{7.14}$$

where ν_0 is the threshold frequency For photons of wavelength shorter than c/ν_0, the excess appears as kinetic energy of the emitted electron

$$h\nu = (W_a - W_m) + \tfrac{1}{2}mv^2 \tag{7 15}$$

A complete description of the ν_p effect due to photons requires a knowledge of the spectral distribution of the photoelectric effect, its threshold frequency ν_0, and the velocity distribution function of the

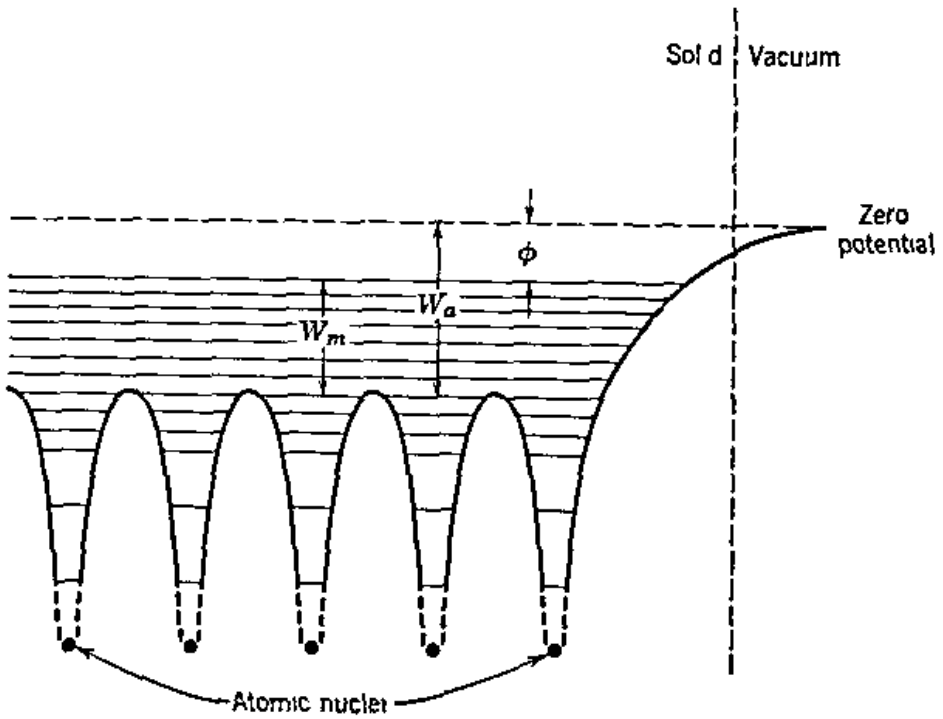


Fig 7 15 Energy level diagram at a metal-vacuum interface

ejected electrons for each wavelength These factors are in general not known

A separation of the γ from photons acting on the walls and the γ from positive ions can be achieved by studying the breakdown in nonuniform fields This has been done for a helium discharge between coaxial cylinders[11] In this geometry the discharge occurs very close to the axial electrode If a high resistance is in series with the discharge, the voltage across the discharge can be depressed so long that ions produced in the discharge are collected before a breakdown field is re-established Under these conditions, the secondary electrons produced under photon bombardment control the effect The investigation proceeded in the following manner First, experimental curves of the breakdown voltage for cylinders of various radii were obtained Second, the mathematical forms for n and γ which would fit the experimental breakdown curves were sought

Third, a reasonable physical interpretation was given for these theoretical functions.

The Townsend theory tells us in Eq. (7.13) that to support a self-maintaining discharge, the voltage between the electrodes must be of sufficient magnitude so that the ionization by collision of an entering charged particle is large enough to create enough ion pairs in the avalanche in order that the photons created in the initial avalanche strike the cathode surface, and produce at least one electron to keep the discharge going.

We may determine the total number of ion pairs formed in a discharge from the relation that $dn = \alpha n\, dr$ and therefore

$$n = \exp \int_{r_c}^{r_a} \alpha\, dr. \tag{7.16}$$

For cylindrical geometry, α depends on r and the straightforward method for computing n for a particular case would be to plot a curve of α as a function of r for values of the field E computed from the equation $E = 2q/r$, where

$$2q = \frac{V}{\ln (r_c/r_a)}. \tag{7.17}$$

The curve of α as a function of r could be graphically integrated to determine n.

This is a very tedious procedure, and to avoid it the following method was introduced. Since $E = 2q/r$ we may write

$$\int -\alpha\, dr = 2q \int \left(\frac{\alpha}{E^2}\right) dE.$$

Actually we would prefer this expression in terms of η and E/p and we therefore have

$$\int -\alpha\, dr = 2q \int \frac{\eta p}{E}\, d\left(\frac{E}{p}\right).$$

We may therefore define a function u such that

$$u = \int_0^{E/p} \frac{\eta p}{E}\, d\left(\frac{E}{p}\right) \tag{7.18}$$

and we may rewrite Eq. (7.16) in the form

$$n = e^{2q(u_a - u_c)}, \tag{7.19}$$

where u_a refers to the value of u at the anode, and u_c its value at the cathode. The usefulness of this function u comes from the fact that it is a function of E/p alone for a particular gas and can therefore be computed

once and for all for this gas No further integrations need be made in determining n for any cylindrical geometry as long as Eq (7 17) holds Figure 7 16 shows the form of u, plotted as a function of E/p for helium

Having determined a form for n, we must now look for a functional form of γ that will be equal to $1/n$ at breakdown for all cylindrical geometries If we assume that the secondary electrons are liberated from the

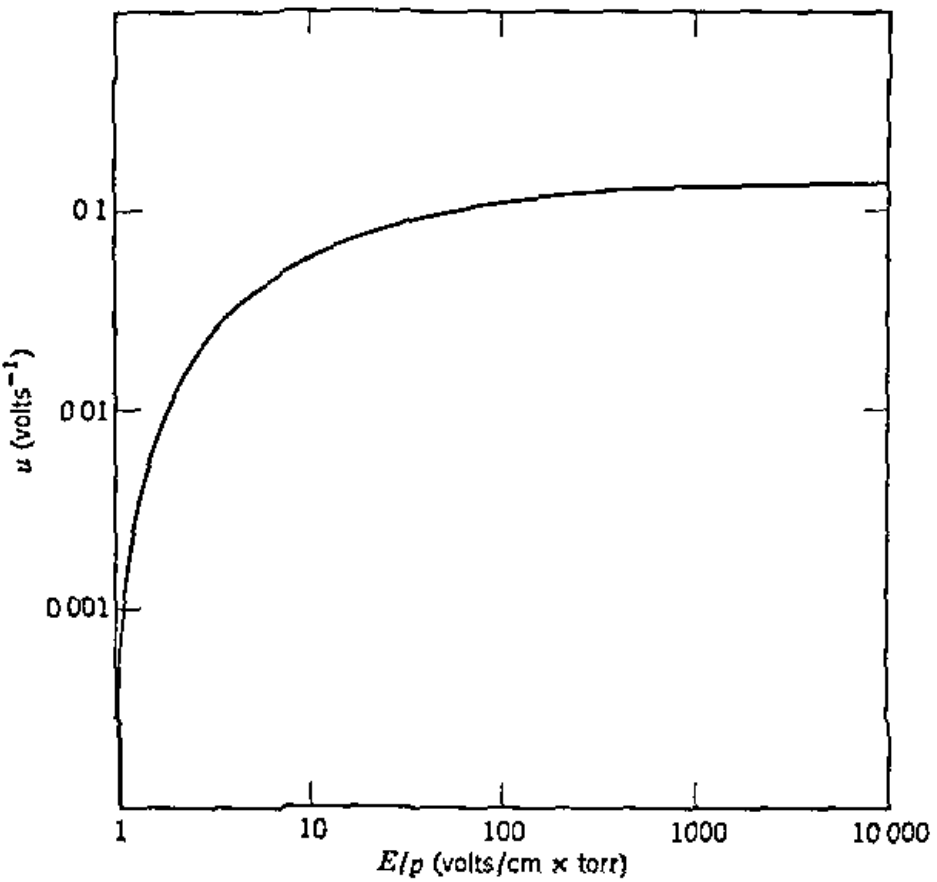


Fig 7 16 u as a function of E/p

cathode through the action of photons, we may set γ equal to the number of photons produced in the gas per ionization, less those absorbed on the way to the cathode, times their photoelectric efficiency Let us designate the corresponding probabilities as ϕ, χ, and ψ ϕ is then a function of the value of E/p (and therefore of u) in the gas where the ionization occurs, as E/p determines the energy of the electronic impacts on the gas atoms Since three fourths of the ions are formed in the last two ionization free paths, only the value of E/p at the anode need be considered χ is a function of the amount of gas the photon must traverse, which can be represented by pd, the pressure times the distance the photons must travel ψ depends only on the nature of the photon and of the cathode surface In general there may be several kinds of photons with different absorption

and photoelectric properties so that γ is in general a sum of terms such as

$$\gamma = \sum_i \phi_i \chi_i \psi_i. \tag{7.20}$$

The experimental data can be represented by using two such terms, which in general would have the form

$$\gamma = \gamma_{\mathrm{I}} + \gamma_{\mathrm{II}} = e^{-\lambda_1 - \mu_1 pd - k_1 u_a} + e^{-\lambda_2 - \mu_2 pd - k_2 u_a}.$$

Actually, fitting the constants to the data shows that some of these constants are negligible, and the form that appears to be satisfactory is

$$\gamma = \gamma_{\mathrm{I}} + \gamma_{\mathrm{II}} = e^{-\lambda_1 - \mu pd} + e^{-\lambda_2 + k u_a}, \tag{7.21}$$

where the constants for helium have the following values:

$$\lambda_1 = 6; \qquad \lambda_2 = 64;$$

$$\mu = \frac{0.25}{\text{cm} \times \text{torr}}; \qquad k = \frac{400 \text{ volts}}{\text{effective ionization}}.$$

It appears from the form of Eq. (7.21) that there are two types of photons involved. The excitation of the first is independent of the energy of impact, but it is strongly absorbed in the gas. The second is more strongly excited by higher-energy impacts, but is not absorbed in the gas.

Let us consider for a moment the energy level scheme of the helium atom, a sketch of which is shown in Fig. 7.17. The most easily excited line in the He I spectrum is its resonance radiation at 585 Å. In the process of ionization, the He I spectrum would be fully developed, and therefore the intensity of the lines would be independent of the mode of excitation and therefore independent of the energy of impact of the electrons in the gas. On the other hand, its outstanding characteristic would be its very high absorption in the helium gas surrounding the discharge, since the photons originating from transitions between excited levels would not be effective photoelectrically. These characteristics are just what is observed in the case of γ_{I}.

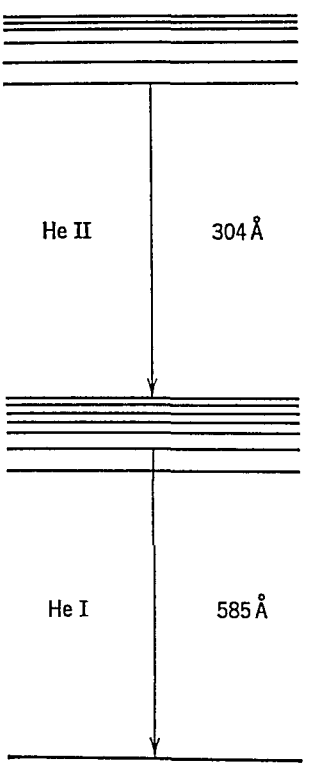


Fig. 7.17 Energy level diagram for helium atom and ion.

The ionization potential of He II is 54 volts The average drop per ionization in the avalanche discharges lies between 30 and 50 volts This is the average voltage, of course, so that a fairly large percentage would have energies considerably higher than this For example, for a mean free path value of 34 volts, 19% will go over 57 volts and 14% will go over 68 volts *This means that although there is no trouble in exciting He II* in the discharge, the spectrum will not be fully developed, and therefore the intensity of its lines will depend on its mode of excitation and hence on E/p On the other hand, the 304 Å resonance radiation of He II would not be expected to be highly absorbed in the normal helium gas These are just the characteristics shown by the fundamental form of γ_{II}

It seems reasonable, therefore, to adopt the following picture of the photoelectric effects in the initial discharge in helium In general both the He I and He II spectrum are excited Since the resonance radiation of He I is the most copiously produced, it will have the dominating effect

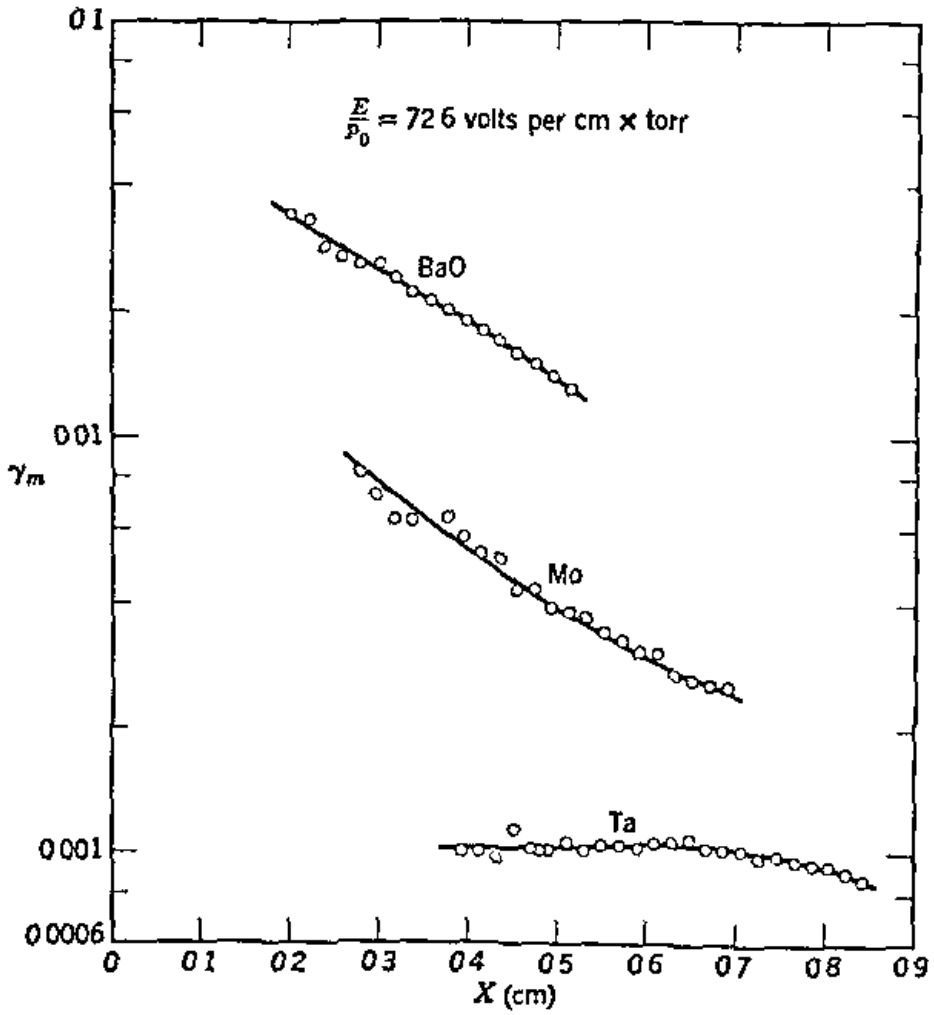


Fig 7 18 γ_m as a function of electrode separation for various cathode materials

as long as the pressure is low enough to allow these photons to reach the cathode. For pressures around 10 mm of He, however, the path length in the gas is too long for these photons to reach the cathode, and thereafter the photoelectric effect depends upon the resonance radiation of He II.

Metastable Effects

The tube illustrated in Fig. 3.3 has been used[12] to separate the effects of metastable atom collisions with the walls γ_m, which have a time constant of milliseconds, from those of γ_i, which have a time constant of tens of microseconds. By interrupting the light source with a sectored disk run by a 30-c/s synchronous motor light pulses of 5 to 10 milliseconds duration with a rise time of 30 μsec illuminated the space between the electrodes. Two secondary electrons collected as a result of the metastable atom collisions with the electrodes led to direct determinations of γ_m, which is illustrated for various cathodes in argon gas in Fig. 7.18.

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Chapter 8

Attachment and Recombination

Electron Attachment

Electron attachment to a neutral atom is a common occurrence for gases whose outer electronic shells are nearly filled An electron that makes ν_a impacts per second under the action of an electric field E moves with a drift velocity of μE cm/sec In the time it takes to move 1 cm, it makes $\nu_a/\mu E$ impacts per cm The number dn out of n electrons that attach in going a distance dx will be

$$dn = -hn \frac{\nu_a}{\mu E} dx,$$

where h is a proportionality constant called the probability of attachment Thus we may write

$$n = n_0 e^{(h\nu_a/\mu E)x} \tag{8 1}$$

The probability of attachment per collision h is not the only common attachment quantity The probability of attachment per centimeter of travel, β is used by analogy to the ionization coefficient α Thus $\beta/p = f(E/p)$ as $\alpha/p = f(E/p)$

$$\beta = \frac{\nu_a}{\mu E} \tag{8 2}$$

where ν_a is the frequency of attachment And

$$h = \frac{\nu_a}{\nu_e} = \frac{\beta \mu E}{\nu_e} \tag{8 3}$$

Electron attachment probabilities can be measured by a method introduced by Cravath,[1] a sketch of which is shown in Fig 8 1 Electrons

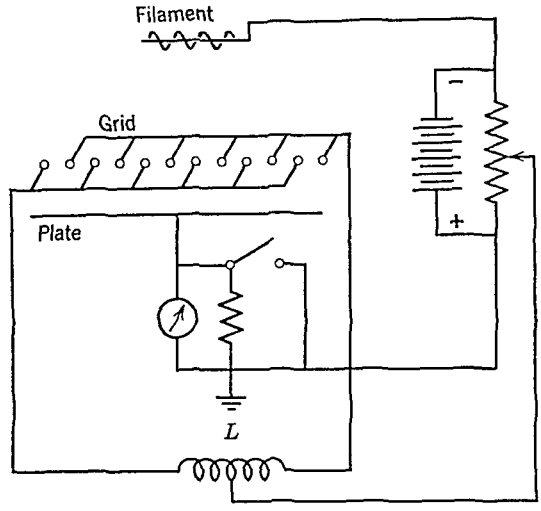


Fig. 8.1 Electron shutter method of measuring attachment coefficients.

pass through the gas from the filament to the grid, and some of them attach to gas atoms on the way. The alternate wires of the grid are connected to an r-f oscillator with a frequency that traps electrons but not ions. By measuring the current between the filament and plate when the grid voltage is zero and when the grid is effective, the ion current relative to the total current can be obtained as a function of grid potential. From these measurements, h can be determined.

Another common method introduced by Lozier[2] is shown in Fig. 8.2. Electrons produce ions in the region J to Q. The electrons are confined to the axis by a magnetic field of 150 gauss, applied by an external solenoid. A fraction of the ions formed travel radially and pass between the disks L and M to reach the collector O. By the application of suitable electric field in the intervals L-M and O-L, negative and positive ions can be separated, and the kinetic energy of the ions can be measured. By

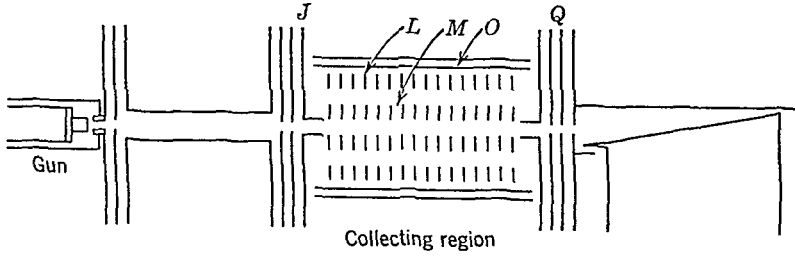


Fig. 8.2 The Lozier method for measuring attachment coefficients.

measuring the magnitude of the negative-ion current relative to the electron current, the electron attachment coefficient can be determined

Measurements of attachment coefficients have been successfully carried out by making careful measurement of the first ionization coefficient α

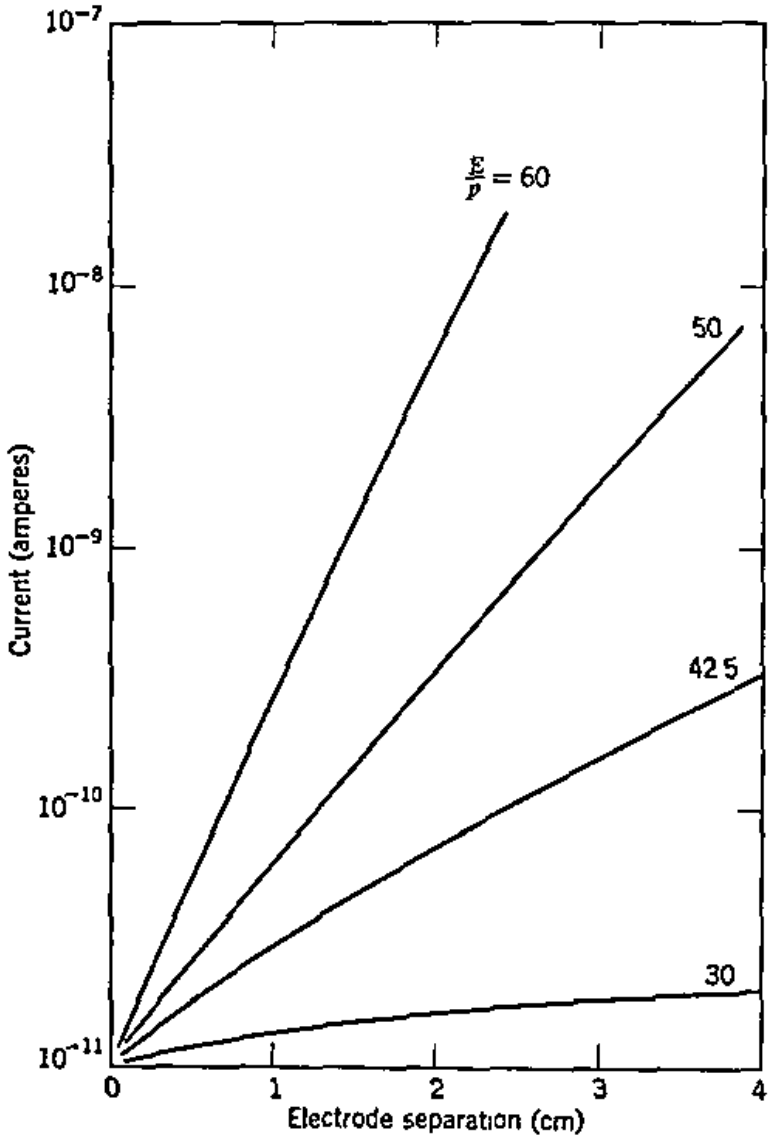


Fig 8 3 Townsend coefficient data in an electron attaching gas

If electron multiplication alone takes place, the log current versus electrode separation curves should be linear Departures from linearity can be related to the electron-attachment phenomena, as has been successfully done by Harrison and Geballe [3] Data of this type in oxygen, from which attachment coefficients were derived are shown in Fig 8 3

Electron attachment to a neutral atom can be most easily accomplished for those neutral atoms whose outer electronic shells are nearly filled A

measure of the ease with which this can be accomplished is given by the electron affinity energy. This energy varies from about 4 volts for so called "attaching" gases like chlorine, fluorine, and oxygen to negative values characteristic of atoms with closed outer electronic shells, like the noble gases.

Collision processes that lead to electron attachment are controlled to a large extent by the conservation of energy which must account for all the energy involved in the collision. In principle, the simplest type of process is the capture of a free electron by an atom and the loss of energy by radiation. Measurements show that a free electron must be in the field of the atom for about 10^{-8} sec for a radiative capture to be at all probable. Thus if an electron is moving with a velocity of 10^7 cm/sec, the duration of the collision will be about 10^{-14} sec, and the capture probability will be only about 10^{-16}, so that it is not a likely event. Radiative capture of an electron with a given kinetic energy T results in a continuous spectrum extending from a long wavelength limit $\lambda = hc/A$ where h is Planck's constant and A is the electron affinity.

If a free electron is captured by an atom in the presence of a third body, this third body can take up the excess energy. To do this, the third body must make an effective collision with the capturing atom when the latter is within an atomic radius of the electron. We would therefore expect this type of attachment phenomena to be pressure-dependent. A typical example[4] of this is $NO + NO + e \rightarrow NO^- + NO$ shown in Fig. 8.4.

Another possible mechanism of electron attachment is the capture of a free electron by a molecule leading to vibrational excitation of the molecule. Subsequently the molecule settles down to the ionic ground state, as illustrated in Fig. 8.5. Two gases that show this effect are the low-energy attachment coefficient measurements in SO_2 and O_2. Characteristics of this type of effect are shown[5] in Fig. 8.6 for SO_2. The decrease in h as E/p increases occurs because the more rapidly the electrons move, the less likely they are to interact with molecules to cause attachments.

A very common type of attachment process is the capture of a free electron by a molecule with the dissociation of the molecule, $O_2 + e \rightarrow O^- + O$. This corresponds to the transition to the repulsive state illustrated in Fig. 8.5. Gases showing this kind of electron attachment have very marked energy thresholds and the attachment coefficient increases with increasing E/p. NH is typical[4] of this behavior and is illustrated in Fig. 8.7.

Of all the electronegative gases that have been studied in detail, electron attachment to oxygen has been reported most often. One of the best of these studies is that reported by Chanin, Phelps, and Biondi.[6] They measured two types of rate coefficients, one for the two-body process χ,

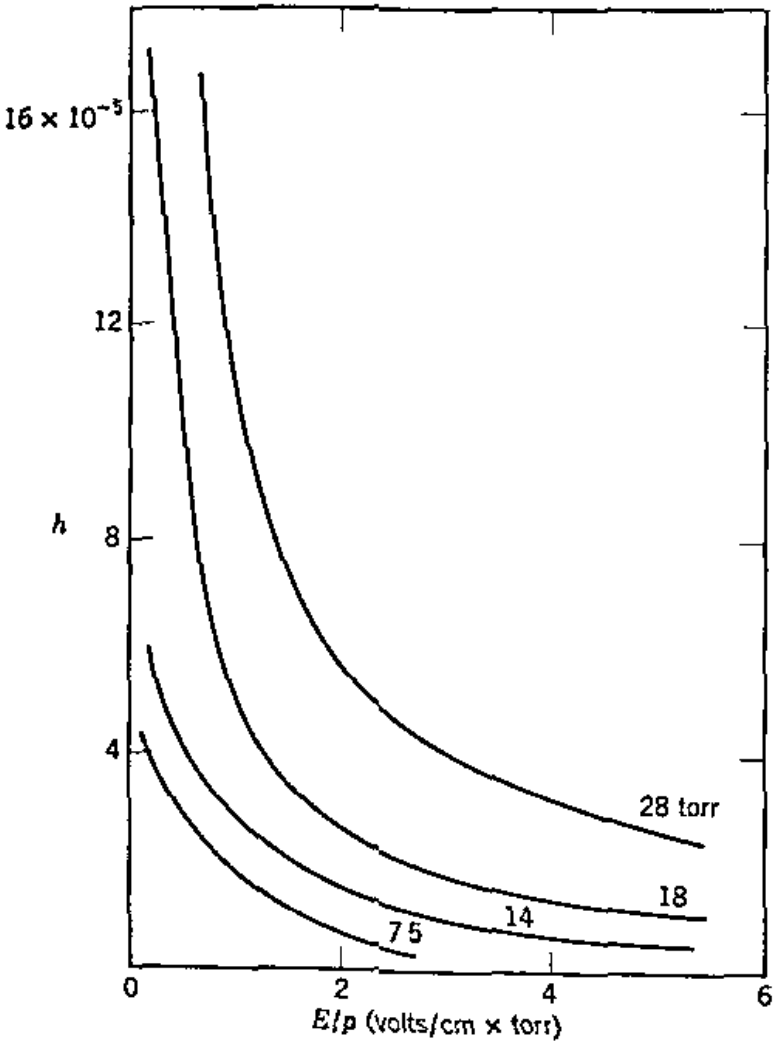


Fig 8 4 Electron attachment in nitric oxide

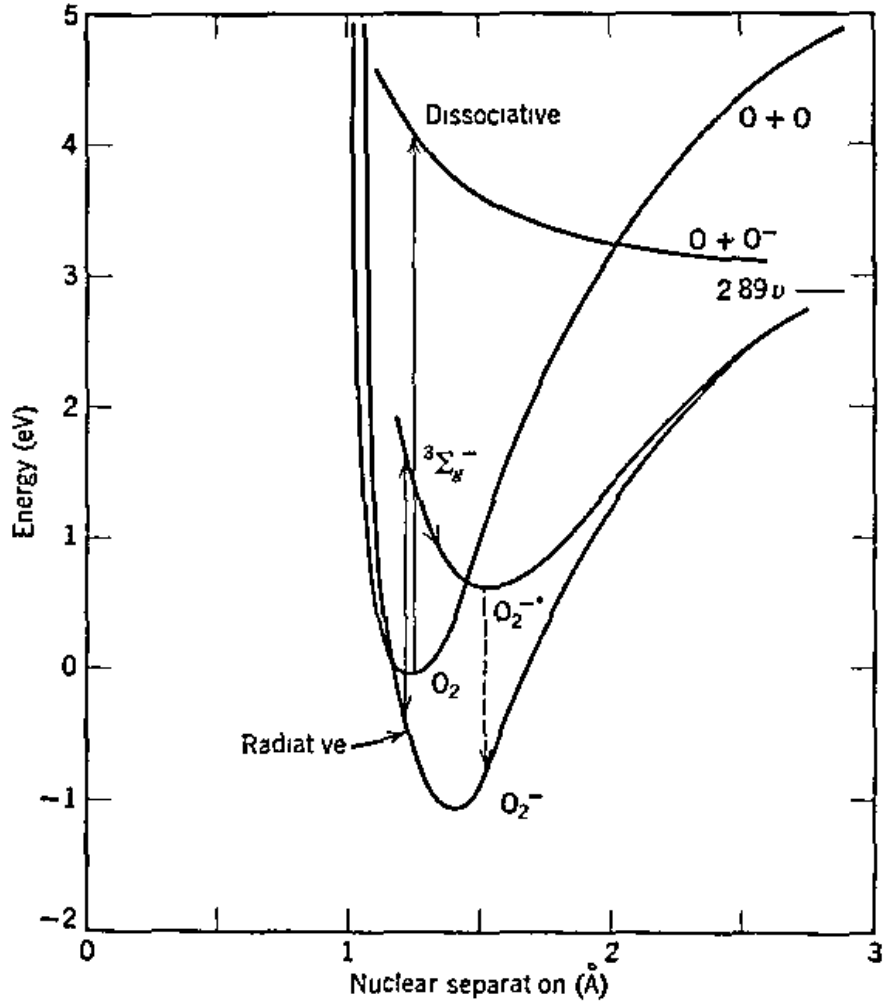


Fig 8 5 Potential energy curves for oxygen

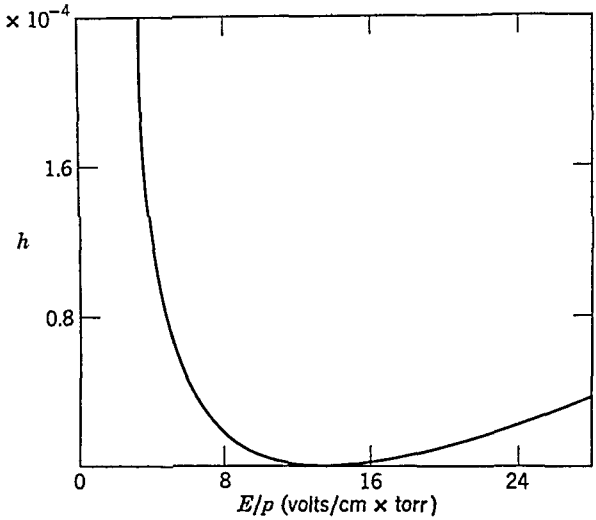


Fig. 8.6 Probability of attachment for SO_2.

and a three-body rate coefficient K. Here χ referred to the process $O_2 + e \rightarrow O^- + O$ and K to the process $O_2 + O_2 + e \rightarrow O_2^- + O_2^*$ so that

$$\frac{dn_e}{dt} = -\nu_a n_e$$

became

$$\frac{dn_e}{dt} = -[\chi n + Kn^2]n_e$$

using n as the gas density.

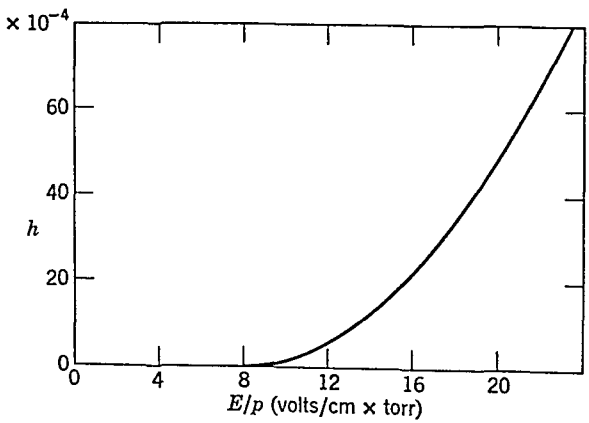


Fig. 8.7 Probability of attachment for ammonia.

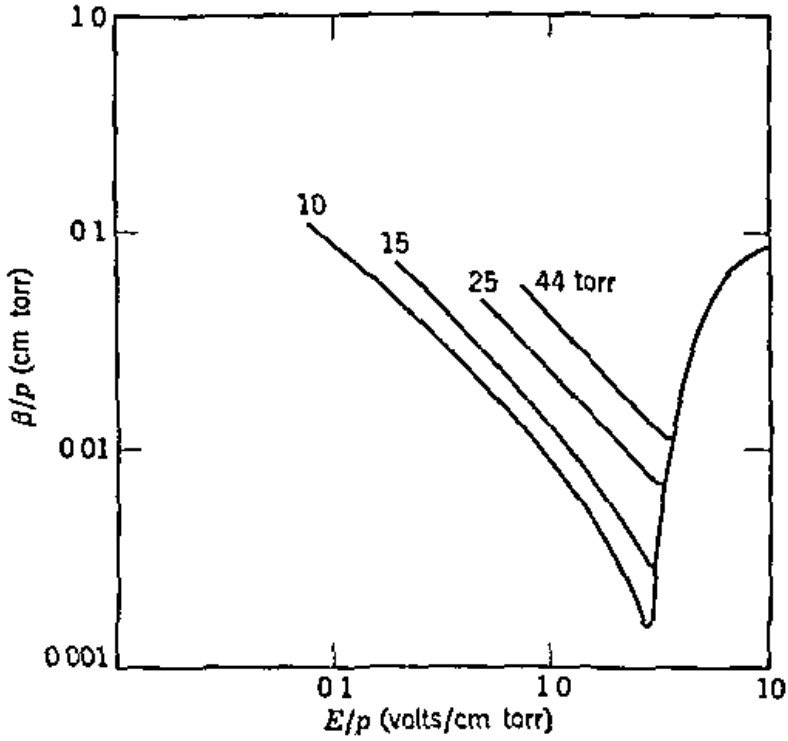


Fig 8 8 Electron attachment in oxygen

They found that, for oxygen the two body process appeared around an E/p of 2 and for values of E/p above this, χ was the controlling coefficient For $E/p < 2$ the three body process predominated In terms of the coefficient β, defined in Eq (8 2) their results are shown in Fig 8 8

Recombination of Ions

A very common loss mechanism for ions is the recombination of negative ions with positive ions The loss of ions due to this effect is proportional to the ion concentration so that

$$\frac{dn_+}{dt} = \frac{dn_-}{dt} = -\alpha n \; n_+$$

The proportionality constant α is called the recombination coefficient If we assume that the positive and negative particle concentrations are equal,

$$\frac{dn}{dt} = -\alpha n^2 \tag{8 4}$$

By solving for n we find

$$\frac{1}{n} = \frac{1}{n_0} + \alpha t \tag{8 5}$$

where n_0 is the initial ion concentration at $t = 0$ Recombination phenomena are usually recognized experimentally by the linear dependence of $1/n$ on t

There are two theories for recombination between ions that have been known for a long time, one introduced by Langevin,[7] applicable to high pressures, and the other due to Thomson,[8] which is useful in the low-pressure region.

The high-pressure theory assumes that the ions make many collisions as they move under the influence of the Coulomb forces between them so that the drift velocity is

$$\bar{v} = (\mu_- + \mu_+)E = \frac{e^2(\mu_- + \mu_+)}{r^2}. \tag{8.6}$$

The number of negative ions per second approaching a positive ion through a sphere of radius r is $4\pi n_- e^2(\mu_- + \mu_+)$. If every collision leads to recombination

$$\frac{dn}{dt} = -n_+ n_- 4\pi e^2(\mu_- + \mu_+), \tag{8.7}$$

and

$$\alpha_i = 4\pi e^2(\mu_- + \mu_+). \tag{8.8}$$

Since the radius r must be very much greater than a mean free path, this theory is applicable only to high pressures. The shape of the pressure dependence can be seen in the following way. For a dc drift $\mu = e/m\nu_m = e/m(\text{const})p$. Thus α is inversely proportional to p, a fact illustrated in Fig. 8.9 for the high-pressure region.

For the low-pressure region the ions' motion is considered to be thermal, and the ions must approach each other to within a critical radius r before

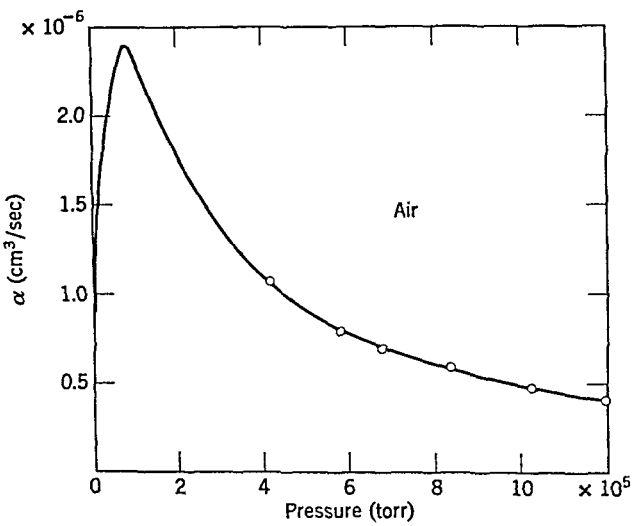


Fig. 8.9 Recombination coefficient in air.

recombining such that

$$\frac{e^2}{r} = \tfrac{3}{2}kT \tag{8.9}$$

The probability of collision of either ion is

$$p = 1 + 2\left\{\frac{e^{(-2r/l)} - 1}{(2r/l)^2} + \frac{e^{(-2r/l)}}{2r/l}\right\}$$

The recombination rate is thus given by

$$\frac{dn}{dt} = \pi r^2 n_+ n_- \sqrt{(\bar{v}_+)^2 + (\bar{v}_-)^2}\,(p_+ + p_- - p_+p_-)$$

so that

$$\alpha = \pi r^2 \sqrt{(\bar{v}_+)^2 + (\bar{v}_-)^2}\,(p_+ + p_- - p_+p_-)$$

For

$$\bar{v}_+ \approx \bar{v}_- = \bar{v} \quad \text{and} \quad p_+ = p_-,$$

$$\alpha = \pi r^2 \sqrt{2}\,\bar{v}(2\bar{p} - \bar{p}^2)$$

$$2p - p^2 = \frac{4}{3}\frac{2r}{l} - \frac{26}{36}\left(\frac{2r}{l}\right)^2 +$$

For low pressure $l \gg r$ and

$$2p - p^2 \approx \frac{8}{3}\frac{r}{l}$$

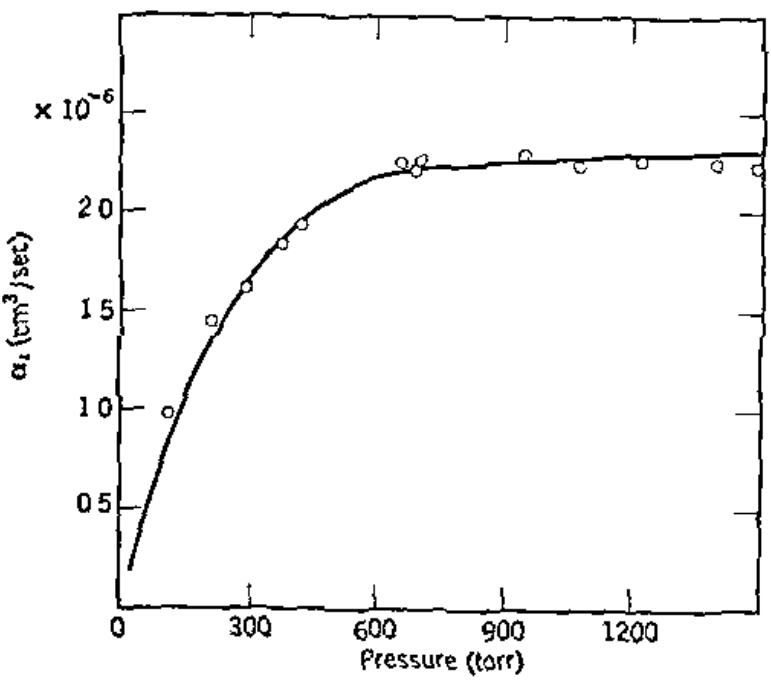


Fig. 8.10 Recombination coefficient in air

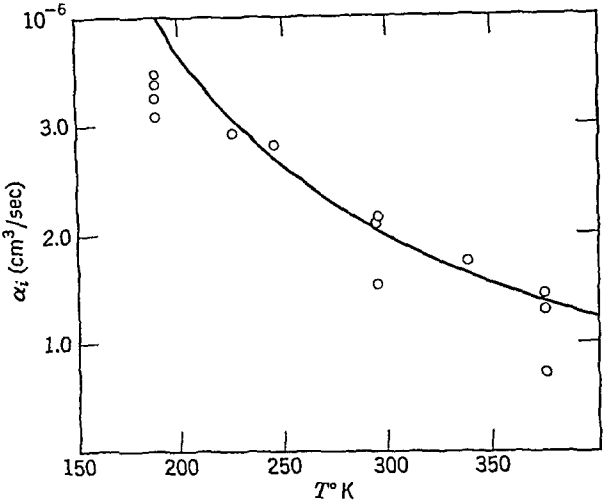


Fig. 8.11 Temperature variation of the recombination coefficient in oxygen at constant pressure.

so that

$$\alpha = \frac{4\pi e^4}{9(kT)^2}\sqrt{2}\,\bar{v}\left(\frac{8}{3}\frac{r}{l}\right) \tag{8.10}$$

for low pressures. Thus the recombination coefficient at these low pressures is inversely proportional to l or directly proportional to p. This is shown in Fig. 8.10.

The recombination coefficient is not independent of the gas temperature. equation (8.10) shows a $1/T^2$ dependence, $\bar{v}$ varies as $T^{1/2}$, and r varies as $1/T$ from Eq. (8.9). At constant pressure for constant ν_c, l varies directly with T. This gives the total temperature variation of α_i as $1/T^{1/2}$. To within rather crude experimental accuracy, this temperature variation is borne out by the experiments of Gardner[10] shown in Fig. 8.11.

Electron-Ion Recombination

Studies of electron-ion recombination have been particularly difficult because of the sensitivity of the recombination coefficients to impurities. As in electron attachment, the processes are governed by the mode of energy dissipation. A common type of electron-ion recombination is called *radiative*, because an electron coming within a short distance of an ion drops into a low-lying electronic orbit and radiates its excess energy. This is recognized spectroscopically by the emission of a continuum lying above the ionization potential of the atom or molecule. Recombination coefficients for this process have been measured by determining the rate of electron density decay by use of Langmuir probes and by the decrease of

intensity of spectral lines after the passage of a discharge through a gas

Positive molecular ions may recombine with an electron resulting in the dissociation of the molecule, leading to what is known as dissociative recombination A particularly nice experiment to demonstrate some of the processes involved in dissociative recombination was carried out by *Biondi,*[11] *who used a microwave technique In pure argon a molecular* species is formed by the collision $A^+ + 2A \rightarrow A_2^+ + A$ Dissociative recombination of the A_2^+ ion and an electron leads to the experimental

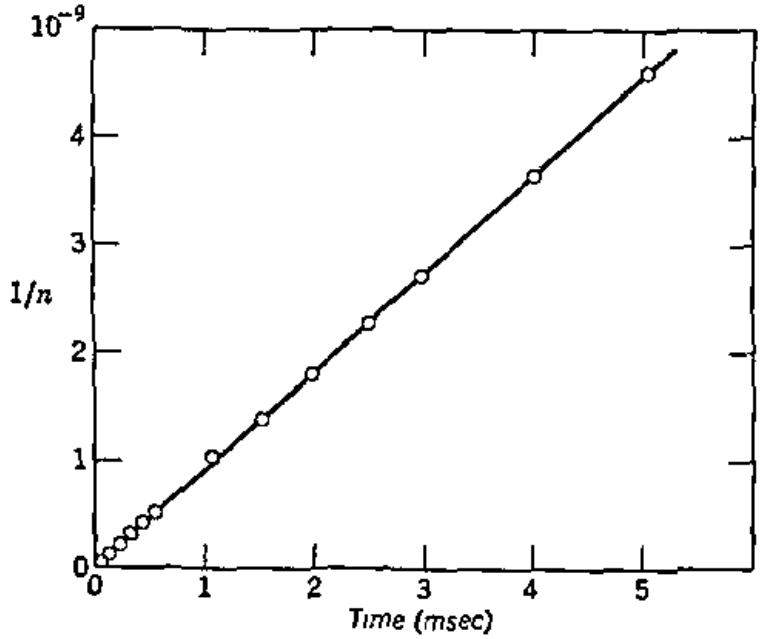


Fig 8 12 Electron density decay as a function of time in argon afterglow

plot of Fig 8 12 If the concentration of argon is very much smaller than that of helium, the argon ions are formed by collisions with metastable helium atoms, and only atomic argon ions are possible $He^m + A \rightarrow A^+ + He + e$ In this case where no molecule is formed, dissociative recombination is obviously impossible, and it is observed, Fig 8 13, that the decay of electron density shows the logarithmic decay characteristic of diffusion, and not recombination

A third form of electron ion recombination can occur by $a^+ + e^- + e^- \rightarrow a + e^-$ This is also a three body collision but with an electron removing the necessary momentum and energy Makin and Keck[12] calculate a least upper bound for this recombination employing a variational technique which is applicable to "classical" plasmas, these are plasmas which are cold enough so that

$$\lambda = \frac{h}{p} < r = \frac{2e^2}{3kT},$$

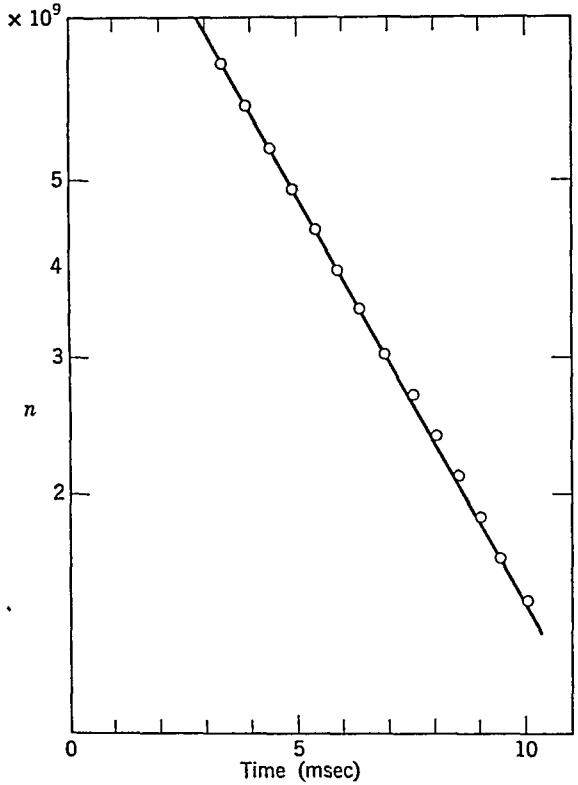


Fig. 8.13 Electron density decay during the afterglow in a helium-argon mixture.

where r is the Thomson radius. This corresponds to an electron temperature of about 1 electron volt. They find

$$\frac{dn}{dt} = 2.3 \times 10^{-8} T^{-9/2} n_+ n_-^2$$

or

$$\alpha = 2.3 \times 10^{-8} T^{-9/2} n_-.$$

This mechanism is important at high electron densities and low temperatures when radiative recombination is unimportant.

COMBINATIONS OF LOSS MECHANISMS

Attachment to Impurities

If attachment is the dominant loss process,

$$\frac{\partial \ln n}{\partial t} = -h\nu_c. \tag{8.11}$$

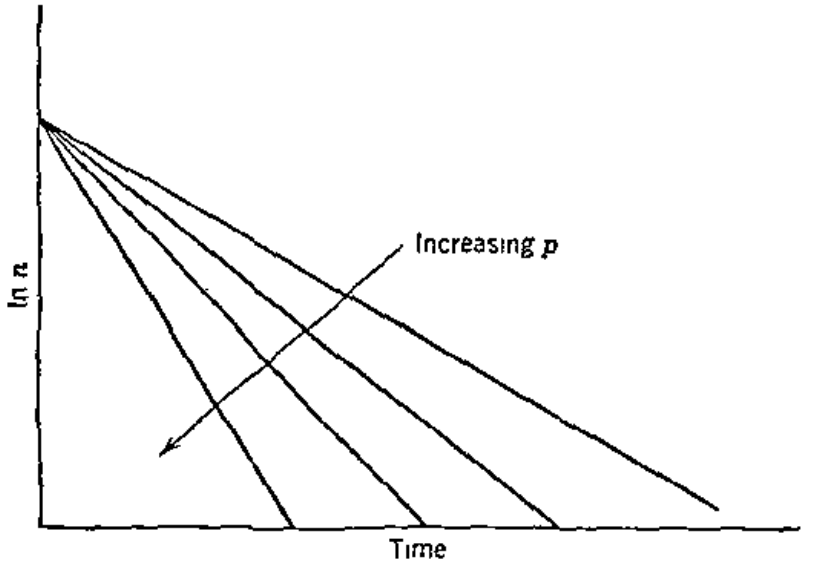


Fig 8 14 Electron attachment with impurities introduced *with the gas*

Electron attachments to impurity atoms are of great practical importance When the impurities are introduced with the gas, $\nu_e = kp$, where p is the total pressure of both the parent and impurity gases, and k is a constant characteristic of the parent gas If this is the case, ln n plotted against t produces the curves of Fig 8 14 On the other hand, if the parent gas is pure and the impurities are coming from the walls, then ν_e(imp) may be independent of the pressure of the parent gas, and the plot is as shown in Fig 8 15 These curves are parallel, and if the initial density is kept

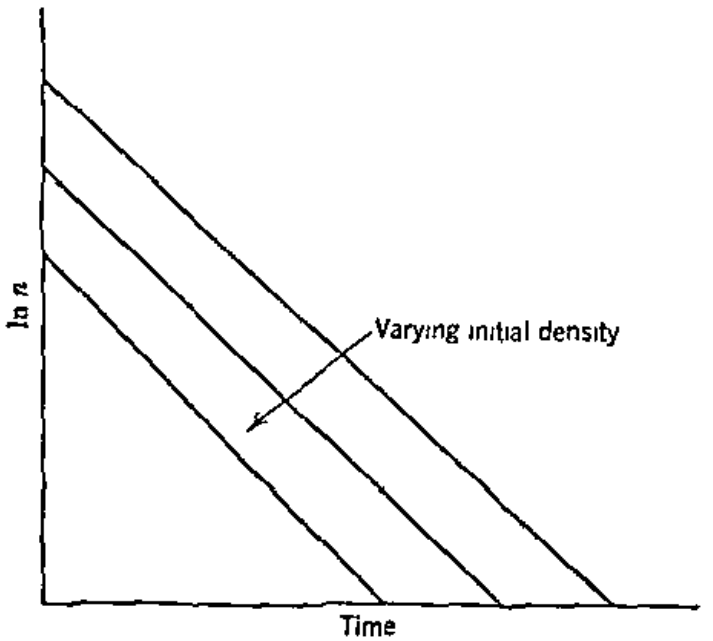


Fig 8 15 Electron attachment with impurities introduced from the walls

constant with pressure the decay curves coincide in spite of varying pressure.

We have a more complicated case if the impurity atoms or molecules vary during the decay. An impurity molecule may be considered lost when an electron attaches to it and a negative ion is formed. We may, therefore, write $\nu_e(\text{imp}) = kN$ where N is the density of impurity particles. The rate of loss may be written as

$$\begin{aligned} \frac{\partial n}{\partial t} &= -hkNn, \\ \frac{\partial N}{\partial t} &= -hkNn, \\ \frac{\partial (n - N)}{\partial t} &= 0. \end{aligned} \tag{8.12}$$

Thus, $N - n = \text{constant} = K$ independent of time, and $N = K + n$, and we obtain

$$\frac{\partial n}{\partial t} = -hk(K + n)n. \tag{8.13}$$

The solution to this differential equation is

$$\left(1 + \frac{K}{n}\right) = \left(1 + \frac{K}{n_0}\right) e^{hkKt}. \tag{8.14}$$

When the density of the impurity molecules is much larger than the electron density (when $K \gg n_0 > n$), Eq. (8.14) reduces to

$$n = n_0 e^{-hkKt}, \tag{8.15}$$

described by Figs. 8.14 and 8.15.

Diffusion and Attachment Combined

We consider next how attachment may influence the diffusion decay. We assume that the diffusion loss is dominant but that attachment is large enough to influence the data. We then have

$$\frac{\partial n}{\partial t} = -\frac{D_a}{\Lambda_e^2} n - hkNn, \tag{8.16}$$

$$\frac{\partial N}{\partial t} = -hkNn. \tag{8.17}$$

When we solve the second equation with respect to N, we have

$$N = N_0 \exp\left(-hk \int_0^t n \, dt\right). \tag{8.18}$$

Table 8 1. Values of hkN in Eq (8 20)

				$hknt$	
	k	t	n/m^3	$h = 1$	$h = 10^{-3}$
O_2	10^{-14}	$p \times 10^{-3}$	10^{15}	$p \times 10^{-2}$	$p \times 10^{-5}$
H_2	2×10^{-14}			$p \times 2 \times 10^{-2}$	$p \times 2 \times 10^{-5}$
H_2O	10^{-12}			p	$p \times 10^{-3}$

Putting this into Eq (8 16), we get

$$\frac{\partial \ln n}{\partial t} = -\frac{D_a}{\Lambda_e^2} - hkN_0 \exp\left(-hk\int_0^t n\,dt\right) \tag{8 19}$$

In terms of the proper time variables for diffusion, this becomes

$$\frac{\partial \ln n}{\partial(t/p\Lambda_0^2)} = -D_a p\left(\frac{\Lambda_0}{\Lambda_e}\right)^2 - hkNp\Lambda_0^2 \tag{8 20}$$

If $hk\int_0^t n\,dt$ is much less than unity in the time range in which we are working, we may neglect the time dependence in the last term, which is valid when

$$hk\int_0^t n\,dt < hkn_{\max}t \ll 1$$

Values of these parameters for O_2, N_2, and H_2O are given in Table 8 1

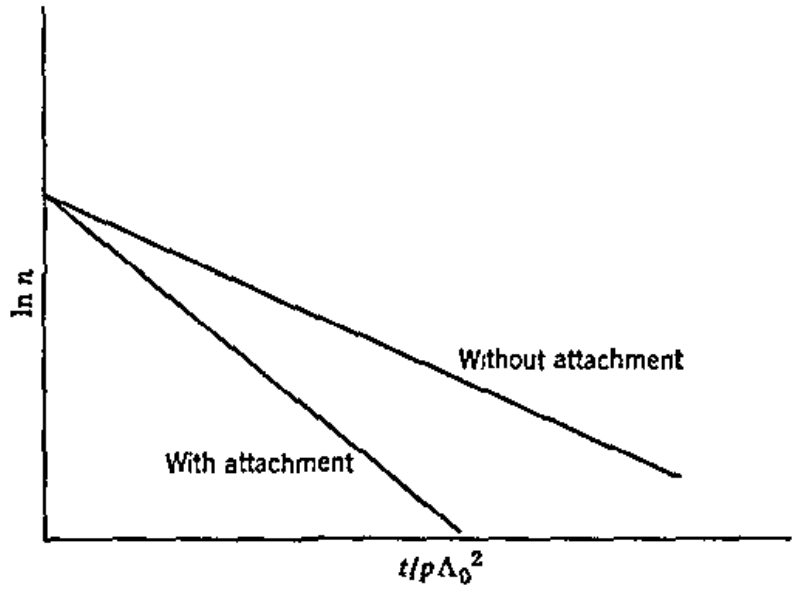


Fig 8 16 Electron density decay with diffusion and attachment combined

We can see from these data that neglecting the time dependence in the last term leads to a reasonable approximation. Thus we find from Eq. (8.20) that a plot of ln n versus $t/p\Lambda_0^2$ changes its slope as a result of attachment. This is illustrated in Fig. 8.16. Figure 8.17 shows how the attachment becomes more and more important as the pressure increases. The pressure

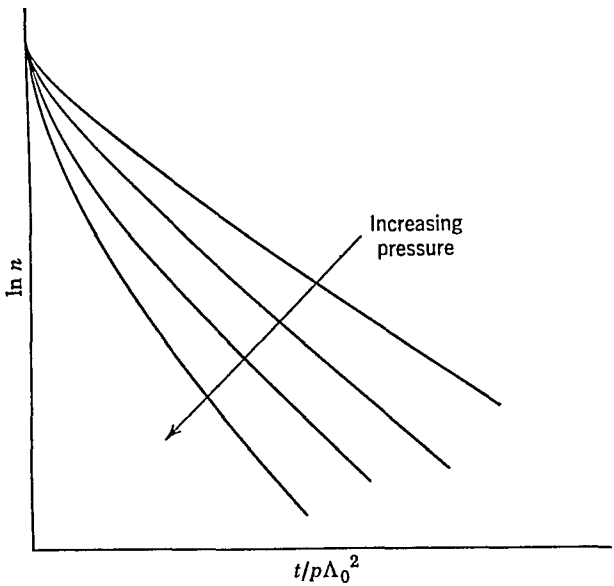


Fig. 8.17 Importance of attachment with changing pressure.

dependence is best visualized by plotting the slope late in the afterglow $\Lambda_e = \Lambda_0$,

$$\frac{\partial \ln n}{\partial(t/p\Lambda_0^2)} = -D_a p - hkN(p\Lambda_0^2) = -(D_a p)_{\text{eff}}. \tag{8.21}$$

In terms of an effective diffusion coefficient thus defined, the result of attachments to impurities is illustrated in Fig. 8.18.

Experimental Determination of Recombination

The solution of Eq. (8.4) in the form of Eq. (8.5) provides the usual method of recognizing a recombination-loss process. The obvious way to plot this is $1/n$ against t, but for relatively short time ranges this process is not unique. For example, let us take the case of an impure gas. The solution to the differential equation, Eq. (8.13), came out in the form of Eq. (8.14). A very interesting case arises when $hkKt \ll 1$. This is a case

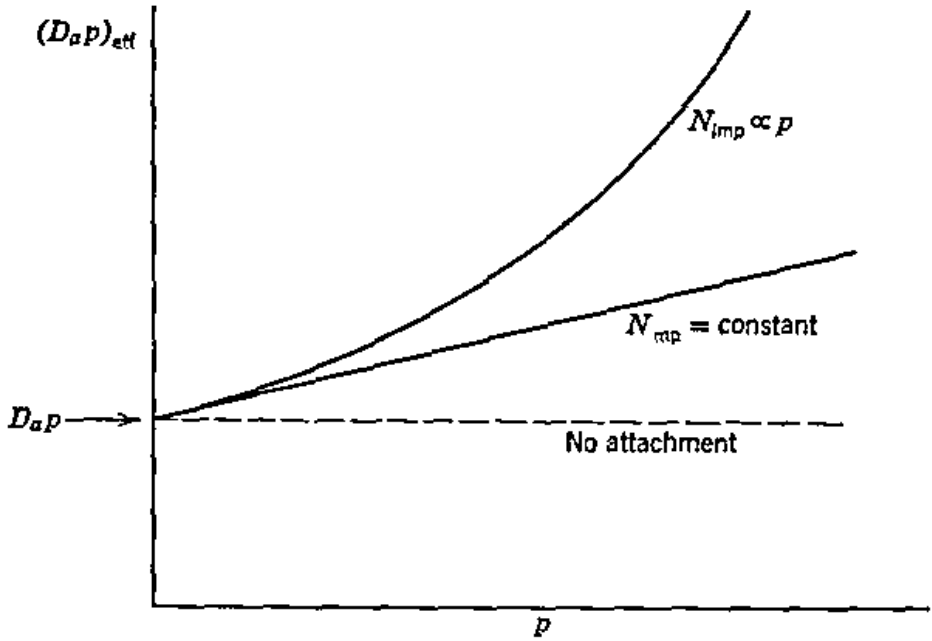


Fig 8 18 Effective diffusion coefficient as a function of pressure

that is quite possible experimentally Under these conditions, Eq (8 14) reduces to

$$\frac{1}{n} = \frac{1}{n_0} + \left(1 + \frac{K}{n_0}\right) hkt, \tag{8 22}$$

which we recognize as being of the same form as Eq (8 5), characteristic of recombination The apparent recombination coefficient is

$$\alpha_a = \left(1 + \frac{K}{n_0}\right) hk \tag{8 23}$$

Since Eq (8 5) is by no means unique for recombination, a somewhat different approach is to plot $\ln n$ versus t, using the relation

$$\frac{\partial \ln n}{\partial t} = -\alpha n \tag{8 24}$$

The slope of $\ln n$ versus t is proportional to n, and the decay curves characteristic of recombination are illustrated in Fig 8 19 Another method is to take the logarithm of the solution and find

$$\ln \frac{n}{n_0} = -\ln (1 + n_0 \alpha t) \tag{8 25}$$

This gives a family of curves that diverge with time If true electron-ion

recombination is effective, α should be essentially independent of the pressure.

When diffusion and recombination are both influencing the decay curves, we cannot obtain an exact solution, but we may use the differential equation to interpret our results. Let us compare decay curves at the same

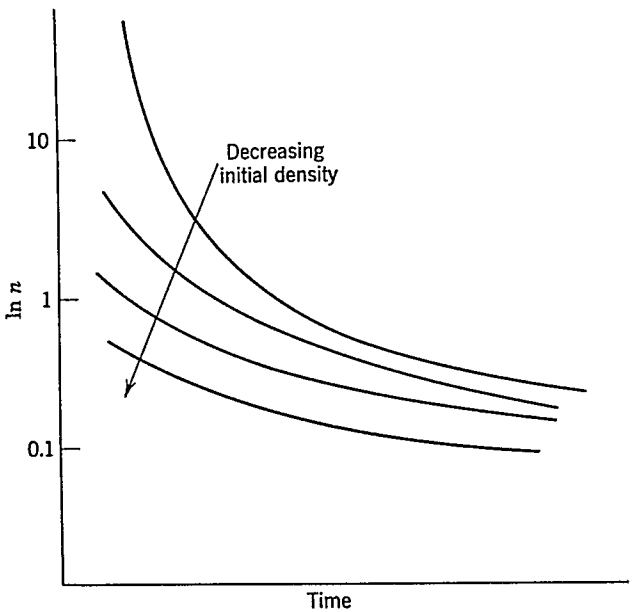


Fig. 8.19 Decay curves characteristic of recombination.

pressure under the same conditions except for differences of initial density n_1 and n_2. Then,

$$\frac{\partial(\ln n_1)}{\partial(t/p\Lambda_0^2)} = -D_a p\left(\frac{\Lambda_0}{\Lambda_e}\right)^2 - \alpha p\Lambda_0^2 n_1$$
$$\frac{\partial(\ln n_2)}{\partial(t/p\Lambda_0^2)} = -D_a p\left(\frac{\Lambda_0}{\Lambda_e}\right)^2 - \alpha p\Lambda_0^2 n_2. \tag{8.26}$$

Subtracting the first equation from the second yields

$$\frac{\partial \ln (n_1/n_2)}{\partial(t/p\Lambda_0^2)} = -\alpha p\Lambda_0^2(n_1 - n_2). \tag{8.27}$$

If the two decay curves are parallel, Eq. (8.27) is zero, and there is no

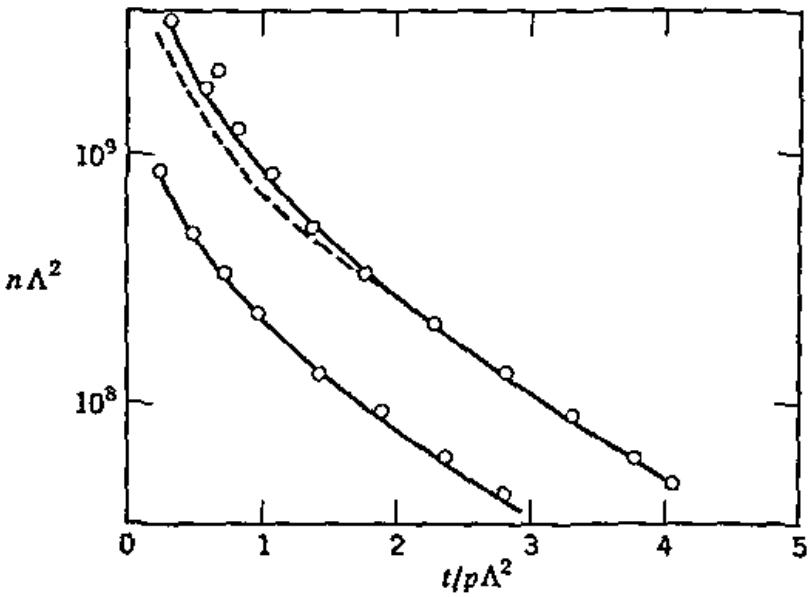


Fig 8 20 Decay of electron density in hydrogen

recombination present If recombination is present, Eq (8 27) provides a means for estimating its magnitude [13] Such a result is shown in Fig 8 20, illustrating a case in which the deviation from parallelism is within the experimental error A serious experimental difficulty in untangling diffusion and recombination by the use of Eqs (8 26) and (8 27) is the requirement that the initial electron spatial distribution in the two cases be identical This is a difficult experimental condition to fulfill, yet the magnitude of the effect is large This is illustrated for a particular case in Fig 8 21

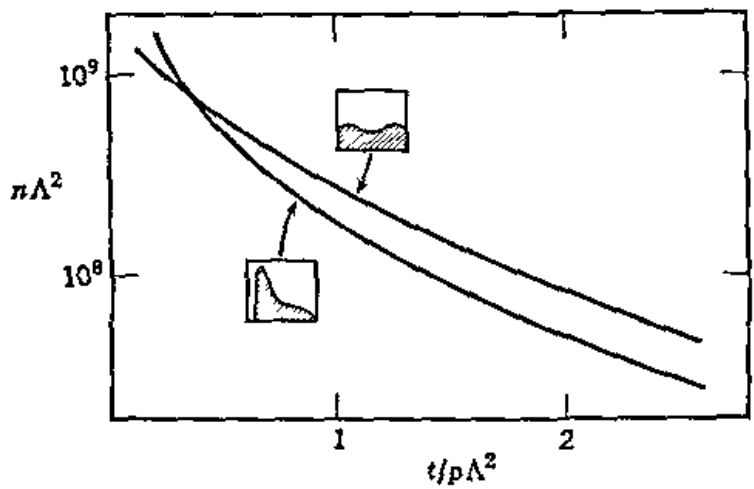


Fig 8 21 The influence of the initial spatial distribution on the decay of electron density The little cartoons illustrate the density distribution across the cavity

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Chapter 9

Radiation from Low-Pressure Discharges

Optical Emission

The technique that has given us the greatest knowledge of the physics of low-pressure plasmas has been spectroscopic analysis The frequencies of the lines emitted from an excited atom or molecule are characteristic of the atoms or molecules themselves while the shape of the line signals the environmental effects on the emitter and therefore tells us something of the plasma itself The line shape is usually characterized by the line width, and there are a number of mechanisms to which broadening can be attributed It is, of course, true that spectral lines have a natural breadth, since the emitted radiation forms wavetrains of finite energy and hence are never of infinite duration or continuous phase, and the environmental broadening effects are always superimposed on the natural width There are two types of so-called impact broadening phenomena If there is a permanent displacement of the energy levels of an atom by the fields of neighboring atoms, the effect is called "statistical" broadening Because a great deal of the work on this effect was done by Holtsmark, this effect sometimes goes by his name also The probability distribution of electric fields experienced at any given point in a randomly distributed swarm of static field centers can be analyzed and it is found that this random field is a function of the average field existing at a given distance from a perturbing field center Such a formulation allows us to calculate the line broadening resulting from collisions from this static electric field point of view and leads to information about the internal electric fields in the plasma

Another type of collision broadening is called "Doppler" broadening, and results from the Doppler effect of the thermal motion of the radiators along the line of observation

The fate of optical radiation emitted by excited atoms in the plasma is well illustrated[1] by Fig. 9.1. Most of the physics illustrated by this figure is self-evident. Resonance radiation is often hindered in its passage through the gas by repeated reabsorption and is said to be "blockaded." This successive absorption and emission has been extensively studied by observing the time for a pulse of incident radiation to be released from the gas. The long effective duration of these excited states is sufficient for

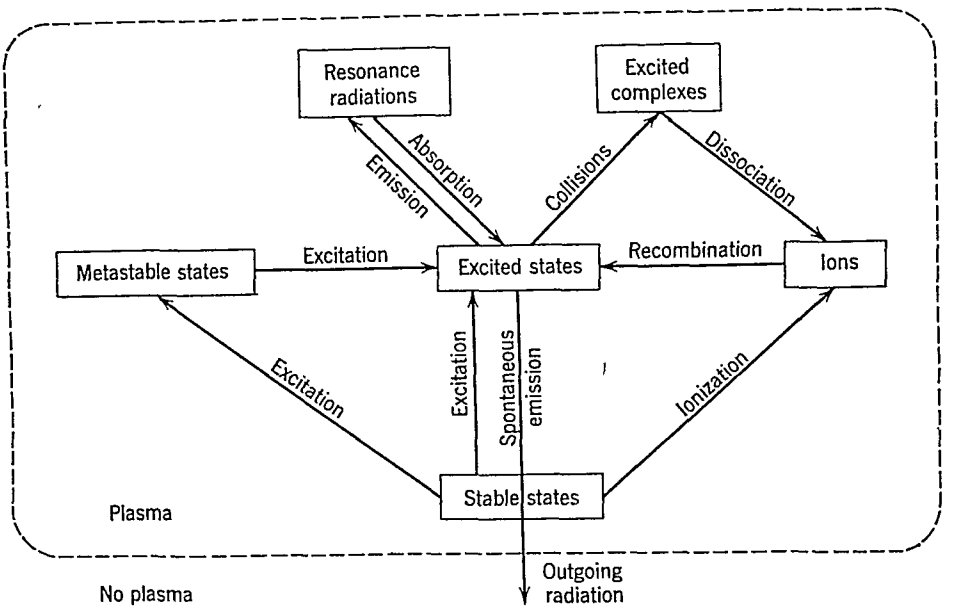


Fig. 9.1 Optical radiation from a plasma.

collisions with neutrals to remove the energy. An example of this is molecular formation in mercury and its resonance radiation. The decay time of this radiation is shown[2,3] in Fig. 9.2. At low densities, the line shape is due to Doppler and the theory can be worked out to give the straight-line portion of the graph. At higher pressures the effects of pressure broadening and hyperfine structure make the calculation much more difficult, The onset of these calculations is illustrated in Fig. 9.2 by the deviation of the experimental points from the calculated line.

Conservation of linear momentum prohibits a free electron from radiating. However, in the presence of another body, such as an ion or an atom, radiation may be emitted when an electron suffers an inelastic collision, and the change in kinetic energy is transferred to radiant energy. The radiation resulting from the acceleration of an electron in the field of an atom is called *bremsstrahlung*. The electron makes a transition between two states of the continuous spectrum in the field of the atom or

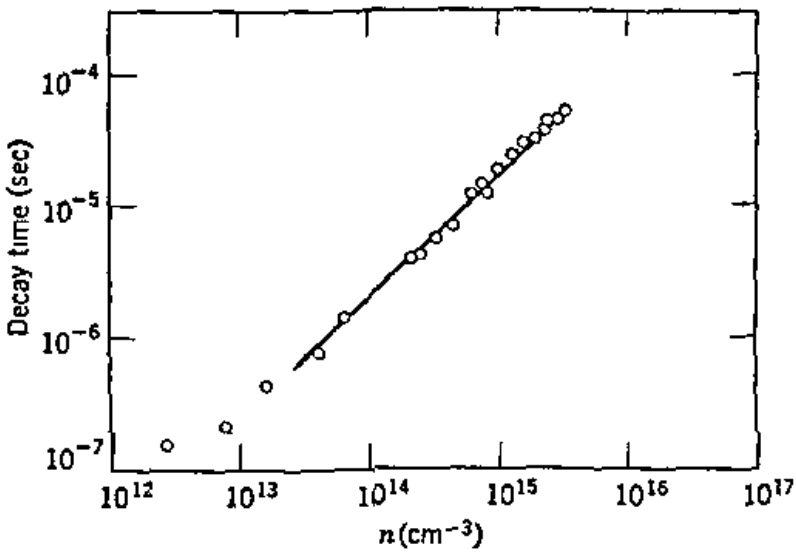


Fig 9 2 The decay time of mercury resonance radiation as a function of gas density

ion, much like the process of radiative capture $H + h\nu \rightleftharpoons H + e$ Since the positive energy states in the atomic or ionic field have a continuous distribution, the electron may lose any fraction of its initial energy up to a maximum frequency ν_0 corresponding to the loss of the entire electronic energy

$$h\nu = (V_0 - V)e$$

These so called free-free transitions in the neutral atom fields are also of interest with the inverse process, the absorption of light by an electron of positive energy in the neighborhood of an atom This may be an important mechanism in the production of plasma by laser beams[4] and is particularly obvious as the absorption mechanism in the atmosphere of stars The sun is a good example of this At wavelengths above 12,000 Å the free free absorption predominates over the free bound absorption of the neutral hydrogen atoms The effect is illustrated[5] in Fig 9 3

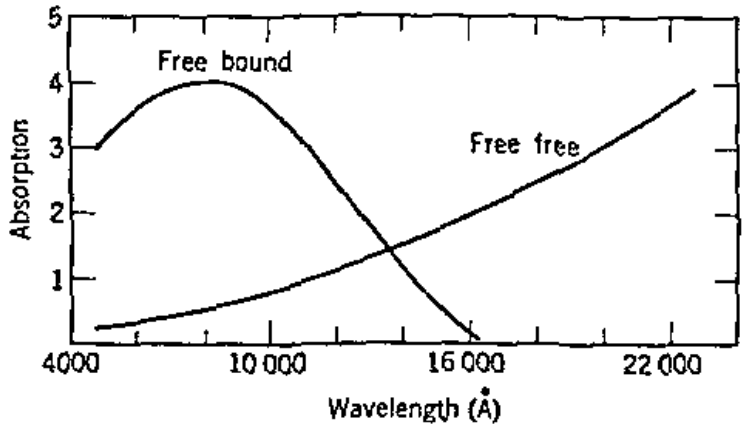


Fig 9 3 Absorption mechanisms observed in the light from the sun

In using the spectroscopic method for determining charged-particle densities and temperature, satisfactory answers are obtained only if local thermal equilibrium occurs. This is equivalent to saying that the kinetic temperature of the electrons, ions, and atoms are alike, as must be the temperature characterizing the distribution over bound states (the Boltzmann factors, $e^{-mv^2/2kT}$) and free states [characterized by the Saha equation, (6.13)].

These conditions set rather specific requirements for density and temperature. As far as charged-particle density is concerned, the electron density must exceed about 10^{16} cm^{-3} in laboratory plasmas to insure excitation and de-excitation as well as ionization and recombination by electron collision. The temperature must be less than about 10 electron volts. If the temperature is higher than this, three-body recombination and collisional de-excitation from multiple ionization will be smaller than the radiative counterpart and detailed balancing will no longer apply. Also above 10 electron volts the cross sections will be too small to allow the establishment of immediate thermal equilibrium; that is, the relaxation times will be large.

Three basic spectroscopic measurements are used to study the details of a plasma:[6] continuum intensities, line intensities, and line profiles.

Continuum intensities can either be from free-free transitions in the electric field of an ion (the so-called *bremsstrahlung*) or the free-bound transitions from radiative recombination. The emitted intensity is proportional to the product of the electron density and the ion density. As we have just seen, the transition between the free-free and the free-bound transitions is smooth, corresponding to the transition from broadened lines to the continuum near the series limits.

From the intensities of individual lines one can calculate the density of atoms in upper states of the lines if the transition probabilities are known. Little is known about the dependence of oscillator strengths on the electron densities, but probably the dependence is negligible.

In plasmas, most the line broadening is due to Stark effect resulting from the electric microfields produced by electrons and positive ions surrounding the emitted atoms or ions. The Stark line profiles are essentially independent of temperature, being given by:

$$I = I_0 \frac{\delta}{2\pi(\nu - \nu_0)^2 + (\delta/2)^2} \tag{9.1}$$

where the line width is δ and ν_0 is the center frequency of the line. The line widths obviously must depend on the type of electric field. For dipole fields $\delta \sim \mu n$, where μ is the dipole moment. For the fields produced by ions $\delta \sim e n^{2/3}$ where e is the ionic charge.

Next to the Stark broadening in importance is the broadening due to the

thermal Doppler effect Here the line intensity is given by

$$I(\nu) \sim \exp\left[-\frac{c^2}{\nu^2}(\nu - \nu_0)^2 \frac{M}{2kT}\right] \tag{9 2}$$

where M is the molecular weight This leads to a half-width of the line

$$\delta = 1\ 67 \frac{\nu_0}{c}\sqrt{\frac{2kT}{M}} \tag{9 3}$$

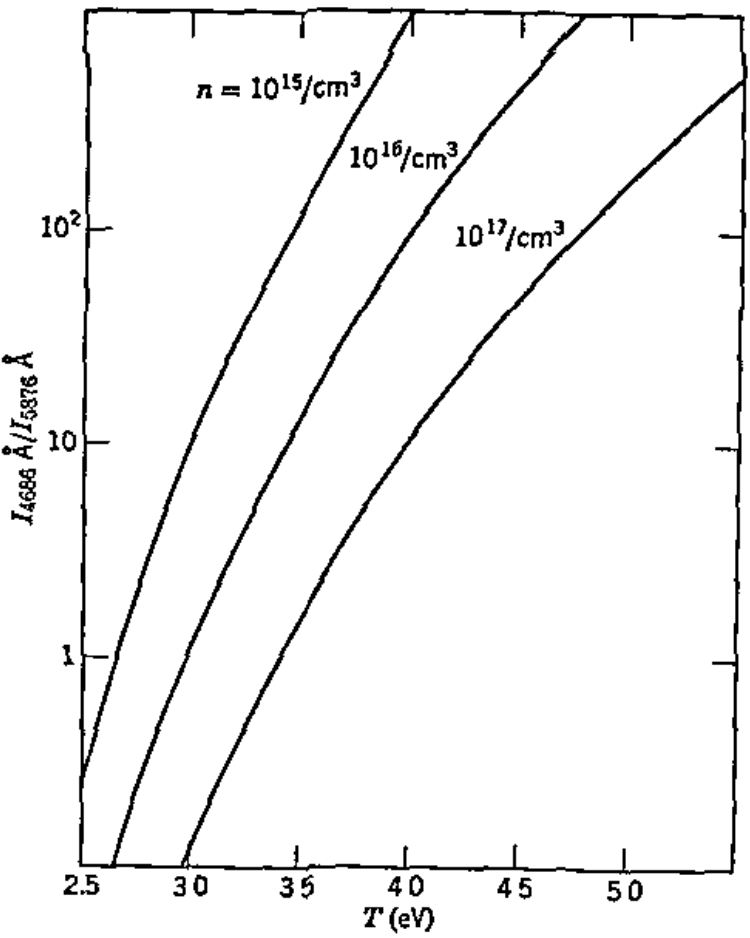


Fig 9 4 Ionized He to neutral He lines for various electron densities

The Doppler effect can be separated from the Stark effect since for Stark broadening electron collisions are important, whereas with the Doppler effect the motions of the emitted atoms or ions are governed by collisions with heavy particles

A considerable amount of information can be obtained by measuring the relative intensities of various lines from subsequent states of ionization in a given gas If the relative oscillator strengths and statistical weights are known, this determines the ratio of the Boltzmann factors for the upper states of the lines from which the temperature can be determined This is illustrated for a particular case in Fig 9 4

Absolute line intensities also allow one to measure temperature if the total gas pressure is known. This is illustrated in Fig. 9.5. For ion lines, the transition probabilities are accurately known so that we can obtain the ion density in the upper states of the lines with an accuracy limited only to the radiation standard. At low temperatures one can determine the

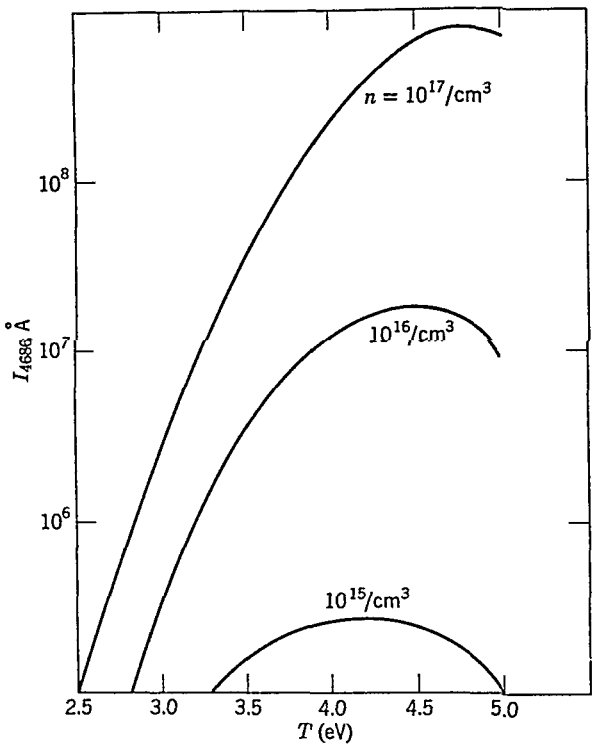


Fig. 9.5 Line intensities in ergs/cm³ sec steradians for various electron densities in ionized He.

electron density from the absolute intensity of neutral lines using the Saha equation and the appropriate Boltzmann factors. This neutral line intensity method is very close to the continuum intensity method.

Relative continuum intensities may also be used to measure plasma temperatures, since the frequency dependence of the *bremsstrahlung* and the recombination continuum is given by $e^{-h\nu/kT}$. This method has reasonable accuracy only in the vacuum ultraviolet region.

Absolute continuum intensity measurements can be used to obtain electron densities from which the temperature follows from the Saha equation if the total pressure or gas density is known. This is only successful if no other continua are present, since in the visible region the

intensity of the recombination and *bremsstrahlung* continua are rather weak

Relative line and continuum intensity measurements yield similar types of information The density of atoms in various excited states, and therefore the corresponding line intensities, according to the Saha equation, is proportional to the square of the electron density, and so is the continuum intensity Therefore the ratio of the total line intensity to the intensity of

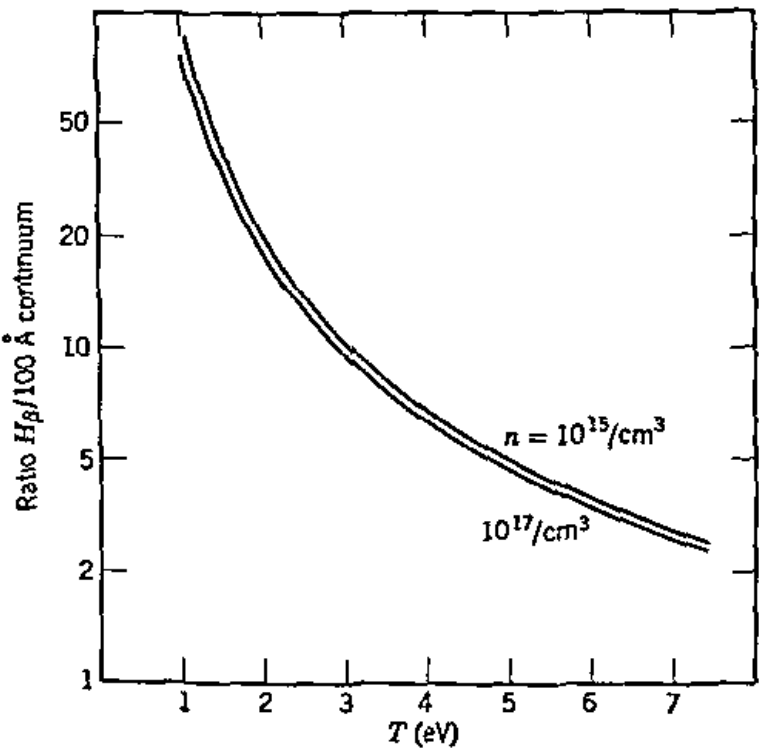


Fig 9 6 Ratio of line intensity to continuum intensity in hydrogen at two electron densities

an adjacent continuum band is a function of the plasma temperature Results of measurements of this sort are illustrated in Fig 9 6

Measurements of Stark widths give n_e, which can be converted to temperature by the Saha equation provided the pressure is known The Stark width method has the advantage that only relative intensities over small wavelength intervals are necessary, which eases the calibration problem Also, the widths are large enough so that good resolution is not required On the other hand, Stark widths are slightly temperature dependent, so the process is complicated

Although the Doppler profile method is only useful if the Stark broadening is negligible, it is the only spectroscopic method that measures the temperature of the atoms or ions directly All other methods determine directly only the electron temperature

Microwave Emission

A satisfactory interpretation of the optical radiation emitted from a plasma rests on the validity of the assumption that local thermodynamic equilibrium exists between all of the plasma components. The assumption of local thermodynamic equilibrium is made necessary by the fact that most of the optical radiation arises from a mechanism which depends on the temperature of the electrons as well as that of the atoms. For instance, the intensity of radiation arising from recombination of a free electron into a certain atomic state will depend on the volume density of singly ionized atoms present in their ground state, and this density will depend on the ion temperature.

Microwave emission from a plasma arises almost entirely from electron emission mechanisms which do not depend strongly on atom and ion temperatures, such as *bremsstrahlung* (the radiation emitted due to the acceleration of the electron during a collision with an ion or atom) or cyclotron emission (due to the acceleration of an electron spiraling in a dc magnetic field). Thomson scattering, which is the re-radiation of electromagnetic energy due to the acceleration of an electron in an electromagnetic field, will be negligible compared to the previously mentioned mechanisms at microwave frequencies.

We have mentioned the two important emission mechanisms for radiation. The processes which are the inverses of these two will be the dominant absorption mechanisms: collisional and cyclotron absorption. It is necessary, of course, that there exists an absorption process which is the inverse of each emission process in order that an equilibrium may be reached between the radiating system (in our case the electrons) and the electromagnetic field.

An approximate expression for the absorption of radiation due to electron-atom collisions may be obtained by assuming the momentum-loss frequency, ν_m, to be independent of electron energy and including its effect in the electron motion by writing it as an effective viscous drag term in the single electron equation of motion in an electromagnetic field:

$$m\left(\frac{dv}{dt} + \nu_m v\right) = e(E_0 e^{i(\omega t - kx)}). \tag{9.4}$$

Neglecting the spatial dependence of the electric field for the moment, we calculate the induced sinusoidal velocity of the electron, and from this the current density $J = nev$ is calculated, where n is the electron density. This current density is then substituted into Maxwell's equations for our electric field $E_0 e^{i(\omega t - kx)}$ and the dependence of the propagation constant,

k, on ω is determined

$$k = \frac{\omega}{c}\left(1 - \frac{\omega_p^2}{\omega(\omega - i\nu_m)}\right)^{1/2}, \tag{9 5}$$

$$k = \frac{\omega}{c}\left[\left(1 - \frac{\omega_p^2}{\omega^2}\right)^{1/2} - \frac{i\omega_p^2\nu_m}{2\omega^3(1 - \omega_p^2/\omega^2)^{1/2}}\right] \tag{9 6}$$

Having assumed $\nu_m \ll \omega$, we define

$$\omega_p^2 \equiv \frac{ne^2}{m\epsilon_0} \tag{9 7}$$

This quantity ω_p is called the plasma frequency, and is a very useful parameter It was first calculated by Lord Rayleigh[7] as a possible solution to energy states in his "pumpkin" atom theory of atomic structure As applied to a plasma, a physical picture of the resonant condition can be obtained by considering the motion of individual electrons and positive ions The plasma region of a discharge contains an equal number of positive and negative charges uniformly distributed so that the resultant space charge is zero If the random motions of the ions and electrons are neglected, the ions and electrons may be considered to have fixed spatial positions such that the resultant space charge of any given regions of the plasma is zero The electrons, which are much lighter than the positive ions, will oscillate about their equilibrium positions if the neutrality of charge is disturbed in any manner Thus an elementary theory of plasma electron oscillations can be derived by considering the positive ions to have fixed positions and the electrons to be displaced an amount ζ from their equilibrium positions Consider the parallel planes X_1 and X_2 and the electrons displaced into the volume bounded by $X_1 + \zeta_1$ and $X_2 + \zeta_2$ of Fig 9 7 If n is the electron density, the change in the electron density from the displacements is

$$\Delta n = -n\frac{\zeta_2 - \xi_1}{X_2 - X_1}$$

since the displacement ζ is small compared to the distance between the two boundary planes Considered as a limiting process, then

$$\Delta n = -n\frac{\partial \zeta}{\partial X}$$

The resultant space charge creates an electric field which can be calculated from Poisson's equation

$$\nabla \cdot \mathbf{E} = -\frac{e\,\Delta n}{\epsilon_0} \rightarrow \frac{\partial E}{\partial X} = +\frac{ne}{\epsilon_0}\frac{\partial \zeta}{\partial X}$$

Since the space charge field is zero when the displacement is zero,

$$E = \frac{ne}{\epsilon_0}\zeta.$$

This field, which acts on the electrons, arises solely from their displacement and is a longitudinal field in the direction of the displacement.

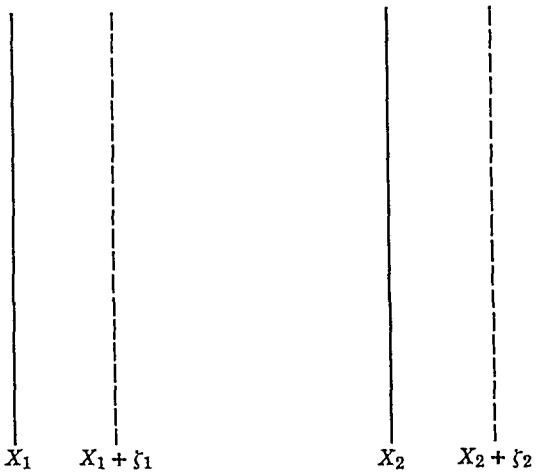


Fig. 9.7 Displaced electrons in a plasma.

Applying the force law

$$m_e \frac{d^2\zeta}{dt^2} = -Ee = -\frac{ne^2}{\epsilon_0}\zeta,$$

we obtain

$$\frac{d^2\zeta}{dt^2} + \frac{ne^2}{m_e\epsilon_0}\zeta = 0.$$

This is the equation of motion of the displaced electrons and shows that the electrons execute simple harmonic motion about their equilibrium position with a frequency, called the plasma frequency, given by

$$\omega_p^2 = \frac{ne^2}{m_e\epsilon_0}$$

$$f \simeq 8980 n^{1/2} \qquad (n = \text{number of electrons per cm}^3).$$

The frequency of the plasma electron oscillations is a function of the electron concentration. A typical value of n is 10^{10} cm^{-3}, which gives a frequency of the order of 1000 megacycles per sec.

Equation (9.6) points to the fact that ν_m gives rise to attenuation of the

wave as it moves through the medium If $\nu_m = 0$,

$$k = \frac{\omega}{c}\left(1 - \frac{\omega_p^2}{\omega^2}\right)^{1/2},$$

or, equivalently, we may say the index of refraction of the plasma is

$$\mu = \left(1 - \frac{\omega_p^2}{\omega^2}\right)^{1/2}$$

It is seen for tenuous plasmas ($\omega_p^2/\omega^2 \ll 1$) that the index of refraction is near unity, and as the electron density increases the index of refraction approaches zero and then becomes imaginary A zero or pure imaginary index of refraction means that the wave can no longer propagate through the medium Such a wave incident on the plasma boundary would therefore be reflected Physically, this reflection occurs because the induced electron current density is equal to or greater in magnitude than the wave's displacement current density, and since the two currents are opposite in phase (for $\nu_m = 0$), the electron current density can cancel or even overpower the displacement current

To study a plasma using electromagnetic techniques, it is necessary to operate in a regime where the wave can penetrate (and also escape) from the plasma For a typical microwave frequency of 9000 megacycles, we are limited to electron densities less than 10^{12} cm ^{3}

We can now begin to discuss the problem of the transport of radiation in the plasma, and let us first calculate a thermodynamic model for the thermal emission of radio frequency noise We consider a plasma in free space of dimensions large compared to the wavelength of the radiation and of index of refraction near unity Under such assumptions the geometrical optics approximation is valid and the radiation may be assumed to travel in rays whose trajectories are straight lines and reflections at the plasma boundaries may be neglected Furthermore, we assume that the radiation is isotropic and travels in all directions within the plasma with equal probability The radiation is defined in terms of the specific intensity, I_ω, which is the radiation crossing a unit area and directed into a unit solid angle per unit time and radian frequency interval

Referring to Fig 9 8 we see that the amount of energy, ΔE_ω, crossing da and contained in $d\Omega$ for the time and radian frequency intervals dt and $d\omega$ is

$$\Delta E_\omega = I_\omega(da \cos\theta)\, d\Omega\, dt\, d\omega \tag{9 8}$$

The specific intensity as it traverses the medium is attenuated by absorption through the net absorption coefficient per unit length, α_ω, and increased by spontaneous emission through the spontaneous emission

per unit volume of the plasma, per unit solid angle, time, and radian frequency interval, j_ω.

The net change in the energy in traversing a distance ds along its path through the plasma will be

$$dI_\omega(da\cos\theta)\,d\Omega\,dt\,d\omega = +j_\omega(ds\,da\cos\theta)\,d\Omega\,d\omega - \alpha_\omega\,ds\,I_\omega(da\cos\theta)\,d\Omega\,dt\,d\omega$$

or

$$\frac{dI_\omega}{ds} = j_\omega - \alpha_\omega I_\omega. \tag{9.9}$$

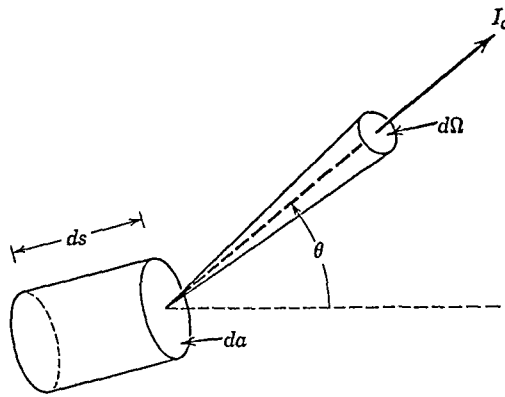


Fig. 9.8 Radiation from a plasma.

This is the equation for the transport of radiation of one polarization. In general, there will be coupling between the two possible polarizations of the radiation, but for negligible coupling the transfer equations may be written separately.

If L is the total thickness of the plasma, and α_w and j_w are constants, the solution to this transfer equation may be written

$$I_\omega = I_{\omega_0}e^{-\alpha_\omega L} + \left(\frac{j_\omega}{\alpha_\omega}\right)(1 - e^{-\alpha_\omega L}). \tag{9.10}$$

Equation (9.10) would describe the situation shown in Fig. 9.9 where I_{ω_0} is the specific intensity at one surface of the plasma due to another radiating body and I_ω is the resultant specific intensity at the other surface of the plasma.

Consider $I_{\omega_0} = 0$. It is seen then that I_ω increases towards a limiting value, j_w/α_w, as the plasma depth, L, is increased. No matter how thick the plasma is made, I_ω cannot exceed j_ω/α_ω. This limit occurs when the absorption of the wave equals the spontaneous emission, as can be seen from

Eq (9 9) For $\alpha_\omega L \ll 1$, $I_\omega = j_\omega L$, which is the limit where I_ω is composed of the emission from each small volume element within the plasma, and the absorption is negligible

The opaque limit of I_ω, j_ω/α_ω, must correspond to the blackbody limit for the case where the electrons have a defined temperature, that is when they have a Maxwell-Boltzmann velocity distribution

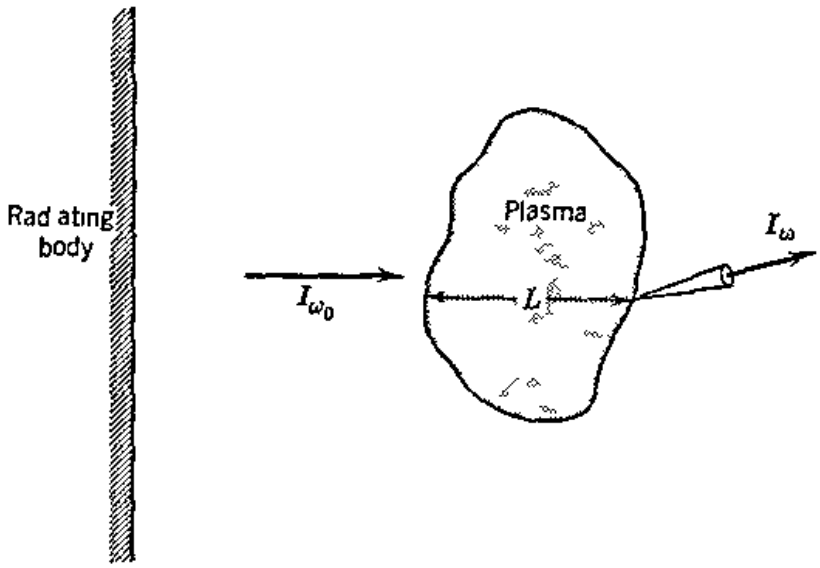


Fig 9 9 Radiation passing through a plasma

For an opaque plasma in free space, and for one polarization of radiation, we must have

$$I_\omega = \frac{j_\omega}{\alpha_\omega} = \frac{\hbar\omega^3}{8\pi^3 c^2}\left(e^{\frac{\hbar\omega}{kT}} - 1\right)^{-1} \tag{9 11}$$

For microwave frequencies $\hbar\omega/kT \ll 1$, giving

$$\frac{j_\omega}{\alpha_\omega} = \frac{\omega^2}{8\pi^3 c^2} kT = \frac{kT}{2\pi\lambda^2} \tag{9 12}$$

Equations (9 11) and (9 12) give a relationship which must exist between the emission and absorption of radiation within a body in thermodynamic equilibrium This relation is Kirchhoff's radiation law

For the case where the electrons of the plasma have a defined temperature, an absolute measurement of the specific intensity emitted from a plasma will yield a measurement of the electron temperature As an example of this let us discuss briefly the experiments of Bekefi, Hirshfield, and Brown [8] Their plasma was a short section of the positive column of a hot-cathode dc glow discharge oriented with its axis at right angles to the propagation direction of the wave in a rectangular wave guide One end of the wave guide terminated in a matched load and the other went to a

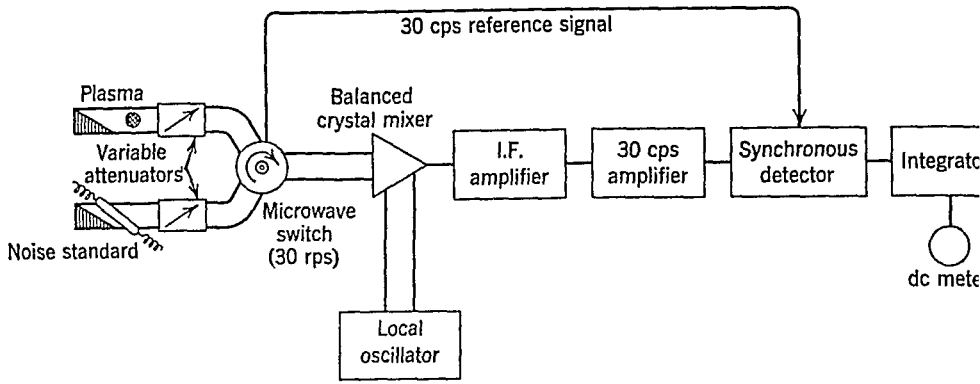


Fig. 9.10 Microwave radiometer.

3000 megacycle/sec receiver. A block diagram of the setup is given in Fig. 9.10. The noise emitted from the plasma was alternated with the noise from a standard source by means of a microwave switch. The radiometer was capable of measuring powers as low as 5×10^{-17} watts. The standard source was a fluorescent lamp calibrated to have a radiation temperature of 10,700°K.

Results of typical measurements of the *bremsstrahlung* power, P, radiated from a helium discharge at a pressure of 0.5 torr are shown in Fig. 9.11. We see that the dependence of the noise power in electron

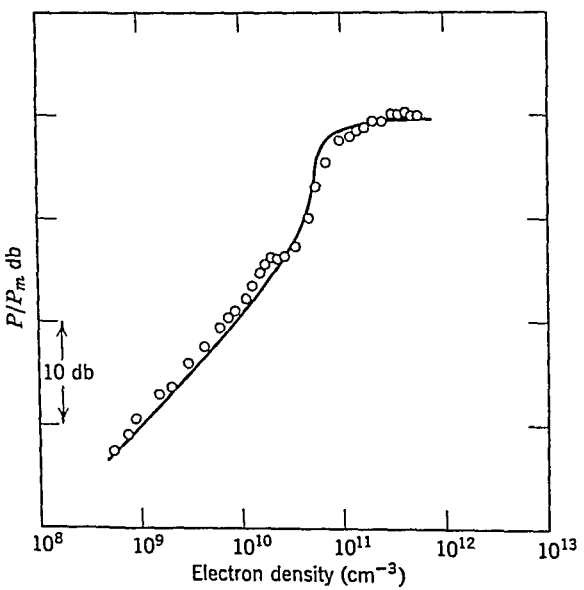


Fig. 9.11 Normalized noise power as a function of electron density for a pressure of 0.5 torr.

density falls to three regions The linear increase of P/P_{max} at low electron densities corresponds to the region where the plasma is transparent The abrupt transition occurs at $\omega_p = \omega$ for the given density at the frequency of 3000 megacycles/sec of the experiment At the density-independent region at high electron density the plasma is opaque to its radiation and is behaving as a black body

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Chapter 10

High-Frequency Breakdown

As one supplies more and more energy to electrons in a gas, in general, more and more ion pairs are produced and the process of electron multiplication takes place. When the rate of gain of electrons exceeds their rate loss by any mechanism such as diffusion, attachment, or recombination, the process runs away with itself and the gas is said to have "broken down." This is observed experimentally by a great increase in conductivity where the current is then limited by the capability of the power supply, and the gas often glows visibly.

Diffusion-Controlled Breakdown

In a high-frequency gas-discharge breakdown, the primary ionization caused by the electron motion is the only production phenomenon that controls the breakdown, and for this reason it is the simplest type to consider. If one calculates the maximum kinetic energy in the oscillatory motion of an electron at the minimum field intensities for breakdown experimentally determined, one finds that this energy corresponds to about 10^{-3} volt. It is, therefore, obvious that the energy of oscillation is insufficient to account for breakdown.

It is well known that a free electron in a vacuum under the action of an alternating field oscillates with its velocity 90° out of phase with the field, and thus takes no power, on the average, from the applied field. The electron can gain energy from the field only by suffering collisions with the gas atoms, and it does so by having its ordered oscillatory motion changed to random motion on collision. The electron gains random energy on each collision until it is able to make an inelastic collision with a gas atom. The fact that the average electron can continue to gain energy in

the field, despite the fact that it may move either with or against the field, can be seen by showing that the energy absorbed is proportional to the square of the electric field and, hence, independent of its sign The rate of gain of energy of the electron from the electric field E (*rms*) is $P = eE\bar{v}$ The power gain expressed in terms of the ac mobility previously discussed is

$$P = e\mu E^2 = \frac{e^2E^2}{m\nu_m}\left(\frac{\nu_m^2}{\nu_m^2 + \omega^2}\right) \tag{10 1}$$

This equation may be written in terms of an effective field E_e

$$P = \frac{e^2E_e^2}{m\nu_m}, \tag{10 2}$$

where E_e is the effective field that would produce the same energy transfer as a steady field

The gas-discharge breakdown occurs when the gain in electron density caused by ionization of the gas becomes equal to the loss of electrons by diffusion, recombination, or attachment When the loss is by diffusion, the problem becomes simple Referring to our previous discussion of diffusion where the current density $\Gamma = -\nabla Dn$, we have at breakdown

$$\frac{\partial n}{\partial t} = \nu_i n - \nabla \cdot \Gamma = 0,$$

where ν_i is the frequency of ionization By writing the diffusion equation as

$$\nabla^2 Dn + \left(\frac{\nu_i}{D}\right) Dn = 0,$$

it follows from our previous work that we may write $\nu_i/D = 1/\Lambda^2$

Typical of the behavior of the diffusion-controlled breakdown field[1] at high frequency with changes in gas pressure are the results shown in Fig 10 1 Data shown in this figure were taken in helium containing small admixtures of mercury vapor, Heg gas, which, as we have pointed out before, has the theoretical advantage of acting as an atom without excitation levels

Although an accurate description of the breakdown measurements can be given theoretically only by taking into account the electron energy distribution,[2] a physical picture of the mechanisms involved can be seen qualitatively in the following way Let us consider first a gas at high pressure ($\nu_m^2 \gg \omega^2$) Here the power that goes into the electron from the electric field is dissipated in elastic collisions between the electrons and the

gas molecules so that for this case, Eq. (10.2) can be written:

$$\text{Energy/collision} = \frac{e^2E^2}{m\nu_m^2} = e\frac{2m}{M}\bar{u}. \tag{10.3}$$

From this we may calculate the electric field to be

$$E = \nu_m\left(\frac{m}{e}\frac{2m}{M}\bar{u}\right)^{1/2} = 0.94p(\bar{u})^{1/2} \tag{10.4}$$

for the case of Heg gas in which nearly all the loss goes into non-ionized collision. Taking the average energy equal to one-third of the ionization

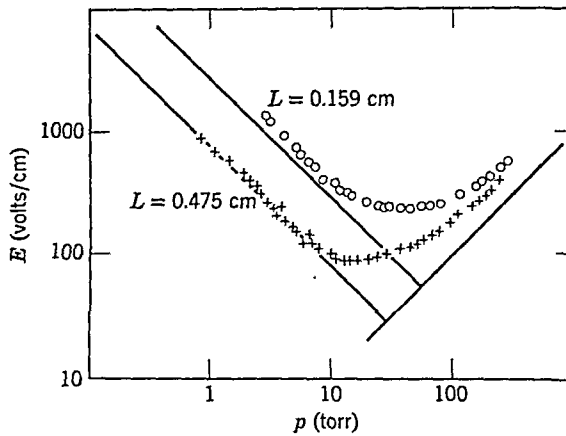


Fig. 10.1 Microwave breakdown in Heg gas.

potential, $E = 2.4p$, it can be seen that the high-pressure breakdown measurements of Fig. 10.1 tend to approach this line at high pressures.

In the low-pressure region, the electrons make many oscillations per collision, and the breakdown field may be determined by equating the number of collisions to ionize to the number of collisions to diffuse out of the tube. Since we are discussing Heg gas, where all the inelastic collisions are ionizing ones, let us consider that all the input power goes into ionization. We may write

$$P = eu_i\nu_i$$

and

$$\nu_i = \frac{P}{eu_i} = \frac{eE^2}{mu_i\nu_m}\left(\frac{\nu_m^2}{\nu_m^2 + \omega^2}\right). \tag{10.5}$$

We have just seen that the breakdown condition is $\nu_i = D/\Lambda^2$, and substituting the value of the diffusion coefficient $D = lv/3$ we find $\nu_i = lv/3\Lambda^2$. If we multiply numerator and denominator of this expression by the

velocity, we obtain an expression in terms of the previous expression for ν_i, in which $\omega^2 \gg \nu_m^2$, to obtain

$$\nu_i = \frac{\overline{v^2}}{3\Lambda^2\nu_m} = \frac{eE^2\nu^m}{u_i m\omega^2} \tag{10 6}$$

If we solve this expression for the electric field, writing ω in terms of λ and $2\bar{u} = mv^2/e$, we get

$$E = \frac{2\pi c}{\Lambda\lambda\nu_m}\left(\frac{2}{3}\bar{u}u_i\right)^{1/2} \tag{10 7}$$

For the case of helium, we find that the collision frequency is a constant independent of energy over almost the entire energy range and equal to $\nu_m = 2\,37 \times 10^9 p$ For Heg, the ionization potential is 19 8 volts, and we assume a very low pressure where all the input power goes into ionization Calculating the electric field under these approximations leads to a relation $E = 1284/p\Lambda\lambda$, which agrees fairly well with the low-pressure measurements in Heg as shown in Fig 10 1

If a gas contained in a vessel is placed in an alternating electric field *for a certain value of the electric field, the gas will break down into an* electrical discharge This breakdown field may be expressed as

$$E_b = E(u_i, \Lambda, \lambda, p),$$

where E is the electric field intensity, u_i is the ionization potential, Λ is a parameter describing the vessel, λ is the wavelength of the exciting field, and p is the pressure The field E is measured in volts per cm, and u_i is measured in volts The term Λ has the units of length, and its appearance in explicit calculations also involves various known dimensionless ratios to describe the shape of the vessel It is customary to measure pressure in torr, and the mean free path, which is inversely proportional to pressure, is measured in centimeters A relation between pressure and mean free path is obtained by introducing the quantity P_m, which is the number of collisions per unit length at 1 torr pressure Thus, p may be regarded as having the units of reciprocal length, even though this is not its true dimension

In treating the breakdown problem dimensionally, there are five variables with only two fundamental dimensions, volts and centimeters This leads to three independent dimensionless variables among which there is a functional relation It is often convenient in physical problems to introduce variables that are not dimensionless but are nevertheless proper variables for dimensionless analysis The completely dimensionless variables contain one or more physically invariant quantities that need not be carried along in a practical discussion There are a number of sets

of such proper variables in a gas discharge problem which may be transformed into one another, and their relative convenience depends on the purpose for which they are to be used.

One very useful set of proper variables is

$$E\Lambda, \quad p\Lambda, \quad p\lambda.$$

The advantage of these variables lies in the fact that p, Λ, and λ are the experimentally independent parameters that determine the dependent variable E, the observed breakdown field.

Another set of proper variables which we shall use is obtained by dividing the first variable by the second and so obtaining

$$E\Lambda, \quad E/p, \quad p\lambda.$$

A particular advantage of this set, in a discussion of breakdown phenomena, is that we may define an ac ionization coefficient $\zeta = 1/E^2\Lambda^2$ which is a function of E/p and $p\lambda$, and is similar in concept to the first Townsend coefficient.

Certain basic assumptions are made in the calculations of breakdown as a balance between the ionization rate and the loss of electrons by diffusion. We examine here the limits that these assumptions place on the way the theory applies to various experimental conditions. These can be discussed in terms of the proper independent variables $p\Lambda$ and $p\lambda$. One can plot on the $p\Lambda$-$p\lambda$ plane the conditions for all breakdown data for a single gas (the case here illustrated is hydrogen) and derive limits of these variables which will define the applicability of the diffusion theory.[3]

At low frequencies, the experimental measurements of breakdown are always taken in vessels whose dimensions are small compared to a wavelength of the exciting power. For this case, the uniform-field assumption may be very good. At very high frequencies, there exists a limit to the size of the discharge tube consistent with the uniform-field assumption of the diffusion theory. This can be written in terms of the size of vessel allowable to sustain a single loop of a standing wave of the electric field. The relation between the parallel plate separation, the wavelength, and the diffusion length may be written as

$$\frac{\lambda}{2} = L = \pi\Lambda.$$

Thus one arrives at a limit that may be written in terms of $p\lambda$ and $p\Lambda$ as

$$p\lambda = 2\pi(p\Lambda). \tag{10.8}$$

This limiting line is plotted in Fig. 10.2 and designated as the *Uniform-Field Limit*.

The diffusion theory will not apply when the electron mean free path becomes comparable with the tube size In the limiting case, this can be expressed as the mean free path l, being equal to Λ The probability of collision P_m is equal to $1/pl$ To plot this condition in Fig 10 2, we may write

$$p\Lambda = 1/P_m \tag{10 9}$$

The value of P_m is not a constant but depends upon the electron energy. If we assume that the average electron has an energy equal to one-third

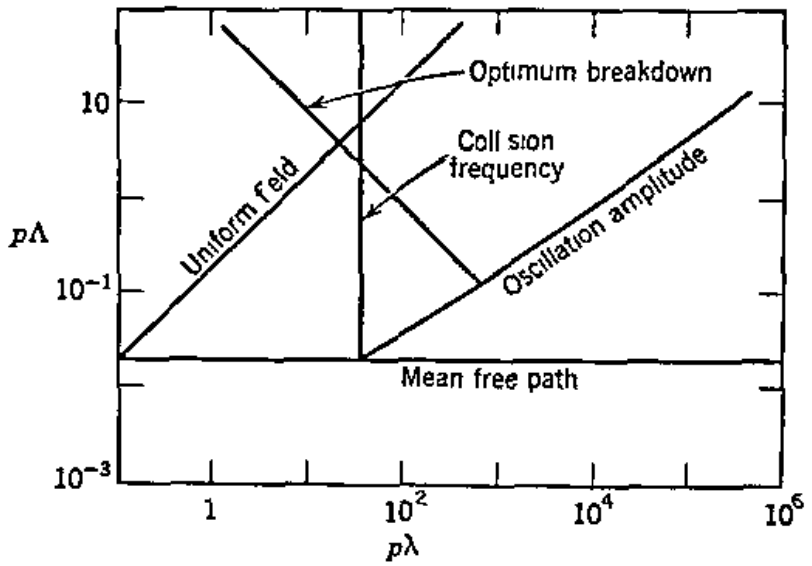


Fig 10 2 Microwave breakdown limits for hydrogen

of the ionization potential, the average electron energy would be 5 volts for hydrogen Using measured values for the probability of collision in hydrogen for the average electron, we get $P_m = 49$ (cm-torr)$^{-1}$ With this value, we obtain the horizontal line in Fig 10 2 marked *Mean Free Path Limit*

When the amplitude of the electron oscillation in the electric field is sufficiently high, the electrons can travel completely across the tube and collide with the walls on every half-cycle

By writing Newton's second law for the motion of the electrons in the form

$$m(dv/dt) + (mv_m)v = -eE_p e^{j\omega t},$$

the velocity may be determined as

$$v = -[e/(j\omega m + mv_m)]E_p e^{j\omega t},$$

and the displacement of the electron oscillation as

$$x = \frac{eE_p e^{j\omega t}}{j\omega(j\omega + \nu_m)m}. \tag{10.10}$$

The amplitude of oscillation is

$$\frac{eE_p}{m\omega(\omega^2 + \nu_m^2)^{1/2}} = \frac{\sqrt{2}\, eE_e}{m\omega\nu_m}.$$

The limiting case of the diffusion mechanism in which all of the electrons will hit the walls would be calculated by setting the oscillation amplitude equal to one-half of the electrode separation. Thus, the oscillation amplitude limit becomes equal to

$$\frac{\sqrt{2}\, eE_e}{m\omega\nu_m} = \frac{L}{2}. \tag{10.11}$$

Substituting λ in terms of ω, v/l in place of ν_m, and $1/pP_m$ for l, we obtain

$$p\lambda = \left(\frac{\pi mc}{e} vP_m\right)\frac{pL}{\sqrt{2}\,(E_e/p)}. \tag{10.12}$$

Putting in numerical values and using the parallel plate relation $L = \pi\Lambda$, we have

$$p\lambda = 10^6\left(\frac{p\Lambda}{E_e/p}\right). \tag{10.13}$$

This equation can be solved numerically when the experimental values of the breakdown field are available. Experiments of this sort have been carried out for parallel plate geometry using hydrogen gas,[4] and numerical values can therefore be determined. This calculation yields the *oscillation amplitude limit* of Fig. 10.2. When the motion of the electron in the ac field corresponds to the amplitude of oscillation limit for diffusion, electrons are lost by colliding with the walls, and the breakdown field must be sharply increased to make up for this new loss, as illustrated in Fig. 10.3. The breakdown field in the mobility-controlled region is a function of the type of material of which the walls are composed, as we would expect.

In contradistinction to the low-pressure type of secondary emission-controlled breakdown (the electron resonance breakdown) illustrated by Fig. 7.6, the high-pressure mobility controlled case is very much a function of the type of gas. This is evident since the velocity of the oscillating electrons is mobility controlled, and the mobility is a function of the type of gas. The dependence of the amplitude of oscillation on the electron mobility provides a good substantiation of the factors controlling the critical conditions for crossing the amplitude of oscillation transition.

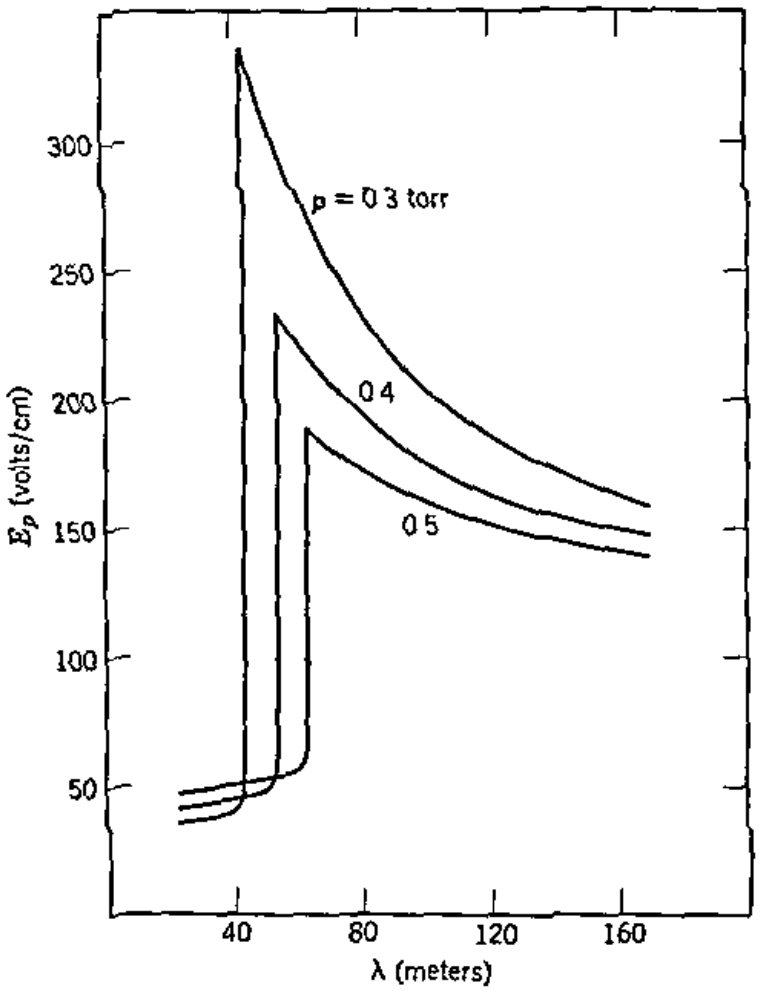


Fig 10 3 Breakdown at the oscillation amplitude limit

Let us illustrate by taking the case in hydrogen where the drift velocity of electrons is approximately linearly proportional to E/p (Fig 3 1) This should predict that the critical electric field at the transition value should be proportional to the gas pressure This is found to be approximately the case by Gill and von Engel,[4] as illustrated in Fig 10 4

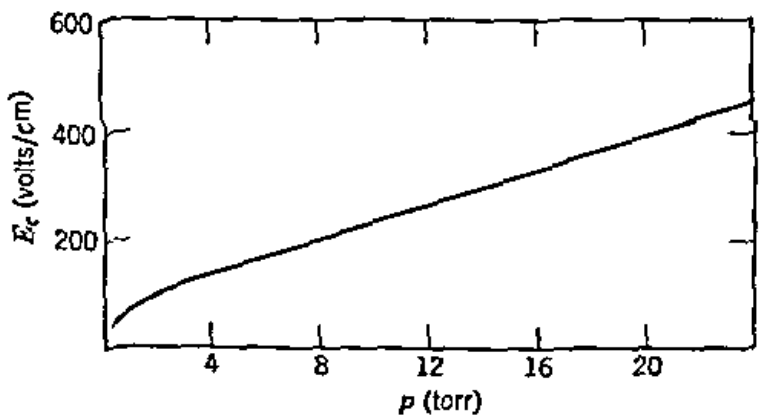


Fig 10 4 Critical field as a function of pressure

One of the most striking characteristics of the mobility controlled breakdown is the fact that after the electron oscillation amplitude is greater than the tube dimensions—that is, after all the electrons are lost to the walls—breakdown is still observed. This is illustrated in Fig. 10.3 by the measured breakdown fields on the longer wavelength side of the oscillation amplitude transition. These data also show that the longer the wavelength of the applied ac field, the more easily the discharge starts. The observation that it is such a strong function of the frequency of the electric field requires the effect to be associated with charged particles. The fact that the electrons are being lost every half-cycle to the walls suggests that the only particles left to account for this increase in ionization must result from the positive ions.

Gill and von Engel[4] used their data in hydrogen to prove that the breakdown beyond the amplitude of oscillation transition was controlled by the positive ion mobility. The amplitude of the oscillation of the velocity is

$$v_{\max} = \frac{eE_p}{m^*(\omega^2 + \nu_m^2)^{1/2}} = \frac{\sqrt{2}\, eE_e}{m^*\nu_m} \tag{10.14}$$

where m^* is the mass of whatever particle is effective in controlling the discharge. If breakdown occurs when the maximum velocity has reached a constant value v, which is a reasonable assumption when the electrons are produced by secondary emission from the walls, we may write

$$e^2E_p^2 = (m^*)^2v^2(\omega^2 + \nu_m^2), \tag{10.15}$$

and E_p^2 plotted against ω^2 should be a straight line. This is illustrated for the data of Gill and von Engel for hydrogen in Fig. 10.5.

For any pressure, the straight lines are seen to cut the E axis at a point E_0 where $\omega = 0$. Here $eE_0 = m^*v\nu_m$, but since $(e/m^*\nu_m) = \mu$, the mobility, we have the possibility of identifying the particles by determining what values of e/m^* lead to the correct relation between E and ω at the various pressures measured in 10.5. The e/m^* ratio for the molecular hydrogen ion agrees better than one would expect from the elementary nature of the analysis, and leaves little doubt that the ion mobility is the controlling mechanism on the long wavelength side of the oscillation amplitude transition. In this region, therefore, the loss of electrons at the walls resulting from their large oscillation amplitude is more than replaced by the secondary electron emission by positive ion bombardment of the walls, and the breakdown field is shown to decrease as the wavelength increases.

Within the limits of experimental conditions in which diffusion theory adequately explains the breakdown behavior, several different phenomena may occur. One of the phenomenological changes that is important is

the transition from many collisions per oscillation of the electron to many oscillations per collision This can be written as the condition that $\nu_m = \omega$, where ν_m, the collision frequency is the ratio of the average velocity v to the mean free path, and ω is the radian frequency of the applied field For hydrogen, $\nu_m = 5\,93 \times 10^9 p$ Putting this in terms of the proper variables, we obtain

$$p\lambda = 32 \tag{10 16}$$

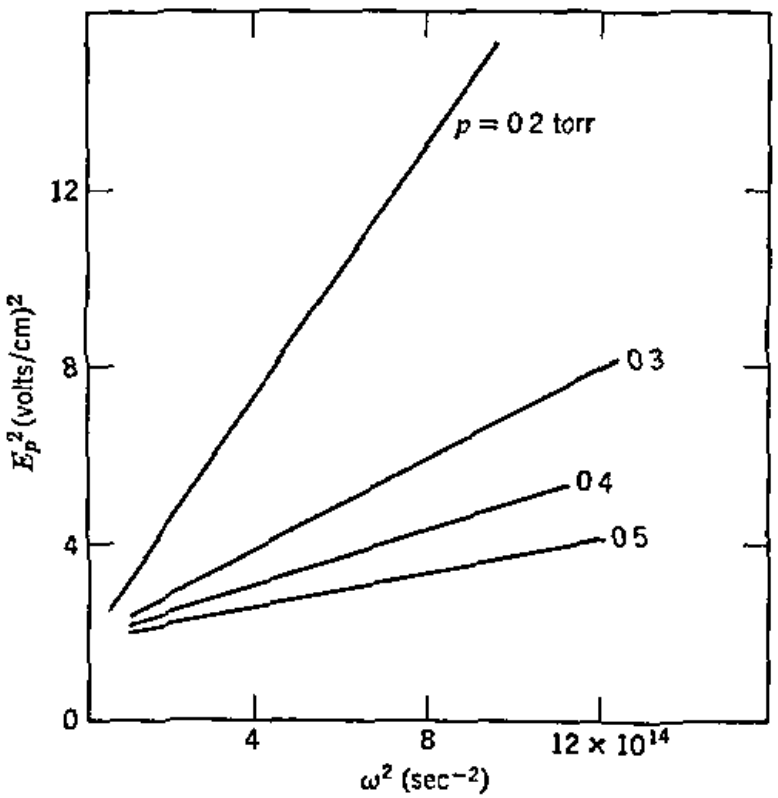


Fig 10 5 Breakdown controlled by ion mobility

This relation is plotted in Fig 10 2 as the dotted line marked *Collision Frequency Transition*

We can calculate a line on the $p\Lambda$-$p\lambda$ plane corresponding to the minimum breakdown field for any given container size as the locus of the intersection of the limiting lines illustrated in Fig 10 1 At low pressure we have seen that the breakdown field approaches the condition that

$$E = \frac{2\pi c(2\bar{u}u_i/3)^{1/2}}{\Lambda\lambda\nu_m}$$

For hydrogen, $\nu_m = 5\,93 \times 10^9 p$, $u_i = 15\,4$ volts, and we consider $\bar{u} = u_i$ This leads to a value of $E = 400/\Lambda\lambda p$ Using Eq (10 4) for hydrogen leads to $E = 5\,3p$ Eliminating the field between these two equations will allow us to calculate the pressure at which breakdown will occur most

easily. In terms of the variables of Fig. 10.2, this leads to the equation

$$p\lambda = \frac{75}{p\Lambda}. \tag{10.17}$$

This relation is plotted in Fig. 10.2 as the line marked *Optimum Breakdown.*

Many of the breakdown curves which have been obtained experimentally have complicated shapes, and it is difficult to visualize the

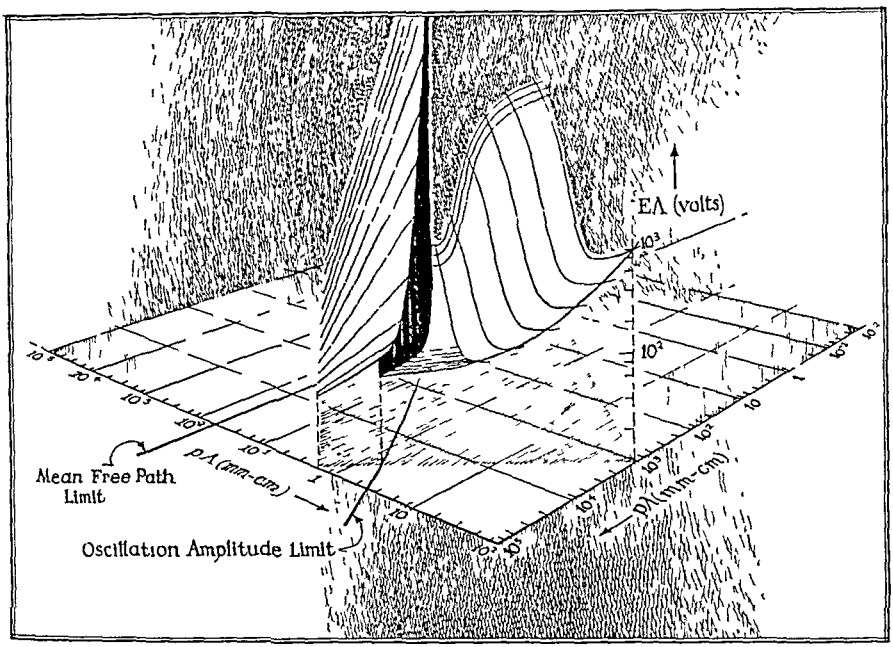


Fig. 10.6

correlation between two-dimensional plots constructed by different works. On the other hand, if we make a three-dimensional plot in which the horizontal axes are $p\lambda$ and $p\Lambda$, and the vertical axis is the effective breakdown voltage $E\Lambda$, all the available data may be represented on a single surface. Such a plot is shown in Fig. 10.6.

Short Pulse Breakdown

We can define a set of limit criteria for diffusion controlled breakdown for short dc pulses in just the same way as we have for the ac microwave case.

The Mean Free Path Limit will be identical in the two cases, that is, the diffusion length must not exceed the mean free path. For the case of air

$$p = 0.03 \text{ cm-torr}. \tag{10.18}$$

For the Uniform Field Limit, the length of the dc pulse, τ, must be shorter than the length of time it takes the wave front to move between the electrodes If the length of the container

$$L = \pi\Lambda, \quad C\tau = \pi\Lambda$$

or

$$p\tau = \pi p\Lambda/C = 10^{-10}p\Lambda \tag{10 19}$$

The equivalence to the ac Oscillation Amplitude Limit, we could call a Transit Time Limit for dc pulses An electron drifting across the inter-electrode space, must cause a gas breakdown before it collides with an electrode For this limiting condition $\tau = L/v$ where $L = \Lambda\pi$ We know that $v = eEp/mv_m$ where $v_m = \iota pP_m$ Combining these, we obtain

$$p\tau = \left(\frac{m\pi\bar{v}P_m}{e}\right)\frac{p\Lambda}{\sqrt{2}\,E_e/p} = 3 \times 10^{-6}\,\frac{p\Lambda}{E_e/p}, \quad \text{sec-torr} \tag{10 20}$$

for the Transit Time Limit

Although for the dc case there is no Collision Frequency Transition we can define an analogous Collision Frequency Limit where the pulse is so short that the average electron makes no collision while the dc is applied For this case, $\tau = 1/v_m$, and taking $v_m = 5 \times 10^9 p$,

$$p\tau = 2 \times 10^{-10} \text{ sec-torr} \tag{10 21}$$

Microwave Breakdown from the Boltzmann Equation

We wrote the Boltzmann transport equation in Eq (3 5) as

$$\frac{\partial F}{\partial t} + \mathbf{v}\cdot\nabla_r F + \frac{e\mathbf{E}}{m}\cdot\nabla_v F = \left(\frac{\partial F}{\partial t}\right)_{\text{coll}} \tag{10 22}$$

To evaluate F it is customary to expand it in spherical harmonics in velocity space and in Fourier series in time [5] When we do this for the range of p, Λ and λ outlined by Fig 10 2, we can arrive at the expression

$$F = \frac{F_0^0 + \mathbf{v}\cdot[\mathbf{F}_0^1 + \mathbf{\Gamma}_1^1 e^{j\omega t}]}{v} \tag{10 23}$$

where F_0^0 is the spherically symmetric term The axis is taken along the direction of grad n with rational symmetry about the axis leading to the F_0^1 term, and F_1^1 term comes from the first term of the Fourier expansion

In evaluating the collision term in Eq (10 22) we deal with the two collision frequencies $v_x = \iota pP_x$ and $v_i = \iota pP_i$ and treat the processes mathematically as though fast electrons disappeared at the rate $(v_x + v_i)F_0^0$ and slow electrons appeared at the rate qF_0^0 Using these ideas in evaluating the collision term, putting Eq (10 23) into Eq (10 22)

and equating coefficients, we get one scalar and two vector equations:

$$(\nu_x + \nu_i - q)F_0^0 =$$

$$-\left(\frac{v}{3}\right)\nabla_r \cdot \mathbf{F}_0^1 + \frac{1}{v^2}\frac{\partial}{\partial v}\left[\left(\frac{ev^2}{6m}\right)(\mathbf{E}_p \cdot \mathbf{F}_1^1)_{\text{real}} + \left(\frac{m}{M}\right)\nu_m v^3 F_0^0\right] \quad (10.24)$$

$$\nu_m \mathbf{F}_0^1 = -v\nabla_r F_0^0 \quad (10.25)$$

$$(\nu_m + j\omega)\mathbf{F}_1^1 = \left(\frac{e\mathbf{E}_p}{m}\right)\frac{\partial F_0^0}{\partial v}. \quad (10.26)$$

The components $\mathbf{F}_0^1$ and $\mathbf{F}_1^1$ can be eliminated from Eq. (10.26) by substituting these terms from the two preceding equations to give the differential equation for F_0^0 as:

$$(\nu_x + \nu_i - q)F_0^0 =$$

$$\left(\frac{v_2}{3\nu_m}\right)\nabla_r^2 F_0^0 + \frac{1}{v^2}\frac{\partial}{\partial v}\left[\frac{eu_c}{3m}\nu_m v^2\frac{\partial F_0^0}{\partial v} + \left(\frac{m}{M}\right)\nu_m v^3 F_0^0\right]. \quad (10.27)$$

In this equation,

$$u_c = \frac{eE_p^2}{2m}\frac{1}{(\nu_m^2 + \omega^2)} \quad (10.28)$$

is the average energy transferred from the field to an electron between collisions.

The total excitation and ionization rates may be defined by

$$n\bar{\nu}_x = \int_0^\infty \nu_x F_0^0 4\pi v^2\, dv$$

$$n\bar{\nu}_i = \int_0^\infty \nu_i F_0^0 4\pi v^2\, dv.$$

Since every exciting collision yields one slow electron and every ionizing collision yields two electrons

$$\int_0^\infty qF_0^0 4\pi v^2\, dv = n(\bar{\nu}_x + 2\bar{\nu}_i).$$

Using these relations in multiplying Eq. (10.27) by $4\pi v^2\, dv$ and integrating over all velocities, we obtain the diffusion equation (the term in the brackets vanish at both limits),

$$n\bar{\nu}_i + \nabla_r^2 Dn = 0. \quad (10.29)$$

This expresses the fact we have already discussed that at breakdown the ionization rate equals the diffusion rate.

For a uniform electric field, the diffusion equation has a solution which is everywhere positive only if

$$\bar{\nu}_i = \frac{D}{\Lambda^2} \tag{10 30}$$

This is often called the breakdown condition

If the function F_0^0 is assumed to be the product of a function $n(x)$ of space, and a function $f(u)$ of energy, we can make use of Eq (10 29) to replace $\nabla_r^2 F_0^0$ by $-nf/\Lambda^2$ and write for $f(u)$

$$\left(\nu_x + \nu_i - q + \frac{2eu}{3m\nu_m\Lambda^2}\right)f = \frac{2}{3\sqrt{u}}\frac{d}{du}\left[(\nu_m u^{3/2})\left(u_c\frac{df}{du} + \frac{3m}{M}f\right)\right] \tag{10 31}$$

The terms on the left hand side represent the electrons leaving unit volume of phase space through excitation ionization, and diffusion, and their reappearance at low energy at the rate qf On the right are the terms describing the energy gained from the field and lost to recoil, the electric field being contained in u_c according to Eq (10 28)

The solution of Eq (10 31) is complicated in general but has been solved for a number of specific cases Let us discuss the solution for helium The inelastic collision frequency term can be taken from the experimental values[6] shown in Fig 10 7 The analytic approximations also shown in this figure are

$$\frac{\nu_x + \nu_i}{\nu_m} = h_0 u - h_1 - \frac{h_2}{u}$$

where $h_0 = 0\,0175$ volts^{-1}, $h_1 = 0\,665$ and $h_2 = -6\,6$ volts

$$\frac{\nu_i}{\nu_c} = h_i(u - u_i)$$

with $h_i = 5\,48 \times 10^{-3}$ volts $^{-1}$ and u_i is the ionization potential of 24 48 volts

The elastic collision frequency, ν_m, can be taken as a constant independent of energy as shown in Fig 1 7 down to an energy corresponding to $u_c = u_i/5$ The distribution function can be calculated for the case of ν_m not constant below this value although it can be shown to make no difference in the value of the breakdown potential as long as E_e/p does not become smaller than 5 volts/cm torr The distribution function $f(u)$ has been calculated a number of times[7] and its dependence on electron energy for two different values of E_e/p is shown in Fig 10 8 The value for u_c is given in general by Eq (10 28), and is assumed here to be $u_c = (e/m)(E_e/\nu_m)^2$

The diffusion equation given as Eq (10 29) has a solution for a uniform electric field which is only positive everywhere if $\nu_i = D/\Lambda^2$ This is the breakdown condition Both the diffusion coefficient and the ionization

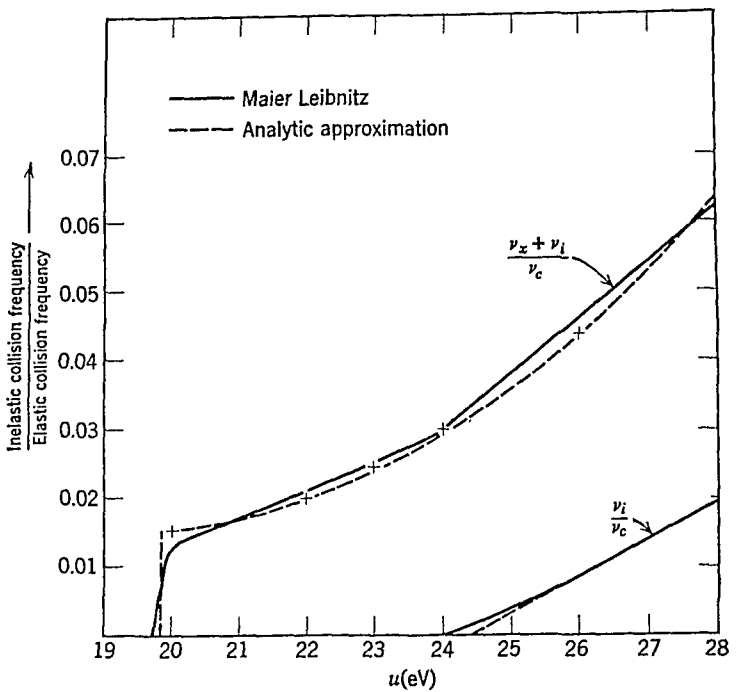


Fig. 10.7 Inelastic collision frequencies of electrons in helium as a function of electron energy.

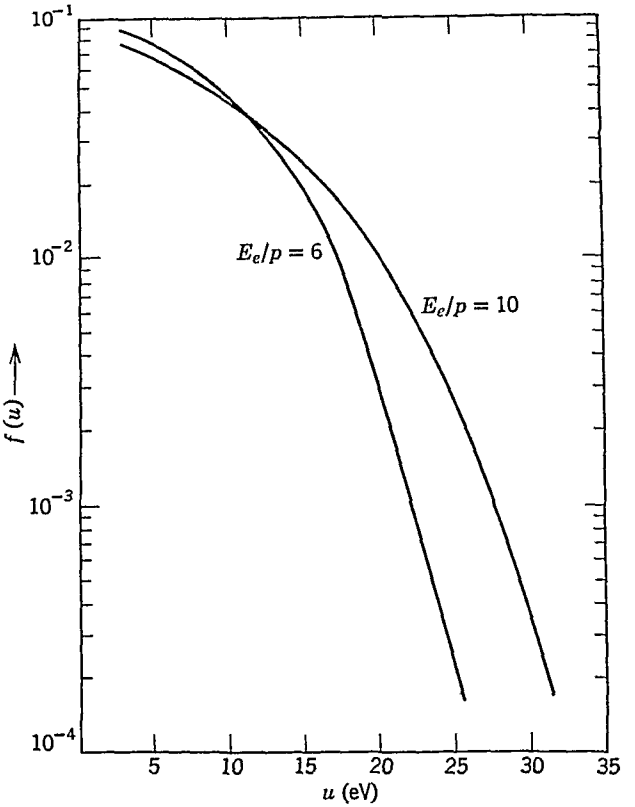


Fig. 10.8 Energy distribution functions for electrons in helium.

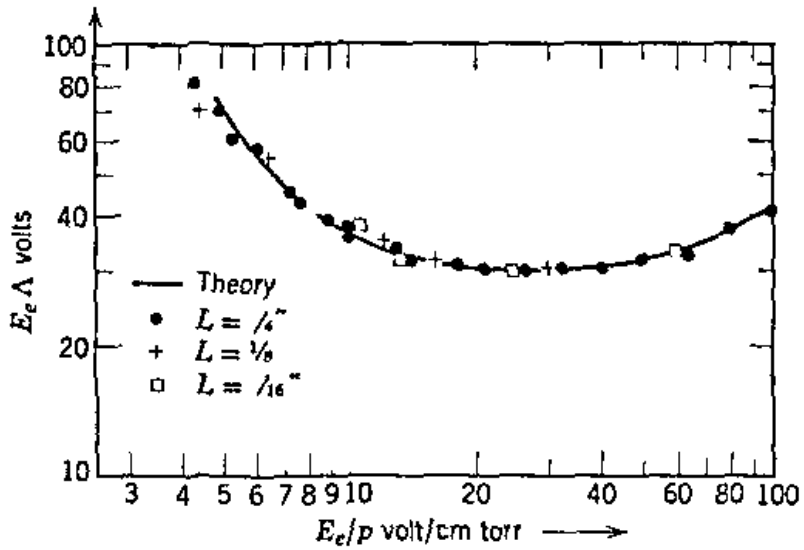


Fig 10 9 High frequency gas discharge breakdown in pure helium

frequency depend on the distribution function in the following manner Writing Eq (3 11) in terms of the distribution function

$$D = \int_0^\infty \left(\frac{2e}{3m\nu_m}\right) uf(u)\, du$$

The average ionization frequency in the same terms, becomes

$$\nu_i = h_i \int_{u_i}^\infty \nu_m(u - u_i) f(u)\, du$$

If we know f as illustrated in Fig 10 8, we can determine D and ν_i and inserting these in the breakdown equation yields a transcendental equation for $E_e\Lambda$ and E_e/p which may be compared with experiment The agreement between theory and experiment is excellent,[7] as illustrated in Fig 10 9 In contemplating the agreement between theory and experiment it is important to realize that the theoretical equations contain no adjustable parameters, and the empirical equations fitted to gas-discharge data on excitation potentials and collision cross sections similarly had no arbitrarily adjustable constants The fit is real both in magnitude and shape

Magnetic Fields

When diffusion is the controlling loss mechanism, the effect of the magnetic field on the rate of electron loss is described in Chapter 2 by introducing an effective diffusion length Λ_b, so that

$$\frac{1}{\Lambda_b^2} = \frac{\nu_m^2}{\nu_m^2 + \omega_b^2}\frac{1}{\Lambda_\perp^2} + \frac{1}{\Lambda^2} \qquad (10\ 32)$$

where $\Lambda_\perp$ and $\Lambda_\parallel$ are the diffusion lengths in the absence of the magnetic field in directions perpendicular and parallel to the magnetic field With

the microwave field at right angles to the magnetic field, the mean energy gained between collisions, u_c, is given by

$$u_c = \frac{eE_p^2}{4m}\left[\frac{1}{(\omega + \omega_b)^2 + \nu_m^2} + \frac{1}{(\omega - \omega_b^2) + \nu_m^2}\right] \equiv \frac{eE_e^2}{m\nu_m^2} \quad (10.33)$$

where E_p is the peak value of the microwave field, and ω is its radian frequency. When the microwave field is parallel to the magnetic field, Eq. (10.33) is valid with $\omega_b = 0$. Equation (10.33) defines an effective field E_e which is the root-mean-square field at high pressure.

Let us examine the behavior of the breakdown field when we superimpose a magnetic field at right angles to the microwave electric field. When the pressure is low, the mean-energy gain between collisions exhibits a resonance at frequencies near the cyclotron frequency ($\omega \approx \omega_b$). Since the transfer of energy from the field to the electrons becomes so efficient at cyclotron resonance, we expect that a lower electric field would be sufficient to cause breakdown at $\omega \approx \omega_b$. That this is a fact is illustrated in Fig. 10.10. These data[8] were taken in a flat cylindrical cavity whose length was short compared with the radius, and with the magnetic field

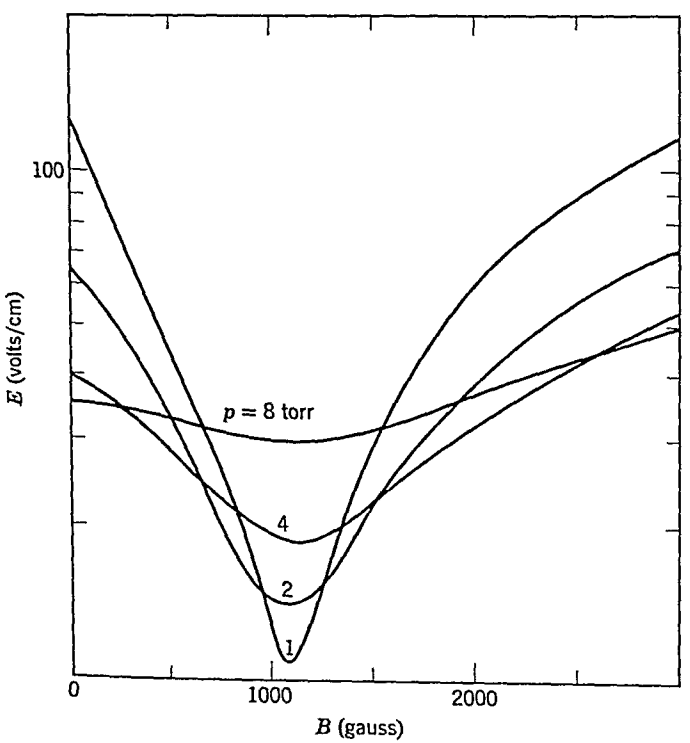


Fig. 10.10 Microwave breakdown in transverse electric and magnetic fields.

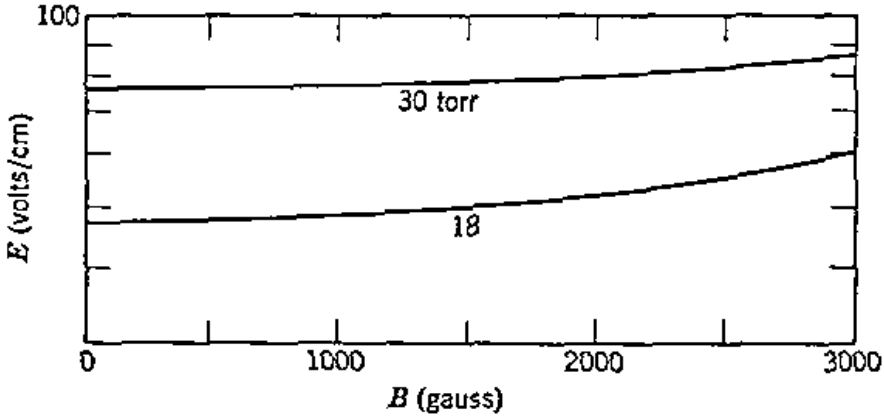


Fig 10 11 Microwave breakdown at high pressure in transverse electric and magnetic fields

transverse to the electric field (oriented along the axis of the cylindrical cavity) whose diffusion length is given by Eq (2 29)

When the pressure is high ($\nu_m > \omega$) the mean energy gained per collision is smaller than it is in the absence of a magnetic field (even at $\omega \approx \omega_b$) with the result that the breakdown field may be larger This behavior has been observed as illustrated in Fig 10 11

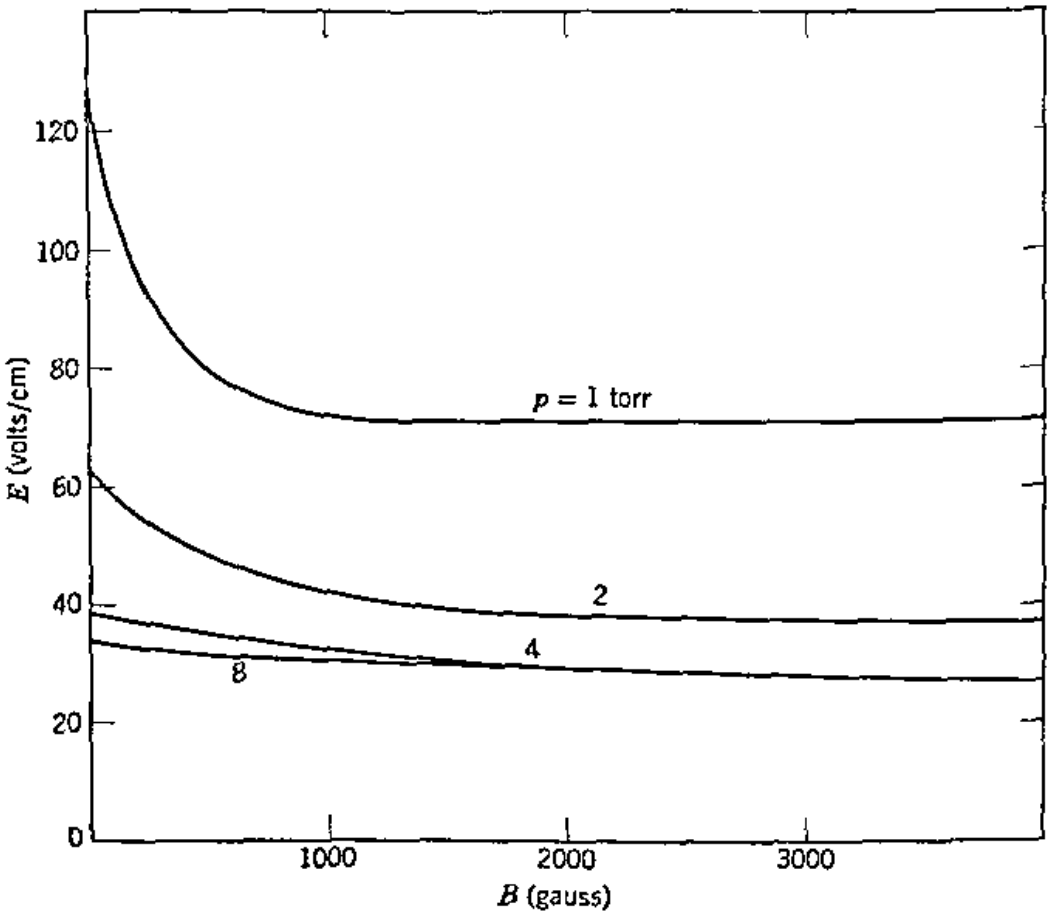


Fig 10 12 Microwave breakdown in parallel electric and magnetic fields

To study the effect of the magnetic field on diffusion alone, the electric and magnetic fields are oriented in the same direction. With a longitudinal magnetic field, the equivalent diffusion length of a cylinder is given by Eq. (2.28). In order to have a large reduction in the diffusion loss by the magnetic field, experiments were carried out in a cavity whose height was greater than the radius. The effect of orienting the magnetic field so that it affects the diffusion alone, is illustrated in Fig. 10.12.

Oscillation-Amplitude Limit

Gill and von Engel[4] have reported in detail on the behavior of the breakdown field as a function of frequency in the absence of a magnetic field near the amplitude of oscillation limit. A typical curve at a given pressure is shown in Fig. 10.3. Under these conditions, the electrons make many collisions per oscillation, so that the mean-energy gain between collisions to the left of the critical wavelength is given by Eq. (10.33) in terms of the effective field E_e, while the amplitude of oscillation limit is given (for a constant pressure) by

$$\frac{E_c \lambda_c}{d} = \text{constant} \tag{10.34}$$

where λ_c is the critical wavelength, E_c is the critical breakdown field, and d is the distance between the ends of the container.

It is not difficult to predict[9] the modification of the oscillation-amplitude limit by a magnetic field. Just to the left of the oscillation-amplitude limit the condition for breakdown is the balance between the gain of electrons from ionization, and the loss of electrons from diffusion. Under these conditions, the breakdown field is given by

$$E_b^2 = \frac{2}{3} \frac{\bar{\nu} u_i}{\Lambda_b^2} \tag{10.35}$$

where u_i is the ionization potential of the gas.

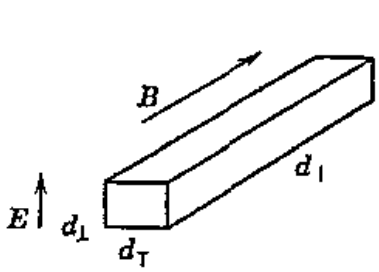
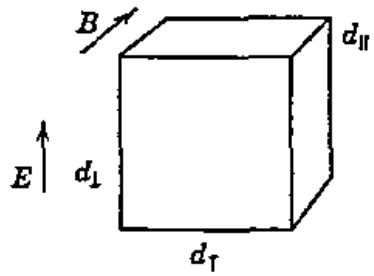
The effect of the magnetic field makes the dimensions of the cavity at right angles to the field appear larger to the electron but does not affect the dimensions parallel to the magnetic field. Thus we define two Λ's: $\Lambda_\perp$, where E and B are perpendicular, and $\Lambda_\parallel$, where E and B are parallel. In terms of these Λ's, by using Eqs. (10.32) and (10.33) for the ratio of the critical breakdown field with magnetic field, E_{bc}, to the critical breakdown field without magnetic field, E_c, we can write

$$\frac{E_{bc}^2}{E_c^2} = (\text{constant}) \frac{\dfrac{1}{\Lambda_\perp^2} \dfrac{\nu_m^2}{\omega_b^2 + \nu_m^2} \dfrac{1}{\Lambda_\parallel}}{\dfrac{\nu_m^2}{(\omega + \omega_b)^2 + \nu_m^2} + \dfrac{\nu_m^2}{(\omega - \omega_b)^2 + \nu_m^2}} \tag{10.36}$$

where the constant has such a value that E_{bc}/E_c is unity when ω_b is zero Since the oscillation-amplitude limit invariably occurs when $\nu_m/\omega \gg 1$, Eq (10 36) can be simplified as follows

$$\frac{E_{bc}^2}{E_c^2} = (\text{constant})\left(\frac{1}{\Lambda_\perp^2} + \frac{1}{\Lambda_\parallel^2}\frac{\omega_b^2 + \nu_m^2}{\nu_m^2}\right) \tag{10 37}$$

Table 10 1 Critical wavelength as a function of geometry

	
$d_\parallel >> \begin{cases} d_T \\ d_\perp \end{cases}$	$d_\parallel << \begin{cases} d_T \\ d_\perp \end{cases}$
$\lambda \sim \dfrac{d_\perp}{\dfrac{E_{po}}{\nu_c^2 + \omega_b^2}\nu_c}$	$\lambda \sim \dfrac{d_\perp}{\dfrac{E_{po}}{\sqrt{\nu_c^2 + \omega_b^2}}}$
$\lambda \sim \dfrac{d_T}{\dfrac{E_{po}}{\nu_c^2 + \omega_b^2}\omega_b}$	$\lambda \sim \dfrac{d_T}{\dfrac{E_{po}}{\sqrt{\nu_c^2 + \omega_b^2}}\dfrac{\omega_b}{\nu_c}}$

We shall now discuss the modifications of relation (10 34) that result from the application of a magnetic field at right angles to the ac electric field for several values of $\Lambda_\perp$ and $\Lambda_\parallel$ Another complication arises Because of the curved path, an electron can strike and get lost to the wall parallel to the electric field With reference to Table 10 1, we define distances $d_\perp$ and d_T as shown These distances must be suitably modified if the discharge container is not parallelepiped

For discharge vessels in which $\Lambda_\perp \gg \Lambda_\parallel$, the magnetic field modification of the diffusion length is negligible compared with the effect on the mean-energy gain of the electrons Thus, in Eq (10 37) the $\Lambda_\perp$ term is negligible, and

$$E_{bc} \approx E_c\left(\frac{\omega_b^2 + \nu_m^2}{\nu_m^2}\right)^{1/4} \tag{10 38}$$

The critical wavelength is obtained by setting the shorter of the electron-drift times across $d_\perp$ or d_T equal to one half-cycle of the ac field That is,

the critical wavelength is given either by

$$t = \frac{d_\perp}{v_\perp} \leq \frac{1}{2f} = \frac{\lambda}{2c} \tag{10.39}$$

or by

$$\frac{d_\top}{v_\top} = \frac{\lambda}{2c}, \tag{10.39b}$$

whichever is the smaller. The drift velocities $v_\perp$ and $v_\top$ are given by[8] the tensor mobility $\overleftrightarrow{\mu}$:

$$\overleftrightarrow{\mu} = \mu \begin{Vmatrix} \dfrac{\nu_m^2}{\nu_m^2 + \omega_b^2} & \dfrac{\nu_m^2}{\nu_m^2 + \omega_b^2}\dfrac{\omega_b}{\nu_m} & 0 \\ -\dfrac{\nu_m^2}{\nu_m^2 + \omega_b^2}\dfrac{\omega_b}{\nu_m} & \dfrac{\nu_m^2}{\nu_m^2 + \omega_b^2} & 0 \\ 0 & 0 & 1 \end{Vmatrix} \tag{10.40}$$

where μ is the mobility in the absence of the magnetic field. From Eqs. (10.39) and (10.40), we obtain the two relations for the critical wavelength when $\Lambda_\perp \gg \Lambda_\parallel$. We have either

$$\lambda_{bc} = \text{constant}\,\frac{d_\perp}{\dfrac{\nu_m}{\nu_m^2 + \omega_b^2}E_{bc}} = \text{constant}\,\frac{d_\perp}{\dfrac{E_c}{(\nu_m^2 + \omega_b^2)^{1/2}}} \tag{10.41a}$$

or

$$\lambda_{bc} = \text{constant}\,\frac{d_\top}{\dfrac{\omega_b}{\nu_m^2 + \omega_b^2}E_{bc}} = \text{constant}\,\frac{d_\top}{\dfrac{E_c}{(\nu_m^2 + \omega_b^2)^{1/2}}\dfrac{\omega_b}{\nu_m}}. \tag{10.41b}$$

The expected behavior is shown in Fig. 10.13, when $d_\perp$ is the governing length and $\omega_b < \nu_m$.

For tubes that are shaped so that $\Lambda_\perp \ll \Lambda_\parallel$, the modification of the diffusion length offsets the decrease in the mean-energy gain, with the result that the critical breakdown field is, to a first approximation, unaltered. The two critical wavelengths, in this case, are given by

$$\lambda_{bc} = \text{constant}\,\frac{d_\perp}{\dfrac{\nu_m}{\nu_m^2 + \omega_b^2}E_{bc}} = \text{constant}\,\frac{d_\perp}{\dfrac{\nu_m}{\nu_m^2 + \omega_b^2}E_c} \tag{10.42a}$$

and

$$\lambda_{bc} = \text{constant}\,\frac{d_\top}{\dfrac{\omega_b}{\nu_m^2 + \omega_b^2}E_{bc}} = \text{constant}\,\frac{d_\top}{\dfrac{\omega_b}{\nu_m^2 + \omega_b^2}E_c} \tag{10.42b}$$

The expected behavior in this case is illustrated by Fig. 10.14 when $d_\perp$ is the governing length and $\omega_b < \nu_m$.

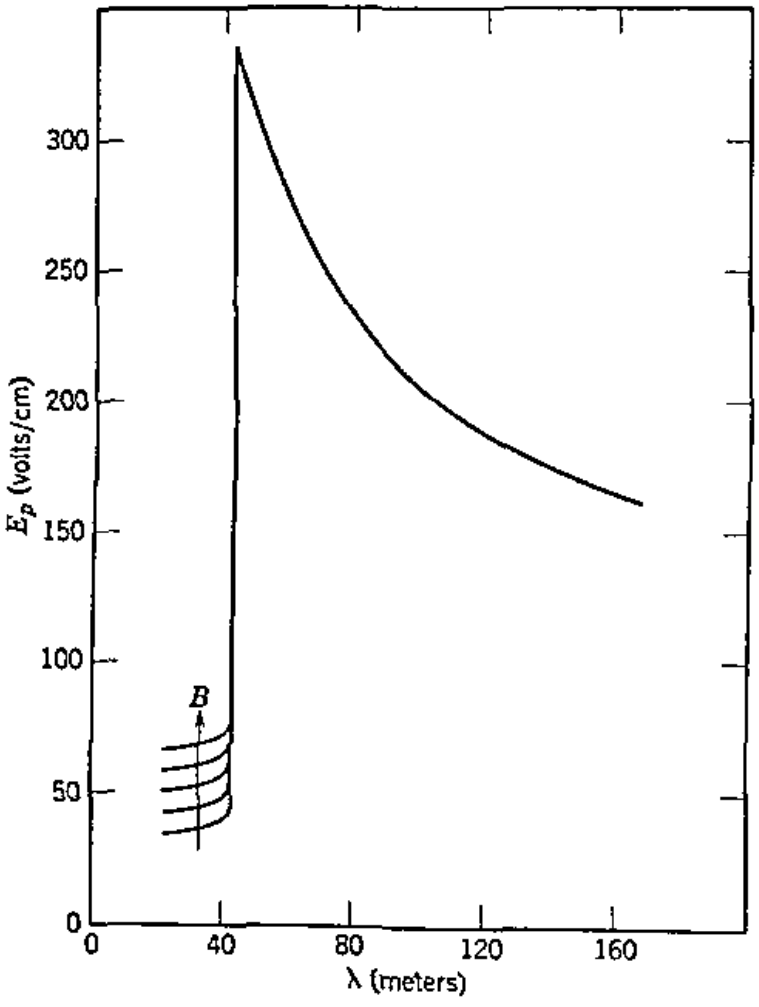


Fig 10 13 ac breakdown in presence of magnetic fields for $\lambda_{\parallel} \ll \lambda_{\perp}$

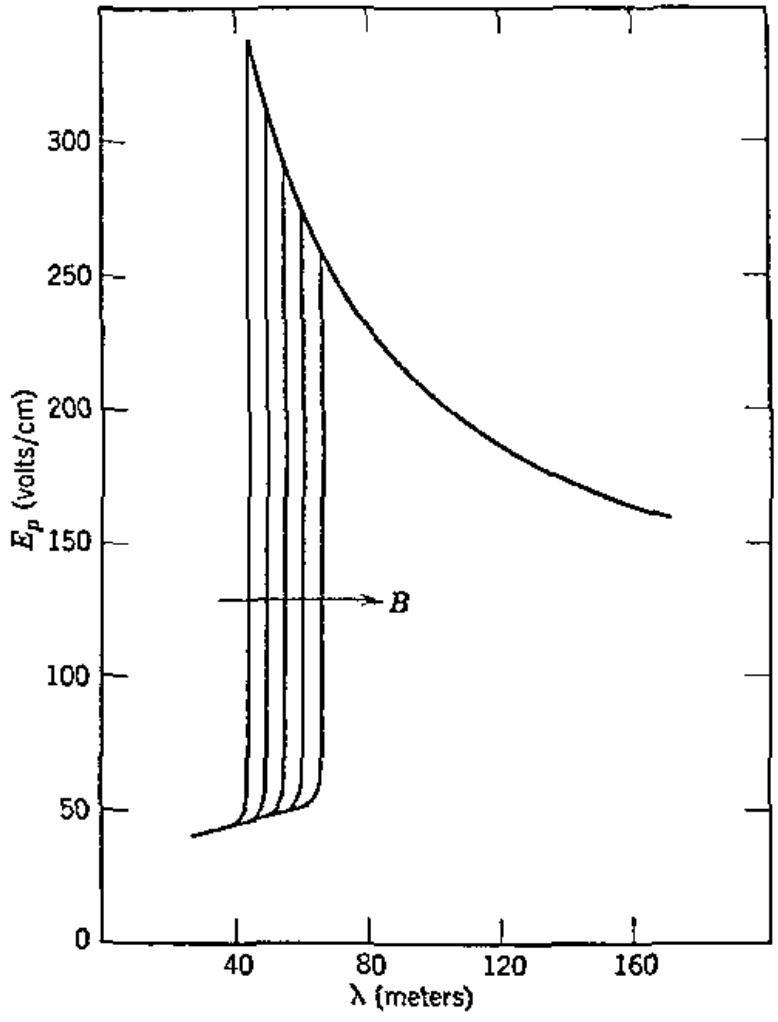


Fig 10 14 ac breakdown in presence of magnetic fields for $\lambda_{\perp} \ll \lambda_{\parallel}$

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Chapter 11

DC Breakdown

The initiation of a dc discharge based directly on the mechanisms discussed in terms of the two Townsend coefficients is called a Townsend breakdown

A simplification of Eq (7 11) results from the fact that $e^{\alpha x} \gg 1$ and

$$i = \frac{i_0 e^{\alpha x}}{1 - \gamma e^{\alpha x}} \tag{11 1}$$

When the value of x approaches a value d for which the denominator approaches zero, the current tends to infinity independent of i_0 This condition defines the breakdown of the gas into a conductor and for this condition, we find

$$\gamma e^{\alpha d} = 1 \quad \text{or} \quad \gamma e^{\eta V_B} = 1, \tag{11 2}$$

where V_B is the breakdown voltage The physical picture of breakdown is that a sufficient number of ion pairs are formed by the α mechanism so that through the γ process at least one electron is produced to maintain the current The breakdown condition provides one of the common ways of determining γ experimentally If we take the natural logarithm of $e^{\eta V_B} = 1/\gamma$, we obtain $\eta V_B = \ln (1/\gamma)$ Taking the common logarithm of both sides, we obtain

$$\log [\ln (1/\gamma)] = \log V_B - \log (1/\eta) \tag{11 3}$$

If the logarithm of the breakdown voltage is plotted as a function of E/p, and on the same plot, the logarithm of $1/\eta$ is also plotted, subtracting these curves yields the value of γ as a function of E/p This is illustrated[1] in Fig 11 1 The shape of these γ curves is explained by a shift in the γ mechanism At high values of E/p (low pressure) the controlling γ process

is electron emission by positive ion bombardment. As the pressure is increased, the back scattering of the emitted electrons results in a decrease of γ from this cause. At low values of E/p, the controlling process is photoelectric emission of electrons from the cathode.

The concept of proper variables to describe the breakdown phenomena is as useful in dc breakdown as in ac. We have just discussed the problem

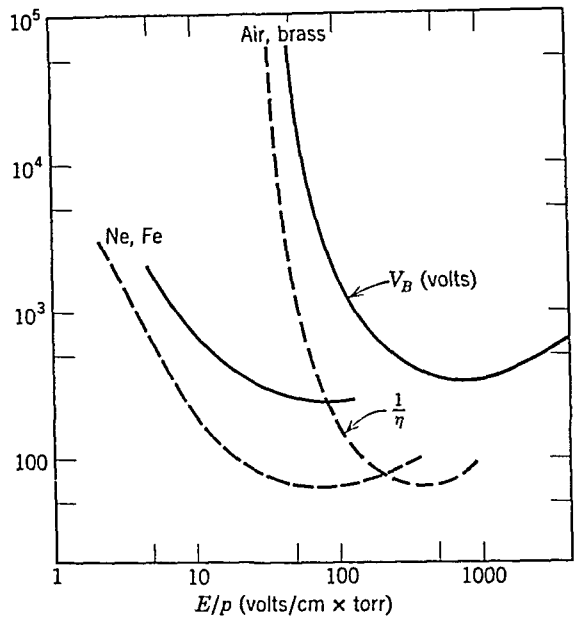


Fig. 11.1 Values of breakdown potential and $1/\eta$ between parallel plates for air and neon.

in terms of one set of proper variables, namely, V_B and E/p. Another set often found useful is V_B and pd where d is the electrode separation. When described in this way, the resulting relation is known as Paschen's law. Typical examples of these curves[2] are shown in Fig. 11.2. The minima in these curves are the result of two competing factors and occur at about the pressure where the mean free path is equal to the electrode separation. To the left of the minimum the necessary voltage breakdown increases since the electrons can get across the interelectrode space with less and less chance of producing an ionizing collision as the pressure decreases. The slower increase to the right of the minimum results from the fact that as the pressure increases, E/p decreases. This means that the energy per mean free path is decreasing and more voltage is required to produce the necessary ionization to cause breakdown.

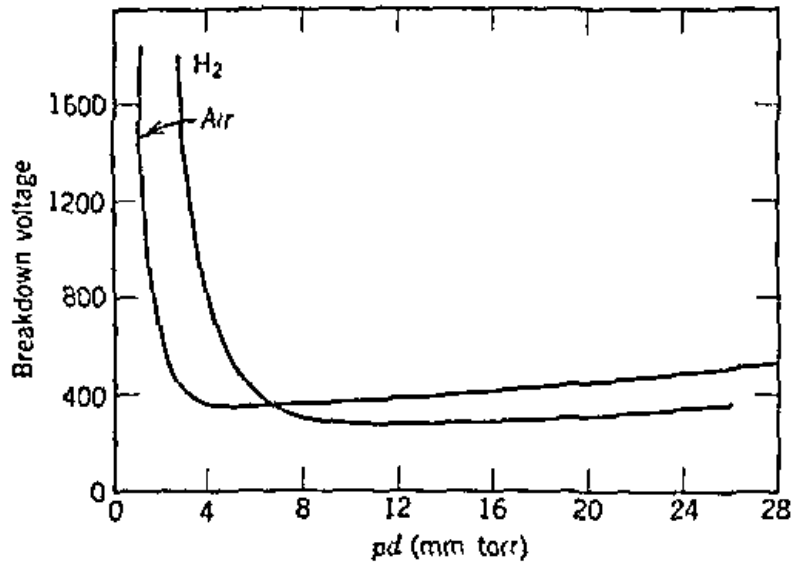


Fig 11.2 Paschen curves for hydrogen and air

Since the value of γ is a function of electrode surface, one would expect a variation of breakdown voltage with type of electrode, other conditions being constant This is shown[1] in Fig. 11 3 It can be seen that the electrode effect is large at low pressure, becoming negligible at high pressure This results from the fact that the breakdown is the product of γ and $e^{\eta l}$ Gamma does not change much with E/p, but η varies very rapidly at low E/p (high pressure) and, therefore, becomes the predominant effect in this region At low pressures a minimum occurs in the $V_B - pd$ curves Since γ does not change rapidly with E/p, the minimum voltage should be found for different cathode materials at the same value of E/p This would predict that the minima for different metals in the same gas should

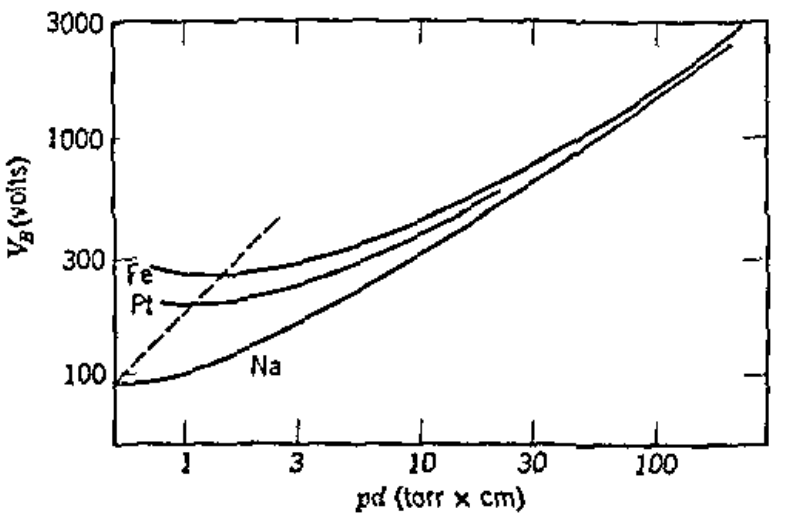


Fig 11.3 The effect of electrode material on breakdown in argon

lie on a straight line making a 45° angle with the log V_B — log pd axes. This behavior is observed.[1]

An abnormal form of the Paschen curve at low values of pd has been found[3] for He which illustrates some of the details of the interaction of the form of γ and η. The effect is shown in Fig. 11.4. Here, over a certain

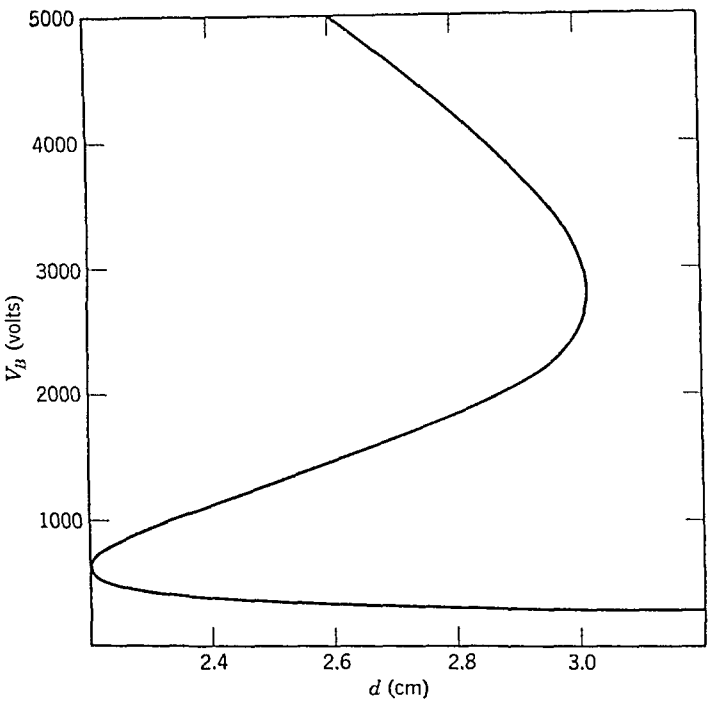


Fig. 11.4 Abnormal Paschen curve in helium. Pressure is 0.84 torr.

region of potential difference, breakdown occurs only when the voltage is lowered to a definite value. An explanation lies in the fact that the ionization probability for electrons in helium has a maximum at about 100 volts energy. For a value of E/p where the electron energy in the gas is higher, the decrease of the potential would not oppose but would favor the breakdown of the discharge, provided γ remains constant. The minimum in pd for the breakdown voltage V_B occurs when $dV_B/d(E/p) = V_B/(E/p)$. This corresponds to the maximum of η as a function of E/p, γ remaining constant. At high values of E/p, γ increases and the V_B curve in Fig. 11.4 bends back again at $d = 3$ and $V_B = 3000$ volts. In most gases the increase in γ masks the influence of the decrease in η, and this peculiar form of the Paschen curve is rarely found.

The breakdown of a gas discharge under the action of a steady field is more complicated than the high-frequency breakdown previously discussed With a dc field the electrons are being continuously swept out of the volume between the electrodes, and secondary mechanisms are required to make up for this electron loss

The α process alone cannot produce breakdown, and secondary mechanisms must be responsible for the further multiplication of the electron current These may occur in the gas (β processes) or at the cathode (γ processes) They may result from metastable atoms (β_m, γ_m), from ionic collisions (β_i, γ_i), and from photons (β_p, γ_p) At high electron and ion densities, space charge effects (σ processes) may play an important role

Just which of the secondary processes predominates under certain conditions is open to argument, and a great many experiments have been undertaken to resolve this question The β_m process is the Penning effect previously discussed It shows up as an increase in α The γ_m process occurs with a considerable time delay, and although it may be important under some conditions we shall not be concerned with it here The ionization mechanisms that Townsend originally postulated was the β_i process, but it is now known to be ineffective at the energies found in the usual gas discharges The γ_p process is proportional to the number of photons per ion, it is, therefore, proportional to the number of excitations per ionization and varies as $e^{Bp/E}$ Similarly, back scattering, which diminishes any γ process though not a β process, increases with p/E Thus one might expect a transition in the dominant secondary process from γ_i to γ_p to β_p as p/E is increased It is obvious that the state of the cathode will effect any γ process considerably

It is difficult to interpret, on the basis of the secondary mechanisms just considered, some of the phenomena observed in dc breakdown At high pressure, the breakdown always takes place in the form of a thin, well-defined spark originating from some one spot on the cathode Experiments with "overvolted" gaps ($V > V_B$) show that the time of formation of the breakdown discharge is so short (10^{-8} seconds and less) that the mechanism previously described cannot operate The formative time is a function of the overvoltage and increases with decreasing overvoltage Here there is difficulty in interpreting Eq (11 1) correctly as defining breakdown It has already been noted that the expression is mathematically indeterminate when $1 = \gamma e^{\alpha d_B}$ and $t_0 = 0$ From a physical viewpoint, however, at least one electron is necessary to initiate a spark It is not sensible to identify such an electron with t_0 The difficulty lies in the fact that although the above expression correctly predicts the steady-state current increase by primary and secondary processes, it does not correctly describe the transient phenomena of the spark formation The

difficulty may be resolved if we consider that, as d or V is increased, i increases to some value i_s which marks the beginning of the current instability. At i_s, space charge becomes important, with the result that, for some constant potential V, i increases rapidly to some value controlled by the external circuit, and continues at this value in the absence of i_0. The condition $1 = \gamma e^{\alpha d_B}$ represents the condition for a self-maintaining discharge following the spark, but it has little to do with the current equation, since, if $1/\gamma = e^{\alpha d_B}$, for an ion created in the gap through ionization by electron impact, there are created at the electrode or in the gas enough electrons to maintain $\gamma e^{\alpha d}$ at its existing value.

It can thus be concluded that space charge plays a definite role in the creation of the spark. We shall now consider this in greater detail.

The Streamer

Ionization produced by electrons driven by the electric field is governed by the Townsend mechanism $n = n_0 e^{\alpha X}$. This holds as long as space charge does not modify the electric field. If there is such a modification, the electron multiplication is very much controlled by the space charge field.

Assume the field takes the form illustrated[4] in Fig. 11.5, where the field is enhanced ahead and behind the avalanche but is reduced in the middle.

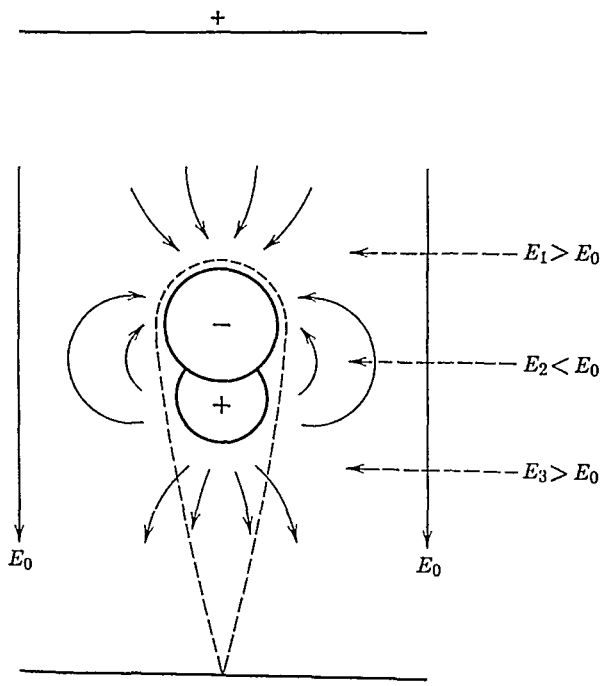


Fig. 11.5 Schematic representation of a streamer.



Fig 11 6

Making the assumption that the space charge takes the form of a sphere of radius r_a, the electric field of such a charged sphere will be

$$E_s = \frac{ee^{\alpha d}}{4\pi\epsilon_0 r_a^2} = 1.5 \times 10^{-7} \frac{e^{\alpha d}}{r_a^2}. \tag{11.4}$$

Measurements of r_a have been made in a cloud chamber by Raether[5] who found shapes for these streamers as shown in Fig. 11.6. For his particular measured values in methane at $E_0 = 3.9$ kV/cm, a pressure of 102 torr and a length of 3 cm, $r_a = 0.08$ cm. For this case $E_s = 2.3 \times 10^{-5} e^{\alpha d}$ so that the field distortion has a very large effect.

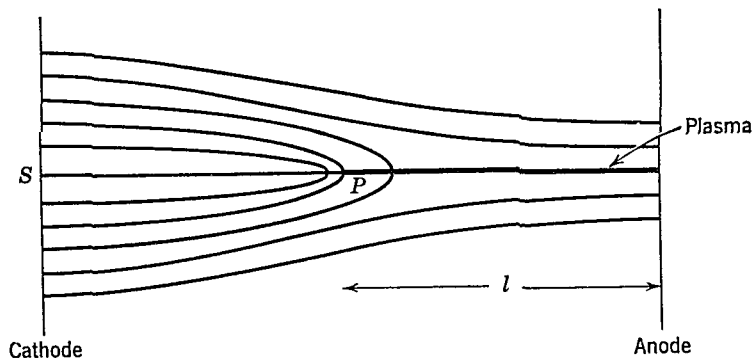


Fig. 11.7 Schematic representation of a streamer.

An important effect of the space charge is to create an electrostatic field E_s in a direction opposite to that of the applied field E, thus tending to cancel the applied field. Therefore, the avalanche grows much more slowly than without space charge, and, when $E_s = E$, the avalanche stops increasing. It should be noted that the electron drift velocity is unaffected by E_s, because the electrons in that part of the sphere which is near the anode are shielded from E_s by the electrons that are in the rear. The action of the space charge, however, is to remove more and more electrons from the avalanche, and finally, when $E = E_s$, just as many electrons are left behind per unit time as there are positive ions. The result is that a streak of neutral plasma is left behind the avalanche on its journey to the anode. This plasma, being a conductor, has a marked effect on the applied field E. It enhances the field in the vicinity of the beginning of the streak, as sketched in Fig. 11.7. Any secondary electrons left or produced behind the avalanche, or a second avalanche, will tend to travel toward the point P at a greatly enhanced speed. This increases the streak of plasma in the direction of the cathode. A *streamer* is formed which travels toward the

cathode Streamers are observed at low values of E/p, high pressure, and hence, β_p is the main mechanism producing the tributory avalanches

A more quantitative analysis of the streamer can be made if we compute the behavior when and where $E = E_s$ Assume that, after ionizing once, electrons in the avalanche travel a distance $1/\alpha$ before ionizing again As a result, we are left with a sphere of positive ions separated by a distance $1/\alpha$ from a similar sphere of electrons The two spheres will overlap if $(1/\alpha) < r_0$ where r_0 is the mean radius Effects of space charge will be important when $n_0 \sim 10^8$ This means that the transition for overlap of the positive and negative charge distribution is given by

$$(\alpha r_0)^2 = \frac{(\alpha X)^2 4\bar{u}}{V} \simeq \frac{4000}{V}$$

The electric field created by the space charge is given by

$$E_s = \frac{n_0 e}{4\pi\epsilon_0 r_0^2} \tag{11 5}$$

In this equation it is assumed that there is no overlap and that all electrons are contained in a sphere of radius r_0 Hence, $E = E_s$ when

$$\frac{n_0 e}{4\pi\epsilon_0} = E r_0^2 = \frac{4\bar{u} E X^2}{V} = 4uX$$

Should conditions be such that an overlap exists, Eq (11 5) must be modified to read

$$\frac{n_0 e}{4\pi\epsilon_0} = 4uX(\alpha r_0) \tag{11 6}$$

Ambipolar diffusion sets in long before cancellation of fields owing to space charge We can correct for the diffusion becoming ambipolar by writing

$$\frac{n_0 e}{4\pi\epsilon_0} \geq \frac{4\bar{u}\, d\mu_-}{\mu_+} \tag{11 7}$$

as the condition on the distance d for a streamer to form The time t_s that it takes for the streamer to form is given by

$$t_s = \frac{X}{v_-} = \frac{X}{\mu_- E}, \tag{11 8}$$

where X is the solution to Eq (11 6) Transforming Eq (11 8) into proper variables yields

$$t_s = \frac{(1/p)pX}{(\mu p)(E/p)} = \frac{1}{p} f\left(\frac{E}{p}\right),$$

$$pt_s = f(E/p) \tag{11 9}$$

Fletcher[6] measured the time necessary for a spark to form and found that experiment agreed closely with the above theory, provided the distance between the cathode and anode was greater than some critical distance d_c. For $d < d_c$, the times for the spark to form were greater than t_s, suggesting that several avalanches were needed for the streamer to form. Fletcher's measurements are shown in Fig. 11.8.

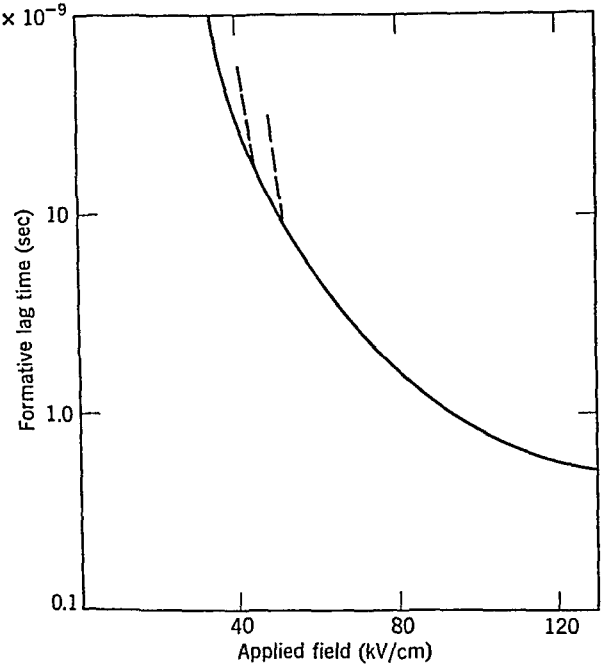


Fig. 11.8 Formative lag times as a function of applied field.

Fletcher found fairly abrupt deviations from his calculated formative lag times at certain critical applied fields for different electrode spacing. If we write Eq. (11.8) as

$$p\tau = \frac{p\Lambda}{\mu_- E} = \frac{p\Lambda}{\bar{v}} \approx \frac{p\Lambda}{E/p}, \tag{11.10}$$

we see that this is the form of Eq. (10.20) defining the Transit Time Limit. Using the actual data presented by Fletcher, it can be shown that the experimental points which disagree with the calculated formative lag times fall beyond the Transit Time Limit and, therefore, in fact should not agree with his calculations.

The most common streamer-produced sparks at atmospheric pressure

are lightning discharges. The charges are caused mainly by spray electrification The strong upward air currents at the head of the storm cloud carry moisture up to cooler regions where condensation into drops occurs When these drops become big enough, they fall The limiting size of the drops is about 0 5 cm in diameter Larger drops than this break up and release negative ions to the air, while the smaller drops that result are positively charged The newly formed small drops are carried upward and grow by condensation until they too fall, and the process is repeated. The rain at the head of the storm is found to be positively charged, while at the end of the storm it is negative This process of cloud charging builds up potentials as high as 10^9 volts so that the field at the earth's surface under the cloud may reach several hundred kilovolts per meter

Schonland[7] has developed the following theory for the progress of a stroke of lightning The discharge is initiated from a negative cloud by a negative streamer, called a pilot streamer, proceeding from the cloud into un-ionized air This streamer travels at a speed of about 10^7 centimeters per second It travels toward the ground for about 50 microseconds and then is stopped by its own space charge A much faster streamer, called a stepped-leader stroke, periodically catches up with the pilot streamer. This stepped-leader stroke is a negative streamer advancing along the ionized path with a speed of about 10^9 centimeters per second There

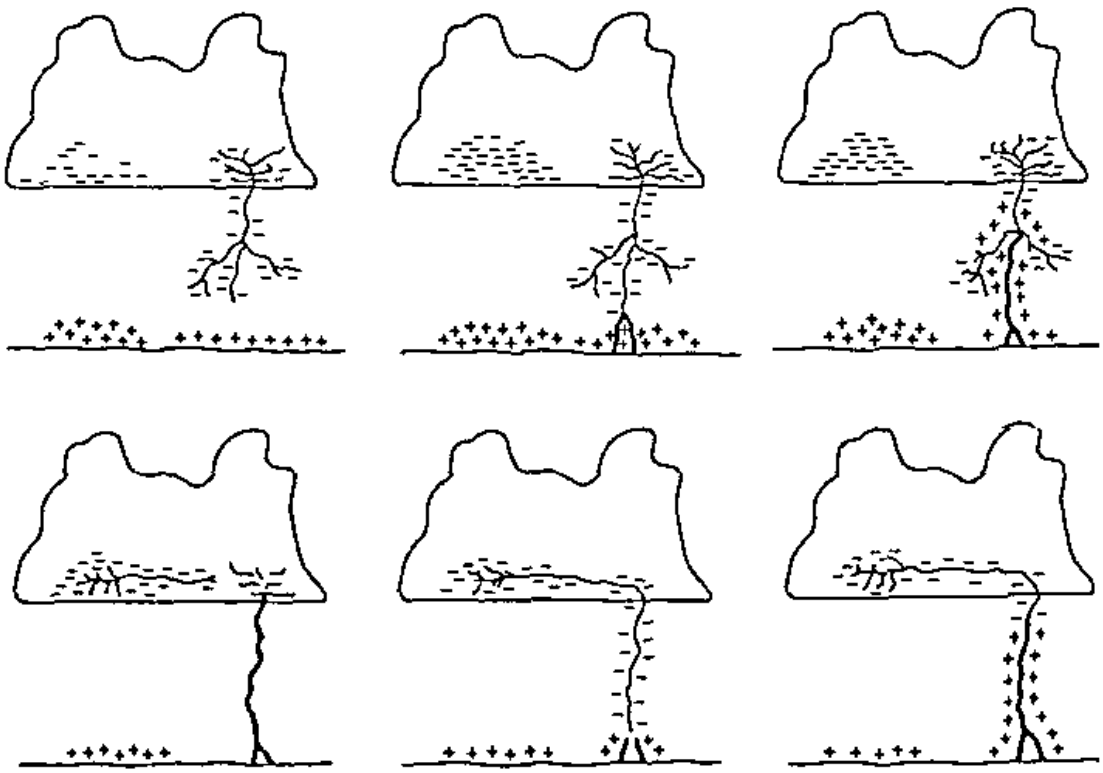


Fig. 11.9 Schematic representation of the build-up of a lightning discharge

follows redistribution of the charge from the cloud along the ionized channel, with a resulting initiation of another pilot streamer and a repetition of the whole process. When the ionized streamer approaches the ground, a positive streamer starts from the ground, and the main stroke is a return stroke passing up through the ionized channel at a speed of about 10^{10} centimeters per second. Many subsequent strokes may travel through the initially produced ionized channel. These consist of fast negative streamers starting from the cloud to be met by the main stroke in the form of a positive streamer from the ground. This behavior of the build-up of a lightning discharge is illustrated schematically in Fig. 11.9.

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Chapter 12

Sheaths

Up until now we have been discussing an infinite plasma In any real laboratory case it is necessary to enclose the discharge in some kind of a tube, and the presence of the walls modifies the plasma Let us examine the plasma regarding its various simple boundary conditions

Consider the problem of a semi-infinite volume of positive and negative charges at temperatures of T_+ and T_- respectively in contact with an infinite plane as shown in Fig 12 1 Assume, contrary to reality, that the charges have perfect reflection from the plane and that their averaged

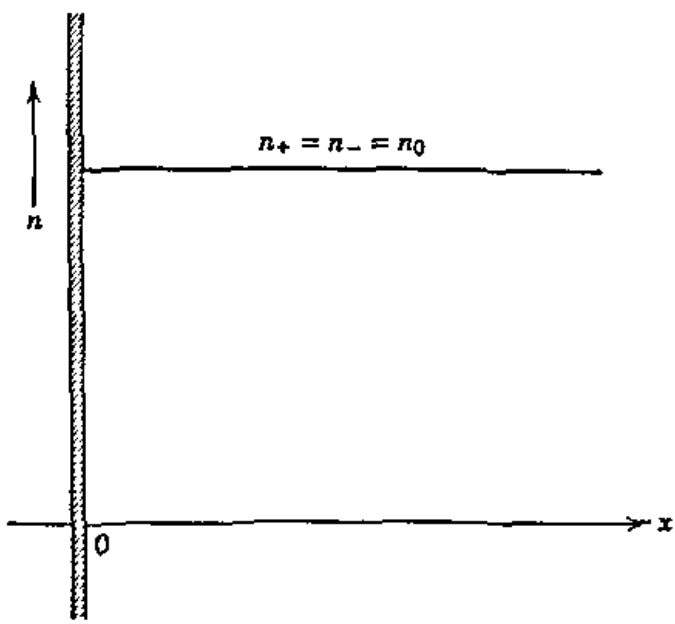


Fig 12 1 Distribution of plasma particles with no surface charge and the electric field zero everywhere

densities are equal, $n_+ = n_- = n_0$. With no surface charges on the plate the electric field will be zero everywhere, and the electrostatic potential can be assumed equal to zero also. Let us now place a positive surface charge density σ_0 on the plate as shown in Fig. 12.2. Since this surface charge tends to establish a field equal to σ/ϵ_0 for positive x, a flow of negative charges is induced toward the plate and positive charges away

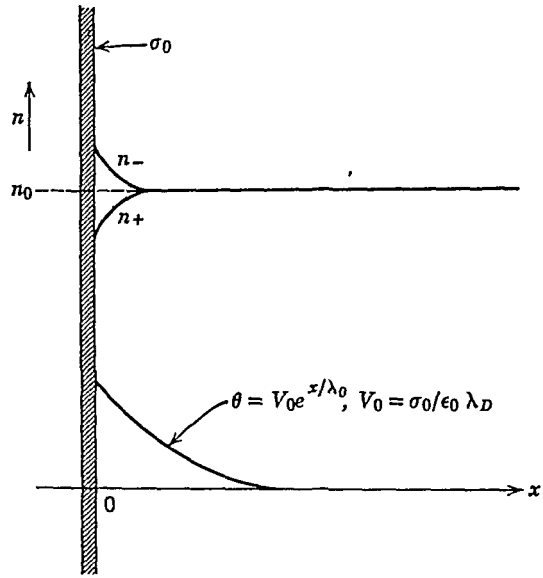


Fig. 12.2 Distribution of plasma particles near a charged plate.

from the plate. This separation of charge produces a divergence of the field and a variation of potential, ϕ, away from the plate. Rather than solve the details of the flow problem, the results of the new equilibrium condition for particle densities can be obtained from what is called the Boltzmann factor, $e^{eu/kT}$, as follows:

$$\begin{aligned} n_- &= n_0 e^{eu/kT} = n_0 e^{\phi/T_-} \\ n_+ &= n_0 e^{-\phi/T_+} \end{aligned} \tag{12.1}$$

where ϕ is assumed to remain zero for large x since the field will not extend to infinity. Substituting into Poisson's equation, we have

$$-\nabla^2\phi = \frac{e}{\epsilon_0}(n_+ - n_-) = \frac{n_0 e}{\epsilon_0}(e^{-\phi T_+} - e^{\phi/T_-}). \tag{12.2}$$

We approximate the exponential factors by $e^{\phi/T} = 1 + \frac{\phi}{T}$ Thus

$$-\nabla^2\phi \simeq \frac{n_0 e}{\epsilon_0}\left(1 - \frac{\phi}{T_+} - 1 - \frac{\phi}{T_-}\right) = \frac{-n_0 e}{\epsilon_0 T}\phi \tag{12 3}$$

where the reduced temperature is

$$T = \frac{T_+ T_-}{T_+ + T} \tag{12 4}$$

Rearranging, we get

$$\nabla^2\phi - \phi/\lambda_D{}^2 = 0 \tag{12 5}$$

where λ_D is the Debye length defined by Eq (4 7) Solving Eq (12 5) with the boundary conditions that at $x = \infty$, $\phi = 0$, and at $x = 0$, $E_0 = \sigma_0/\epsilon_0$, we get

$$\phi = \frac{\sigma_0}{\epsilon_0}\lambda_D e^{-x/\lambda_D} \tag{12 6}$$

The potential and field decay exponentially with a decay length λ_D, and the densities are obtained by substituting Eq (12 6) into Eq (12 1) The maximum value of the potential occurs at $x = 0$ and is given by

$$V_0 = \frac{\sigma_0}{\epsilon_0}\lambda_D = E_0\lambda_D \tag{12 7}$$

For typical values of $T = 1$ volt and $n_0 = 10^{12}$ cm^{-3}, $V_0 = 7\,4 \times 10^{-6}E_0$ and we see that very large fields at the surface are generated with very small voltages Thus our assumption that $\phi/T < 1$ is valid even for large fields

Wall Sheaths

When the ion and electron density becomes large, their densities are almost equal and the plasma is quasi-neutral If in addition we apply the boundary condition of zero density at the walls of a diffusing plasma, the electric field, potential, and space charge density all become infinite at the wall This singularity arises because the boundary condition violates the high density assumption A closer look at the details of the boundary reveals the formation of a "sheath" which acts as a transition region of the plasma in a high energy state to the wall which is in a relatively low-energy state Within this region there is a considerable amount of charge separation and a corresponding density variation, potential drop, and electric field

From the solution of the diffusing plasma in a plane parallel geometry,

the density and charge density are given as follows:

$$n = n_0 \cos \frac{x}{\Lambda} \tag{12.8}$$

$$\rho = \frac{\epsilon_0 T_-}{\Lambda^2} \sec^2 \frac{x}{\Lambda}. \tag{12.9}$$

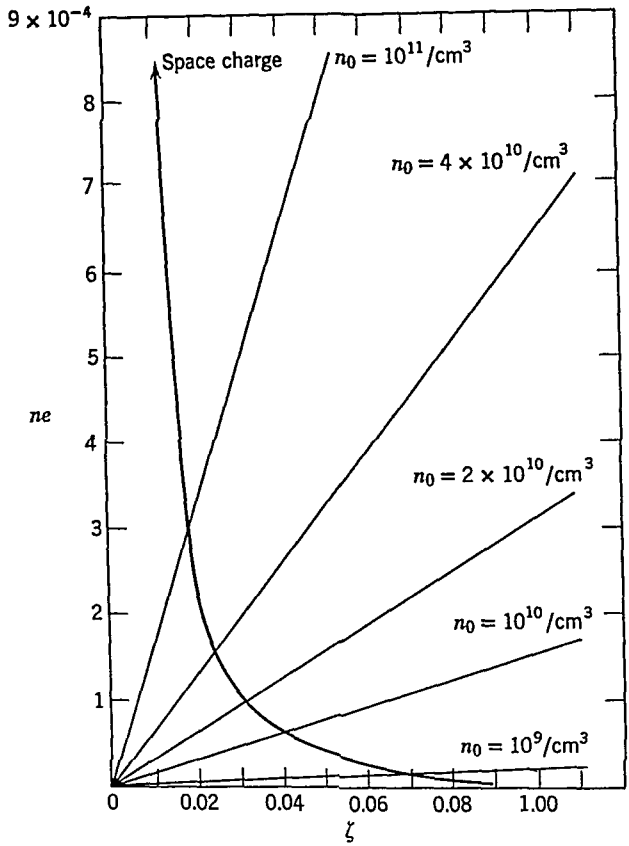


Fig. 12.3 Density profiles as a function of distance from a wall.

Figure 12.3 shows a sketch of Eqs. (12.1) and (12.2) near the boundary at $x = L/2$ and since we are interested in small displacements from the wall we have defined the parameter $\xi = (L/2 - x)/\Lambda$. The density profiles for five different central densities are shown along with a space charge curve for $T_- = 1$ volt and $\Lambda = 1$ cm calculated from Eq. (12.9). For any given central density there is a position near the wall where (ne) is equal to the required space charge. For higher densities this position is closer

to the wall At this critical distance the ambipolar solution must break down because the space charge density can certainly not become larger than the local density of ions or electrons Since the space charge is positive, one concludes that the electrons no longer exist between this critical position and the wall This picture is consistent with electrons having a large random thermal velocity escaping from their ion cloud in the vicinity of the wall

From this simple derivation we have been able to define a rather precise boundary for the occurrence of a sheath The distance l_s will be defined as the sheath thickness For high densities, this distance is fairly well-defined but for low densities, the sheath merges into the plasma and the definition is somewhat meaningless The magnitude of l_s can be obtained from equating (ne) to ρ and using the small argument approximation to the trigonometric functions

$$l_s = \left[\frac{\epsilon_0 T_-}{n_0 e}\right]^{1/2} = [\lambda_D^2]^{1/2} \tag{12 10}$$

The potential and the electric field will no longer be infinite on the wall but this derivation is not precise enough to determine their exact value

Mobility Limited Sheath

If the ion and electron mean free path is small compared to the thickness of the sheath, it is relatively easy to determine an approximate solution for the sheath thickness First it is important to note that the electrons are flowing against an electric field such that the resultant current is a negligible fraction of its random current and the electron density is approximately related to the electrostatic potential through the Boltzmann factor, even in the sheath Next, the ions are being accelerated toward the wall, and the ion current equation can be written as

$$\Gamma_+ = -D_+ \nabla n + n\mu E_s$$

where

$$E_s = -\left(\frac{D_- + D_+}{\mu_- + \mu_+}\right)\frac{\nabla n}{n} = -T_- \frac{\nabla n}{n},$$

$$\Gamma_+ = -D_+ \nabla n - \mu_+ T_- \nabla n \tag{12 11}$$

The last term is the effect of the electric field and dominates the density gradient term by the ratio of T_-/T_+ This ratio is large and we will neglect the density gradient force on the ions

Figure 12 4 shows an expanded view of the ion and electron densities near one wall of a parallel plane geometry which contains a plasma in the steady state The densities are finite at the wall and their values are

determined by the considerations of the current. The quasi-neutrality assumption is violated in the sheath and separate equations must be written for the ion and electron densities. One of the distinguishing features of this wall sheath is that the current of both electrons and ions is determined by the net ionization rate from the center of the plasma to the wall and the sheath must adjust itself to accommodate this current.

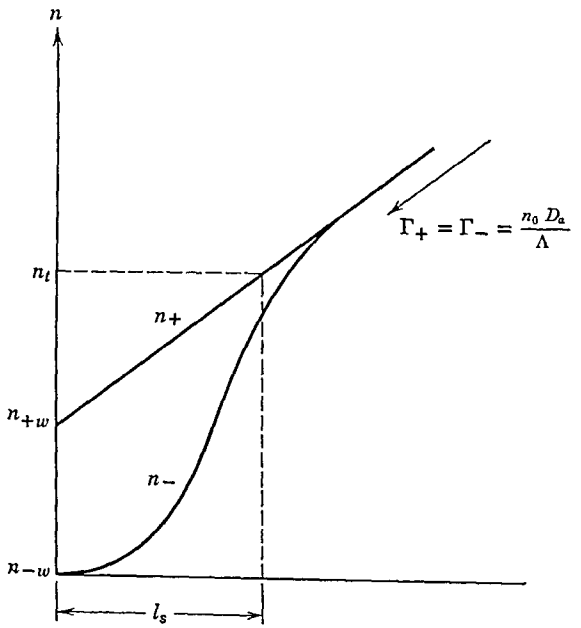


Fig. 12.4 Expanded view of the ion and electron densities near the wall.

The current distribution is given by

$$\Gamma_+ = \Gamma_- = \Gamma = -D_a \nabla n = \frac{n_0 D_a}{\Lambda} \sin \frac{x}{\Lambda} \tag{12.12}$$

and for positions near the wall, $\sin(x/\Lambda) \simeq 1$ and the current is therefore equal to $n_0 D_a/\Lambda$.

The equations which relate to the sheath are thus given by

$$\Gamma_+ = n_+ \mu_+ E = n_0 D_a/\Lambda = \Gamma_w \tag{12.13}$$

and from Poisson's equation:

$$n_+ - (n_-) = \frac{\epsilon_0}{e} \bar{\nabla} \cdot \bar{E}$$

assuming the Boltzmann factor

$$n_- = n_t e^{(\phi - \phi_t)/T_-}.$$

The subscript t stands for the value of the variables at an approximate transition plane between the plasma and the sheath and for this solution we will fix the origin of the coordinate system at this plane The values of the variables at any position in the plasma can be written in terms of the local density and at the transition plane where $n = n_t$ and $n_t \ll n_0$ the other variables can be approximated by

$$E_t = \frac{T_-}{\Lambda}\frac{n_0}{n_t}, \qquad \phi_t = -T_- \ln \frac{n_0}{n_t} \tag{12 14}$$

The sheath thickness, l_s and hence the values n_t, E_t, and ϕ_t are unknown and for their solution we must specify the boundary values at the wall

First the ion and electron densities at the wall will be determined by setting the random current at the wall equal to $n_0 D_a/\Lambda$

$$n_{+w} = \frac{4\Gamma_w}{\langle v_+ \rangle}, \qquad n_{-w} = \frac{4\Gamma_w}{\langle v_- \rangle} \tag{12 15}$$

Next the electric field at the wall is determined from Eqs (12 15) and (12 13)

$$E_w = \frac{n_0 D_a/\Lambda}{n_{+w}\mu_+} = \frac{\langle v_+ \rangle}{4\mu_+} \tag{12 16}$$

The total potential drop across the sheath can be determined from the net decrease in electron density which must occur

$$\phi_w - \phi_t = -T_- \ln \frac{n_t}{n_{-w}} \tag{12 17}$$

With these equations we can proceed with the solution Let us neglect n_- in Poisson's equation within the sheath, and substituting for n_+ from Eq (12 13) in Poisson's equation, we get

$$\frac{dE}{dx} = \frac{e\Gamma_+}{\epsilon_0 \mu_+ E}$$

Integrate from the transition plane at $x = 0$,

$$E^2 - E_t^2 = ax \quad \text{where} \quad a = \frac{2e\Gamma}{\epsilon_0 \mu_+} \tag{12 18}$$

Integrating the electric field to get the potential, we have

$$\phi - \phi_t = -\frac{2}{3a}\left[(ax + E_t^2)^{3/2} - E_t^3\right] \tag{12 19}$$

If we let $x = l_s$ in Eqs. (12.18) and (12.19) and substitute for E_w and ϕ_w from (12.16) and (12.17), we get the following two relations:

$$\left(\frac{\langle v_+ \rangle}{4\mu_+}\right)^2 = E_t^2 + al_s \tag{12.20}$$

$$\frac{2}{3a}(al_s + E_t^2)^{3/2} - E_t^3 = T_- \ln \frac{n_t \langle v_- \rangle}{4\Gamma_w}. \tag{12.21}$$

E_t and n_t are related through (12.14) and hence we have three equations and three unknowns, n_t, E_t, and l_s. Equation (12.21) is transcendental and must be solved graphically.

Free Fall Sheath

The phenomena that occur when a conducting or insulating probe is placed into the region of a plasma is of great practical importance since they lead to a determination of the plasma properties. A probe must not perturb the plasma outside of a sheath thickness from its surface. This, of course, does not eliminate the possibility of collecting current on the probe, but the total current must be small compared to the random current in the plasma.

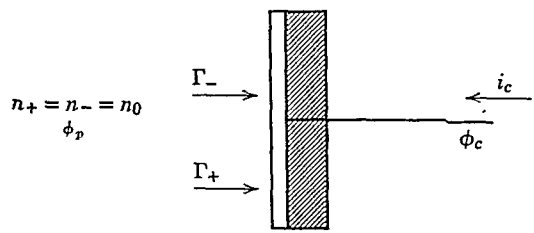


Fig. 12.5 Sketch of a collector probe.

Consider the plasma and collector pictured in Fig. 12.5. For simplification the collector is a plane conductor insulated on the back side so the current will be collected from only one side. Figure 12.6 shows a sketch of collector current i_c as a function of the collector voltage ϕ_c which we can vary with some external power supply. For a collector potential in the vicinity of plasma potential ϕ_p, two saturation levels are observed. When $\phi_c \geq \phi_p$, the current is positive and is produced by the electrons flowing to the collector since the ions will be repelled. For a negative or "retarding potential" ($\phi_c < \phi_p$) the current decreases and saturates at a small negative value that results from the ions flowing to the collector, since the electrons are now repelled. At the "floating potential," ϕ_F, the net current is zero. This is the potential that the collector will assume if it is left with an open circuit.

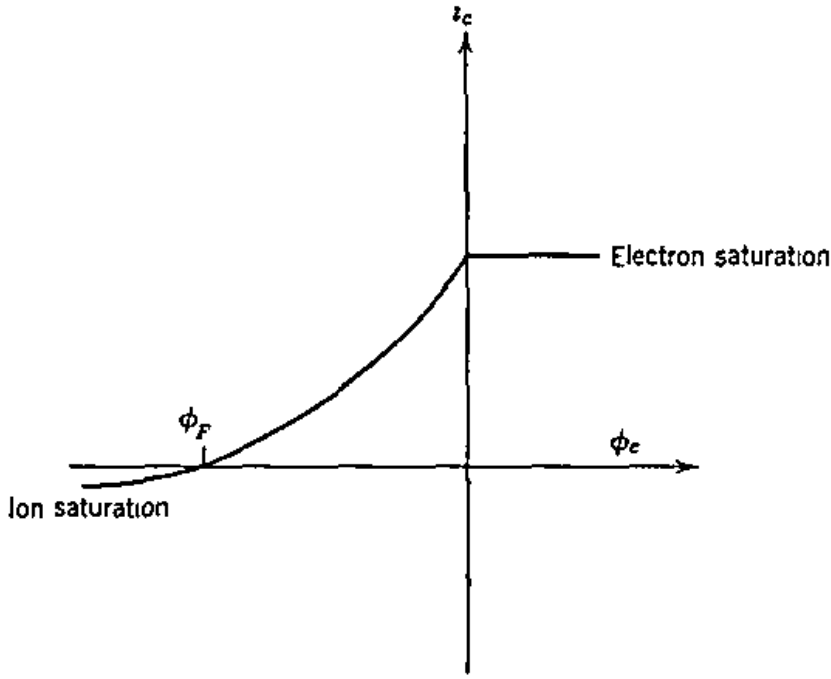


Fig. 12.6 Collector characteristics

In general, a sheath will develop in front of the collector and its size and potential will depend on the collector voltage Let us discuss the region of the characteristic for which the collector is negative with respect to the plasma potential This is sometimes called the "retarding" region since the flow of electrons is retarded to the collector

Figure 12 7 shows the approximate density profiles in front of the collector for these conditions A sheath of thickness l_s develops and the

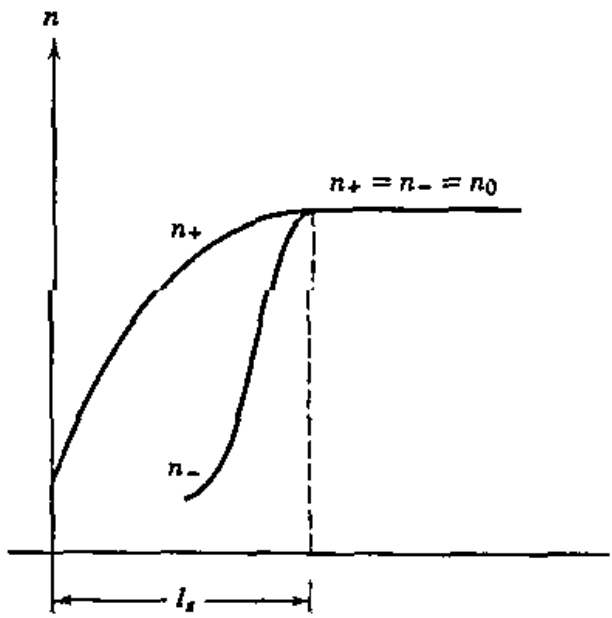


Fig. 12.7 Free fall sheath

sheath potential $\phi_s = \phi_c - \phi_p$ accelerates the ions and repels the electrons. If we assume that the ion mean free path is long compared to l_s, then the ions fall freely to the collector. The electron density, which is assumed to have a Maxwellian velocity distribution, is reduced in the sheath by the Boltzmann factor

$$n_- = n_0 e^{(\phi - \phi_p)/T_-}.$$

A simplified solution for the sheath is obtained by neglecting the electron space charge in the sheath and using the Child-Langmuir law for the positive ion current[1] with $E = 0$ and $v_+ \simeq 0$ at $x = 0$

$$j_+ = \frac{4\epsilon_0}{9}\left[\frac{2e}{M}\right]^{1/2} \frac{\phi_s^{3/2}}{l_s^2}. \tag{12.22}$$

Since ions do not suffer collisions in the sheath, any ion that falls over the edge of the sheath will be collected and hence the current is the independent variable given by the rate at which they arrive at the sheath

$$j_+ = \frac{en_0\bar{v}_+}{4} = en_0\left[\frac{eT_+}{2\pi M}\right]^{1/2}. \tag{12.23}$$

Equation (12.22) therefore is an equation for determining l_s as a function of ϕ_s and j_+. Substituting (12.23) into (12.22), we have

$$l_s^2 = \frac{8\pi^{1/2}}{9}\left(\frac{T_-}{T_+}\right)^{1/2}\left(\frac{\phi_s}{T_-}\right)^{3/2}\lambda_D^2. \tag{12.24}$$

Thus for a sheath potential equal to the electron temperature, the sheath thickness is about three times the Debye length.

The electron current to the collector probe as a function of the sheath potential is obtained from the Boltzmann relation, assuming $\phi_s \leq 0$,

$$j_- = -en\frac{\bar{v}_-}{4} = -en_0\frac{\bar{v}_-}{4}e^{\phi_s/T_-}.$$

The saturation value of j_- at $\phi_s = 0$ is

$$j_{-\text{sat}} = -en_0\frac{\bar{v}_-}{4} = -en_0\left[\frac{eT_-}{2\pi m}\right]^{1/2}. \tag{12.25}$$

The net current to the probe therefore is given by

$$j = j_+ + j_- = en_0\left[\left(\frac{eT_+}{2\pi M}\right)^{1/2} - \left(\frac{eT_-}{2\pi M}\right)^{1/2}e^{\phi_s/T}\right]. \tag{12.26}$$

This is essentially the curve shown plotted in Fig. 12.6 with $i_c = -jA_c$. For positive values of ϕ_s, the current is given by $j_{-\text{sat}}$, Eq. (12.25), since

the ion temperature is so small and only a small positive voltage will repel the ions

The picture of the probe sheath presented above is correct in principle but the details must be considered more carefully as was done by Langmuir[2] in his classic treatment of "Electron and Positive Ion Space Charges"

Probe curves can be analyzed to give the following information

1 By assuming that the ion current is negligibly small, the logarithm of the current versus potential is a straight line whose slope is equal to the electron temperature providing the distribution is Maxwellian
2 The electron saturation current is proportional to $n_0 T^{1/2}$ and hence the density in the plasma can be determined
3 The plasma potential is determined by the potential at the knee or break in the current curve
4 The ion mass can be determined either from the ratio of saturation currents or from the floating potential

REFERENCES

1 See, for example L B Loeb, *Basic Processes of Gaseous Electronics*, University of California Press Berkeley (1955) p 623
2 I Langmuir, *Phys Rev*, **33**, 954 (1929) See section beginning on p 976

Chapter 13

Glow Discharges

In a low-pressure discharge, a large number of phenomena may occur, and there are two common ways of classifying these phenomena. The first is by observing the changes in the voltage-current characteristics. These characteristics, as well as a schematic representation of the tube, are shown in Fig. 13.1. Each electron that leaves the cathode moves toward the anode producing ionization on the way. The ions thus produced drift

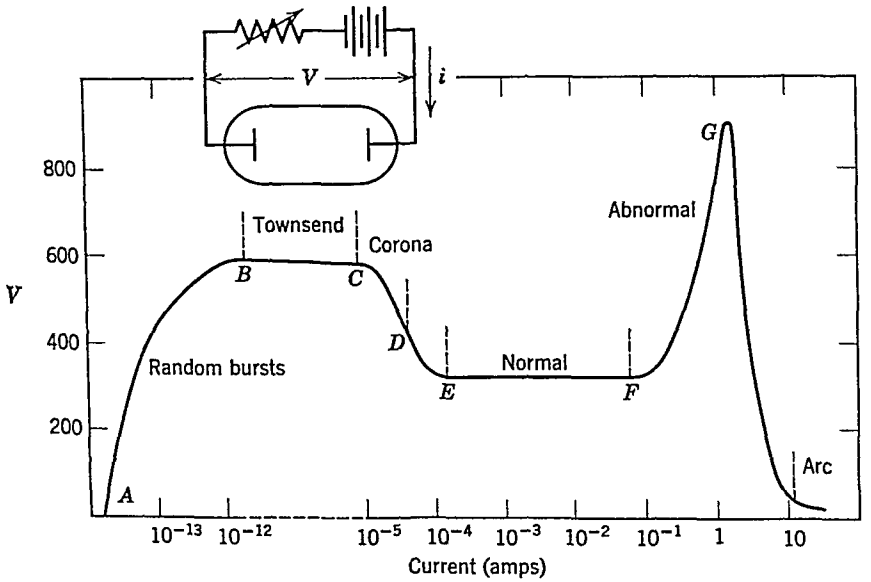


Fig. 13.1 Voltage-current characteristics of a dc discharge.

toward the cathode, becoming sources of more electrons by secondary emission when the ions hit the cathode At very low voltages and currents, the multiplication is insufficient to lead to a selfsustaining discharge and the current is measured in random bursts Where an external supply of electrons is present from a photoelectric source, the current is controlled by the intensity of the illumination Breakdown occurs at the beginning of what is called the Townsend discharge region, *BC* As the current increases, the discharge begins to glow visibly and the potential across the tube drops to a constant value This is region *CDE* If the discharge takes place in a very inhomogeneous field, such as between a point and

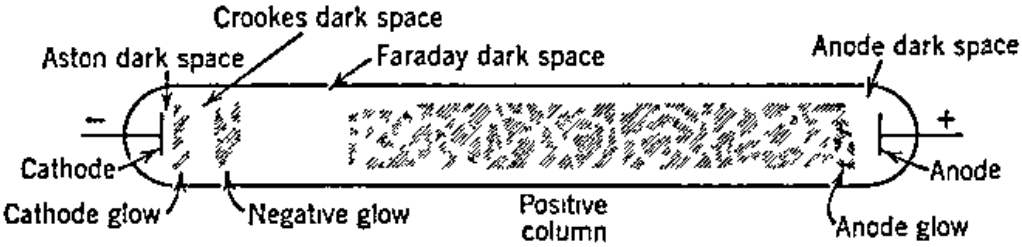


Fig 13 2 Classical nomenclature for various parts of a discharge

a plane or between coaxial cylinders, the discharge is called a "corona" discharge The region in which the potential difference is constant, *EF*, is called the "normal glow " Here the discharge covers only a part of the cathode, the area covered being proportional to the current The current density remains constant When the whole cathode is covered with glow, increasing the current further causes the voltage to rise, this is the region *FG* called the abnormal glow The high-current operation of the discharge is termed the arc

The early work in gas discharges centered around the study of the visual behavior of the glow discharge, and led to a classification of the various visible regions illustrated in Fig 13 2

The Aston dark space is a region where the electrons, produced by positive ion bombardment of the cathode, accumulate to be accelerated through the Crookes dark space

The cathode glow can take on three different characteristics the "first cathode layer," the "cathode glow," and the "cathode light " The first cathode layer is a low-pressure, low-voltage phenomenon associated with the loss of electron energy to gas excitation Sometimes two or three such layers can be seen At higher pressures and voltages, the cathode surface is covered with a velvety coating called the "cathode glow " This results from the loss of excitation energy of the positive ions on neutralization When the cathode is covered with a thin metallic or oxide layer, atoms of

the layer may be sputtered off, and the excitation of these sputtered atoms gives rise to what is called the "cathode light."

The Crookes (or cathode) dark space is a space in which the electrons gain energy sufficient to cause the negative glow on inelastic impact with the gas atoms. This space is also called the "cathode" dark space and the potential drop across it is called the cathode fall.

In the Faraday dark space, the electrons once more gain their energy lost in the negative glow, and when they again rise to ionization energy, they cause the positive column. The positive column itself is a passive region. Its primary function is to connect the cathode and anode end of the discharge. In general its length is dictated by the length of the tube and the gas pressure. If the tube length is too short or the pressure too low, the positive column can be completely absent without affecting the operation of the discharge.

Langmuir[1] was at one time quite interested in the function and behavior of the positive column. Being impressed by its passive character and the fact that it molded itself to the shape of the tube containing it, in 1929 he proposed to name the discharge in the positive column a "plasma" using for its root the Greek word "to mold." This word is now in general use.

The anode glow is produced by electrons that may have gained enough energy in their last few mean free paths to produce a slight excess of ionization because of the space charge field disturbance by the positive ions that are forced from the anode.

The anode dark space has a thickness of the same order of magnitude as the electron mean free path.

The general appearance of the glow discharge shown in Fig. 13.2 is characteristic of pressures around 1 torr. If the pressure is increased, all the negative zones—cathode dark space, negative glow, and Faraday dark space—compress toward the cathode. The positive column expands to fill the space. At around 100 mm Hg the negative glow is apparently on the surface of the cathode and the only negative space left is the Faraday dark space. Also at about 100 torr the positive column contracts radially and no longer fills the tube.

As the pressure is decreased, the negative zones expand and the boundaries between the zones become diffuse; the positive column is driven into the anode and finally disappears. With further reduction in pressure, the Faraday dark space and the negative glow also disappear into the anode. As the Faraday dark space moves in, the anode glow disappears and the discharge goes out unless the voltage across the tube is increased. When this occurs, it is called an "obstructed" discharge.

With further decrease in pressure, the negative glow disappears when the cathode-to-anode distance is less than the normal length of the dark

space This kind of a discharge occurs around a pressure of 10^{-3} torr and with voltages around 2 to 5 kilovolts across the tube is called a "canal ray" discharge

As the distance between the electrodes is increased, more voltage is required across the discharge to maintain it The positive column increases in length but there is no effect on the negative zones

Table 13.1 Characteristic Colors for a Large Number of Gases

Gas	First Cathode Layer	Negative Glow	Positive Column
Air	pink	blue	
Hydrogen	brownish red	pale blue	pink
Nitrogen	pink	blue	red
Oxygen	red	yellowish white	pale yellow with pink center
Helium	red	green	red to violet
Argon	pink	dark blue	dark red
Neon	yellow	orange	brick red
Krypton		green	
Xenon		olive-green	
Bromine		yellowish-green	reddish
Chlorine		yellowish green	whitish-green
Iodine		buff	reddish-blue
Lithium	red	light red	
Sodium	pink to orange	whitish	yellow
Potassium	green	pale blue	green
Rubidium	pink	blue	pinkish-red
Cesium	pink	milky green	yellowish brown
Mercury	green	green	greenish
Calcium	violet blue	violet-red	
Magnesium		green	green
Aluminium	violet-blue	violet-blue	
Thallium		green	whitish blue
Cadmium		red	bluish green
Arsenic		bluish	greenish
Silver		pink	bluish green
Lead		yellowish red	violet
Zinc	violet-red	blue to violet	red
$SnCl_4$		green	sky blue
CCl_4		pale green	whitish green
HCl		green	pink
NO		bluish-white	
NO_2		bluish-white	
NH_3		yellowish-green	

As the distance between the electrodes is decreased, it takes less voltage to operate the glow. As the anode moves through the Faraday dark space, the anode glow disappears accompanied by a slight fall of voltage. As the anode moves near to the cathode edge of the negative glow, the discharge again becomes "obstructed" and the voltage rises rapidly.

As the potential difference across the tube is increased, the length of the cathode dark space decreases and the length of the negative glow increases slightly. As the current increases, the discharge gets brighter. At higher voltages, the cathode glow becomes brighter and may take on a different color from the gas glow. Also the zones become more sharply defined.

As one changes from gas to gas the colors of the negative zones and of the positive column are characteristic of the gas. These characteristic colors for a large number of gases are given in Table 13.1.

Crookes Dark Space and the Cathode Fall of Potential

The cathode or Crookes dark space corresponds to the distance in the tube which electrons must travel before they produce excitation and ionization. The potential drop across this space is called the cathode fall. There are two common types of glow discharges, which are called "normal" and "abnormal" glow. In the normal glow the cathode is not completely covered by the discharge, and, as the current in the tube is increased, the current density in the cathode falls and the voltage across it remains constant. Thus the area of the discharge increases with increasing current. In the abnormal glow the discharge completely covers the cathode, so that neither the current nor the current density remains constant with variations of applied voltage.

A glow discharge is maintained by the electrons produced by the positive-ion bombardment of the cathode. If n_0 electrons come from the cathode for n electrons that reach the plasma at the end of the cathode drop, $(n - n_0)$ is the number of ions produced and $n_0 = \gamma(n - n_0)$. Thus,

$$\frac{n}{n_0} = \frac{1 + \gamma}{\gamma} = \exp \int \alpha \, dx.$$

The cathode fall satisfies the relation

$$\exp \int_0^d \alpha \, dx = 1 + \frac{1}{\gamma},$$

where d is the width of the Crookes dark space. It is therefore to be expected that the cathode fall is a function of both the gas and the cathode material.

Since the thickness of the Crookes dark space depends upon electron collision phenomena, the thickness d times the pressure p should be a constant if no complicating factors enter The relation dp = constant is found to be roughly true, the constant depending upon the gas but not on the cathode material Typical values[2] of dp are shown in Table 13 2

A number of workers[3] have measured the magnitude of the electron field in the Crookes dark space as a function of the distance from the

Table 13 2 Normal Cathode Fall Thicknesses
dp in cm-torr at Room Temperature

Cathode	Air	A	H_2	He	Hg	N_2	Ne	O_2
Al	0 25	0 29	0 72	1 32	0 33	0 31	0 64	0 24
C			0 9		0 69		I	
Cu	0 23		0 8		0 6			
Fe	0 52	0 33	0 9	1 30	0 34	0 42	0 72	0 31
Mg			0 61	1 45		0 35		0 25
Hg			0 9					
Ni			0 9		0 4			
Pb			0 84					
Pt			1 0					
Zn			0 8					

cathode They find that the field and the distance are linearly dependent, and can be written

$$\frac{dE}{dx} = \frac{E_0}{d} \tag{13 1}$$

where E_0 is the field at the cathode and d is the thickness of the cathode dark space

This rather surprising experimental result does not agree with the calculation of the field dependence on distance as determined from Poisson's equation

$$\frac{dE}{dx} = \frac{1}{\epsilon_0}\left(\frac{j_+}{v_+} - \frac{j}{v_-}\right) \approx \frac{1}{\epsilon_0}\left(\frac{j_+}{v_+}\right)$$

Since $j = j_+(1 + \gamma)$, this equation may be written

$$\frac{dE}{dx} = \frac{j}{\epsilon_0 v_+(1 + \gamma)}, \tag{13 2}$$

and obviously Eqs (13 1) and (13 2) are not equal

The glow discharge is maintained by Townsend ionization

$$\int_0^d \alpha \, dx = \ln\left(1 + \frac{1}{\gamma}\right) \equiv \lambda \tag{13.3}$$

where, as we have seen before,

$$\frac{\alpha}{p} = Ae^{-Bp/E} \equiv Ae^{-y}. \tag{13.4}$$

A solution for the characteristics of the cathode fall has been arrived at under the various assumptions[4] that either Eqs. (13.1) or (13.2) are valid, that the gas either be a molecular gas, or a rare gas, and that either $v_+ = \mu_+(E/p)$ or $v_+ = k_+(E/p)^{1/2}$ as described in Fig. 3.11.

Table 13.3 Constants for Approximations

Case	Gas	Mobility	Field Dist.	k	m	n	K	M	y_n	$S_n(y_n)$
1	Rare	μ_+	Lin.	4	2	3	$B^2/4A$	$2\mu_+AB^4$	1.560	2.161×10^{-2}
2	Rare	μ_+	Pois.	6	0	5	$B^2/6A$	$2\mu_+AB^4$	1.687	3.813×10^{-3}
3	Rare	k_+	Lin.	4	1	3	$B^2/4A$	$2k_+AB^3$	1.560	2.161×10^{-2}
4	Rare	k_+	Pois.	5	0	4	$B^2/5A$	$2k_+AB^3$	1.633	8.969×10^{-3}
5	Mol.	μ_+	Lin.	2	1	2	$B/2A$	μ_+AB^2	0.610	4.453×10^{-1}
6	Mol.	μ_+	Pois.	3	0	3	$B/3A$	μ_+AB^2	0.708	3.278×10^{-1}
7	Mol.	k_+	Lin.	2	$\frac{1}{2}$	2	$B/2A$	$k_+AB^{3/2}$	0.610	4.453×10^{-1}
8	Mol.	k_+	Pois.	$\frac{5}{2}$	0	$\frac{5}{2}$	$2B/5A$	$k_+AB^{3/2}$	0.67	3.79×10^{-1}

The general solution for the various assumptions is that

$$j = \frac{M(1 + \gamma)p^2\epsilon_0 S_n(y_0)}{\lambda {y_0}^m} \tag{13.5}$$

and

$$d = \frac{\lambda}{[Ap(n - 1)y_0^{(n-1)}S_n(y_0)]} \tag{13.6}$$

where the values of M, m, and n are given in Table 13.3, and $S_n(y_0)$ is the exponential integral

$$S_n(y_0) = \int_y^\infty \frac{e^{-t}}{t^n} \, dt. \tag{13.7}$$

For the specific case of hydrogen, Eq. (13.5), in terms of the proper variable (j/p^2), becomes

$$\frac{j}{p^2} = \frac{\mu_+AB^2(1 + \gamma)\epsilon_0 S_n(y_0)}{\ln(1 + 1/\gamma)y_0} \tag{13.8}$$

and Eq. (13.6) in terms of the proper variable pd becomes:

$$pd = \left(\frac{E_0}{p}\right)\frac{\ln(1 + 1/\gamma)}{ABS_n(y_0)}. \tag{13.9}$$

The cathode fall of potential, V_c, is related to the field at the cathode E_0 by the relation

$$V_c = \int E\,dx = \int_0^d \frac{E_0 x}{d}\,dx,$$

thus

$$E_0 = \frac{2V_c}{d} \tag{13 10}$$

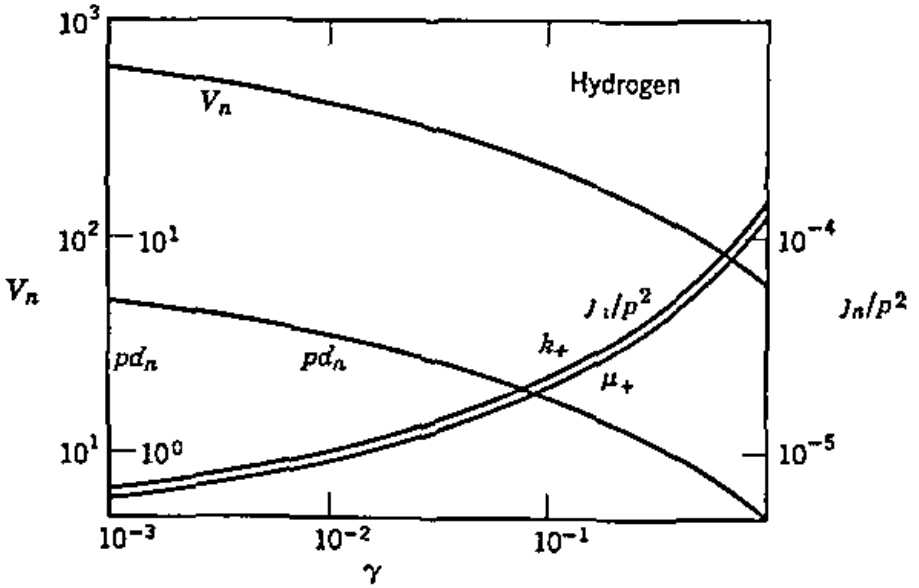


Fig 13.3 Variation of normal cathode fall, V_n, normal cathode fall thickness d_n, and normal current density j_n as a function of the second ionization coefficient γ

so that we may write an equation for the cathode fall as

$$V_c = \left(\frac{E_0}{p}\right)^2 \frac{\ln(1 + 1/\gamma)}{2ABS_n(y_0)} \tag{13 11}$$

The normal cathode fall V_n is defined as the minimum value of V_c or the condition that $dV_c/dy_0 = 0$ The condition for minimum V_c is that $y_0 = y_n$ where

$$S_n(y_n) = e^{-y_n/A\,y_n^{n-1}}$$

and A is given in Table 13 3

The variations of pd_n, j_n/p^2, and V_n all as a function of γ are illustrated in Fig 13 3

The Negative Glow

The negative glow is produced by the excitation of the gas atoms by the electrons which have gained energy in the Crookes dark space There is a good correlation between the length of the negative glow and the

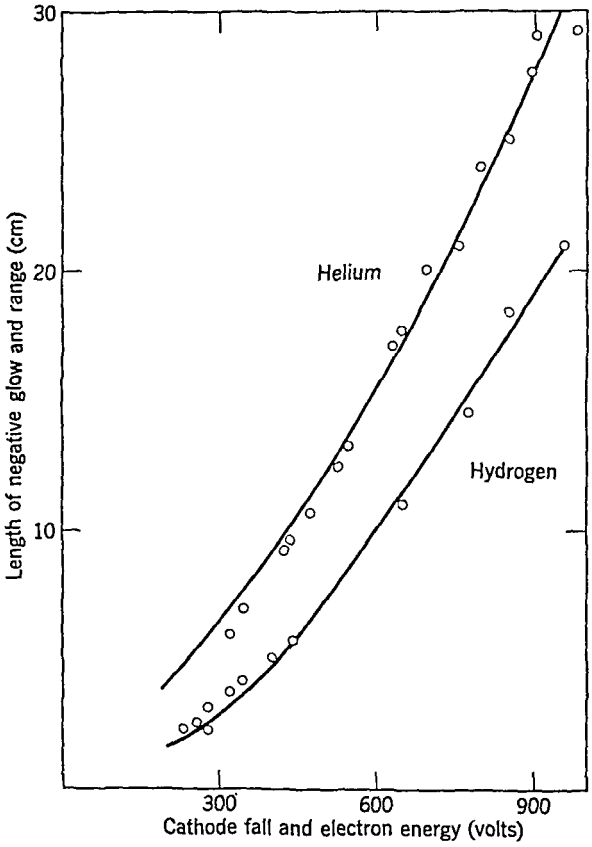


Fig. 13.4 The points represent measurements of the length of negative glow as a function of cathode fall and the line are the range of electrons as a function of electron energy.

range of electrons that have been accelerated by a potential difference equal to the cathode fall. The excellence of this correlation is shown in Fig. 13.4 from the work of Brewer and Westhaver.[5]

The Positive Column

The positive column is a very common source for plasma research and has been extensively studied. Let us first calculate the radial potential distribution. The particle flow equations as given before [Eq. (4.1)] are

$$\Gamma_r = -D_+ \frac{dn}{dr} + \mu_+ n E_r$$

$$= -D_- \frac{dn}{dr} - \mu_- n E_r = nv.$$

Solving these equations for E_r under the assumption that $D_+ \ll D_-$ and $\mu_+ \ll \mu_-$, we find

$$E_r = -\frac{D_-}{\mu_-}\frac{1}{n}\frac{dn}{dr} \tag{13 12}$$

If we use the relation that $D/\mu = kT/e$, we get

$$E_r = -\frac{kT_e}{e}\frac{1}{n}\frac{dn}{dr}, \tag{13 13}$$

and from the relation that $E_r = -dV/dr$, we find that

$$\frac{n_r}{n_0} = e^{-e(V_0-V_r)/kT_e},$$

where n_0 and V_0 refer to the values on the axis of the positive column and n_r and V_r to the values at the radius r

The dominant loss mechanism is by ambipolar diffusion to the walls, and we can calculate the wall current per unit length of plasma by writing

$$D_a = 2D_+ = 2\mu_+\left(\frac{kT_e}{e}\right)$$

and

$$n = n_0 J_0(2\,4r/R)$$

The radial current to the wall is

$$\imath = 2\pi R e\Gamma = -2\pi R e D_a \left.\frac{dn}{dr}\right)_{r=R}$$

$$\left.\frac{dn}{dr}\right)_{r=R} = n_0\left(\frac{2\,4}{R}\right)J_0'(2\,4),$$

and is given by

$$\imath = 7\,85 n_0 \mu_+ kT_e \tag{13 14}$$

It is also of considerable interest to calculate the electron temperature in the positive column. Assuming that the ionization function varies linearly with the energy above the ionization potential, the number of ion pairs produced per second by electrons of velocity v is

$$F(u) = a'(u - u_i)\frac{v}{l} = ap(u - u_i)v, \tag{13 15}$$

where a' and a are appropriate constants

For a Maxwellian distribution of velocities, we find

$$\frac{dn}{n} = \frac{4}{(\pi)^{1/2}}\frac{v^2}{c_0^2}e^{-v^2/c_0^2}d\left(\frac{v}{c_0}\right)$$

$$= \frac{2}{(\pi)^{1/2}}\frac{(eu)^{1/2}}{(kT_e)^{3/2}}e^{-eu/kT_e}d(eu) \tag{13 16}$$

where c_0 is the most probable velocity calculated from $\frac{1}{2}mc_0^2 = kT_e$ whereas the effective or rms velocity v comes from $\frac{1}{2}mv^2 = \frac{3}{2}kT_e$.

Using Eqs. (13.15) and (13.16), the number of ion pairs formed per second is

$$\nu_i = \int_{u_i}^{\infty} F(u)\frac{dn}{n} = ap\frac{m}{e}\frac{4}{(\pi)^{1/2}}\left(\frac{2kT_e}{m}\right)^{3/2} e^{-eu_i/kT_e}\left[1 + \frac{1}{2}\frac{eu_i}{kT_e}\right].$$

Simplifying this equation, it can be written in the form

$$\left(\frac{eu_i}{kT_e}\right)^{-1/2} e^{+eu_i/kT_e} = 1.16 \times 10^7 c^2 p^2 R^2 \tag{13.17}$$

where

$$c^2 = \frac{a\sqrt{u_i}}{\mu_+ p}.$$

The values of c are given in Table 13.4 and a universal curve for all gases from Eq. (13.17) is given in Fig. 13.5.

Table 13.4 Values of the Constant c for Calculating the Electron Temperature of the Positive Column

Gas	c
He	$3.9(10)^{-3}$
Ne	$5.9(10)^{-3}$
A	$5.3(10)^{-2}$
Hg	$1.1(10)^{-1}$
N_2	$3.5(10)^{-2}$
H_2	$1.05(10)^{-2}$
H_1 in H_2	$5.7(10)^{-3}$
O_2	$2.9(10)^{-2}$

To check experimentally whether this theory agrees with observation, the temperature of the plasma must be measured. This is usually carried out by means of Langmuir probes. Measuring temperatures in this fashion leads to the experimental checks[6] of Eq. (13.17) as shown in Fig. 13.6.

We can also calculate the longitudinal field in the positive column in the following way. If x is the fractional energy loss for an electron traveling in the field, we find

$$dP = \frac{xmv^2}{2} dN_c$$

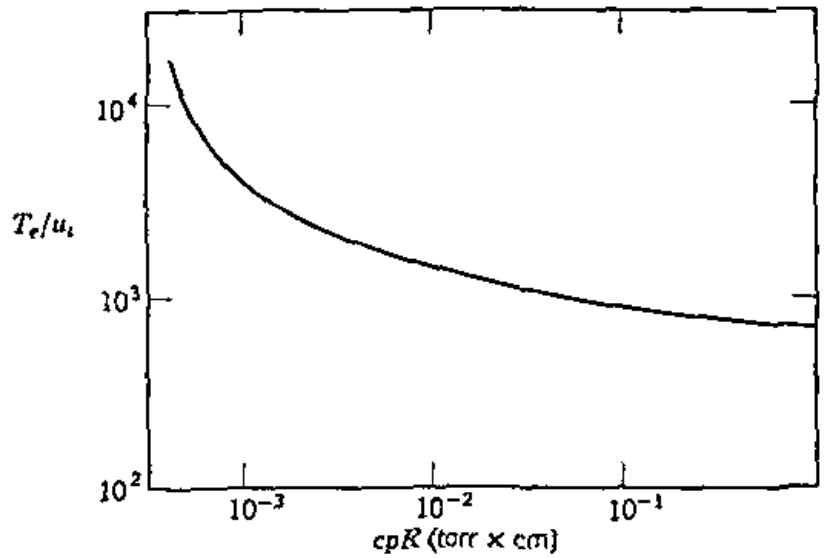


Fig 13 5 Ratio of electron temperatures to ionization potential as a function of the parameters in Eq 13 17

is the energy loss by the electron per second where $dN_e = 4\pi v^2 \nu_c f\, dv$ is the number of collisions per second for a Maxwellian distribution of velocities

$$dP = \frac{2m\varkappa v^5}{(\pi)^{1/2} l c_0^3} e^{-v^2/c_0^2}\, dv$$

and the rate of energy transfer is

$$P = \frac{2m\varkappa c_0^3}{l(\pi)^{1/2}} \tag{13 18}$$

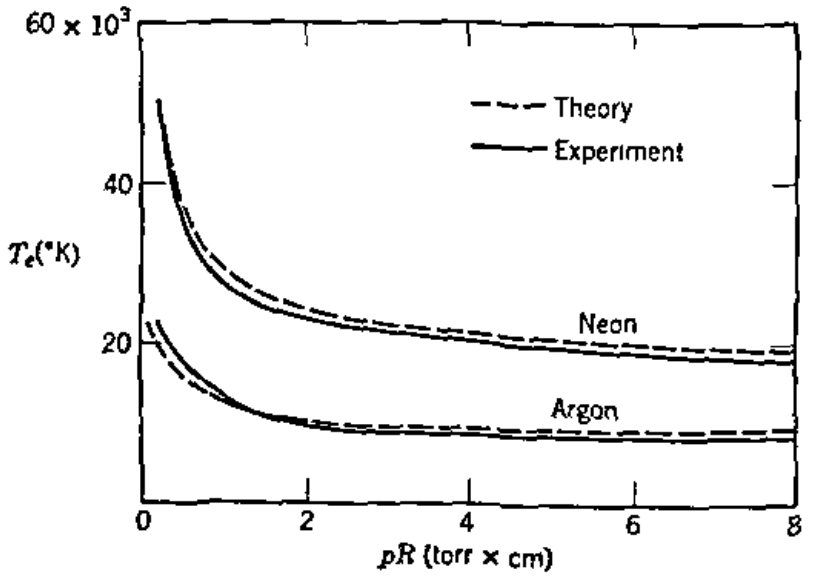


Fig 13 6 Electron temperature in the positive column as a function of pressure and tube radius in neon and argon

Since the energy gained per second is $P = evE$, where $v = elE/mc_0$:

$$P = \frac{2mxc_0^3}{l(\pi)^{1/2}} = \frac{e^2lE^2}{mc_0}$$

$$E^2 = \frac{2m^2xc_0^4}{(\pi)^{1/2}e^2l^2}$$

$$E = 1.83 \times 10^{-4} \frac{(x)^{1/2}}{l} T_e \tag{13.19}$$

$$\frac{E}{p} = \frac{1.83 \times 10^{-4}(x)^{1/2}}{pl} T_e. \tag{13.20}$$

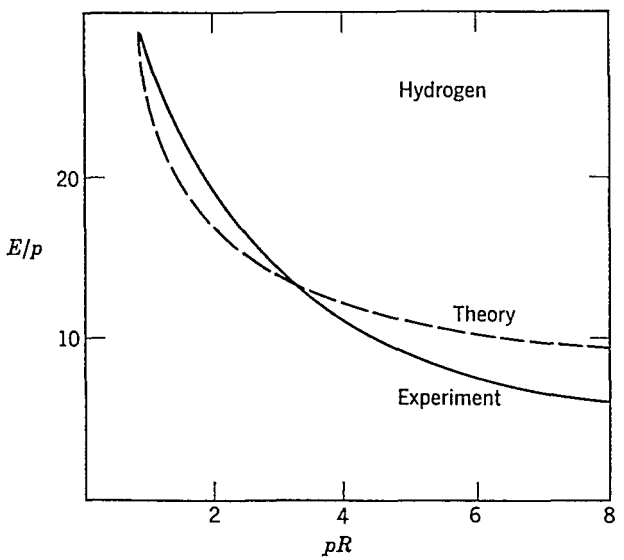


Fig. 13.7 Experimental verification of Eq. (13.20).

Since T_e is a function of (pR), E/p is also a function of (pR), and the agreement between Eq. (13.20) and similar experimental determination[7] is shown in Fig. 13.7.

Positive Column in a Magnetic Field

The problem of diffusion in a magnetic field has been discussed in detail. This mechanism controls the behaviour of the positive column in a magnetic field, and we will discuss briefly two of the most often referred to experiments.

Bickerton and von Engel[8] made probe measurements in a positive

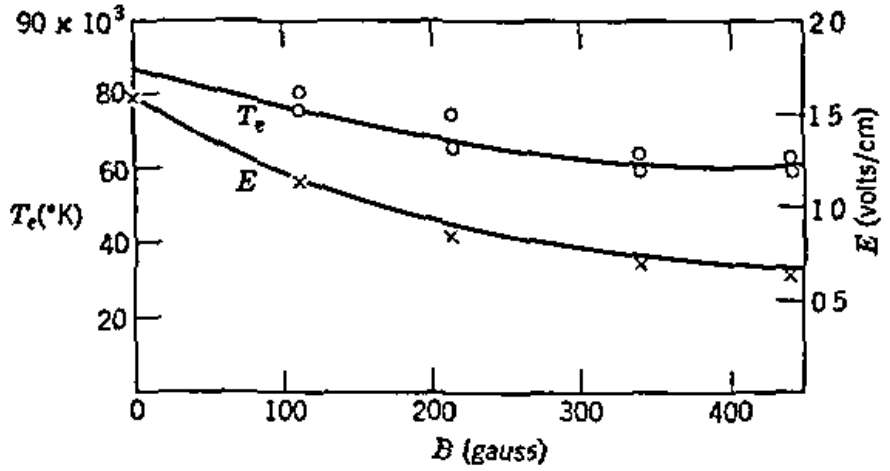


Fig 13 8 Decrease in radial electric field and electron temperature as a function of magnetic field

column immersed in a longitudinal magnetic field and have reported results on the electron temperatures and radial electric fields

As we have seen in Chapter 4, when a longitudinal field is applied to a plasma column the radial electron diffusion is reduced Since the rotary component of the electron motion reduces the flow of particles to the walls, the radial component of the electric field is decreased, and this was

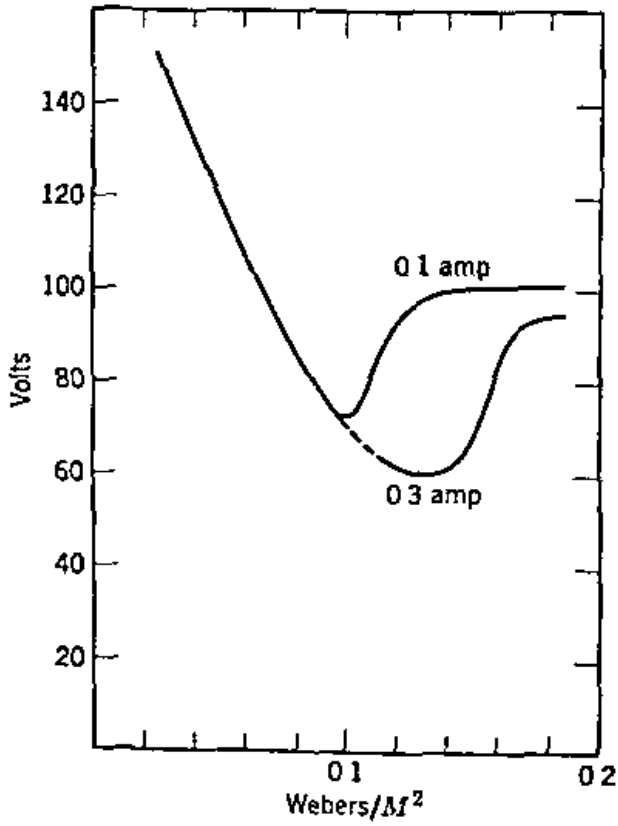


Fig 13 9 Anomalous behavior of the positive column at high magnetic fields

measured by Bickerton and von Engel as shown in Fig. 13.8. This will also result in the electron temperature going down as the magnetic field goes up, a result which they also measured by a probe technique, the results also being given in the same figure.

Positive column experiments carried to much higher magnetic fields introduce an entirely different phenomenon.[9] The axial electric field quite closely follows what one would expect up to some pressure and current-dependent critical field, but above this critical magnetic field, the electric field rises sharply. This behavior is illustrated in Fig. 13.9. The diffusion mechanism is incapable of explaining this fact, and the increased rate of particle loss is ascribed to some type of plasma instability which results in mass motion of the whole plasma column.

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Chapter 14

Steady-State Microwave Discharges

Thus far most of the microwave discussion has dealt with the phenomena associated with breakdown The next step is to introduce the problems of the operating ac discharge An operating discharge is characterized by the presence of positive ions These do not diffuse as rapidly as the electrons, and hence their concentration builds up in excess of the electron concentration A space charge field results which enhances the diffusion of the positive ions while reducing the diffusion of the electrons, and to describe this we have already introduced the space charge controlled diffusion coefficient D_s

In our development of the Boltzmann equation derivation for ac breakdown, we wrote in Eq (10 31)

$$\left(\nu_x + \nu_i - q + \frac{2eu}{3m\nu_m\Lambda^2}\right) f = \frac{2}{3\sqrt{u}} \frac{d}{du}\left[(\nu_m u^{3/2})\left(u_c \frac{df}{du} + \frac{3m}{M} f\right)\right]$$

The only difference between the breakdown and steady-state cases lies in the dc space charge field E_s which must be superimposed on the high-frequency field[1] so that

$$E = E_s + E_p e^{j\omega t}, \tag{14 1}$$

where E_s varies with position and E_p is assumed to be constant

To put Eq (10 31) in the form applicable to the steady state we assume that

$$\mathbf{E}_s = -u_s \frac{\nabla n}{n} \tag{14 2}$$

and Eq. (10.31) becomes

$$(\nu_x + \nu_i - q)f + \frac{2eu}{3m\nu_m\Lambda^2}\left(f + u_s\frac{df}{du}\right) = \frac{2}{3\sqrt{u}}\frac{d}{du}\left[(\nu_m u^{3/2})\left(u_c\frac{df}{du} + \frac{3m}{M}f\right)\right]. \quad (14.3)$$

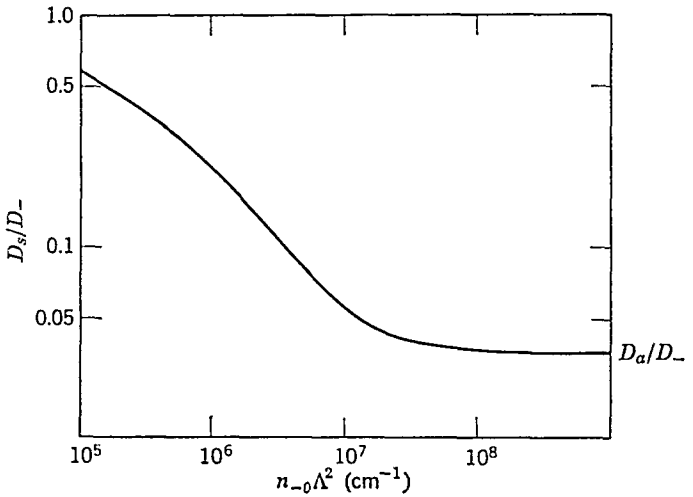


Fig. 14.1 Space charge controlled diffusion.

We define an effective electric field E_e as

$$E_e^2 = \frac{\nu_m E_p^2}{2(\nu_m^2 + \omega^2)}. \quad (14.4)$$

When we now put in values for ν_x, ν_i, and ν_m for a particular case, we may solve Eq. (14.3) in terms of the proper variables pE_e, $E_e\Lambda$, and u_s. If we multiply Eq. (14.3) by $4\pi v^2\,dv$ and integrate over all velocities, the right-hand side vanishes, and we find that

$$D_- - u_s\mu = \langle\nu_i\rangle\Lambda^2 \equiv D_s. \quad (14.5)$$

This defines the space charge controlled diffusion coefficient. From the solution of Eqs. (14.3) and (14.5), we obtain D_-, μ_-, and D_s/D_- calculated as functions of p/E_e and $E_e\Lambda$. Allis and Rose[2] give D_s/D_- as a function of $n_0\Lambda^2\mu_-/D_-$ which was illustrated in Fig. 4.3. We can replot this figure in Fig. 14.1 to show the ratio of D_s/D_- as a function of the proper variable $n_{-0}\Lambda^2$, where n_{-0} is the electron density at the center of the discharge under consideration. Steady-state operation voltage can be computed from the corresponding breakdown curves[1] by using the results of Fig. 14.1 and by knowing the electron density of the operating discharge.

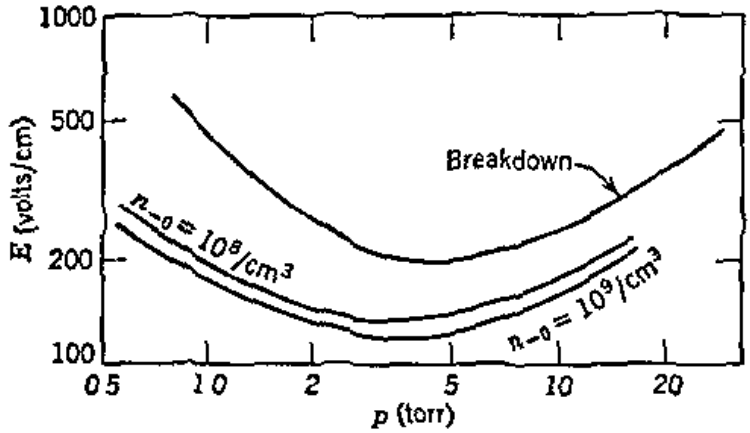


Fig 14 2 Breakdown and steady state microwave electric fields in hydrogen

Experimental curves for hydrogen are given in Fig 14 2 Since D_a is less than D_-, we see that the steady-state field is always lower than the breakdown field Physically, this results from the fact that since the *diffusion loss is less, it takes a lower field to maintain the balance between* loss and gain To be able to compare with the theoretical calculations it is necessary to obtain the data of Fig 14 2 for the steady state discharge at a constant electron *density To achieve this the following experimental* arrangement was employed [1] A block diagram of the microwave system is shown in Fig 14 3 The hydrogen gas was admitted at the desired pressure and the discharge was initiated by adjusting the incident power level with the power divider The light output from the discharge as monitored by a photocell was noted The magnetron frequency was then varied, and the incident power was adjusted so that the light output was the same as before At each frequency the standing wave ratio and position of the

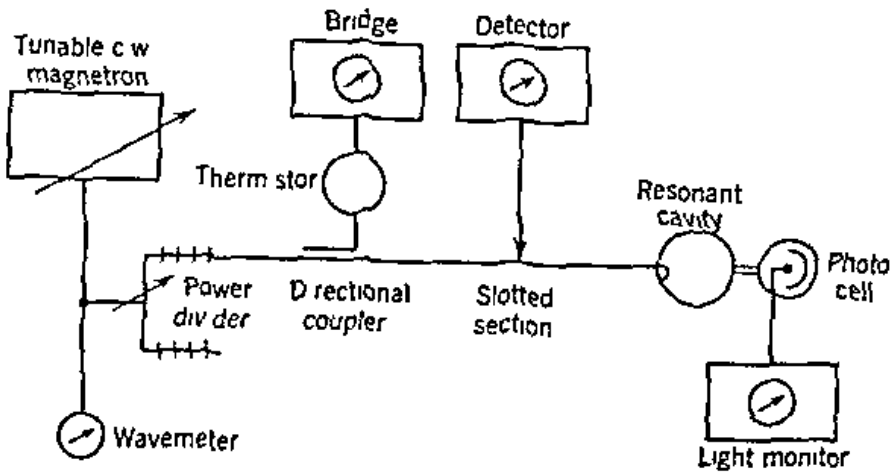


Fig 14.3 Block diagram of microwave system for measuring steady state discharge characteristics

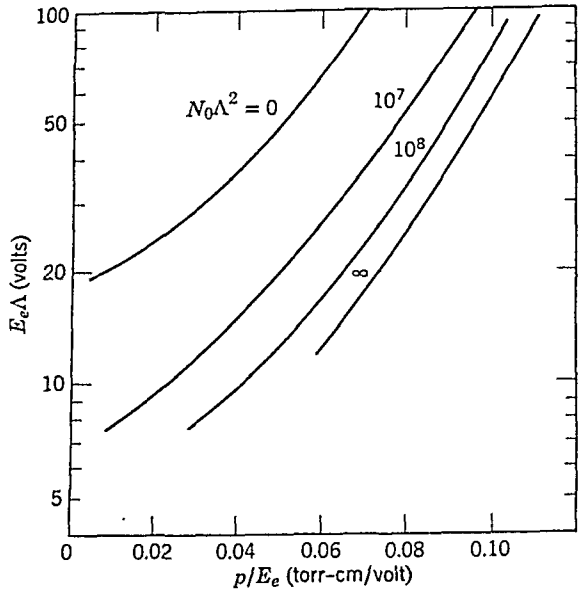


Fig. 14.4 Theoretical calculations for steady-state microwave discharge.

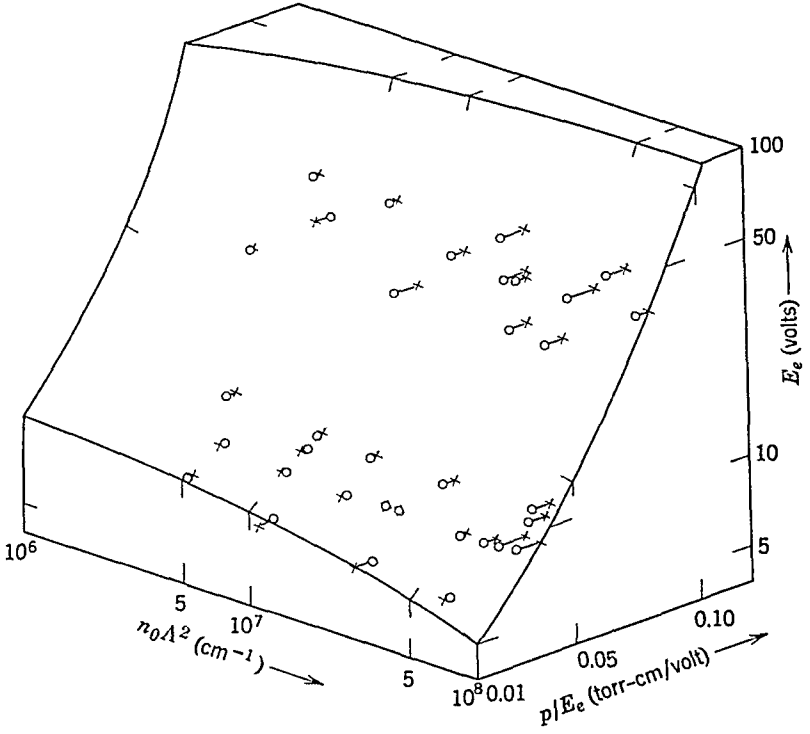


Fig. 14.5 Comparison of experiment and theory for steady-state microwave discharge.

voltage minimum on the slotted section and the incident power as given by the thermistor bridge were measured Such measurements along with the known characteristics of the resonant cavity are sufficient to determine the maintaining electric field and the electron density

The theoretical results for hydrogen are shown in Fig 14 4, where contours of constant $n_0\Lambda^2$ are plotted on the $p/E_e - E_e\Lambda$ plane The contour $n_0\Lambda^2 = 0$ represents breakdown, and the contour $n_0\Lambda^2 = \infty$ represents ambipolar diffusion

Figure 14 5 shows the theoretical results of Fig 14 4 plotted as a surface in $E_e\Lambda$, p/E_e $n_0\Lambda^2$ space The surface in the region of large $n_0\Lambda^2$ and small $E_e\Lambda$ has been obtained by extrapolation of the theoretical calculations of Fig 14 4 The experimental points are shown as dots, their projection in the p/E_e direction onto the theoretical plane, as crosses The average of the experimental points is in agreement with the theory within 6%

Limits

This discussion of the steady state microwave discharge has very definite experimental limits As we have seen before free space is capacitative in nature, and the effect of electrons in a microwave field is inductive Therefore under the proper conditions plasma resonance will occur as

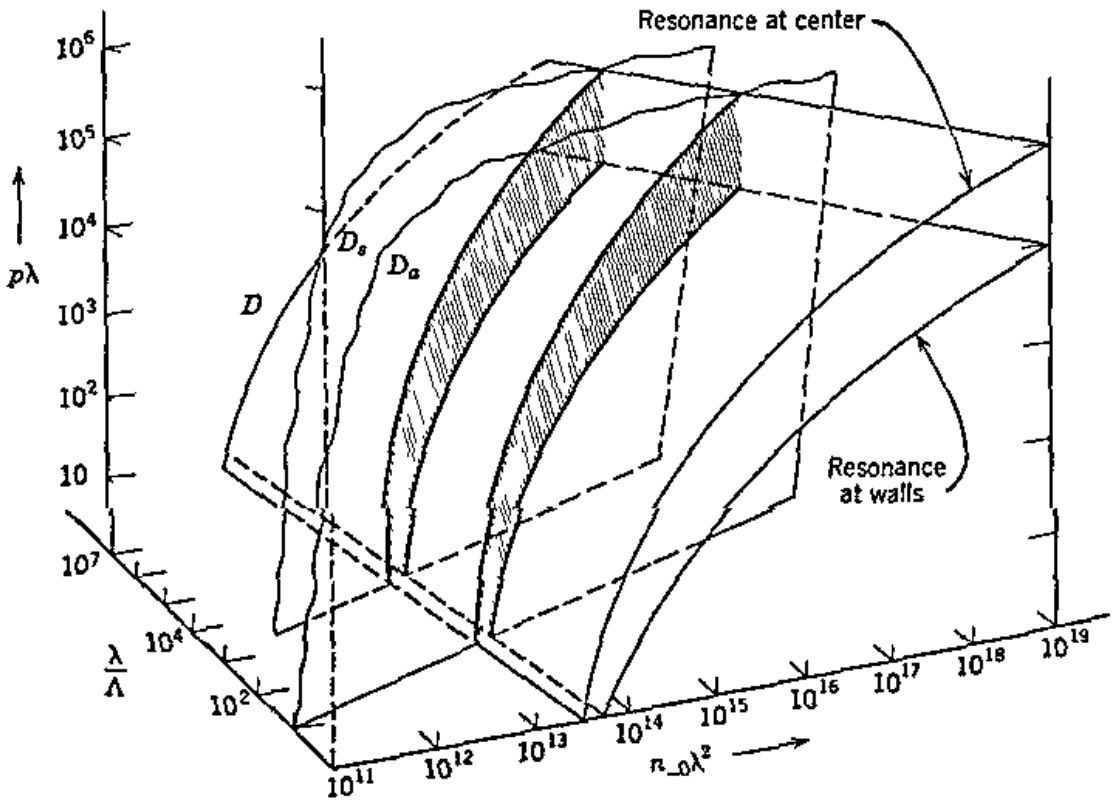


Fig 14 6 Plasma resonance conditions in steady state microwave discharge

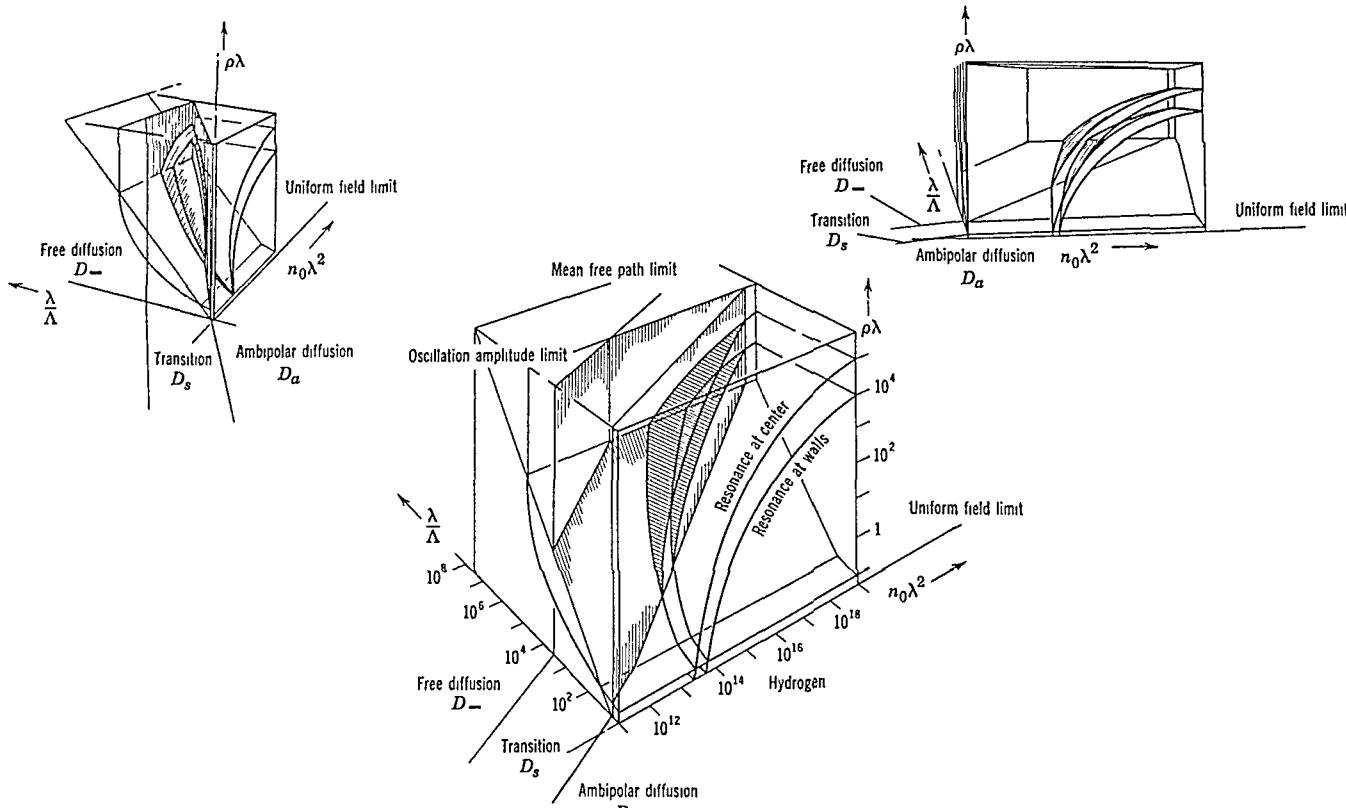


Fig. 14.7 Limits of steady-state microwave discharge theory.

discussed in Chapter 9 In terms of the magnitude of the impedance in a microwave discharge, under the assumption that ν_c is constant, we have

$$\left|\frac{E}{J}\right| = \frac{(\nu_c^2 + \omega^2)^{1/2}}{\omega\epsilon_0[(r-1)^2\omega^2 + \nu_c^2]^{1/2}} \tag{14 6}$$

where

$$r = (ne^2)/(m\omega^2\epsilon_0)$$

It is obvious that a maximum will occur for $r = 1$ at the plasma frequency

$$\omega^2 = ne^2/m\epsilon_0 \tag{14 7}$$

Plasma resonance provides a maximum density above which our methods for calculating microwave electric fields[2] do not apply

In laboratory tubes containing a discharge, the density is not uniformly distributed from the center to the walls Thus it is quite possible for the density at the center of a tube to be controlled by ambipolar diffusion, the density near the walls to be characterized by free diffusion, while in between, D_a is the proper diffusion coefficient In terms of proper variables for microwave discharges these conditions can be calculated, and represented in a three-dimensional coordinate system as $p\lambda$, λ/Λ, and $n_{-0}\Lambda^2$ This is illustrated in Fig 14 6 The nonuniformity of the electron density also leads to the conclusion that as the density is increased, we will come to the plasma resonance condition, first at the center of the tube, and then further out This can also be illustrated in the $p\lambda$, λ/Λ, $n_{-0}\Lambda^2$ volume and is thus shown in Fig 14 6

The steady-state microwave discharges discussed in this chapter are all controlled by diffusion loss and hence the limits of diffusion theory discussed in Chapter 10 are applicable here as well Visualization of all these limits at once is a little complicated but an illustration which attempts to do this is shown in Fig 14 7

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Chapter 15

Arc Discharges

The arc discharge includes a large group of phenomena appearing in the conduction of electricity through gases. The arc discharge corresponds to the region from G onward in the curve given in Fig. 13.1. As shown in this figure, the arc discharge is characterized by a low cathode fall of the order of the ionization potential of the gas, as compared to the cathode falls of about 200 volts in the glow discharges. The current-voltage characteristic is falling, and the current density of the arc can be very high. From these features it is obvious that the electron emission from the cathode must be governed by some mechanism other than positive-ion bombardment of the cathode, and it is not uncommon to classify arcs by the emission process that occurs at the cathode. Four types of arcs may be thus classified: (1) the thermionic arc in which the cathode is heated by the discharge and the arc is self-maintaining, (2) the thermionic arc in which the cathode is heated by an external source and the arc is not self-maintaining, (3) field-emission arcs in which the electron current at the cathode results from a very high electric field at the cathode surface, and (4) liquid metal arcs.

There are various ways in which an arc can be initiated, the most obvious of which is to follow the course plotted in Fig. 12.1 by increasing the discharge current while supplying the appropriate voltage difference between the electrodes. The types of transitions shown in Fig. 12.1 are characteristic of gas pressures in the range of a few millimeters.

If the pressure is in the vicinity of one atmosphere, breakdown into an arc occurs by way of a transient spark. Essentially all the maturation steps of Fig. 12.1 are compressed into a very short time span and the current-voltage characteristics are represented schematically in Fig. 15.1.

Another transient method of initiating an arc is to put the electrodes first in contact and then to draw them apart so that the arc will burn between Because of this mechanical technique of striking the plasma, it is often referred to as a "drawn" arc The characteristic electrical behavior of the drawn arc is that its initiation requires no overvoltage, the only

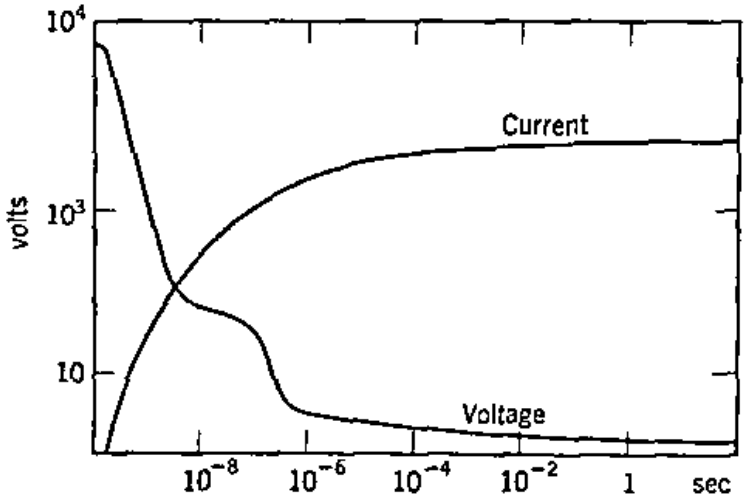


Fig 15 1 The time variation of current and voltage on the initiation of an arc

voltage required being the running voltage It is obvious that it is this type of arc that is so destructive in breaking the contacts in switches of all sorts

Low Pressure Arcs

An example of the thermionic arc that is self-maintained is the tungsten-nitrogen arc studied by Wehrli [1] The cathode has a constant temperature over its surface, and the area of the cathode is known Using the Richardson thermionic emission equation, Wehrli has shown that the electron emission is equal to or greater than the current of the arc and that the cathode is surrounded by a double charge sheath that may be regarded as a dark space since it is so much smaller than the mean free path of an electron As a result, very few collisions occur in this region Immediately in front of the cathode the net space charge is negative while at the outer edge of the dark space the net space charge is positive The potential difference across this region is called the cathode fall and is the most important part of the discharge The energy given the positive ion heats the cathode while the energy given the electrons leads to the ionization necessary to maintain the arc The anode phenomena of the low-pressure arc are essentially the same as those of the glow discharge If the arc pressure is very low (0 02 mm in argon), the cathode is surrounded by a dark layer 0 35 mm thick, and this in turn is surrounded by a luminous layer about 2 mm thick followed

by the usual positive column. A field exists in the dark space, and the primary electrons from the cathode are accelerated here as a uniform beam. In the luminous or scattering layer this electron beam is transformed into an isotropic Maxwellian distribution in energy by collisions with the gas atoms and electron interaction. The electrons in the final distribution are referred to as ultimate electrons. Frequently an intermediate Maxwellian distribution of electrons with a higher energy occurs in the scattering layer, and these electrons are referred to as secondary electrons.

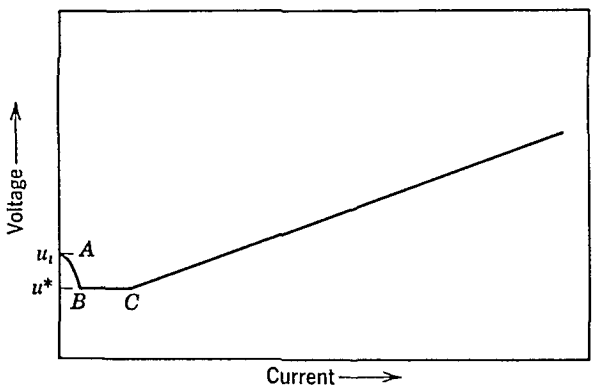


Fig. 15.2 Current-voltage characteristics externally heated thermionic arc.

In the externally heated thermionic arc the transition region from a glow discharge to an arc discharge does not occur, and the current-voltage characteristic is simplified. A typical characteristic is given in Fig. 15.2 for an argon discharge at 0.2 mm pressure, using an oxide-coated cathode.[2] The cathode is maintained at a constant temperature. Since the cathode is heated externally, the cathode fall will be lower than the cathode falls found in self-maintained thermionic arcs, and the arc drop will be essentially that of the cathode fall of potential. Throughout the range of currents from zero up to 0.5 ampere, $(A - B)$, the voltage necessary to maintain the arc decreases. Initially the arc voltage is equal to the ionization potential of the gas, and ionization by single electron collisions occurs. As the current through the tube increases, the time interval between successive collisions of one atom with individual electrons decreases. Since argon possesses metastable excitation levels, the atoms may be ionized by two or more electron collisions. Thus at high currents the electrons need be given only the metastable excitation energy of the gas. This process is termed cumulative ionization and results in the arc voltage dropping to the value of the metastable energy level, $(B - C)$.

The current voltage characteristic then remains essentially flat until the arc current equals the thermionically emitted current If the voltage is increased beyond this point a field will exist at the cathode Many years ago Schottky[3] showed that the potential barrier of a metal was lowered by the application of an external field so that by means of the field the arc current may be increased by this Schottky effect, and the current-voltage characteristic will rise sharply

The Hollow Cathode Arc

The hollow cathode arc is a dc source of highly ionized plasma, which was originally developed by J S Luce[4] at the Oak Ridge National Laboratory as a source of dissociated molecular hydrogen ions for the thermonuclear machine DCX It is distinguished from normal arc by an unusual cathode configuration and mechanism which allow a high degree of ionization (25–75%) to be achieved in the dense plasma column The cathode is a hollow cylinder of refractory material which carries a flow of neutral gas into the discharge chamber A typical cathode is a tantalum tube of 0 125 in diameter, 0 010 in wall thickness, and several inches length The discharge runs from a hot spot inside the cathode, which reaches temperatures on the order of 2500°K Typical background gas pressures are on the order of 10^{-3} torr, with a corresponding number density of $10^{13}/cm^3$ A magnetic field of several hundred to over a thousand gauss, aligned with the cathode axis, is applied to collimate the plasma, and also to aid in starting The plasma column runs from the cathode along the field lines to the anode The position and geometry of the anode are not critical to arc operation and anode material has no apparent significance

The primary source of plasma appears to be ionization within the hollow cathode Electrons are furnished by thermionic emission, probably field enhanced, from the cathode hot spot The cathode interior serves as a trap for positive ions, as well as for radiation from energy transition therein Ions which hit the cathode wall and recombine give up energy to the cathode, and indirectly enhance electron emission by heating the cathode The hollow cathode geometry thus feeds energy back into the discharge that would be lost to container walls in a plane cathode geometry The ionization processes in the cathode are further enhanced by the generous flow of neutral gas atoms directly into the ionization volume As this gas flows through the cathode, it falls to a pressure of about a torr, suitable for operation of a high current discharge with a mean free path for ionization on the order of the cathode diameter The hot spot forms at the corresponding area of the cathode An atom inside the cathode has a high probability of being ionized before escaping into the

discharge volume. Thus, effectively, the flow of gas atoms is at least partially converted to a flow of ions, which, coupled with electron emission, forms a plasma source feeding the discharge. Ions are able to flow against the electric field due to their pressure gradient in the cathode. Calculation of diffusion rates[5] indicates an ion density on the order of $10^{16}/cm^3$ in the cathode. An estimate of the cathode sheath thickness, the Debye length, gives about 3×10^{-5} cm with resulting average sheath fields on the order of 10^{16} volts/cm. This supports the field enhanced thermionic emission assumption; however, no particular cathode mechanism has been definitely established.

The external plasma of the hollow cathode arc consists of the central column of highly ionized, high-density and -temperature plasma, surrounded by less dense plasma through which particles are diffusing to the discharge walls. Microwave diagnostics and probe studies indicate central column densities of 10^{13} to $10^{14}/cm^3$ at 10^{-3} torr gas pressure, 20–40 amps arc current, and 1.7 atm-cc/sec flow. These densities correspond to ionization ranging up to 75% at this pressure. Langmuir probe studies of the surrounding, less dense plasma indicate electron temperatures from 1 to 18 electron volts, the higher figure corresponding to the edge of the plasma column, and ion temperatures on the order of 1 electron volt.

Low-Voltage Arcs

If the pressure of the gas in a discharge tube is increased to a few torr and the temperature of the cathode made high enough to ensure a thermionic emission current greater than the arc current, the voltage necessary to maintain the arc drops well below even the excitation potential of the gas. Such a low-voltage arc was studied by Compton and Eckart[6] and by Druyvesteyn.[7] A region of intense lumination is observed between the cathode and the anode. By means of probes, the space potential and electron concentration were measured and found to possess maximum values in the bright region. These measurements are illustrated in Fig. 15.3. Druyvesteyn found in a neon tungsten arc that the potential maximum could be as high as 18.7 volts (the excitation energy of neon is 18.5), while the voltage across the arc was only 12.3 volts. The maximum electron concentration was $2.2 \times 10^{12}\ cm^{-3}$. With the discovery of the potential maximum, the process of ionization is easily understood. In this region intense ionization can occur by the process of cumulative ionization. Since the electrons diffuse outward much more rapidly than the positive ions, a positive space charge exists to produce the necessary high potential region. The electrons are produced in the region, and thus their concentration is also a maximum. If the energy of the electron is 18.7 volts and this is lost in the ionization process, where does the electron

gain energy to reach the anode against a retarding potential of 18 7 — 12 3 = 6 4 volts? From a group of possible mechanisms, Druyvesteyn showed that an electron interaction between other high energy electrons

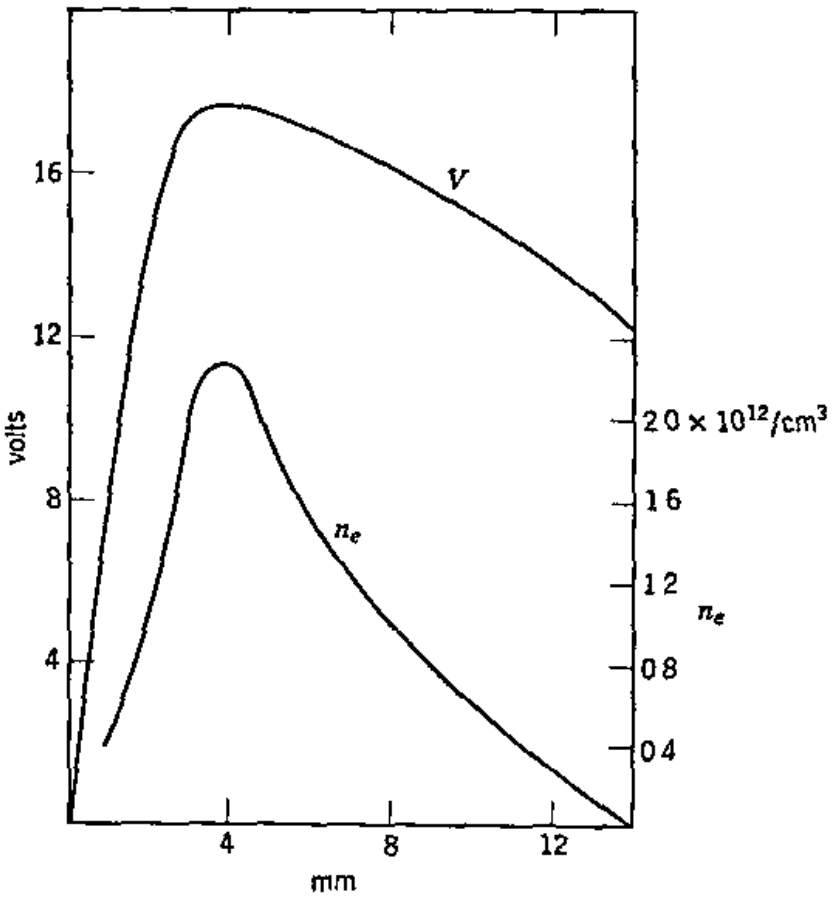


Fig 15 3 Low voltage arc characteristics

and the low energy electrons leads to sharing of energy in such a way that the group is able to diffuse against the retarding field to the anode

Insulated Cathodes

Electron emission caused by a high field at the cathode may occur when the cathode is coated by a thin film of an insulating material The positive ions accumulate on the outside of the film because the rate of leakage through the film is very small This field is capable of drawing emission currents of many thousands of amperes per square centimeter while the thermionic emission current aided by a Schottky effect produces only a few amperes of current per square centimter In addition the field emission process is independent of temperature while thermionic emission is very temperature dependent Thus high field emission currents may be produced by cathodes operating at room temperatures while the therm ionic current is negligible at this temperature

Mercury Arc

The operating cathode area of a mercury arc is called the cathode spot. In the mercury arc with a liquid pool cathode, the spot moves rapidly over the surface of the mercury. If a strip of metal dips into the surface of the mercury, the spot becomes "anchored" and moves rapidly along the boundary between the mercury and the metal. The velocity, in either case, is of the order of 50 to 150 cm/sec. The area of the cathode spot is difficult to measure. In general, it is a region of high luminosity, and this luminous region is assumed to be the current transmitting area of the arc. Earlier estimates of the cathode current densities varied from 1000 to 10,000 amp/cm^2. Cobine and Gallagher[8] have measured the cathode spot size of the mercury arc and found that the current density must be greater than 50,000 amp/cm^2. The measurement was performed by an optical-electrical system. The study of other metallic arcs indicated the current density of the spots was always 50,000 amp/cm^2 or greater.

In early studies of mercury arcs, the cathode process was believed to be a high field emission process in which the required high field at the cathode was created by the space charge limited positive ions. Mackeown[9] developed a theory in contradiction to this by calculating the magnitude of the electric field as an ordinary cathode fall assuming no collisions in the cathode fall region and making the current flow a continuous quantity. As a result of his calculations, it seems unnecessary to introduce anything as complicated as field emission to explain the high current densities observed in the liquid metal cathode spots.

High-Pressure Arcs

One way of differentiating between the high- and low-pressure arcs is in terms of electron and gas temperature. At low pressures, most of the collisions between electrons and atoms are elastic. In an elastic collision, the electron gives only a small fraction of its kinetic energy to an atom. Although the energy lost in a single impact is very small, the total loss may be considerable because the collision frequency is high (of the order of 10^9p). The energy transferred to the gas causes the temperature to rise. At low pressures, the difference between the electron temperature and the gas temperature is very large. With increasing pressure, however, the electrons lose more and more energy to the atom as a result of the increasing frequency of collision. This causes the electron temperature to fall, and the gas temperature to rise. This is illustrated[10] in Fig. 15.4. Beyond a certain pressure, which can be called the beginning of the high-pressure discharge, the gas temperature and the electron temperature are practically the same. Since the electrons gain their energy from the electric field and lose it partially to elastic collisions with the atoms, the electron

temperature must be higher than the gas temperature because energy transfer by collisions between the two groups of particles is only possible from the higher temperature to the lower temperature

If the concentration of atoms in the ground state is n_0, the concentration n_k of atoms excited to a level u_k volts above the ground state is

$$n_k = n_0\left(\frac{g_k}{g_0}\right)e^{-eu_k/kT}, \tag{15 1}$$

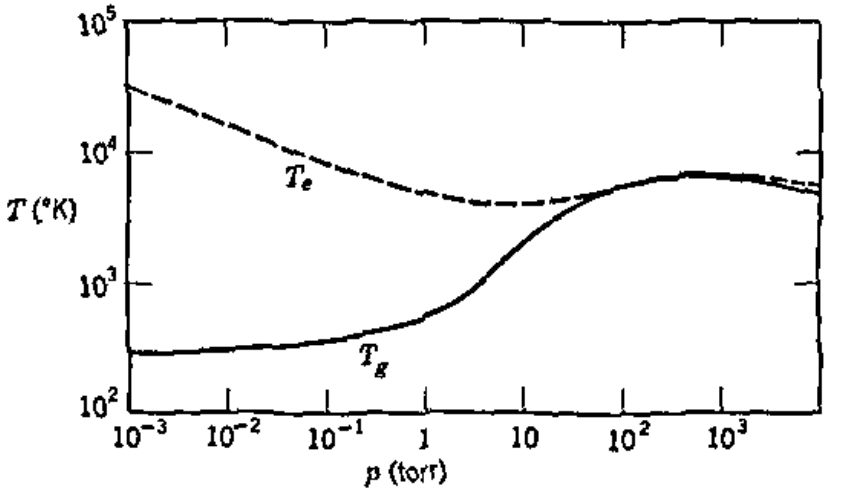


Fig 15.4 Temperatures in arcs as a function of pressure

where g_0 and g_k are the statistical weights of the ground level and the k level, respectively The concentration of electrons and positive ions is determined by Saha's equation [11]

$$\frac{n_+ n_-}{n_0} = \frac{(2\pi m_- kT)^{3/2}}{h^3}\, T^{3/2}\, 2\,\frac{g_0'}{g_0}\, e^{-eu_i/kT} \tag{15 2}$$

For mercury, $g_0'/g_0 = 2$

If the temperature is known, we can determine n_k by Eq (15 1) and $n_- = n_+$ by Eq (15 2) With known transition probabilities, the first gives us the intensity of the various spectral lines, and the second enables us to calculate the field E for a given electron mobility

The temperature on the axis of a high-pressure mercury arc must be of the order of 5000° to 7000°K in order for the population of the Hg levels to become dense enough to produce sufficient radiation It is impossible for this high temperature to exist across the whole tube Not only would the tube walls melt, but also a temperature gradient is necessary near the walls in order that the energy be conducted to the walls A temperature variation actually does exist from about 6000° on the axis to 1000°K at the walls One can calculate the spatial distribution of a

particular line, as is illustrated[10] for the yellow line $u_k = 8.85$ electron volts in Fig. 15.5.

The concentration of electrons and ions decreases according to Eq. (15.2). Here, $n_+ = n_-$ and $n_0 \sim 1/T$.

$$n_- \sim T^{1/4} e^{-eu_i/2kT}. \tag{15.3}$$

The current density j is proportional to $n_- l_- T^{1/2}$ and l_-, the electron mean free path, is proportional to T. The variation of j and T with radius is

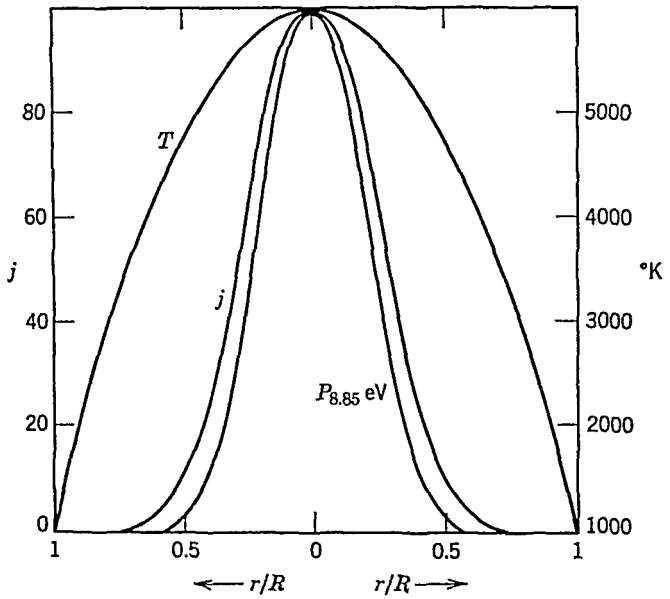


Fig. 15.5 Characteristics of a high pressure arc as a function of radius.

also illustrated in Fig. 15.5. From this figure it is clear that the optical width of the spectral line follows very closely the current density profile in the arc. Furthermore, it is seen that the arc radius is considerably smaller than the tube radius controlling the temperature profile. The arc plasma column is said to be "contracted."

The temperature is a very important quantity in the high-pressure mercury vapor discharge. If the temperature distribution is known, the properties of the discharge can be calculated. Many different attempts have been made to measure the temperature distribution experimentally. These include such techniques as: thermocouples,[11] from measurements of pressure and density,[12] absorption of X-rays,[13] broadening of spectral lines,[14] and from the absolute intensity of a spectral line of known transition

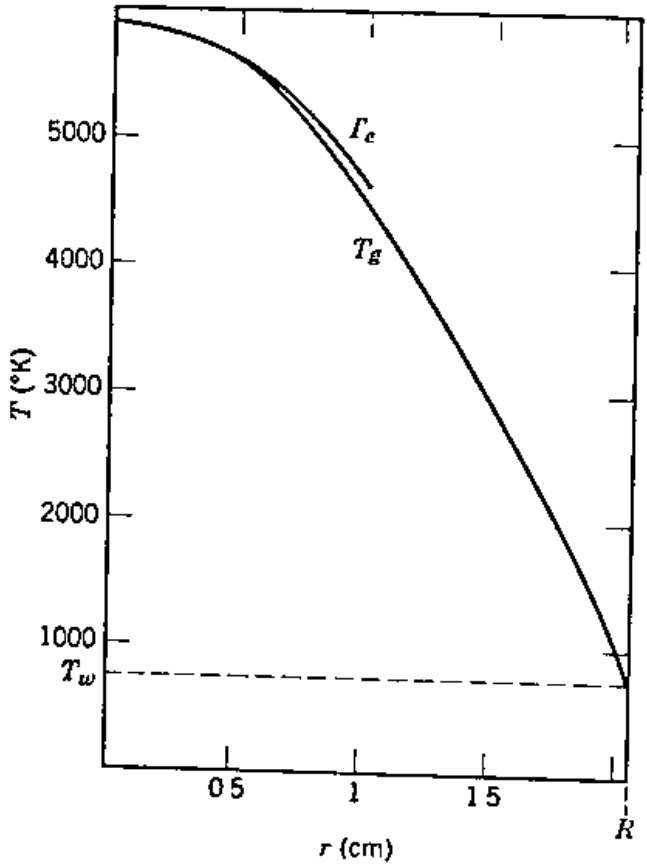


Fig. 15 6 Distribution of temperature in a high pressure mercury arc

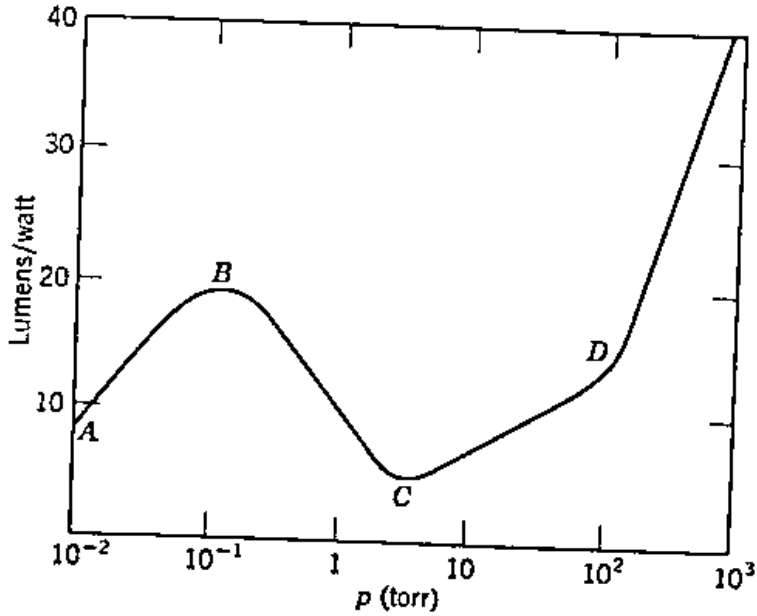


Fig. 15.7 The luminous efficiency of an arc as a function of pressure

probability.[15] The accuracy of these methods is not, in general, sufficient to agree on the details of the radial distribution of temperature, but they do agree in the general magnitude and form. A particular case is illustrated in Fig. 15.6.

Another way to differentiate between various types of discharges is by their luminous efficiencies. Figure 15.7 shows a plot, for a mercury vapor arc tube, of the luminous efficiency as a function of pressure.[10] As pressure is increased from a very low value, the luminous efficiency increases

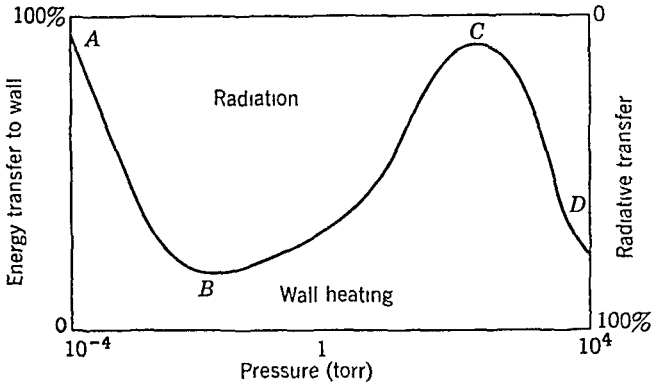


Fig. 15.8 Transitions in an arc plasma column.

initially and passes through a maximum at about 0.1 mm (B). The maximum occurs because, with increasing pressure, the absorption and the number of collisions between electrons and atoms increase and the energy lost by elastic collisions increases. Thus the gas temperature rises, and a considerable part of the energy input is transferred to the tube wall by thermal conduction. At C the temperature of the mercury vapor near the axis has become so high that temperature radiation starts to become important. This is the boundary between the low- and high-pressure ranges. By raising the pressure further, the luminous efficiency increases rapidly, and the radiation is mainly the result of the high temperature.

These same transitions can be described in terms of the energy transferred out of the arc plasma column. This can be illustrated[16] by Fig. 15.8. In the very low-pressure region all the energy in the plasma is dissipated at the wall either as recombination energy or as direct particle impact. As the pressure increases, the electron temperature goes down and the ratio of ionizing to exciting collisions follows the temperature. This means that more and more energy is being radiated out through the walls instead of being transferred to them (A — B). From B to C, more and more energy is being transferred from the electrons to the gas atoms by elastic

collision, and the gas begins to heat up, this heat energy is again transferred to the walls As we go on toward higher pressure, $C - D$, the number of atom-atom collisions increase and thermal excitation and ionization becomes important This thermal radiation again can pass out through the tube walls, so that the curve $C - D$ is a falling one as the pressure increases This latter characteristic depends on whether or not the gas is an efficient radiator under the physical conditions of the arc and it has not been observed for all gases

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Chapter 16

Propagation of Electromagnetic Waves in a Plasma

A detailed discussion of electromagnetic waves is motivated by the many properties of a plasma that can be experimentally studied by such waves, microwaves in particular. These properties include electron densities, electron-atom momentum transfer frequencies, electron temperature, and the electron velocity distribution function.

All of the electromagnetic wave phenomena that will be discussed in this chapter can be described by Maxwell's equations which will be used in the following form:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \tag{16.1}$$

$$\nabla \times \mathbf{E} = -\mu_0 \dot{\mathbf{H}} \tag{16.2}$$

$$\nabla \cdot \mathbf{H} = 0 \tag{16.3}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \epsilon_0 \dot{\mathbf{E}}. \tag{16.4}$$

The charge continuity equation will also be used

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0. \tag{16.5}$$

Consider the motion of an electron in a transverse electric field of angular frequency ω propagating through the plasma and supported by an external source. We shall consider electron-atom collisions as the only loss mechanism and place the momentum loss rate proportional to the electron velocity (as in the case of a frictional force) and also proportional to the collision frequency for momentum transfer, ν_m. From Newton's

second law

$$m\frac{\partial \mathbf{v}}{\partial t} + m\mathbf{v}\nu_m = -e\mathbf{E}_0 e^{i(\omega t - \mathbf{k}\cdot\mathbf{r})} \tag{16 6}$$

The solution to this equation is

$$\mathbf{v} = \frac{-e\mathbf{E}}{m(\nu_m + i\omega)} \tag{16 7}$$

where

$$\mathbf{E} = E_0 e^{i(\omega t - \mathbf{k}\cdot\mathbf{r})},$$

then

$$\mathbf{J} = -ne\mathbf{v} = \sigma\mathbf{E} = \frac{ne^2\mathbf{E}}{m(\nu_m + i\omega)} \tag{16 8}$$

Multiplying the numerator and denominator of Eq (16 8) by ϵ_0 and simplifying it by use of ω_p, we get the following expression for $\mathbf{J}$

$$\mathbf{J} = \sigma\mathbf{E} = \frac{\omega_p^2\epsilon_0}{(\nu_m + i\omega)}\mathbf{E} \tag{16 9}$$

Substituting (16 9) and the equation following (16 7) into (16 4), we find

$$\nabla \times \mathbf{H} = \left[\left(\frac{\omega_p^2\epsilon_0}{\nu_m + i\omega}\right) + i\omega\epsilon_0\right]\mathbf{E} \tag{16 10}$$

The first term in the bracket is the conduction term The second term is the displacement term

Rewriting (16 10), we obtain

$$\nabla \times \mathbf{H} = i\omega\epsilon_0\left[1 + \frac{\omega_p^2}{i\omega(\nu_m + i\omega)}\right]\mathbf{E} \tag{16 11}$$

Let us make the simplifying assumption $\nu_m/\omega \ll 1$ This corresponds to an all-electron gas Then Eq (16 11) becomes

$$\nabla \times \mathbf{H} = i\epsilon_0\omega\left(\frac{1 - \omega_p^2}{\omega^2}\right)\mathbf{E} \tag{16 12}$$

We see that a cut-off occurs at $\omega_p = \omega$ Physically this means that the displacement current is 180° out of phase with the conduction current, but of equal amplitude and thus the two cancel Since for this case

$$\nabla \times \mathbf{H} = 0$$

and

$$\nabla \cdot \mathbf{H} = 0$$

H cannot vary with space or with time (it may however be a constant), so no wave can propagate through the plasma

For $\omega_p > \omega$ the sign of the term on the right of Eq. (16.12) is negative, implying **E** is 90° out of phase with **H**. This further implies that Poynting's vector is zero.

$$\mathbf{S} \equiv \mathbf{E} \times \mathbf{H} = 0$$

and no power goes into the plasma.

If we take a cylindrical cavity and gradually increase the electron density, the power is at first absorbed by an amount proportional to the density. See Fig. 16.1a.

When the density reaches n_p, the density at which $\omega_p = \omega$, power can no longer be absorbed and the situation is as shown in Fig. 16.1b. At

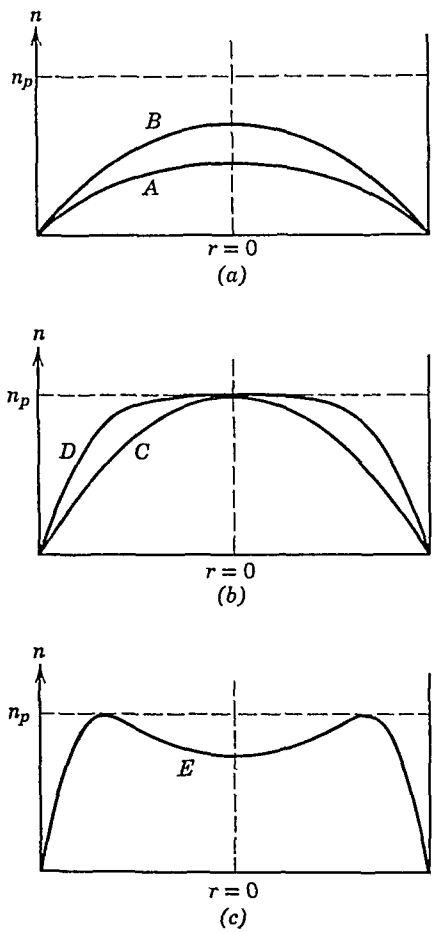


Fig. 16.1 Density distributions for different total densities. The density is assumed to increase in the order *A*, *B*, *C*, *D*, *E*.

even higher densities the interior may be strongly shielded as shown in Fig 16 1c

Let us now calculate the dielectric constant of the plasma Assuming ρ varies as $e^{i\omega t}$, Eq (16 5) may be written as follows

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J}$$

$$i\omega\rho = -\nabla \cdot \mathbf{J}$$

$$\rho = \frac{-\nabla \cdot \mathbf{J}}{i\omega}$$

Substituting for ρ in Eq (16 1) we obtain

$$\nabla \cdot \mathbf{E} = \frac{-\nabla \cdot \mathbf{J}}{i\omega\epsilon_0} = \frac{-\nabla \sigma \mathbf{L}}{i\omega\epsilon_0}$$

or rewritten, we find

$$\nabla \left(1 + \frac{\sigma}{i\omega\epsilon_0}\right)\mathbf{E} = 0 \qquad (16\ 13)$$

Since the dielectric constant K is defined by the equation we find

$$\nabla K\mathbf{E} = 0 \qquad (16\ 14)$$

By comparing (16 13) with (16 14), we obtain

$$K = 1 + \frac{\sigma}{i\omega\epsilon_0} \qquad (16\ 15)$$

The next question to be asked is what is the propagation vector in such a medium?

Again we assume that

$$\mathbf{E} = \mathbf{E}_0 e^{i(\omega t - \mathbf{k}\cdot\mathbf{r})}$$

Using Eq (16 9), we may write Eq (16 11) in the following form

$$\nabla \times \mathbf{H} = i\omega\epsilon_0\left[1 + \frac{\sigma}{i\omega\epsilon_0}\right]\mathbf{E} \qquad (16\ 16)$$

By comparison with (16 15) this may be written in terms of the dielectric constant

$$\nabla \times \mathbf{H} = i\omega\epsilon_0 K\mathbf{E} \qquad (16\ 17)$$

From (16 2) assuming that H varies as $e^{i\omega t}$, we find

$$\nabla \times \mathbf{E} = -i\omega\mu_0\mathbf{H} \qquad (16\ 18)$$

Substituting for **H** in (16.17) from (16.18) we obtain

$$\nabla \times \left(\frac{-\nabla \times \mathbf{E}}{i\omega\mu_0}\right) = i\omega\epsilon_0 K\mathbf{E}$$

$$\nabla \times (\nabla \times \mathbf{E}) = \omega^2\epsilon_0\mu_0 K\mathbf{E}. \tag{16.19}$$

We know that $\epsilon_0\mu_0 = 1/c^2$ where c is the velocity of light in free space. We may therefore define a propagation vector in free space

$$k_0^2 = \frac{\omega^2}{c^2}$$

and

$$\nabla \times (\nabla \times \mathbf{E}) = -i\mathbf{k} \times (-i\mathbf{k} \times \mathbf{E}) = -\mathbf{k} \times (\mathbf{k} \times \mathbf{E}). \tag{16.20}$$

Combining (16.19) and (16.20), we obtain

$$-\mathbf{k} \times (\mathbf{k} \times \nabla) = k_0^2 K\mathbf{E}$$

which may be simplified using the vector identity:

$$\mathbf{V}_1 \times (\mathbf{V}_2 \times \mathbf{V}_3) = \mathbf{V}_2(\mathbf{V}_1 \cdot \mathbf{V}_3) - \mathbf{V}_3(\mathbf{V}_1 \cdot \mathbf{V}_2)$$

to

$$\mathbf{k}(-\mathbf{k} \cdot \mathbf{E}) - \mathbf{E}(-\mathbf{k} \cdot \mathbf{k}) = k_0^2 K\mathbf{E}. \tag{16.21}$$

Since the electromagnetic wave is a transverse wave we find that $\mathbf{k} \cdot \mathbf{E} = 0$. This may also be seen from Eq. (16.14). Since K is a constant, we obtain

$$\nabla \cdot K\mathbf{E} = K(\nabla \cdot \mathbf{E}) = -iK(\mathbf{k} \cdot \mathbf{E}) = 0.$$

Equation (16.21) thus simplifies to

$$k^2\mathbf{E} = k_0^2 K\mathbf{E}.$$

for $E \neq 0$

$$k^2 = k_0^2 K = k_0^2\left(1 + \frac{\sigma}{i\omega\epsilon_0}\right).$$

Substituting σ from (16.9), we obtain

$$k^2 = k_0^2\left[1 + \frac{\omega_p^2}{i\omega(\nu_m + i\omega)}\right]. \tag{16.22}$$

Extracting the real and imaginary parts, we find that

$$k^2 = k_0^2\left[1 - \frac{\omega_p^2}{\omega^2 - i\omega\nu_m}\right] = k_0^2\left[1 - \frac{\omega_p^2(i\omega\nu_m + \omega^2)}{\omega^2\nu_m^2 + \omega^4}\right]$$

$$= k_0^2\left[\left(1 - \frac{\omega_p^2}{\nu_m^2 + \omega^2}\right) - \frac{i\omega_p^2\nu_m}{\omega(\nu_m^2 + \omega^2)}\right] \tag{16.23}$$

Then $k = \sqrt{k^2} = k_r - ik_i$ This is a complicated function The graph for limiting cases is shown in Fig 16 2

Note that at $\nu_m = 0$ (16 23) becomes

$$k^2 = k_0^2\left(1 - \frac{\omega_p^2}{\omega^2}\right) \tag{16 24}$$

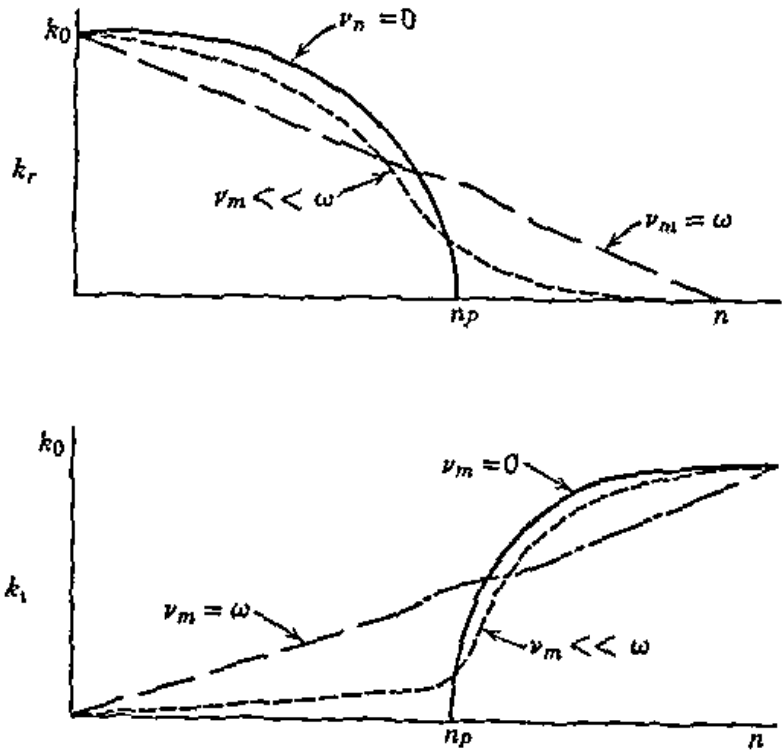


Fig 16 2 Real and imaginary components of the propagation vector

Since $k_r = 2\pi/\lambda$, the plot shows how the wavelength varies as a function of n for constant ω The interesting point is that

$$\frac{2\pi}{k_0} \leq \lambda < \infty$$

as n goes from zero to n_p

Now let us analyze the behavior of the plasma statistically by introducing the distribution function and calculate the macroscopic properties by suitable averages over the distribution The differential equation which gives the changes in the distribution function as a result of the physical mechanisms which are at work is the Maxwell Boltzmann equation In its simplest form this equation may be written

$$\frac{\partial f}{\partial t} = -v\frac{\partial f}{\partial r} - \frac{eE}{m}\frac{\partial f}{\partial v} + \left(\frac{\partial f}{\partial t}\right)_{\text{coll}} \tag{16 25}$$

The rate at which the distribution function f varies with time is given by the three terms on the right-hand side of this equation. The first of these terms represents a loss which has a homogenizing effect on the distribution and represents physically such mechanisms as diffusion. The second term gives the effect of external fields and is an anisotropifying term which would include the collective behavior of the plasma particles. The last term marks the effect of collisions and can be described by the collision frequency ν_m:

$$\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} = -(f - f_0)\nu_m$$

where f_0 is the distribution function in thermal equilibrium and $(f - f_0) \ll f$.

In the steady state,

$$\frac{eE}{m}\frac{\partial f}{\partial v} + v\frac{\partial f}{\partial r} = -(f - f_0)\nu_m$$

$$f = f_0 - \frac{1}{\nu_m}\left(\frac{eE}{m}\frac{\partial f}{\partial v} + v\frac{\partial f}{\partial r}\right)$$

and since $(f - f_0) \ll f$

$$f = f_0 - \frac{1}{\nu_m}\left(\frac{eE}{m}\frac{\partial f_0}{\partial v} + v\frac{\partial f_0}{\partial r}\right). \tag{16.26}$$

f_0 is a function of the energy u

$$\frac{\partial f_0}{\partial v} = \frac{\partial f_0}{\partial u}\frac{du}{dv} = mv\frac{\partial f_0}{\partial u}.$$

The electrical conductivity is defined under the conditions that $dT/dr = 0$ and $dn/dr = 0$ which results in $\partial f_0/\partial r = 0$, so that:

$$f = f_0 - \frac{eE}{\nu_m} v\frac{\partial f_0}{\partial u}.$$

The current density is determined by the relative motion of the electrons and ions in a plasma

$$J = -en\langle v\rangle = -ne\int vf\,dv$$

$$= -\frac{ne^2E}{\nu_m}\int v^2\frac{\partial f_0}{\partial u}\,dv$$

assuming ν_m independent of velocity. For a Maxwellian distribution, we find

$$f_0 \sim e^{-mv/2kT}$$

$$\frac{\partial f_0}{\partial u} = -\frac{f_0}{kT}$$

so that

$$J = \frac{ne^2E}{kTv_m}\int v^2 f_0\, dv$$

Since $\frac{1}{2}m\int v^2 f_0\, dv = \frac{1}{2}kT$

$$J = \frac{ne^2E}{mv_m} \tag{16 27}$$

and

$$\sigma = \frac{J}{E} = \frac{ne^2}{mv_m} \tag{16 28}$$

which is the same equation we derived from the average electron picture for the dc conductivity

Let us now discuss the interaction of the plasma with an electromagnetic field From Maxwell's equations

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \qquad \mathbf{E} = \mathbf{E}_0 e^{j\omega t}$$

$$\begin{aligned}\nabla \times \mathbf{H} &= \mathbf{J} + j\omega_0\epsilon_0\mathbf{E}_0 e^{j\omega t}\\ &= (j\omega\epsilon_c\mathbf{E}_0 + \sigma_{ac}\mathbf{E}_0)e^{j\omega t}\\ &= j\omega\mathbf{E}_0\epsilon_0\left[1 + \frac{\sigma_{ac}}{j\omega\epsilon_0}\right]e^{j\omega t}\end{aligned}$$

where

$$\left[1 + \frac{\sigma_{ac}}{j\omega\epsilon_0}\right] = K = \mu^2, \tag{16 29}$$

which gives us the dielectric coefficient, K, providing we can calculate σ_{ac} Let us do this from the distribution function equation (16 25), splitting f up into two components

$$f = f_0(\text{dc}) + f_1(\text{ac})$$

where

$$f_1 = f_{10}e^{j\omega t}$$

so that

$$\frac{\partial f}{\partial t} + \frac{eE}{m}\frac{\partial f}{\partial v} = -(f - f_0)v_m = -(f_0 + f_1 - f_0)v_m = -f_1 v_m,$$

where $f_1 \ll f_0$ Then with $E = E_0 e^{j\omega t}$,

$$j\omega f_{10} + \frac{eE_0}{m}\frac{\partial f_0}{\partial v} = -f_{10}v_m$$

$$f_{10}(v_m + j\omega) = -\frac{eE_0}{m}\frac{\partial f_0}{\partial v}$$

so that

$$f_{10} = \frac{-eE_0}{\nu_m + j\omega} v \frac{\partial f_0}{\partial u}.$$

The ac current density then becomes

$$J_{ac} = ne\langle v \rangle = ne \int v f_{10}\, dv$$

and again assuming ν_m constant independent of velocity, we find that

$$J = \frac{-ne^2 E}{\nu_m + j\omega} \int v^2 \frac{\partial f_0}{\partial u}\, dv$$

and

$$\sigma_{ac} = \frac{ne^2}{m}\left(\frac{1}{\nu_m + j\omega}\right). \tag{16.30}$$

The conductivity is, of course, complex, $\sigma_c = \sigma_r + j\sigma_i$ and writing each of these separately, we find

$$\sigma_r = \frac{ne^2}{m\omega}\left[\frac{\nu_m/\omega}{1 + (\nu_m/\omega)^2}\right] = \frac{ne^2}{m}\frac{\nu_m}{\omega^2 + \nu_m{}^2} \tag{16.31}$$

$$\sigma_i = -\frac{ne^2}{m\omega}\left[\frac{1}{1 + (\nu_m/\omega)^2}\right] = -\frac{ne^2}{m}\frac{\omega}{\omega^2 + \nu_m{}^2}. \tag{16.32}$$

Writing this in terms of the dielectric coefficient, we obtain

$$K = 1 + \frac{\sigma_{ac}}{j\omega\epsilon_0} = 1 + \frac{ne^2}{m}\frac{1}{\nu_m + j\omega}\frac{1}{j\omega\epsilon_0}$$

$$K = 1 - \frac{ne^2}{m\omega^2\epsilon_0}\frac{[1 + j(\nu_m/\omega)]}{[1 + (\nu_m/\omega)^2]}$$

and since $\omega_p{}^2 = ne^2/m\epsilon_0$

$$K = 1 - \frac{\omega_p{}^2}{\omega^2}\left[\frac{1 + j(\nu_m/\omega)}{1 + (\nu_m/\omega)^2}\right]. \tag{16.33}$$

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Chapter 17

Interaction of Plasmas with Electromagnetic Fields

The square of the dielectric coefficient given in Eq (16 33) is related to the attenuation and phase shift of a plane wave as shown in Fig 17 1 Here the attenuation is given in nepers per wavelength A neper is the natural logarithm of the ratio of two currents $n = \log_e (I_1/I_2) = 8\ 686$ db In the low density region, the attenuation and phase shift are linear functions of the density, but at higher densities this linearity disappears and the functional relation becomes more complicated

If a magnetic field is applied, the dielectric coefficient depends not only on the density of the charged particles and the magnitude of the magnetic

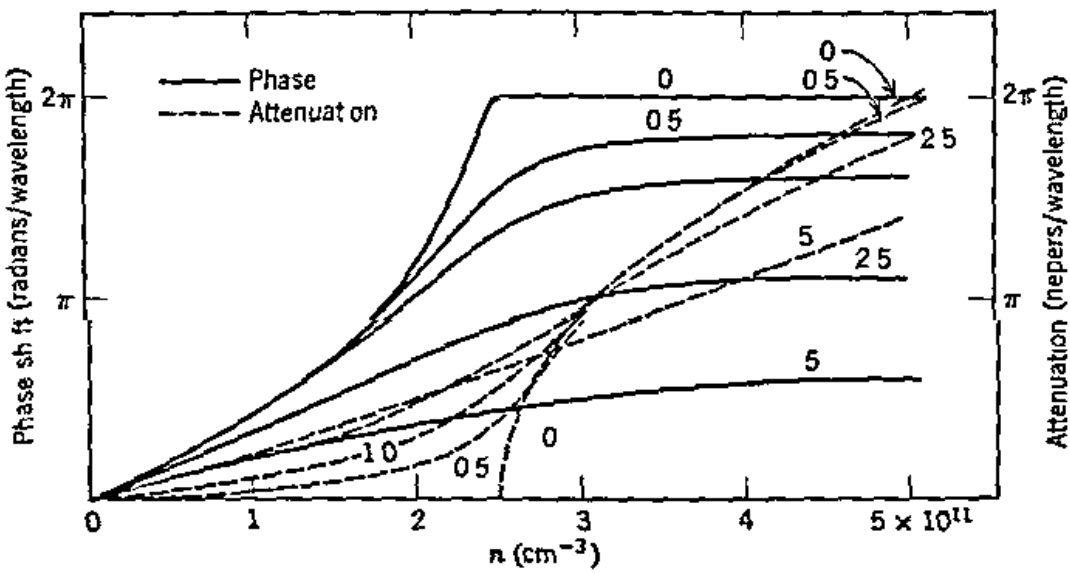


Fig 17 1 Phase shift and attenuation of an electromagnetic wave as a function of electron density

field, but also on the geometrical configuration and the direction of propagation of the electromagnetic wave with respect to the applied magnetic field. Four cases can be distinguished for the purposes of simplifying the discussion.

For propagation of the wave[1,2] along the direction of the magnetic field, we can describe a right-handed circularly polarized wave and a left-handed circularly polarized wave. In the right-handed wave the electron motion is considered to be in the same sense as the electric field and for the left-handed wave the senses are opposite. The dielectric coefficients take the form

$$K_r = 1 - \left\{\frac{\omega_p^2}{\omega^2}\left[\left(1 - \frac{\omega_b}{\omega}\right) - j\frac{\nu_m}{\omega}\right]^{-1}\right\} \tag{17.1}$$

and

$$K_l = 1 - \left\{\frac{\omega_p^2}{\omega^2}\left[\left(1 + \frac{\omega_b}{\omega}\right) - j\frac{\nu_m}{\omega}\right]^{-1}\right\}. \tag{17.2}$$

There are also two cases for the condition that the propagation vector be perpendicular to the magnetic field. These are called the ordinary wave, when the electric field is parallel to B, and the extraordinary wave when E and B are mutually perpendicular.

$$K_0 = 1 - \left[\frac{\omega_p^2}{\omega^2}\left(1 - j\frac{\nu_m}{\omega}\right)^{-1}\right] \tag{17.3}$$

and

$$K_x = 1 - \left[\frac{\omega_p^2}{\omega^2}\frac{\omega_p^2 - \omega^2}{\omega_p^2 + \omega_b^2 - \omega^2}\right]. \tag{17.4}$$

In Eq. (16.4) we have to consider that there is also a component of E field along the B field so that the electric field is not divergenceless. Also in this equation we consider $\nu_m = 0$. A vector representation of the various waves is shown in Fig. 17.2.

When we are confronted with the problems of keeping four polarizations of waves straight, we look for various schemes to systematize our discussion. Such a systematization[1] is represented schematically in Fig. 17.3. This particular representation is restricted to the case of no collisions, for an infinite plasma, and for low-energy electrons. This figure illustrates the behavior of the interaction between the plasma and electromagnetic waves in the domain in which the quantity proportional to the electron density is plotted on the abscissa, and the quantity proportional to the magnetic field intensity is plotted on the ordinate. There are three different regions, each of which shows certain characteristics as to the type of interaction that can exist. The plasma resonant condition ($ne^2/\epsilon_0 m\omega^2 = 1$) and

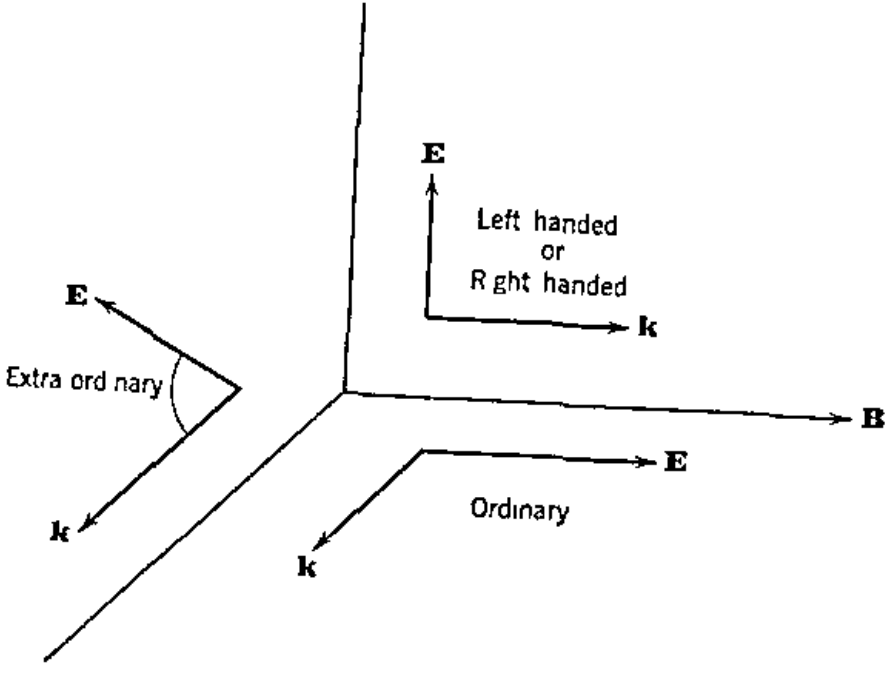


Fig 17 2 Polarization nomenclature

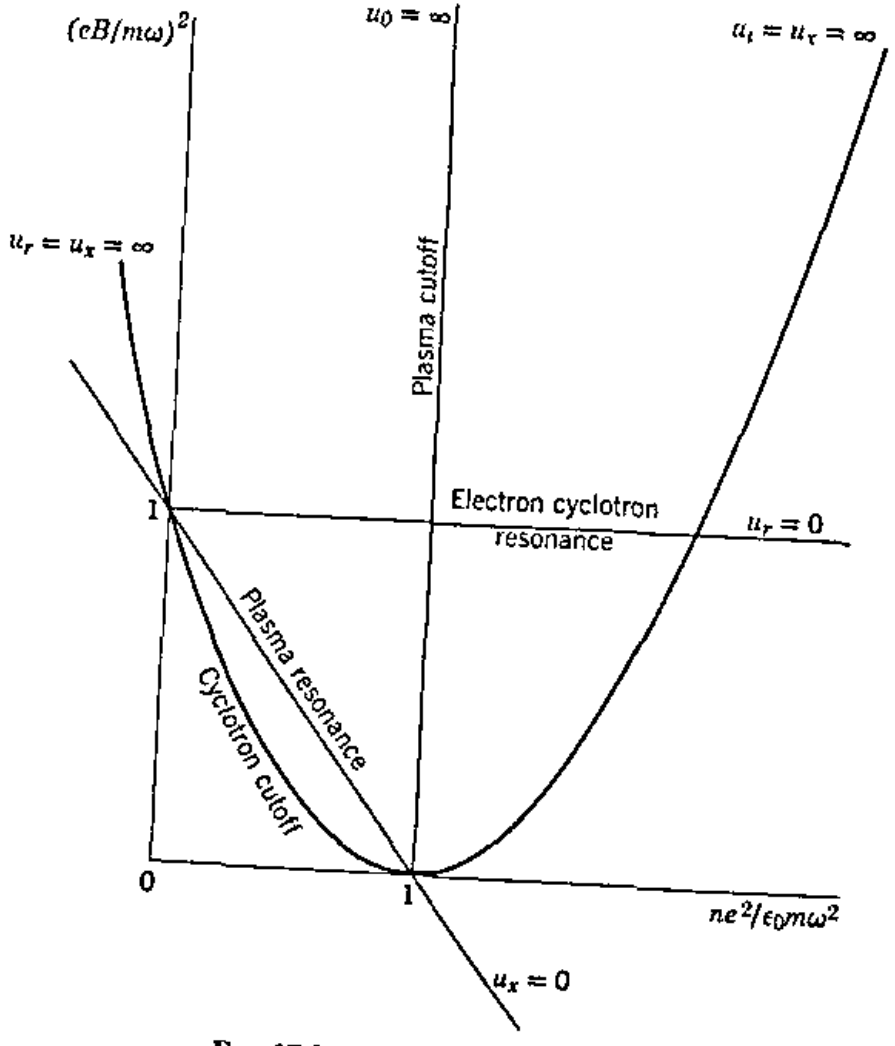


Fig 17 3 Plasma resonances and cut offs

the cyclotron resonant condition ($eB/m\omega = 1$) are easily recognized. We can discuss all four types of waves we have just introduced, and these are represented in the vector diagram of Fig. 17.2. Furthermore it is convenient to discuss the interactions in terms of the phase velocity of the various waves rather than in terms of the dielectric coefficient. The phase velocity $u = c/n$, where n is the index of refraction, is related to the dielectric coefficient by $n^2 = K$.

The ordinary wave is that polarization which is not affected by the magnetic field, where the E and B vectors are parallel, and the propagation is at right angles to the magnetic field. At plasma resonance, ω_p, the phase velocity of this wave, $u_0 = \infty$.

For the case in which the propagation vector along B and E is perpendicular, we have defined the right-handed and left-handed wave. Cyclotron resonance, ω_b, corresponds to the phase velocity of the right-handed wave, $u_r = 0$.

The fourth type of wave, called the extraordinary wave, is made up of electric vectors rotating right- and left-handed about B, describing an ellipse in a plane containing the direction of propagation perpendicular to B. The extraordinary velocity is intermediate between the right- and left-handed velocities.

The radiation and absorption behaviors of a plasma in a magnetic field show discontinuities, not only at the plasma resonances, ω_p and ω_b, but also as $u_x = 0$, $u_r = u_x = \infty$ and $u_l = u_x = \infty$.

In the absence of a magnetic field we expect the following properties of radiation from a plasma as a function of the electron density. The microwave radiation from a volume of ionized gas originates primarily from free-free collisions of electrons with the atoms and molecules of the gas—the so-called *bremsstrahlung* radiation. At very low electron densities, any radiation generated within a small volume of the plasma escapes with negligible attenuation; hence the noise power increases linearly with the electron density. As the electron density increases, the attenuation presented to the wave likewise grows until the energy becomes completely absorbed and re-emitted in the form of blackbody radiation from the skin of the plasma.

To illustrate this on our B-n domain of Fig. 17.3, let us replot this figure with two changes. Whereas Fig. 17.3 is a linear plot, let us change, in the next figure, to a logarithmic scale to cover a wider range of variables. Let us use a three-dimensional surface to represent the absorption in the third dimension in units of nepers/wavelength. This is now illustrated in Fig. 17.4, which is drawn[3] for the "ordinary" wave where the electric field E is parallel to the magnetic field B and perpendicular to the propagation vector k. This surface, as is the case of all our calculations,

represents a zero collision limit in the absence of thermal motion and density gradients Furthermore, this surface is a plot of evanescent absorption and in the laboratory much of the experimentation involves the study of radiation Nevertheless, transitions which mark abrupt changes or resonances in emitted radiation will be marked by similar changes in absorbed power so that we may study the physical behavior of the plasma either way In this we see the rapid increase of absorption at $\omega_p = \omega$

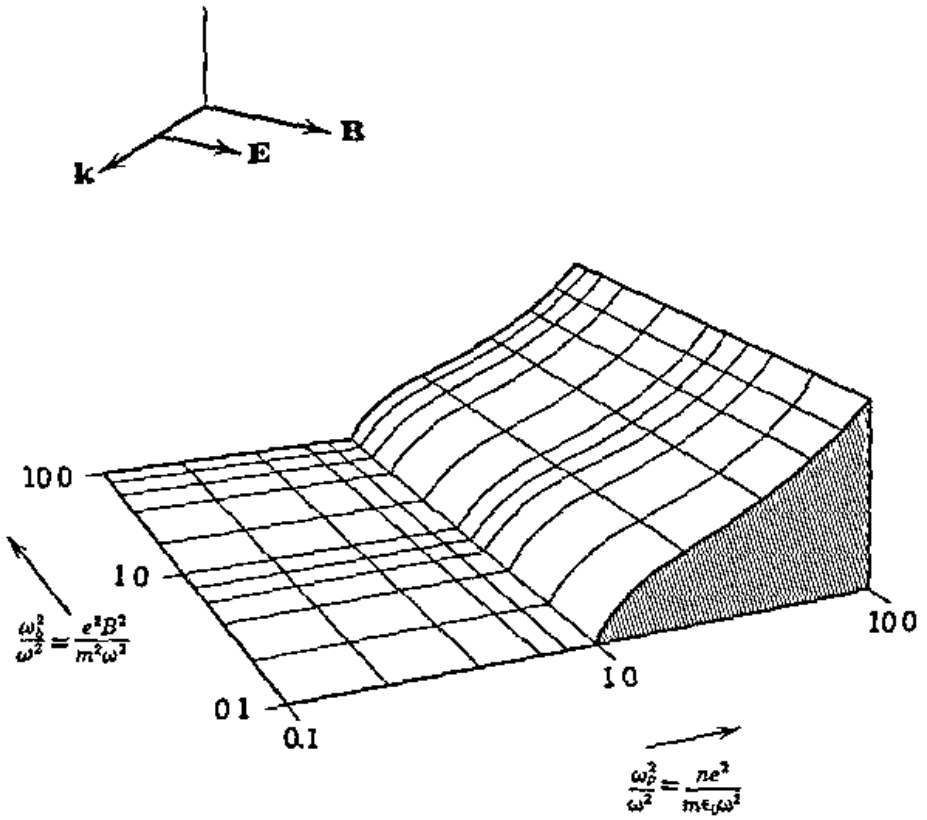


Fig 17 4 Evanescent absorption of the ' ordinary" polarization

independent of the intensity of magnetic field, and most of the microwave diagnostic studies have been carried out at frequencies higher than the onset of this limit Experiments have been carried out with the radiation of this "ordinary" wave in the absence of magnetic field and with increasing plasma density and should be compared with the front face of the three-dimensional surface illustrated in this figure

Some experimental studies[8] have been made to show the reasonableness of this approach, and these are shown in Fig 17 5 Here the measurements are of the noise power radiated from a helium discharge in the form of a plasma cylinder oriented with its axis perpendicular to the electric field of a surrounding wave guide and detected by a superheterodyne receiver Along the abscissa is plotted the electron density Along the ordinate we plot the measured noise power P, corrected for reflections and normalized

to the noise emitted by an opaque plasma. The scale here is in decibels below the maximum power radiated at any given pressure. For convenience of presentation, the curves are displaced vertically 10 db with respect to each other. We see that the dependence of the noise power on electron

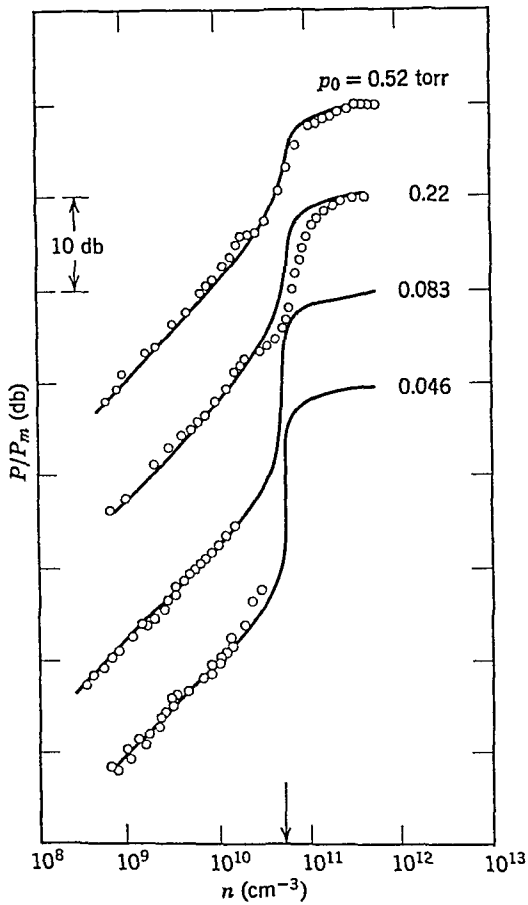


Fig. 17.5 Radiated 3000 mc power from a helium discharge.

density falls into three regions. The linear increase of P/P_m with electron density for low values of n falls into the region in which the plasma is essentially transparent (the horizontal plane of the previous figure). The nonlinear increase falls into the semiopaque region that occurs in the vicinity of $\omega_p = 1$. At high densities the radiation becomes black body.

The particular case just discussed was that of radiation in the absence of a magnetic field, or for the ordinary wave with E parallel to B which is

unaffected by the magnetic field Let us next take the case when the emitted radiation from the plasma is observed along the magnetic field with the electric field perpendicular to B If we take the rotation of the E field in the same direction as that of the electrons, we have the case described above as the "right-hand" wave The absorption calculation for this case is illustrated again as a surface in Fig 17 6

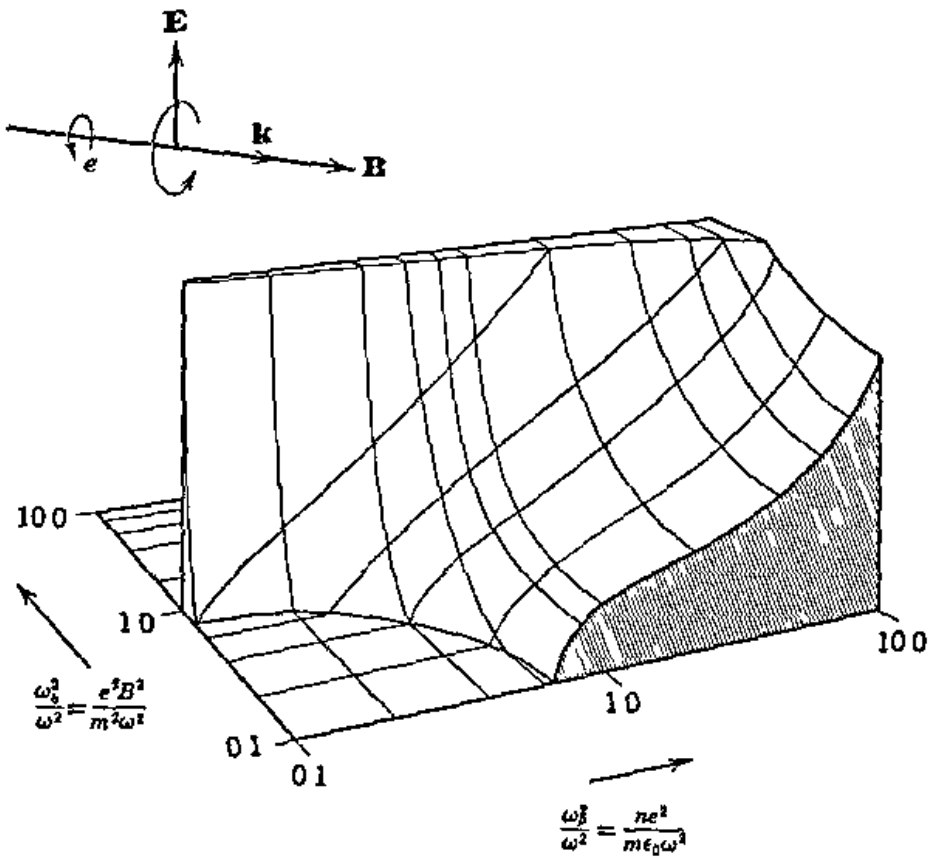


Fig. 17 6 Evanescent absorption of "right hand ' polarization

In the limit of zero magnetic field, which the three dimensional surface approaches at its front face, the shape of the absorption curve as a function of electron density is the same as the front face of the surface of Fig 17 4 At very low electron densities the familiar resonance absorption at cyclotron resonance, $\omega_b = eB/m$ is evident The shape of this absorption peak has been frequently studied in connection with high-frequency discharge breakdown in magnetic fields

The surface shows that as B is increased at constant n, the absorption or radiation increases slowly to infinity and then drops precipitously to zero across the $u_r = 0$ line, the shape of the curve changing as n increases Both the absorption and emission characteristics of a plasma for this right-handed wave have been measured as a function of magnetic field at various constant densities and the experimental data show the same

behavior as that predicted by this surface. These experimental observations[5] are shown in Fig. 17.7. The absorption, A, and the emission, P/P_m, are the fractions which would be absorbed or emitted if the plasma were matched to its surroundings. The plasma was a hydrogen discharge at about 20 microns pressure in the form of a cylinder appropriately mounted in a rectangular wave guide.

The electron cyclotron resonance type of behavior does not appear when the noise radiation emitted from the plasma propagates in a direction

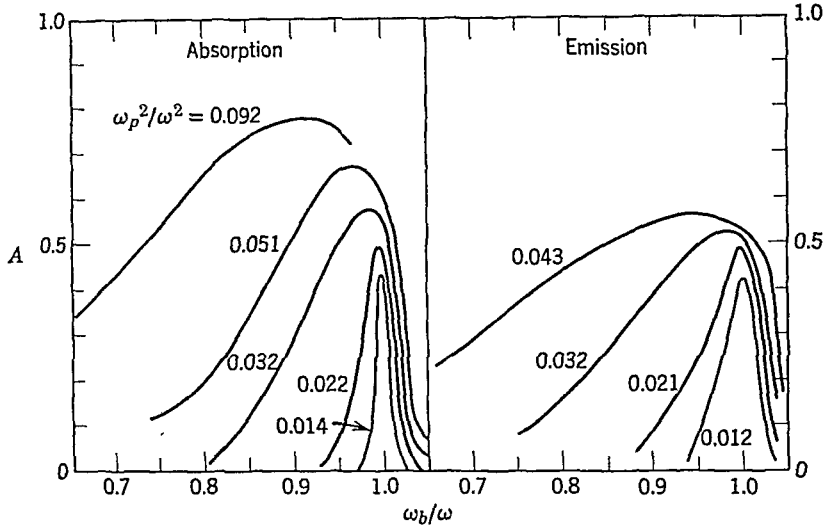


Fig. 17.7 Experimental observations of absorption and emission of the "right-handed" wave.

perpendicular to the magnetic field. However, there are resonances in this "extraordinary" wave which are characterized by their dependence on the electron density. The characteristic absorption of this type of wave is illustrated in Fig. 17.8. This particular surface is not the complete picture of the "extraordinary" wave as it exhibits another branch for values of $\omega_p^2/\omega > 1$. Observations have been made on the resonances illustrated in Fig. 17.8 by studying the emission characteristics of this "extraordinary" wave from plasma columns in magnetic fields. These experiments involve the same technique as was used in studying the *bremsstrahlung* emission in the absence of a magnetic field; but here, as the magnetic field was kept constant the electron density was varied to show the presence of these resonances as illustrated theoretically by the surface shown in this figure. Actual experimental data[4] are shown in Fig. 17.9, the existence of the resonances becoming more and more pronounced as the pressure becomes

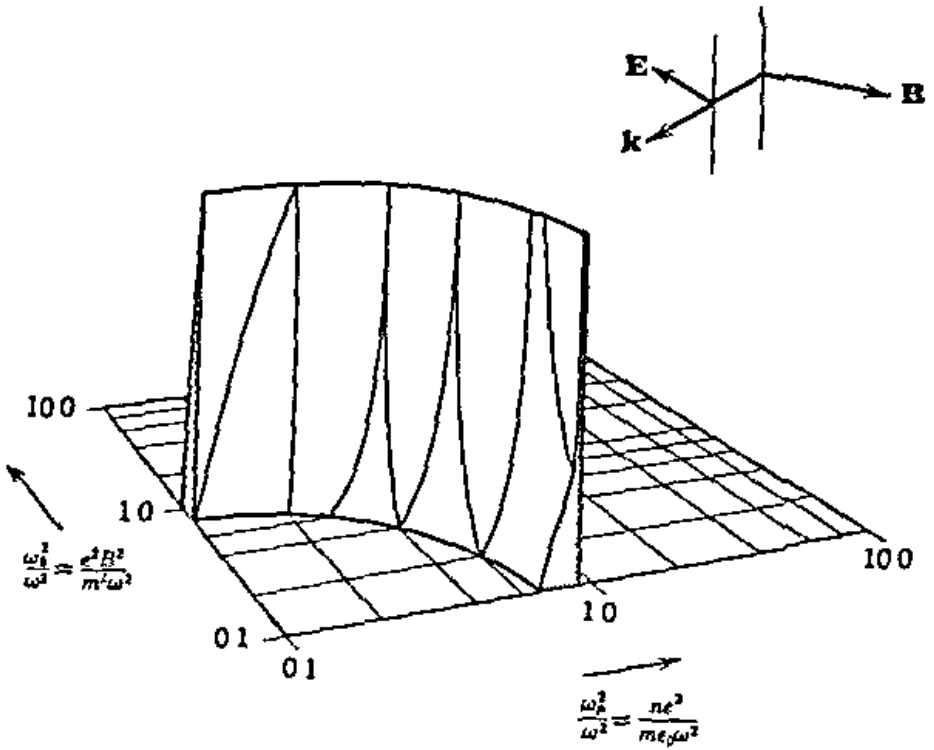


Fig. 17.8 Evanescent absorption of "extraordinary" polarization

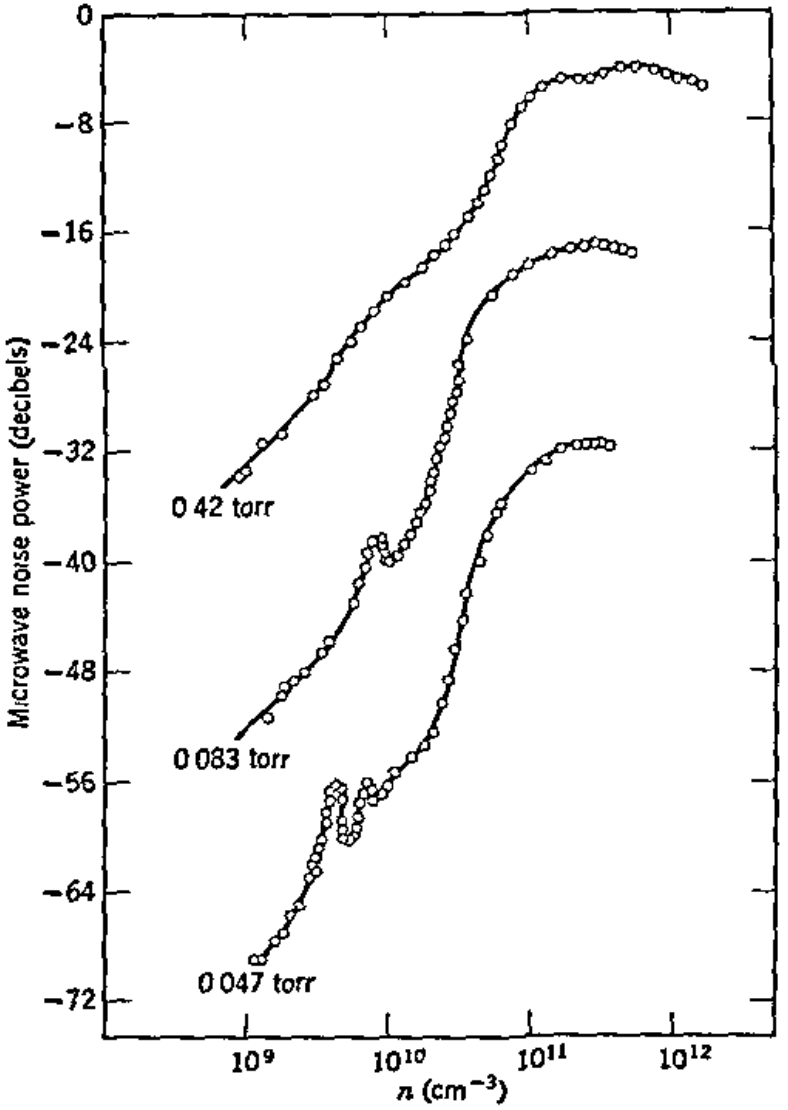


Fig. 17.9 Experimental measurements showing "extraordinary" wave resonances

less and less. The double nature of the peaks is due only to the superheterodyne response of the microwave radiometer, the peaks being separated by twice the frequency of the I.F. Measurements of the variation of position of these resonances in the *B-n* domain were also made and qualitative agreement was obtained with the theoretical calculations. The results of these measurements are shown in Fig. 17.10.

The high-density branch of the "extraordinary" wave absorption turns out to be identical with the "left-handed" polarized wave which rotates in the direction opposite to the motion of the electrons. The surface

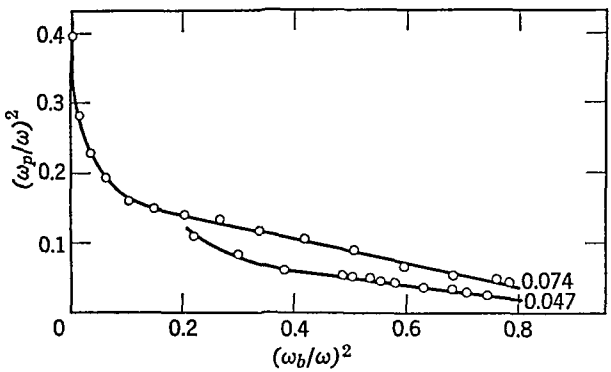


Fig. 17.10 Experimental measurements of "extraordinary" wave resonances.

representing the absorption or emission of these waves is illustrated in Fig. 17.11. Although the "left-handed" wave plays a relatively minor role when we are concerned with the study of the interaction of waves with electrons, it plays a dominant role when we turn our attention to ions, since it is just this wave, for example, which will exhibit ion cyclotron resonance.

To extend the *B-n* domain to include the ion effects we need a change in scale over the domain picture illustrated by Fig. 17.3. The line in the $\omega_b^2 - \omega_p^2$ plane where the absorption surface starts to increase is $u_x = u_l = \infty$. If we change our scale and, for clarity of presentation, carry out the calculations for an m/M which will emphasize the ion effects (taken here as $m/M = 0.25$), we get a plot illustrated by Fig. 17.12. Here we see that the $u_x = u_l = \infty$ line, which on all previous plots bent to the right, is now extended to bend back to the left so that at low densities and high magnetic fields the u_x and $u_l = \infty$ line becomes indistinguishable from both the $u_x = 0$ and the $u_l = 0$ lines.

Several laboratories have been deeply engrossed in studies of the possible ion resonances similar to the well-known resonances for electrons,

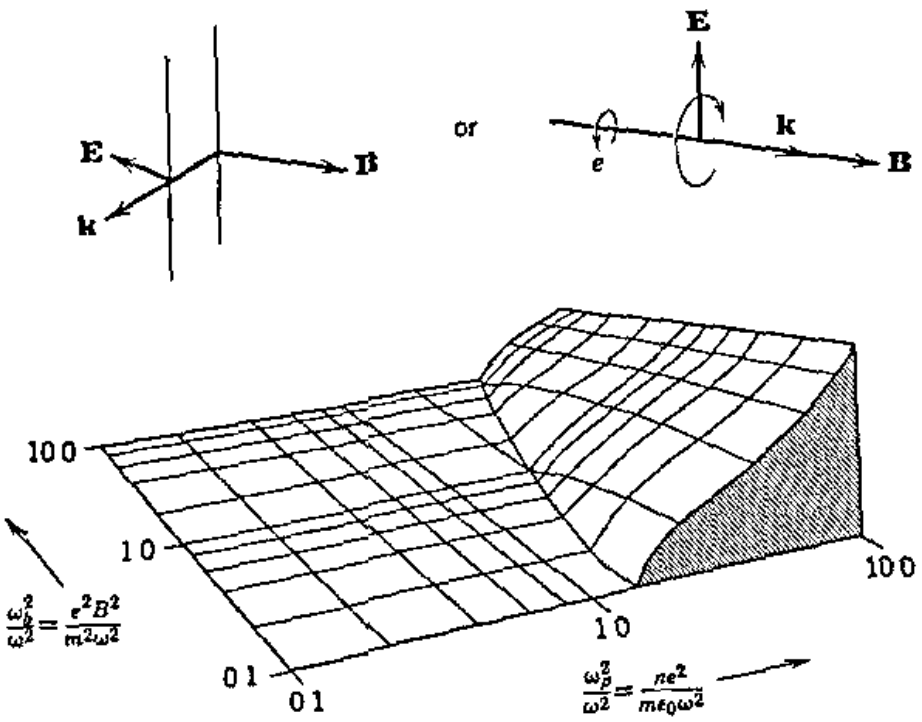


Fig 17 11 Evanescent absorption of "extraordinary" polarization

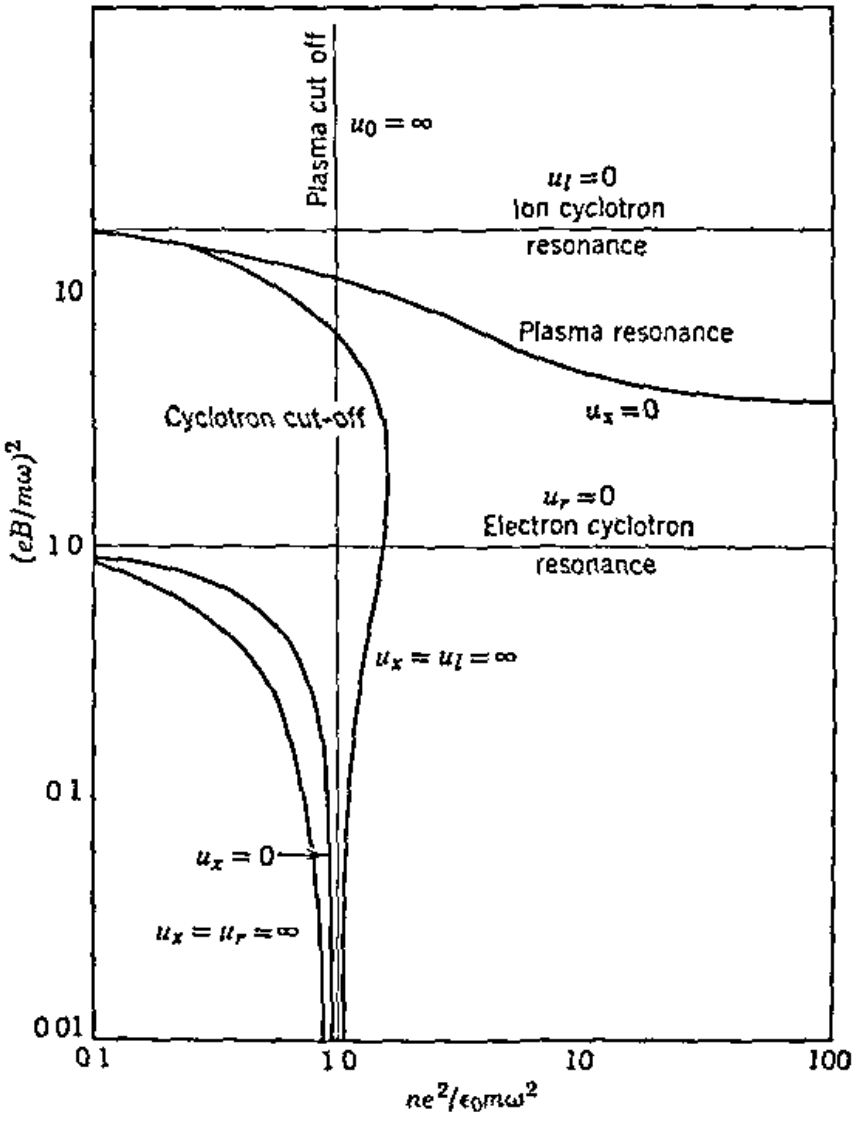


Fig 17 12 Plasma resonances and cut-offs

and have observed strong resonances corresponding to a geometrical mean frequency between the electron and ion cyclotron frequencies. These are called hybrid frequencies. The line corresponding to the hybrid frequency emission is shown on this figure as $u_x = 0$. One can see that at low ion densities these hybrid frequencies are indistinguishable from the pure ion resonances. As the ion density increases, the hybrid resonances separate more and more from the pure ion cyclotron resonance at $u_i = 0$, approaching an asymptotic value illustrated in this figure.

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Chapter 18

Plasma Oscillations

In 1925 Langmuir[1] investigated the velocity scattering properties of a low-pressure dc mercury arc plasma by means of a probe The electron velocity distribution of a discharge plasma is roughly Maxwellian as shown in Fig 18 1a At very low pressures a beam of high-energy electrons should also be present in the plasma region adjacent to the cathode fall as shown in Fig 18 1b The high-energy beam is formed from the electrons emitted by the cathode which are accelerated in the cathode fall of the arc but undergo no collisions with the gas atoms The experimental data, represented in Fig 18 1c, indicated the beam electrons were distributed in velocity over a wider range than expected and that some electrons possessed velocities corresponding to energy gains higher than any dc potential existing in the discharge tube Tonks and Langmuir[2] proposed the mechanism of charged particle oscillations to account for the anomalous scattering and detected the presence of the oscillations experimentally

A more complete experimental study of electron scattering and plasma oscillations was reported by Merrill and Webb[3] The experimental arrangement consisted of a movable thin wire probe suspended in the plasma of a hot cathode low-pressure mercury vapor discharge The electrode arrangement is shown in Fig 18 2 The "temperature" and concentration of the main group of electrons, the velocity distribution of the fast electrons, and the resultant oscillations were studied by moving the probe axially through the discharge from the cathode to the anode The oscillations were detected on a Lecher wire system by means of a crystal rectifier and galvanometer connected to the probe and the anode

The connection between the velocity scattering of the electron beam and the oscillations picked up by the probe can be understood by

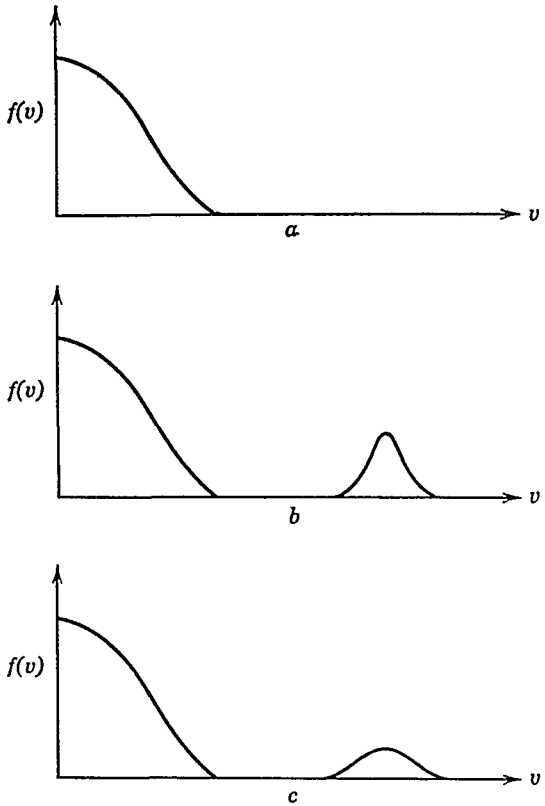
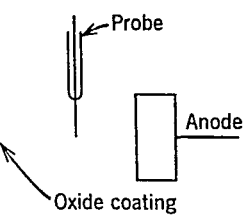


Fig. 18.1 Electron velocity distributions.

g Fig. 18.3. The fast electrons emerge from the oxide coated area
ıode and are accelerated through the cathode fall region of the
he plasma with a sharply peaked velocity distribution. The
elocity of this beam of electrons corresponds to the energy
crossing the cathode fall region of the plasma. The high-energy
nove through the section of the plasma defined by the dotted
ıding from the cathode to the anode. The peaked velocity



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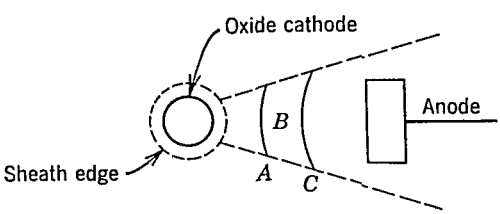


Fig. 18.3 Diagrammatic representation of an oscillating scattering layer.

distribution of the beam of electrons emerging from the cathode fall is maintained through the region of the plasma extending from the edge of the cathode fall sheath to the line marked *A* on Fig 18 3 In a narrow layer of the plasma represented by line *A*, the velocity distribution of the fast electrons is spread over a larger range of velocities This narrow region is called a *scattering layer* The velocity distribution of the fast electrons remains essentially constant throughout the region of the plasma indicated by *B* and in this region oscillations were detected by the probe Usually a second thin scattering layer, line *C*, occurred in which the velocity distribution of the fast electrons was again spread over a larger range and included electrons with energies several volts higher than any dc potential existing in the arc The distribution of the fast electrons remained essentially constant through the rest of the plasma extending to the anode The Langmuir relation for $\omega_p^2 = ne^2/m\epsilon_0$ was verified by measuring the frequency of the oscillation and the electron density of the plasma The fact that the observed frequency was the same as the frequency calculated from plasma electron density indicated that the oscillation originated in the plasma Biasing the probe to repel all but the fast electrons showed that the oscillation which was detected by the probe was carried by the fast electrons The displacement from the scattered layer of the region in which the oscillations were detected arises from the interaction of the oscillations in the plasma and the fast electrons The plasma oscillations of potential occur in the sharp scattered layers The beam of fast electrons passes through this layer and is spread over a wider range of velocities, and in the process the plasma potential oscillations modulate the mean velocity of the fast electron beam After traveling a short distance, the electron density distribution is bunched along the beam length as the electrons which are accelerated by the plasma potential oscillation catch up with the electrons previously retarded by the oscillation Thus the fast electrons are picked up by the probe in groups and produce the oscillation detected by the probe The oscillations had a definite frequency over a limited current range However, as either the arc current or the mercury vapor pressure was increased the scattering layers and oscillation regions moved closer to the cathode The entire turbulent region was always less than 8 mm from the cathode These results suggest that a very narrow region of violent scattering may occur in the immediate vicinity of the cathode when the discharge operates over a higher current or pressure range

In 1942 Sluzkin and Maydanov[4] investigated the factors that influenced the amplitude of the plasma oscillations Several anode-cathode structures were investigated but the strongest oscillations were produced in a low-pressure mercury arc by a cylindrical anode around a hot tungsten filament

The frequency of oscillation and the plasma electron density verified the Langmuir formula. The parameters studied were pressure, discharge current, anode voltage, and anode size. Large oscillations were obtained for particular values of discharge currents and anode voltages. The results indicated that, for the electrode structure used in their investigations, strong oscillations depended on a second relationship.

$$\lambda V_A^{1/2} = \text{constant} \tag{18.1}$$

where λ = free space wavelength of oscillation and V_A = anode voltage. Although these authors accepted the work of Merrill and Webb, they felt that the electron beam emerged from the cathode in bunches and that the entire plasma of their tube was oscillating in standing waves. Thus the transit time of the beam electrons across the tube, determined by $V_A^{1/2}$ must be synchronized with the period of oscillation, determined by λ.

In 1951, Wehner[5] developed a plasma oscillator capable of delivering several watts of power. The tube was divided into two parts by a grid which controlled the flow of electrons to the anode and essentially acted as a "cathode" for the anode half of the discharge. The "cathode fall" voltage, and thus, the beam energy in the arc could be increased to several hundred volts by this structure. A movable probe showed electron scattering similar to that observed by Merrill and Webb. The density of the plasma electrons was measured at the oscillating layer and the Langmuir formula verified. The transit time relationship $\lambda V_A^{1/2} = \text{constant}$ was observed to control the operation of the tube. Strong oscillations appeared in modes in which the constant of Eq. (18.1) assumed different values. Using a second carefully designed tube, Wehner concluded that the tube operated in several modes similar to the modes of a klystron.

In 1949, D. Bohm and E. Gross[6] published a theory of plasma electron oscillations based on a one-dimensional analysis of a uniform infinite plasma. In contrast to the earlier theory of Langmuir and Tonks, the effects of the random velocities of the plasma electrons and of collisional processes were considered. Moreover, the theory showed that a traveling longitudinal potential field could be excited by a beam of high-velocity electrons in the plasma.

The theory of Bohm and Gross is based on a condition of self-consistency. Thus a traveling potential wave is assumed and from the resultant electric field the motions of the individual electrons are calculated. From the knowledge of the particle motions the perturbation of the particle density can be determined. By the use of Poisson's equation this density perturbation is made self-consistent with the assumed potential and this leads to a dispersion law relating the frequency and wavelength of the

oscillation The plasma oscillations reported in previous experiments do not appear to be consistent with the theory of Bohm and Gross

The plasma electron oscillations observed in the experiments just discussed were created by a beam of electrons formed by electrons emerging from the cathode fall region of discharge The disadvantage of this technique is that the plasma itself is controlled by the density and energy of the beam electrons since these electrons maintain the plasma A more flexible arrangement[7] is illustrated in Fig 18 4 The main plasma

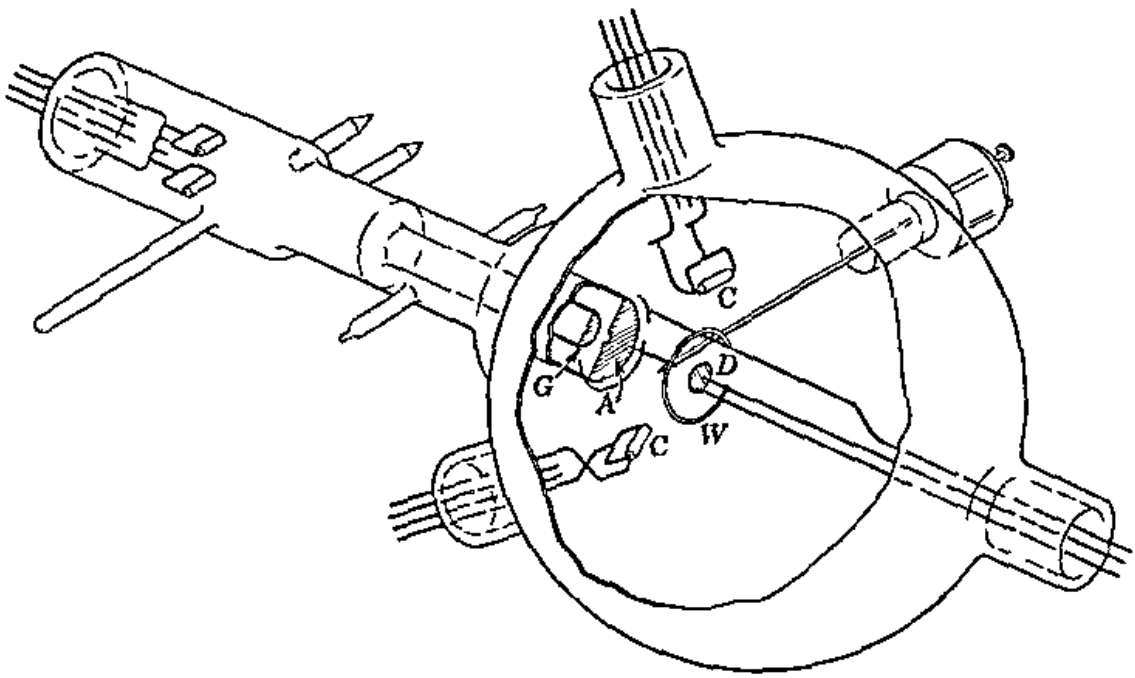


Fig 18 4 Looney and Brown's experimental tube

is produced by two oxide coated cathodes *C-C* A second discharge in the side arm acted as a source of electrons for an electron gun A beam of electrons was accelerated through several hundred volts and impinged on an anode disk *D*, which was electrically connected to the accelerating anode *A*, 1 5 cm away The thickness of the ion sheaths is controlled by the bias on the wire rings *W* A probe, attached to the tube by a stainless steel bellows, could be moved through the interaction region of the plasma and the electron beam

The behavior of the plasma oscillations was studied by varying the electron density and the thickness of the ion space charge sheath The probe picked up the oscillation whether it was in or near the electron beam

The frequency of oscillation was observed to vary in a discontinuous manner with the gun current as shown in Fig 18 5 With the electron density held constant, motion of the probe coupled to a high-frequency receiver showed that the entire plasma between *A* and *D* was oscillating

in a standing wave. As the electron density was varied, the wave pattern changed, each frequency plateau corresponding to a given wave pattern.

An explanation of an experimental oscillator has been given by Wehner.[5] The mechanism of energy transfer from the electron beam to the oscillation of the plasma electrons was described by him as a transit time process similar to that controlling a klystron. The electron beam enters the plasma through an ion sheath and its velocity is modulated at a point near the

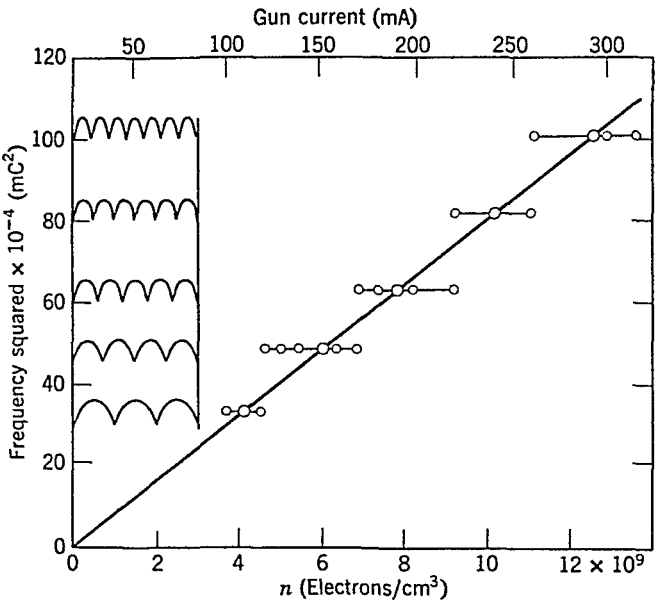


Fig. 18.5 Plasma oscillation frequencies with changes of electron density.

sheath edge. As the beam traverses the plasma, the velocity modulation of the beam causes the electrons to bunch. An electric field of the proper phase for retarding the bunched electrons, and thus extracting energy from the beam, must be assumed. Wehner pointed out that the proper phase relation between the electric field where the beam is modulated and where the energy is extracted requires that the average transit time t of a beam electron across the distance X_t between the modulation and extraction points be

$$t = \frac{X_t}{v_0} = \frac{(4k + 1)/4}{f}, \tag{18.2}$$

where $k = 0.1, 2, \ldots$; v_0 is the average velocity of the beam electrons, and f is the frequency of oscillation. If the distance X_t is assumed constant, Eq. (18.2) is the same as Eq. (18.1).

If the modulation point is taken to be at the sheath edge and the extraction point is fixed by the standing wave pattern, the extraction point is a fixed distance $X_s + X_t$ from the accelerating anode, see Fig 18 6 If the sheath thickness X_s increases, the plasma density remaining constant, X_t must decrease, and the frequency of oscillation increases to satisfy the transit time relation

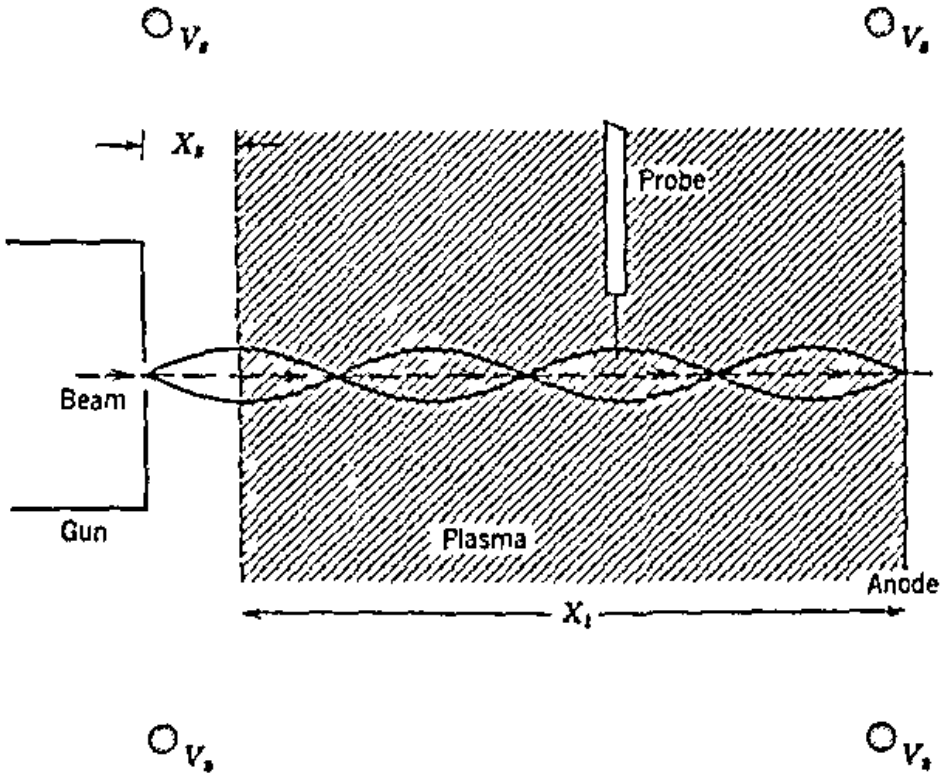


Fig 18 6 Schematic representation of longitudinal standing waves in a plasma

M J Kofoid[8] has carried out an interesting modification of the Looney-Brown experiment Standing waves of coherent longitudinal electron oscillations were excited in a plasma by two independent oppositely directed interpenetrating electron beams, with no sheaths on the electrodes from which the electron streams entered the plasma In this plasma an initial perturbation of thermal fluctuation magnitude in the field at some point may produce a velocity modulation in each beam In time this causes a bunching in space charge in each beam which creates a potential much larger than the original perturbation With a strong feedback effected by the close coupling of beams and with strong coupling of beams with the original plasma, oscillations can build up to large amplitude In these experiments the two beams, each of which produced amplification, were so closely coupled through the plasma that the system produced standing waves without any dependence on reflection at the boundaries

These experimental results substantiated the original Bohm and Gross theory that standing waves of electron oscillations could be excited in a bounded plasma by two independent streams of electrons.

Up to this point we have been discussing plane sheets of oscillating charges as illustrated by the derivation leading to the equation $\omega_p^2 = ne^2/m\epsilon_0$. Other experimental arrangements are obviously possible, and the effect in ionized cylinders has been studied in considerable detail. Consider

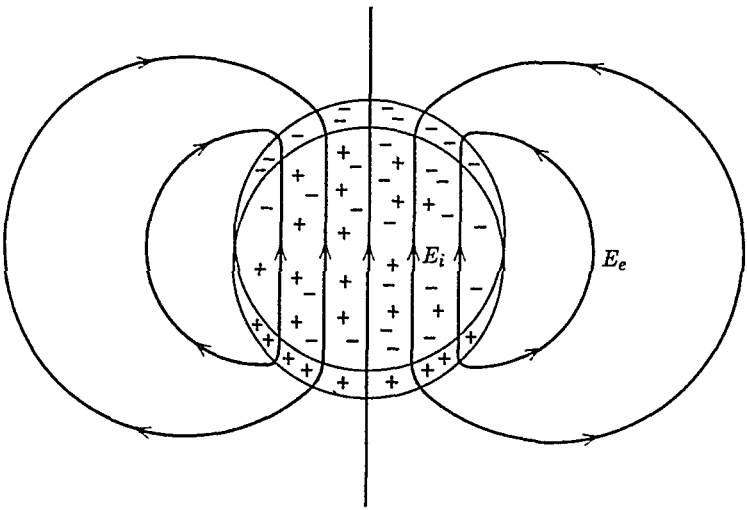


Fig. 18.7 Cylindrical plasma oscillations.

a cylinder with constant electron density n inside and with no electrons outside. If we displace the electrons by a distance x, the cylinder becomes uniformly polarized, and gives rise to an electric field E_i inside and an external field E_e outside the cylinder as illustrated in Fig. 18.7.

The external field E_e can be derived from the potential

$$V = -\frac{\cos\phi}{r} \tag{18.3}$$

and the surface charge density can be determined from the difference between the radial components of the displacement D:

$$q = n_i D_i - n_e \cdot D_e. \tag{18.4}$$

Thus

$$q = \epsilon_0\left(E_i \cos\phi + \frac{c}{r^2}\cos\phi\right). \tag{18.5}$$

Since the tangential component of the electric field is continuous at the

boundary, we have

$$E_i = \frac{c}{r^2} \tag{18 6}$$

and

$$q = 2\epsilon_0 E_i \cos\phi = en\,x\cos\phi \tag{18 7}$$

so that

$$E_i = \frac{enx}{2\epsilon_0} \tag{18 8}$$

From Newton's second law, we obtain

$$m\frac{dx}{dt} = -eE_i = -\frac{e^2 n}{2\epsilon_0}x$$

$$\omega^2 = \frac{en}{2\epsilon_0 m} = \frac{\omega_p^2}{2} \tag{18 9}$$

Langmuir and Tonks found this mode of oscillation and explained their measurements in terms of this mode and experiments have been carried out by various investigators since Romell[9] measured the scattering of a microwave signal from a plasma column when the electric vector was polarized perpendicular to the column He observed the scattering as a function of the current through the discharge and observed a resonance when the frequency of the incident field equaled $\omega_p/\sqrt{2}$

Dattner[10] carried out a similar experiment, using propagation through a waveguide at microwave frequencies In addition to a strong absorption at $\omega_p/\sqrt{2}$, he observed a number of peaks at lower densities as shown in Fig 18 8 R W Gould[11] has produced evidence to show that these lower-density oscillations resulted from a radial oscillation in the plasma cylinder, representing waves that travel in and out from the center of the plasma They are reflected at the origin, as they go in and out some field penetrates outside the plasma When the velocity of these radial waves becomes so low as to travel at particle speeds, what is called Landau damping seemed responsible for their disappearance Thus one would not expect their spectrum to extend very far toward low densities

Landau damping[12] has its origin in the strong interaction between a longitudinal plasma wave and those particles whose velocities are nearly equal to its phase velocity It comes about because these particles in general tend to pick up energy from the wave when they are initially randomly phased with respect to it We can divide the electrons into two groups, the main plasma and the resonant particles The main plasma contains all the electrons except those in a small interval about the wave velocity as shown in Fig 18 9 Since the main plasma has essentially no

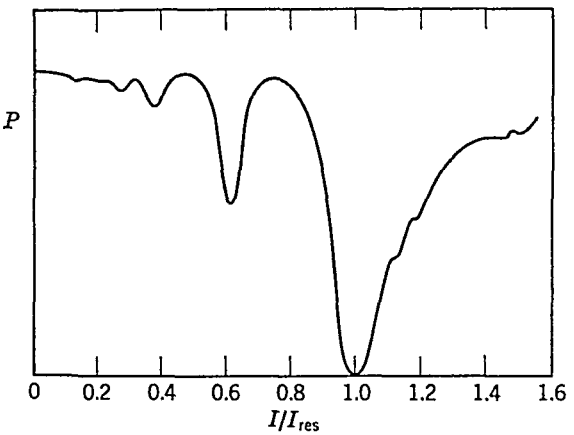


Fig. 18.8 Dattner resonances. Plot of absorbed power as a function of discharge current normalized to current at ω_p.

particles traveling at the wave velocity, there exist undamped waves of the main plasma which have this velocity. If we assume that such a wave has been set up in the main plasma, due to interaction of this wave with the resonant particles there will be an exchange of energy between the wave and these particles. Particles which are traveling faster than the wave tend to give energy to the wave (so-called "slow wave" amplification) while particles traveling slower than the wave tend to take up energy from

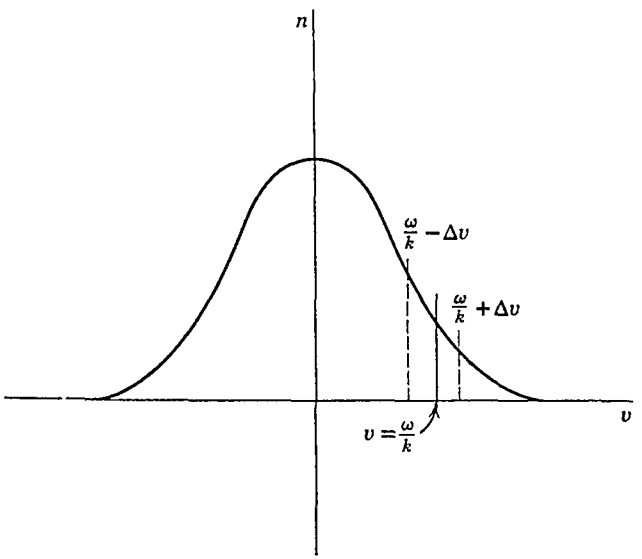


Fig. 18.9 A number density-velocity distribution to produce Landau damping.

it Thus if there are more particles moving slower than the wave than there are particles moving faster than the wave, which is the usual case illustrated in Fig 18 9, there will be a net absorption of energy and the wave will be damped

Another type of oscillation resulting from this dipole mode of oscillation of a cylindrical plasma column is shown in Fig 18 10 The lines with arrows depict the electric field lines generated by the oscillations of charge under the assumption that there is no interior volume charge The electric field generated will tend to pull the electrons back, the force being

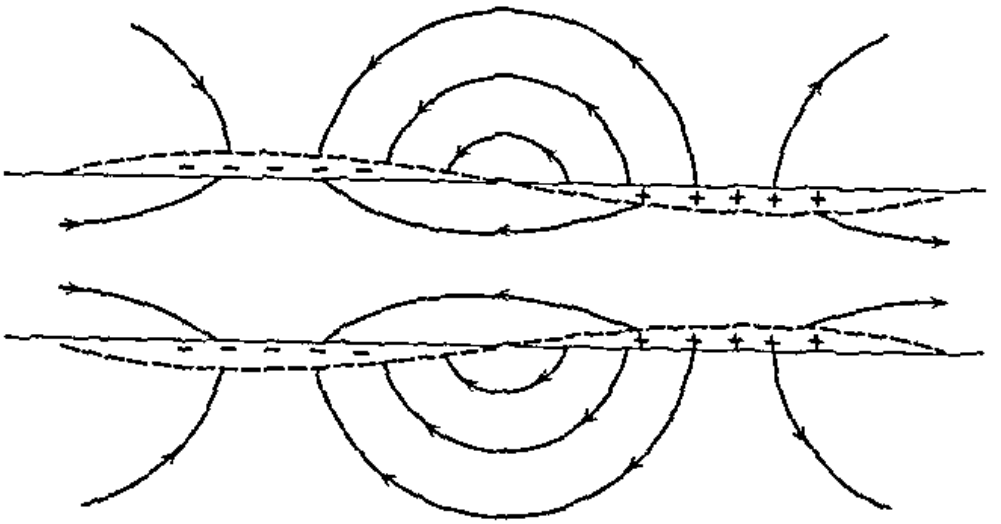


Fig 18 10 Dipole mode of cylindrical plasma oscillation

proportional to the amount of surface charge and hence one obtains simple harmonic motion

Gould verified the existence of these waves experimentally with a plasma column of a mercury arc in a glass tube and a collector acted as an antenna for the radio-frequency fields A movable probe was slid back and forth to observe a wave traveling down the tube and the amplitude and phase of this wave was measured as a function of position by comparing the signal picked up on the probe with a reference signal As the current through the discharge was varied, the plasma frequency was changed The attenuation was largest near the cut off of the mode and in the region where the attenuation was least, the amplitude dropped by a factor of e in about three wavelengths

These surface waves were slow waves, and, therefore, if an electron beam was sent along the axis of the column at the same velocity as one of these waves, a strong interaction between the electron beam and the wave was obtained Waves growing down the plasma were observed under these conditions Gould found that without the electron beam, the signal on a

cavity resonator which was in a position to pick up the wave was down in amplitude by 10 to 20 db at the output cavity. When the electron beam was turned on, the output increased sometimes in excess of the input signal.

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